



Inhoud: **Tentboek (conform NEN-EN 8020-41:2012)**

Object: **10m stretchtent en alternatieven**

Eigenaar object: **Eurostretch B.V.**

Documentcode: **26.03.00030.2**

Auteur: **S. Banga**

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A handwritten signature in black ink, appearing to read 'S. Banga'.

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Geautoriseerd door: ir. R. Houtman



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A. Inleiding

Eurostretch B.V. heeft Tentech B.V. opdracht gegeven om een statische analyse uit te voeren van een stretchtent. Eurostretch B.V. is een Nederlands bedrijf dat gespecialiseerd is in de verhuur en verkoop van tentconstructies van een rekbaar membraan, zogenaamde Stretch Tents of Bedouin tents. De tenten zijn gemaakt van rekbaar membraan, hetgeen een vrije vorm mogelijk maakt. Afhankelijk van de locatie kan gevarieerd worden met het aantal, de lengte en de plaatsing van de masten en afspanningen. Dit resulteert in een op maat gemaakte overkapping.

De vormvrijheid wordt mogelijk gemaakt door een rekbaar materiaal, aangezien de gewenste vorm uit een rechthoekig stuk materiaal wordt gerekt. Het nadeel van deze vormflexibiliteit is dat het moeilijk is om alle verschillende mogelijkheden te onderzoeken en deze te generaliseren voor een statische analyse.

Dit rapport bevat de statische analyse van een 10m stretchtent serie, bestaande uit verschillende afmetingen: **10x10, 10x15, 10x20, 10x12 en 9x12 m**. Het rapport toont de statische analyse van de representatieve afmetingen: 10x10 m, 10x15 m en 10x20 m. De statische analyse van de tent van 10x15 m is geldig voor de kleinere afmetingen van 10x12 m en 9x12 m dankzij de toepassing van het **schaalprincipe**. Alle modellen en varianten worden onderzocht in een ‘floating’ opstelling – de hoeken en zijkanten van het membraan zijn boven de grond verbonden, voornamelijk met palen.

Schaalprincipe:

Een tent met een vergelijkbare opstelling van palen en afspanningen, maar die kleiner is of een gunstigere vorm heeft, wordt automatisch goedgekeurd zodra de berekening van de minder gunstige of grotere tent is goedgekeurd, op voorwaarde dat de afstand tussen de palen en afspanningen van de geschaalde tent gelijk is aan of kleiner is dan die van de oorspronkelijke tent. Bij het bouwen van kleinere tenten met verschillende vormen is het van belang om vergelijkbare (of gunstigere) opstellingen van palen en afspanningen te gebruiken op basis van een verhouding van afstanden (om de juiste afstand tussen palen en afspanningen te garanderen).

In dit document worden de benodigde gegevens voor een tentboek volgens de NEN 8020-41:2012 gebundeld en gepresenteerd voor de 10m stretchtent serie van Eurostretch B.V.

Dit tentboek bevat:

- Eigendom gegevens;
- Tekeningen van de verschillende varianten van de tent inclusief maatvoering onderdelen aanduiding en benodigde verankering en ballast;
- Toelaatbare nuttige belasting;
- Maximaal toelaatbare windsnelheden (conform NEN-EN 1991-1-4:2005);
- Constructieve berekening (conform NEN-EN 13782:2015);
- Materiaalcertificaten (sterkte-eigenschappen en brandeigenschappen).

Utrecht, 10.08.2026, S. Banga

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C. **Normen**

- NEN-EN 1990 Eurocode, Grondslag van het constructief ontwerp
- NEN-EN 1991 Belastingen op constructies; Deel 1-4: Algemene belastingen –
Windbelasting
- NEN-EN 1995 Ontwerp en berekening van houtconstructies; Deel 1-1:
Gemeenschappelijke regels en regels voor gebouwen
- NEN-EN 13782 Tijdelijke constructies – Tenten – Veiligheid
- NEN-EN 12195-2 Banden gemaakt van synthetische vezels
- ISO 1141 Vezeltouwen – polyester
- CTS Franse norm - Chapiteaux, Tentes, Structures

D. Samenvatting

Fabrikant:	Eurostretch B.V. Akkermansbeekweg 15A 7061 ZA Terborg Nederland info@eurostretch.com	
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Dit document geldt voor vijf tentafmetingen: 10x10 m, 10x15 m, 10x20 m, 10x12 m en 9x12 m. Hoewel er geen afzonderlijke berekening wordt gemaakt voor de afmetingen 10x12 m en 9x12 m, kunnen alle elementen, afmetingen en verankeringskrachten worden gebaseerd op het 10x15 m model. Voor gebruik in Nederland zijn twee opbouwopties mogelijk – met en zonder stormbanden.

D.1. Samenvatting - opbouw met stormbanden

Stormbanden zijn altijd nodig.

Algemene informatie:				
Hoofd-afmetingen:		10x10m	10x15m / 10x12m / 9x12m	10x20m
Breedte:		10m	10m / 9m	10m
Lengte:		10m	15m / 12m	20m
Zijhoogte van het membraan:		3m	3m	3m
Max. hoogte:		4m	4m	4m
Hoofdmasten: *		4.0m Ø100 mm [Spruce,C18]	4.0m Ø100 mm [Spruce,C18]	4.0m Ø100 mm [Spruce, C18]
Zijmasten: *		3.0m Ø 70 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]
Hoekmasten: *		2.5m Ø 70 mm [Spruce, C18]	2.5m Ø 75 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
Stormbanden:		50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
Touwen:		Ø 8mm polyester, breeksterkte ≥ 900 daN **	Ø 8mm polyester, breeksterkte ≥ 900 daN **	Ø 8mm polyester, breeksterkte ≥ 900 daN **
Lange zijde – gecombineerd:		4 sneden	4 sneden	4 sneden
Lange zijde - V-touwen:		3 sneden	3 sneden	3 sneden
Korte zijde – gecombineerd:		-----	4 sneden	4 sneden
Korte zijde - V-touwen:		-----	3 sneden	3 sneden
Banden:		50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
Verbindings-middelen:		Carbine 180x14 M16 & M12 Oogbout Ratchet, BL = 3.5 kN	Carbine 180x14 M16 & M12 Oogbout Ratchet, BL = 3.5 kN	Carbine 180x14 M16 & M12 Oogbout Ratchet, BL = 3.5 kN

	Basisplaat voor zij- en hoekmast	25x25cm	25x25cm	25x25cm
	Basisplaat voor hoofdmast	35x35cm	35x35cm	35x35cm
Gebruiks-belasting:	Het is toegestaan om max. 25 kg per hoofdmast aan decoratie, geluid- of lichtinstallaties toe te passen. De belasting moet centrisc h aangebracht worden.			
Sneeuw-belasting:	Er is rekening gehouden met een sneeuwbelasting van 0.1 kN/m ² (4cm) overeenkomstig met de Franse CTS norm.			
Wind-belasting:	Conform NEN-EN 13782 De volgende stuwdrukwaarden zijn gehanteerd: h < 5m: 0.50kN/m ² Deze waarde komt overeen met windkracht 8 beaufort. Boven deze windkracht (bij aanvang windkracht 9) is de veiligheid van de constructie niet meer gewaarborgd en dient deze ontruimd te worden.			

* De gemiddelde diameter, als minimumvereiste in het midden van de paal.

** De vereiste breeksterkte (BL_{tot}) kan worden bereikt door meerdere touwsecties te gebruiken (n): BL_{rope} = BL_{tot}/n

Conform NEN-EN 13782	Bebouwd* ¹ NL cat: III	Onbebouwd* ² NL cat: II	Kust* ³ NL cat: 0
A. Beaufort (indicatief)	< 10 BFT	< 9 BFT	< 8 BFT
B. 10 min. gemiddelde windsnelheid	< 25.1 m/s	< 23.6 m/s	< 17.8 m/s
C. Piekwindsnelheid (windvlaag)	< 113 km/u	< 113 km/u	< 113 km/u

Tabel 1. Toelaatbare windsnelheden

1. Met een bebouwde omgeving wordt bedoeld: binnen de bebouwde kom met voldoende beschutting in Nederland.
2. Met een onbebouwde omgeving wordt bedoeld: buiten de bebouwde kom, of binnen de bebouwde kom met weinig beschutting in Nederland.
3. Met kust wordt bedoeld: de kortste afstand van de constructie tot aan het water bedraagt minder dan 50m en het betreffende water is 2km lang vrij van obstakels.

Veiligheid tegen verschuiven, opwaaien en omwaaien:

Ankers:	Aannames: hoek van 45 graden De volgende ontwerpweerstand van de verankeringskrachten is vereist: (Zie meer over ankertesten volgens EN 13782 onder statische analyse)			
		10x10m	10x15m / 10x12m / 9x12m	10x20m
	Afspanning lange zijde	6.05 kN (γ = 1.2)	6.24 kN (γ = 1.2)	6.92 kN (γ = 1.2)
	Afspanning korte zijde		6.28 kN (γ = 1.2)	6.70 kN (γ = 1.2)
	Afspanning hoek	4.73 kN (γ = 1.2)	5.06 kN (γ = 1.2)	5.32 kN (γ = 1.2)
	Stormbanden:	6.74 kN (γ = 1.2)	7.21 kN (γ = 1.2)	7.12 kN (γ = 1.2)
Vereiste ankers:	Bij dichte, niet-cohesieve grond (bijv. zandgrond). Wanneer ankers Ø30mm x 1000mm (effectieve lengte) worden gebruikt:			
		10x10m	10x15m / 10x12m / 9x12m	10x20m

	Afspanning lange zijde	2 ankers	2 ankers	2 ankers
	Afspanning korte zijde		2 ankers	2 ankers
	Afspanning hoek	1 anker	1 anker	1 anker
	Stormbanden:	2 ankers	2 ankers	2 ankers
Vereiste ballast:	De ballastberekening wordt uitgevoerd voor 2 verschillende situaties: - Beton op beton – er worden betonnen ballasten op het beton geplaatst - Beton op hout of rubber – de betonnen ballasten worden op hout geplaatst, of er wordt een antislipmat onder de ballast gelegd			
	10x10m			
		Beton op beton	Beton op hout of rubber	
	Afspanning lange zijde	1280 kg	1140 kg	
	Afspanning hoek	1000 kg	890 kg	
	Stormbanden:	1430 kg	1270 kg	
	10x15m / 10x12m / 9x12m			
		Beton op beton	Beton op hout of rubber	
	Afspanning lange zijde	1320 kg	1180 kg	
	Afspanning korte zijde	1330 kg	1180 kg	
	Afspanning hoek	1070 kg	960 kg	
	Stormbanden:	1530 kg	1360 kg	
	10x20m			
		Beton op beton	Beton op hout of rubber	
	Afspanning lange zijde	1470 kg	1310 kg	
	Afspanning korte zijde	1420 kg	1260 kg	
Afspanning hoek	1130 kg	1000 kg		
Stormbanden:	1510 kg	1340 kg		

D.2. Samenvatting - opbouw zonder stormbanden

Algemene informatie:				
Hoofd-afmetingen:		10x10m	10x15m / 10x12m / 9x12m	10x20m
	Breedte:	10m	10m / 9m	10m
	Lengte:	10m	15m / 12m	20m
	Zijhoogte van het membraan:	3m	3m	3m
	Max. hoogte:	4m	4m	4m
	Hoofdmasten: *	4.0m Ø100 mm [Spruce,C18]	4.0m Ø100 mm [Spruce,C18]	4.0m Ø100 mm [Spruce, C18]
	Zijmasten: *	3.0m Ø 70 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]
	Hoekmasten: *	2.5m Ø 70 mm [Spruce, C18]	2.5m Ø 75 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
	Touwen:	Ø 8mm polyester, breeksterkte ≥ 900 daN **	Ø 8mm polyester, breeksterkte ≥ 900 daN **	Ø 8mm polyester, breeksterkte ≥ 900 daN **
	Lange zijde – gecombineerd:	3 sneden	3 sneden	4 sneden
	Lange zijde - V-touwen:	2 sneden	3 sneden	3 sneden
	Korte zijde – gecombineerd:	-----	3 sneden	3 sneden
	Korte zijde - V-touwen:	-----	2 sneden	2 sneden
	Banden:	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
	Verbindings-middelen:	Carbine 180x14 M16 & M12 Oogbout Ratchet, BL = 3.5 kN	Carbine 180x14 M16 & M12 Oogbout Ratchet, BL = 3.5 kN	Carbine 180x14 M16 & M12 Oogbout Ratchet, BL = 3.5 kN
	Basisplaat voor zij- en hoekmast	25x25cm	25x25cm	25x25cm
	Basisplaat voor hoofdmast	35x35cm	35x35cm	35x35cm
Gebruiks-belasting:	Het is toegestaan om max. 25 kg per hoofdmast aan decoratie, geluid- of lichtinstallaties toe te passen. De belasting moet centrisch aangebracht worden.			
Sneeuw-belasting:	Er is rekening gehouden met een sneeuwbelasting van 0.1 kN/m ² (4cm) overeenkomstig met de Franse CTS norm.			
Wind-belasting:	Conform NEN-EN 13782 De volgende stuwdrukwaarden zijn gehanteerd: h < 5m: 0.25kN/m ²			

* De gemiddelde diameter, als minimumvereiste in het midden van de paal.

** De vereiste breeksterkte (BL_{tot}) kan worden bereikt door meerdere touwsecties te gebruiken (n): $BL_{rope} = BL_{tot}/n$

Conform NEN-EN 13782	Bebouwd* ¹ NL cat: III	Onbebouwd* ² NL cat: II	Kust* ³ NL cat: 0
A. Beaufort (indicatief)	< 8 BFT	< 7 BFT	< 6 BFT
B. 10 min. gemiddelde windsnelheid	< 17.8 m/s	< 16.7 m/s	< 12.6 m/s
C. Piekwindsnelheid (windvlaag)	< 80 km/u	< 80 km/u	< 80 km/u

Tabel 2. Toelaatbare windsnelheden

1. Met een bebouwde omgeving wordt bedoeld: binnen de bebouwde kom met voldoende beschutting in Nederland.
2. Met een onbebouwde omgeving wordt bedoeld: buiten de bebouwde kom, of binnen de bebouwde kom met weinig beschutting in Nederland.
3. Met kust wordt bedoeld: de kortste afstand van de constructie tot aan het water bedraagt minder dan 50m en het betreffende water is 2km lang vrij van obstakels.

Veiligheid tegen verschuiven, opwaaien en omwaaien:

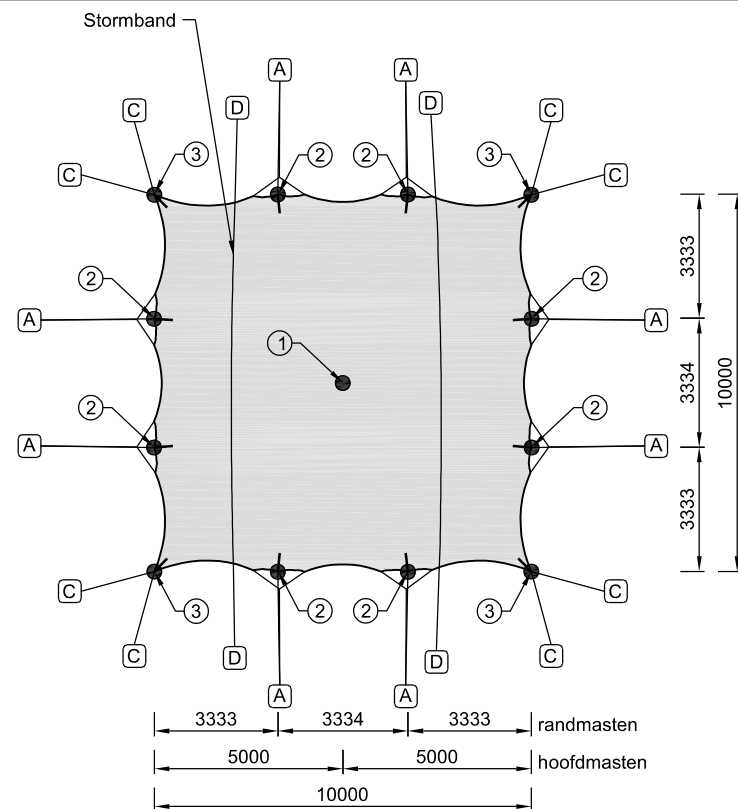
Ankers:	Aannames: hoek van 45 graden De volgende ontwerpweerstand van de verankeringskrachten is vereist: (Zie meer over ankertesten volgens EN 13782 onder statische analyse)		
		10x10m	10x15m / 10x12m / 9x12m
	Afspanning lange zijde	4.56 kN ($\gamma = 1.2$)	5.78 kN ($\gamma = 1.2$)
	Afspanning korte zijde		5.06 kN ($\gamma = 1.2$)
	Afspanning hoek	3.83 kN ($\gamma = 1.2$)	4.21 kN ($\gamma = 1.2$)
Vereiste ankers:	Bij dichte, niet-cohesieve grond (bijv. zandgrond). Wanneer ankers Ø30mm x 1000mm (effectieve lengte) worden gebruikt:		
		10x10m	10x15m / 10x12m / 9x12m
	Afspanning lange zijde	1 anker	2 ankers
	Afspanning korte zijde		1 anker
	Afspanning hoek	1 anker	1 anker
Vereiste ballast:	De ballastberekening wordt uitgevoerd voor 2 verschillende situaties:		
	- Beton op beton – er worden betonnen ballasten op het beton geplaatst - Beton op hout of rubber – de betonnen ballasten worden op hout geplaatst, of er wordt een antislipmat onder de ballast gelegd		
	10x10m		
		Beton op beton	Beton op hout of rubber
	Afspanning lange zijde	970 kg	860 kg
	Afspanning hoek	810 kg	720 kg
	10x15m / 10x12m / 9x12m		
		Beton op beton	Beton op hout of rubber

Afspanning lange zijde	1230 kg	1090 kg
Afspanning korte zijde	1070 kg	960 kg
Afspanning hoek	980 kg	790 kg
10x20m		
	Beton op beton	Beton op hout of rubber
Afspanning lange zijde	1450 kg	1290 kg
Afspanning korte zijde	950 kg	840 kg
Afspanning hoek	880 kg	780 kg

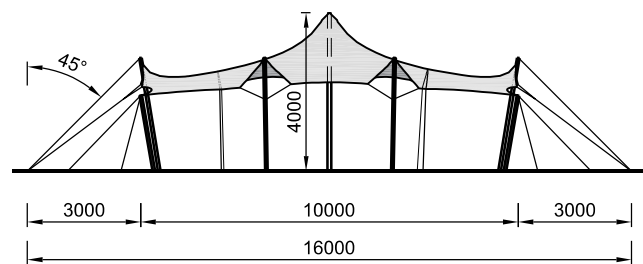
E. Tekeningen: hoofdmaatvoering en verankering

In dit hoofdstuk worden tekeningen voor de volgende afmetingen getoond:

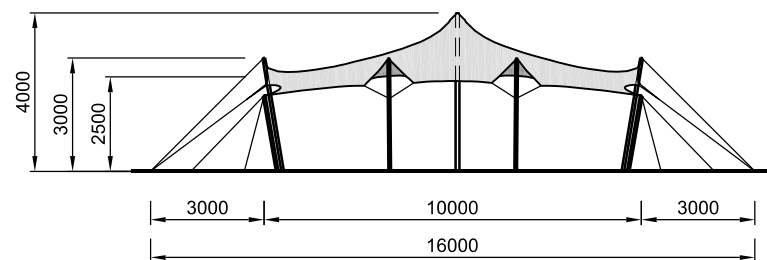
- 10x10m model
- 10x15m model
- 10x20m model
- 10x12m model (gebaseerd op 10x15m model)
- 9x12m model (gebaseerd op 10x15m model)



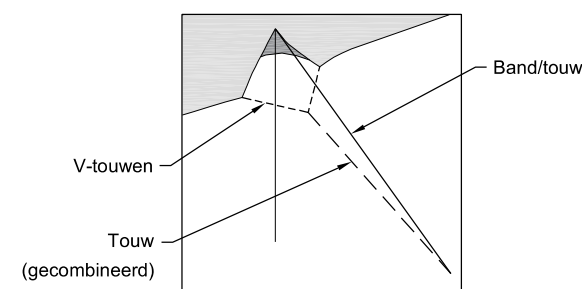
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[1:200]



VOORAANZICHT
[1:200]



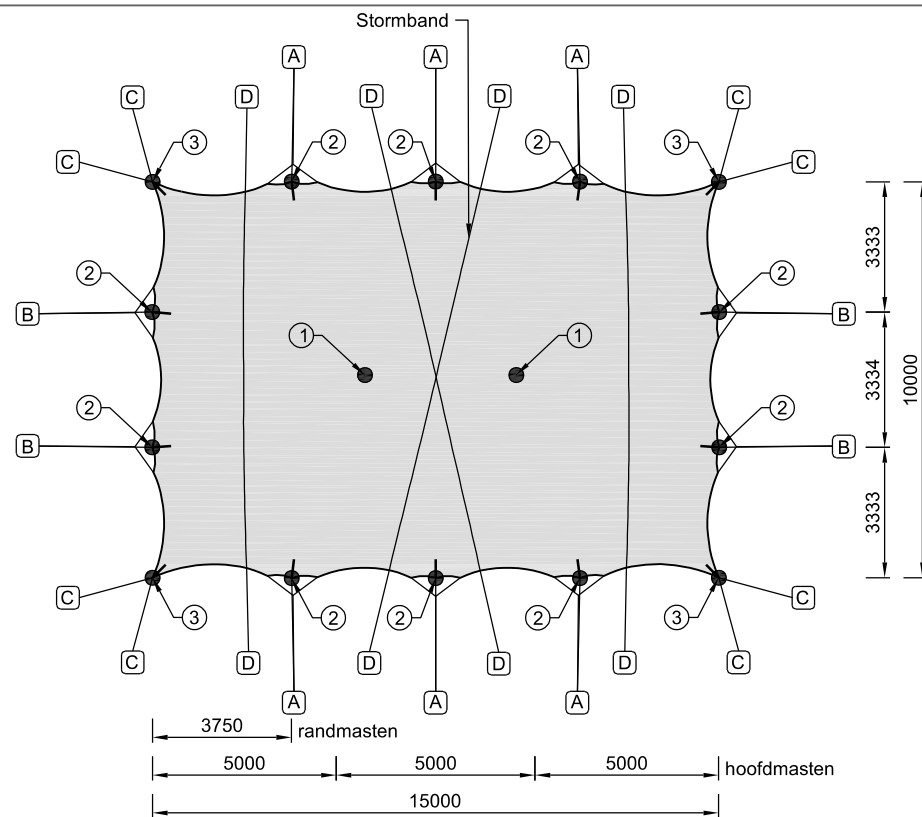
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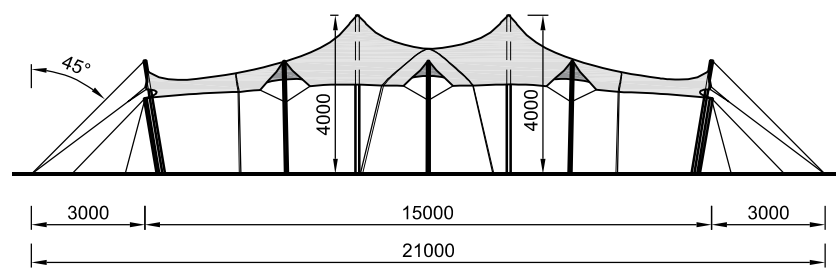
Afspanningsschema
[1:100]

Veiligheid tegen verschuiven, opwaaien en omwaaien			
Voor dichte, niet samenhangende grond.			
	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
Bij opbouw met stormbanden:			
A	Afspanning - lange zijde	2x ankers	1280 kg
	Afspanning - korte zijde	-	-
C	Afspanning - hoek	1x ankers	1000 kg
D	Stormband (Per uiteinde)	2x ankers	1430 kg
			1270 kg
	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
Bij opbouw zonder stormbanden:			
A	Afspanning - lange zijde	1x ankers	970 kg
	Afspanning - korte zijde	-	-
C	Afspanning - hoek	1x ankers	810 kg
			720 kg
Alle spanbanden/touwen en stormbelts van het membraan moeten onder een hoek van ≥ 45° aan de ankers worden bevestigd.			
Masten			
Hoofdmasten 4.0m:			
① - Ø100mm [C18, Spruce]			
Randmasten 3.0m:			
② - Ø70mm [C18, Spruce]			
Hoekmasten 2.5m:			
③ - Ø70mm [C18, Spruce]			
Afspanningen			
Stormband:	[PES] BL 3000 daN = LC 1000 daN		
Afspanning band:	[PES] BL 3000 daN = LC 1000 daN		
Afspanning touw:	Touw: Ø 8mm polyester, breeksterkte ≥ 900 daN		
	Met stormbanden	Zonder stormbanden	
Gecombineerd en v-touwen:	Touw	Touw	
n - aantal secties	n	n	
Afspanning - lange zijde (gecombineerd)	4x	3x	
Afspanning - lange zijde (v-touwen)	3x	2x	
Afspanning - korte zijde (gecombineerd)	-	-	
Afspanning - korte zijde (v-touwen)	-	-	
Banden (alternatief touw):	Band	Touw	
n - aantal secties	(aantal)	n	
Afspanning - lange zijde	1x	1x	
Afspanning - korte zijde	-	-	
Afspanning - hoek	1x	3x	
Alternatieve tentmaten			
10 x 10 m	10 x 12 m		
10 x 15 m	9 x 12 m		
10 x 20 m			
* Basis-afmetingen			

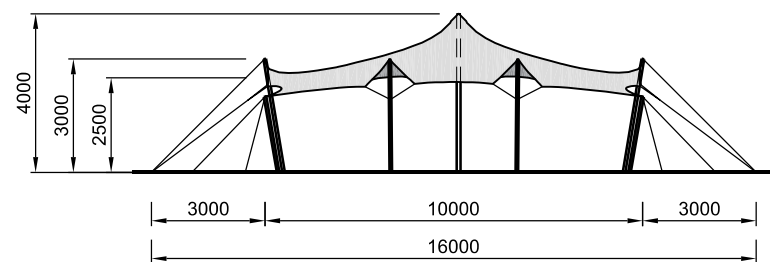
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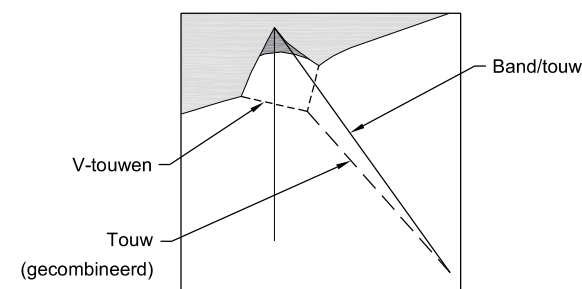
BOVENAANZICHT
[1:200]



VOORAANZICHT
[1:200]



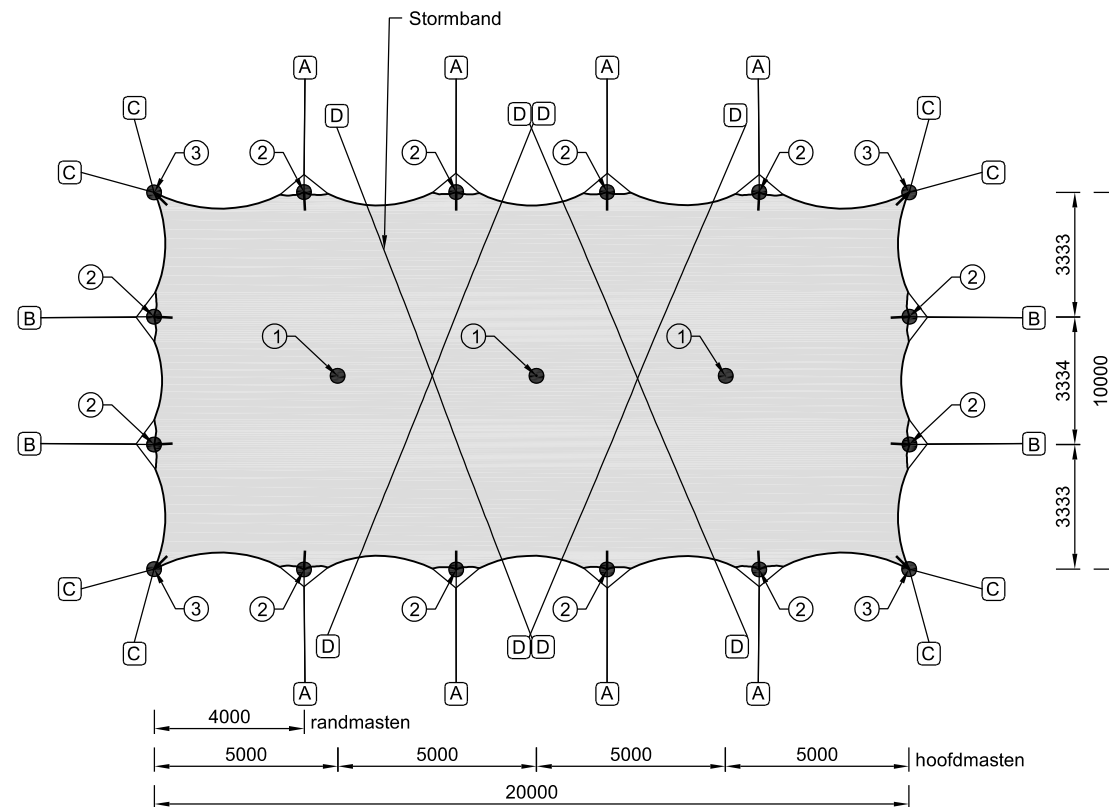
ZIJAAZICHT
[1:200]



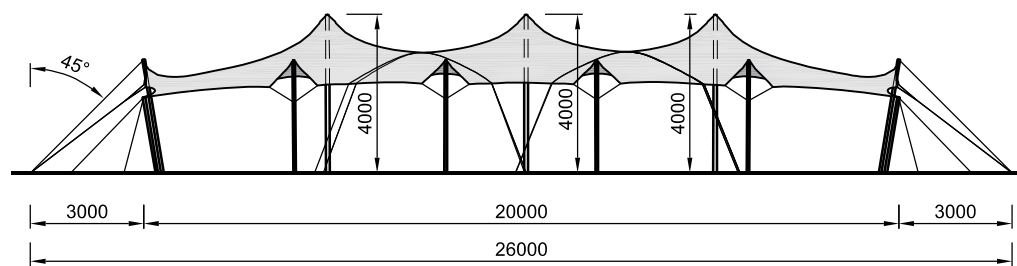
Afspanningsschema
[1:100]

Veiligheid tegen verschuiven, opwaaien en omwaaien			
Voor dichte, niet samenhangende grond.			
	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
Bij opbouw met stormbanden:			
A	Afspanning - lange zijde	2x ankers	1320 kg
B	Afspanning - korte zijde	2x ankers	1330 kg
C	Afspanning - hoek	1x ankers	1070 kg
D	Stormband (Per uiteinde)	2x ankers	1530 kg
	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
Bij opbouw zonder stormbanden:			
A	Afspanning - lange zijde	2x ankers	1230 kg
B	Afspanning - korte zijde	1x ankers	1070 kg
C	Afspanning - hoek	1x ankers	980 kg
Alle spanbanden/touwen en stormbelts van het membraan moeten onder een hoek van ≥ 45° aan de ankers worden bevestigd.			
Masten			
Hoofdmasten 4.0m:			
① - Ø100mm [C18, Spruce]			
Randmasten 3.0m:			
② - Ø75mm [C18, Spruce]			
Hoekmasten 2.5m:			
③ - Ø75mm [C18, Spruce]			
Afspanningen			
Stormband:	[PES] BL 3000 daN = LC 1000 daN		
Afspanning band:	[PES] BL 3000 daN = LC 1000 daN		
Afspanning touw:	Touw: Ø 8mm polyester, breeksterkte ≥ 900 daN		
	Met stormbanden	Zonder stormbanden	
Gecombineerd en v-touwen:	Touw	Touw	
n - aantal secties	n	n	
Afspanning - lange zijde (gecombineerd)	4x	3x	
Afspanning - lange zijde (v-touwen)	3x	3x	
Afspanning - korte zijde (gecombineerd)	4x	3x	
Afspanning - korte zijde (v-touwen)	3x	2x	
	Band	Touw	
Banden (alternatief touw):	Band	Touw	
n - aantal secties	(aantal)	n	
Afspanning - lange zijde	1x	1x	
Afspanning - korte zijde	1x	1x	
Afspanning - hoek	1x	3x	
Alternatieve tentmaten			
10 x 10 m	10 x 12 m		
10 x 15 m	9 x 12 m		
10 x 20 m			
* Basis-afmetingen			

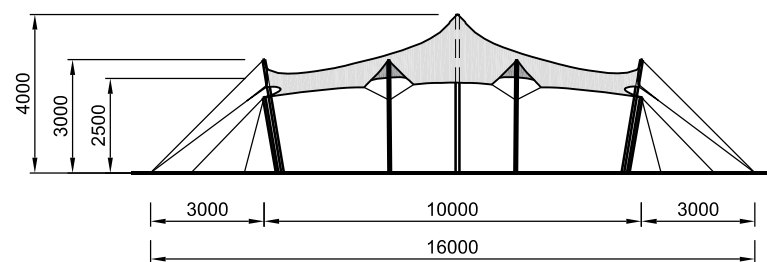
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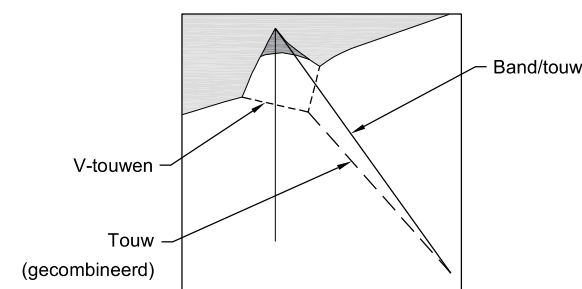
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VOORAANZICHT
[1:200]



ZIJAAZICHT
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Afspanningsschema
[1:100]

Veiligheid tegen verschuiven, opwaaien en omwaaien

Voor dichte, niet samenhangende grond.

	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
Bij opbouw met stormbanden:			
A Afspanning - lange zijde	2x ankers	1470 kg	1310 kg
B Afspanning - korte zijde	2x ankers	1420 kg	1260 kg
C Afspanning - hoek	1x ankers	1130 kg	1000 kg
D Stormband (Per uiteinde)	2x ankers	1510 kg	1340 kg

	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
Bij opbouw zonder stormbanden:			
A Afspanning - lange zijde	2x ankers	1450 kg	1290 kg
B Afspanning - korte zijde	1x ankers	950 kg	840 kg
C Afspanning - hoek	1x ankers	880 kg	780 kg

Alle spanbanden/touwen en stormbelts van het membraan moeten onder een hoek van $\geq 45^\circ$ aan de ankers worden bevestigd.

Masten

Hoofdmasten 4.0m:

① - Ø100mm [C18, Spruce]

Randmasten 3.0m:

② - Ø75mm [C18, Spruce]

Hoekmasten 2.5m:

③ - Ø70mm [C18, Spruce]

Afspanningen

Stormband: [PES] BL 3000 daN = LC 1000 daN

Afspanning band: [PES] BL 3000 daN = LC 1000 daN

Afspanning touw: Touw: Ø 8mm polyester, breeksterkte ≥ 900 daN

	Met stormbanden	Zonder stormbanden
Gecombineerd en v-touwen:		
n - aantal secties	Touw n	Touw n
Afspanning - lange zijde (gecombineerd)	4x	4x
Afspanning - lange zijde (v-touwen)	3x	3x
Afspanning - korte zijde (gecombineerd)	4x	3x
Afspanning - korte zijde (v-touwen)	3x	2x

	Band (aantal)	Touw n
Banden (alternatief touw):		
n - aantal secties	(aantal)	n
Afspanning - lange zijde	1x	1x
Afspanning - korte zijde	1x	1x
Afspanning - hoek	1x	3x

Alternatieve tentmaten

10 x 10 m	10 x 12 m
10 x 15 m	9 x 12 m
10 x 20 m	
* Basis-afmetingen	

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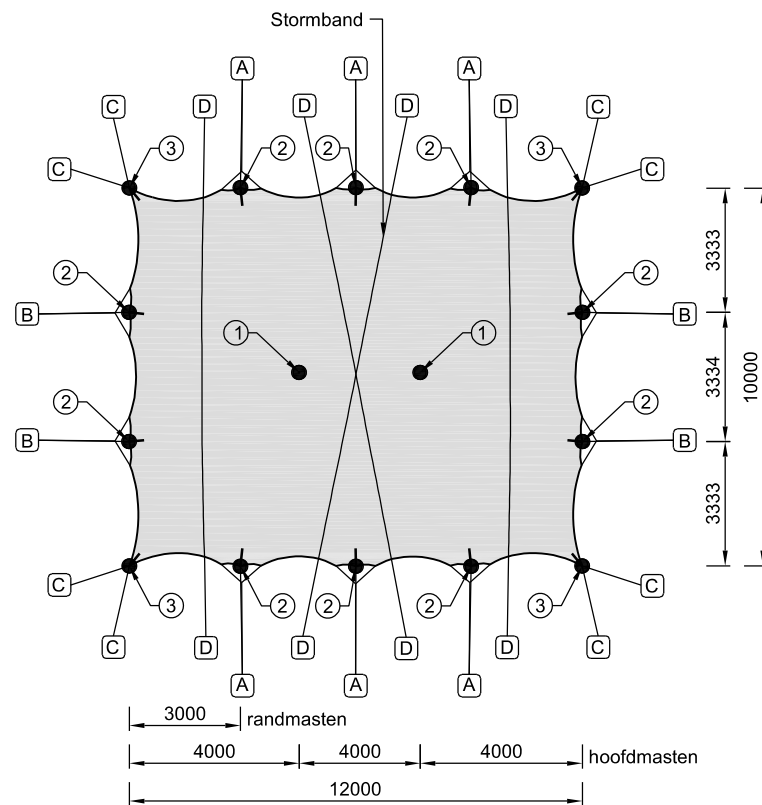
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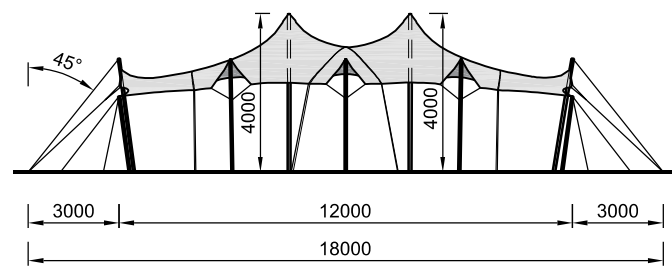
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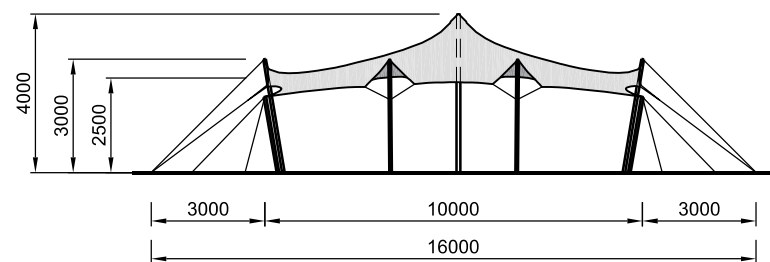
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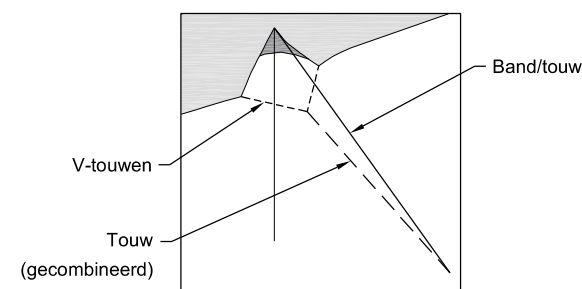
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Afspanningsschema
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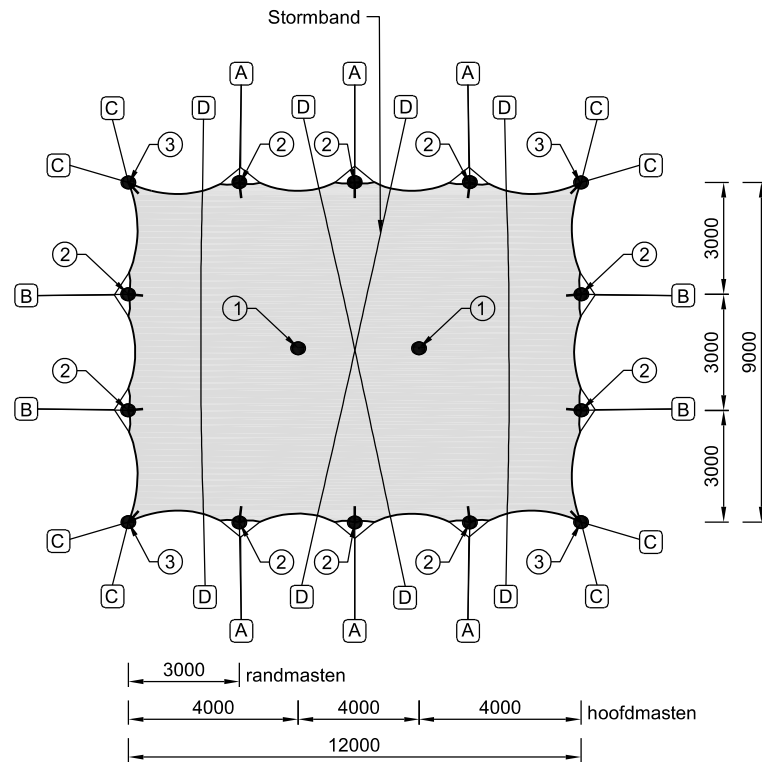
Veiligheid tegen verschuiven, opwaaien en omwaaien			
Voor dichte, niet samenhangende grond.			
	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
Bij opbouw met stormbanden:			
A	Afspanning - lange zijde	2x ankers	1320 kg
B	Afspanning - korte zijde	2x ankers	1330 kg
C	Afspanning - hoek	1x ankers	1070 kg
D	Stormband (Per uiteinde)	2x ankers	1530 kg
	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
Bij opbouw zonder stormbanden:			
A	Afspanning - lange zijde	2x ankers	1230 kg
B	Afspanning - korte zijde	1x ankers	1070 kg
C	Afspanning - hoek	1x ankers	980 kg

Masten	
Hoofdmasten 4.0m:	
①	Ø100mm [C18, Spruce]
Randmasten 3.0m:	
②	Ø75mm [C18, Spruce]
Hoekmasten 2.5m:	
③	Ø75mm [C18, Spruce]

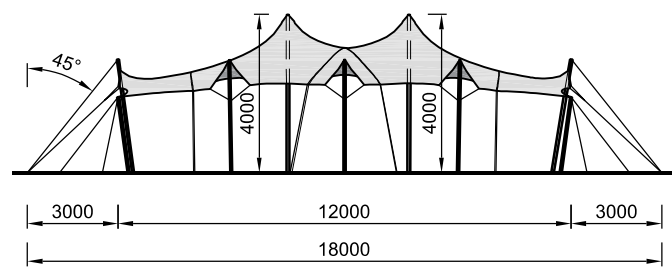
Afspanningen		
Stormband:	[PES] BL 3000 daN = LC 1000 daN	
Afspanning band:	[PES] BL 3000 daN = LC 1000 daN	
Afspanning touw:	Touw: Ø 8mm polyester, breeksterkte ≥ 900 daN	
	Met stormbanden	Zonder stormbanden
Gecombineerd en v-touwen:		
n - aantal secties	Touw n	Touw n
Afspanning - lange zijde (gecombineerd)	4x	3x
Afspanning - lange zijde (v-touwen)	3x	3x
Afspanning - korte zijde (gecombineerd)	4x	3x
Afspanning - korte zijde (v-touwen)	3x	2x
Banden (alternatief touw):		
n - aantal secties	Band (aantal)	Touw n
Afspanning - lange zijde	1x	1x
Afspanning - korte zijde	1x	1x
Afspanning - hoek	1x	3x

Alternatieve tentmaten	
10 x 10 m 10 x 15 m 10 x 20 m * Basis-afmetingen	10 x 12 m 9 x 12 m

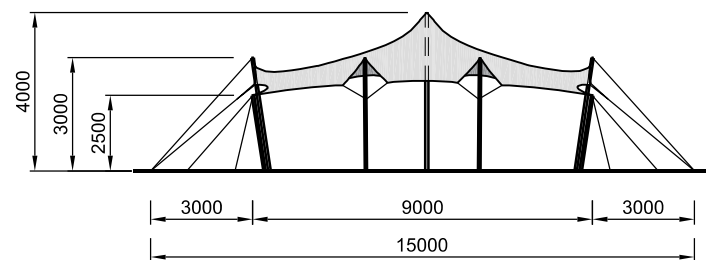
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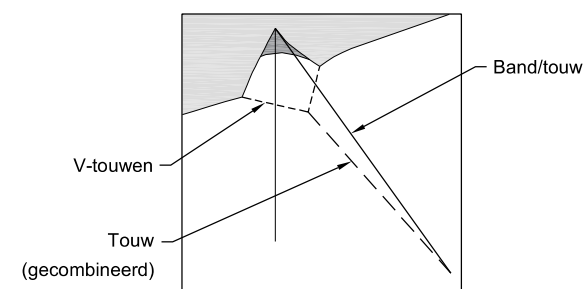
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ZIJAAZICHT
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Afspanningsschema
[1:100]

Veiligheid tegen verschuiven, opwaaien en omwaaien			
Voor dichte, niet samenhangende grond.			
	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
Bij opbouw met stormbanden:			
A	Afspanning - lange zijde	2x ankers	1320 kg
B	Afspanning - korte zijde	2x ankers	1330 kg
C	Afspanning - hoek	1x ankers	1070 kg
D	Stormband (Per uiteinde)	2x ankers	1530 kg
	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
Bij opbouw zonder stormbanden:			
A	Afspanning - lange zijde	2x ankers	1230 kg
B	Afspanning - korte zijde	1x ankers	1070 kg
C	Afspanning - hoek	1x ankers	980 kg
Alle spanbanden/touwen en stormbelts van het membraan moeten onder een hoek van ≥ 45° aan de ankers worden bevestigd.			
Masten			
Hoofdmasten 4.0m:			
① - Ø100mm [C18, Spruce]			
Randmasten 3.0m:			
② - Ø75mm [C18, Spruce]			
Hoekmasten 2.5m:			
③ - Ø75mm [C18, Spruce]			
Afspanningen			
Stormband:	[PES] BL 3000 daN = LC 1000 daN		
Afspanning band:	[PES] BL 3000 daN = LC 1000 daN		
Afspanning touw:	Touw: Ø 8mm polyester, breeksterkte ≥ 900 daN		
	Met stormbanden	Zonder stormbanden	
Gecombineerd en v-touwen:	Touw	Touw	
n - aantal secties	n	n	
Afspanning - lange zijde (gecombineerd)	4x	3x	
Afspanning - lange zijde (v-touwen)	3x	3x	
Afspanning - korte zijde (gecombineerd)	4x	3x	
Afspanning - korte zijde (v-touwen)	3x	2x	
Banden (alternatief touw):	Band	Touw	
n - aantal secties	(aantal)	n	
Afspanning - lange zijde	1x	1x	
Afspanning - korte zijde	1x	1x	
Afspanning - hoek	1x	3x	
Alternatieve tentmaten			
10 x 10 m	10 x 12 m		
10 x 15 m	9 x 12 m		
10 x 20 m			
* Basis-afmetingen			

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F. **Uitgangspunten en randvoorwaarden**

Dit document geldt voor de gebouwde constructie wanneer de volgende uitgangspunten en randvoorwaarden zijn aangehouden:

- De gebruikte materialen, onderdelen en doorsneden (membraan, masten, afspanningen, stormbanden, verankering) zijn in overeenstemming met dit document.
- De afmetingen van de opgebouwde constructie komen overeen met de maten in dit document.
- Er mogen geen onderdelen (masten, afspanningen, ankers) verwijderd worden.
- **Waterophopingen dient onmiddellijk te worden verwijderd.**
- Obstakels dienen op ten minste 0.5m afstand van het membraan geplaatst te worden (gemeten loodrecht op het doek).
- Bij sterke wind (zie samenvatting, onderdeel windbelasting) wordt de tent gesloten voor publieke toegang.
- Er mag maximaal 25 kg per mast aan decoratie, geluid- of lichtinstallaties bevestigd worden aan de hoofdmasten. De belasting dient centrisch aangebracht te worden.
- Alle masten moeten droog worden opgeslagen. Als ze nat worden, moet er een mogelijkheid zijn om ze te drogen.
- De verankering is gebaseerd op dichte, niet-samenhangende grond. Wanneer op andere ondergrond wordt gebouwd, moet mogelijk aanvullende verankering worden voorzien of ankertesten worden uitgevoerd.
- Er wordt uitgegaan van een standaardbelasting van 0,1 kN/m² volgens EN 13782, wat overeenkomt met de vereiste sneeuwbelasting (4 cm) volgens de Franse CTS.

G. Toelaatbare windsnelheden

G.1. Toelaatbare windsnelheden

Wind kan op verschillende manieren uitgedrukt worden:

- 10 minuten gemiddelde windsnelheid – een gemiddelde windsnelheid over 10 minuten gemeten op 10m hoogte in een open terrein (NEN-EN 1991-1-4 NB NL terreincategorie II).
- Piekwindsnelheid – een kortstondige maximale windstoot met een bepaalde snelheid, afhankelijk van de hoogte. Vaak gegeven in km/u.

De stuwdruk waarop een tent berekend is, is bepalend voor de sterkte van de tent. Het gaat er dus om dat op de juiste manier wordt vastgesteld welke windsnelheid moet worden aangehouden om te kunnen bepalen of de stuwdruk overschreden wordt.

G.1.1 Opbouw met stormbanden

In de constructieve berekening is een stuwdruk van **0.50 kN/m² tot 5m** hoogte aangehouden, deze stuwdruk is omgerekend naar windsnelheden. Boven onderstaande windsnelheden dient de structuur ontruimd en gesloten te worden. Boven de gegeven windsnelheid is de constructie niet meer gewaarborgd betreffende sterkte en/of stabiliteit.

Conform NEN-EN 13782	Bebouwd* ¹ NL cat: III	Onbebouwd* ² NL cat: II	Kust* ³ NL cat: 0
A. Beaufort (indicatief)	< 10 BFT	< 9 BFT	< 8 BFT
B. 10 min. gemiddelde windsnelheid	< 25.1 m/s	< 23.6 m/s	< 17.8 m/s
C. Piekwindsnelheid (windvlaag)	< 113 km/u	< 113 km/u	< 113 km/u

Tabel 3. Toelaatbare windsnelheden

1. Met een bebouwde omgeving wordt bedoeld: binnen de bebouwde kom met voldoende beschutting in Nederland.

2. Met een onbebouwde omgeving wordt bedoeld: buiten de bebouwde kom, of binnen de bebouwde kom met weinig beschutting in Nederland.

3. Met kust wordt bedoeld: de kortste afstand van de constructie tot aan het water bedraagt minder dan 50m en het betreffende water is 2km lang vrij van obstakels.

Wanneer de hierboven genoemde windsnelheden bereikt worden dient de structuur ontruimd en voor publiek gesloten te worden.

Bovenstaande waarden (A, B en C) kunnen op verschillende manieren gemeten worden en kunnen alle drie onafhankelijk van elkaar gebruikt worden:

- Dit is een indicatieve Beaufort schaal welke hoort bij de 10 minuten gemiddelde windsnelheid, deze waarde moet afkomstig zijn van het dichtstbijzijnde meteostation.
- 10 min gemiddelde windsnelheid op 10 meter hoogte in een open terrein, deze waarde moet afkomstig zijn van het dichtstbijzijnde meteostation.
- Windvlaagsnelheid op 10 meter hoogte in een open terrein, deze waarde moet afkomstig zijn van het dichtstbijzijnde meteostation.

G.1.2 Opbouw zonder stormbanden

In de constructieve berekening is een stuwdruk van **0.25 kN/m² tot 5m** hoogte aangehouden, deze stuwdruk is omgerekend naar windsnelheden. Boven onderstaande windsnelheden dient de structuur ontruimd en gesloten te worden. Boven de gegeven windsnelheid is de constructie niet meer gewaarborgd betreffende sterkte en/of stabiliteit.

Conform NEN-EN 13782	Bebouwd* ¹ NL cat: III	Onbebouwd* ² NL cat: II	Kust* ³ NL cat: 0
A. Beaufort (indicatief)	< 8 BFT	< 7 BFT	< 6 BFT
B. 10 min. gemiddelde windsnelheid	< 17.8 m/s	< 16.7 m/s	< 12.6 m/s
C. Piekwindsnelheid (windvlaag)	< 80 km/u	< 80 km/u	< 80 km/u

Tabel 4. Toelaatbare windsnelheden

1. Met een bebouwde omgeving wordt bedoeld: binnen de bebouwde kom met voldoende beschutting in Nederland.
2. Met een onbebouwde omgeving wordt bedoeld: buiten de bebouwde kom, of binnen de bebouwde kom met weinig beschutting in Nederland.
3. Met kust wordt bedoeld: de kortste afstand van de constructie tot aan het water bedraagt minder dan 50m en het betreffende water is 2km lang vrij van obstakels.

Wanneer de hierboven genoemde windsnelheden bereikt worden dient de structuur ontruimd en voor publiek gesloten te worden.

Bovenstaande waardes (A, B en C) kunnen op verschillende manieren gemeten worden en kunnen alle drie onafhankelijk van elkaar gebruikt worden:

- D. Dit is een indicatieve Beaufort schaal welke hoort bij de 10 minuten gemiddelde windsnelheid, deze waarde moet afkomstig zijn van het dichtstbijzijnde meteostation.
- E. 10 min gemiddelde windsnelheid op 10 meter hoogte in een open terrein, deze waarde moet afkomstig zijn van het dichtstbijzijnde meteostation.
Windvlaagsnelheid op 10 meter hoogte in een open terrein, deze waarde moet afkomstig zijn van het dichtstbijzijnde meteostation.

G.2. Windsnelheid berekening

G.2.1 Opbouw met stormbanden

De maximale stuwdruk wordt middels de NEN-EN 1991-1-4 teruggerekend naar een basiswindsnelheid voor het kustgebied, een bebouwde omgeving en een onbebouwde omgeving. (uitgaande van terreinruwheid volgens de Nederlandse NB).

Piekwindsnelheid op 10m hoogte

$$605 \times \frac{1}{2} \times \rho \times v^2 = \frac{1}{2} \times 1.25 \times v^2 \rightarrow v = 31.1 \text{ m/s} \rightarrow 113 \text{ km/u}$$

Verg. 4.10 NEN-EN 1991-1-4
Basisstuwdruk

Windsnelheid kust gebied op 10m hoogte

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.005}{0.05}\right)^{0.07} = 0.16$$

Verg. 4.5 NEN-EN 1991-1-4
Terreinfactor

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.16 \times \ln\left(\frac{4}{0.005}\right) = 1.08$$

Verg. 4.4 NEN-EN 1991-1-4
Ruwheidsfactor

$$V_m = C_r \times V_b = 1.08 \times V_b$$

Verg. 4.3 NEN-EN 1991-1-4
Gemiddelde windsnelheid op hoogte

$$\sigma_v = K_r \times V_b = 0.16 \times V_b$$

Verg. 4.6 NEN-EN 1991-1-4
Standaardafwijking van de turbulentie

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.16 \times V_b}{1.08 \times V_b} = 0.15$$

Verg. 4.7 NEN-EN 1991-1-4
Turbulentie intensiteit

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 1.5 \times V_b^2$$

Verg. 4.8 NEN-EN 1991-1-4
Extreme stuwdruk

$$500 = 1.5 \times V_b^2 \rightarrow V_b = 18.3 \text{ m/s}$$

Karakteristieke windsnelheid

Windsnelheid onbebouwd gebied op 10m hoogte

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.2}{0.05}\right)^{0.07} = 0.21$$

Verg. 4.5 NEN-EN 1991-1-4
Terreinfactor

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.21 \times \ln\left(\frac{4}{0.2}\right) = 0.63$$

Verg. 4.4 NEN-EN 1991-1-4
Ruwheidsfactor

$$V_m = C_r \times V_b = 0.63 \times V_b$$

Verg. 4.3 NEN-EN 1991-1-4
Gemiddelde windsnelheid op hoogte

$$\sigma_v = K_r \times V_b = 0.21 \times V_b$$

Verg. 4.6 NEN-EN 1991-1-4
Standaardafwijking van de turbulentie

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.21 \times V_b}{0.63 \times V_b} = 0.33$$

Verg. 4.7 NEN-EN 1991-1-4
Turbulentie intensiteit

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 0.82 \times V_b^2$$

Verg. 4.8 NEN-EN 1991-1-4
Extreme stuwdruk

$$500 = 0.82 \times V_b^2 \rightarrow V_b = 24.7 \text{ m/s}$$

Karakteristieke windsnelheid

Windsnelheid bebouwd gebied op 10m hoogte

$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.5}{0.05}\right)^{0.07} = 0.22$	Verg. 4.5 NEN-EN 1991-1-4 Terreinfactor
$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.22 \times \ln\left(\frac{7}{0.5}\right) = 0.59$	Verg. 4.4 NEN-EN 1991-1-4 Ruwheidsfactor
$V_m = C_r \times V_b = 0.59 \times V_b$	Verg. 4.3 NEN-EN 1991-1-4 Gemiddelde windsnelheid op hoogte
$\sigma_v = K_r \times V_b = 0.22 \times V_b$	Verg. 4.6 NEN-EN 1991-1-4 Standaardafwijking van de turbulentie
$L_v = \frac{\sigma_v}{V_m} = \frac{0.22 \times V_b}{0.59 \times V_b} = 0.38$	Verg. 4.7 NEN-EN 1991-1-4 Turbulentie intensiteit
$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 0.79 \times V_b^2$	Verg. 4.8 NEN-EN 1991-1-4 Extreme stuwdruk
$500 = 0.79 \times V_b^2 \rightarrow V_b = 25.1 \text{ m/s}$	Karakteristieke windsnelheid

G.2.2 Opbouw zonder stormbanden

De maximale stuwdruk wordt middels de NEN-EN 1991-1-4 teruggerekend naar een basiswindsnelheid voor het kustgebied, een bebouwde omgeving en een onbebouwde omgeving. (uitgaande van terreinruwheid volgens de Nederlandse NB).

Piekwindsnelheid op 10m hoogte

$303 \times \frac{1}{2} \times \rho \times v^2 = \frac{1}{2} \times 1.25 \times v^2 \rightarrow v = 22.0 \text{ m/s} \rightarrow 80 \text{ km/u}$	Verg. 4.10 NEN-EN 1991-1-4 Basisstuwdruk
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Windsnelheid kust gebied op 10m hoogte

$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.005}{0.05}\right)^{0.07} = 0.16$	Verg. 4.5 NEN-EN 1991-1-4 Terreinfactor
$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.16 \times \ln\left(\frac{4}{0.005}\right) = 1.08$	Verg. 4.4 NEN-EN 1991-1-4 Ruwheidsfactor
$V_m = C_r \times V_b = 1.08 \times V_b$	Verg. 4.3 NEN-EN 1991-1-4 Gemiddelde windsnelheid op hoogte
$\sigma_v = K_r \times V_b = 0.16 \times V_b$	Verg. 4.6 NEN-EN 1991-1-4 Standaardafwijking van de turbulentie
$L_v = \frac{\sigma_v}{V_m} = \frac{0.16 \times V_b}{1.08 \times V_b} = 0.15$	Verg. 4.7 NEN-EN 1991-1-4 Turbulentie intensiteit
$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 1.5 \times V_b^2$	Verg. 4.8 NEN-EN 1991-1-4 Extreme stuwdruk
$250 = 1.5 \times V_b^2 \rightarrow V_b = 12.62 \text{ m/s}$	Karakteristieke windsnelheid

Windsnelheid onbebouwd gebied op 10m hoogte

$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.2}{0.05}\right)^{0.07} = 0.21$	Verg. 4.5 NEN-EN 1991-1-4 Terreinfactor
--	--

$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.21 \times \ln\left(\frac{4}{0.2}\right) = 0.63$	Verg. 4.4 NEN-EN 1991-1-4 Ruwheidsfactor
$V_m = C_r \times V_b = 0.63 \times V_b$	Verg. 4.3 NEN-EN 1991-1-4 Gemiddelde windsnelheid op hoogte
$\sigma_v = K_r \times V_b = 0.21 \times V_b$	Verg. 4.6 NEN-EN 1991-1-4 Standaardafwijking van de turbulentie
$L_v = \frac{\sigma_v}{V_m} = \frac{0.21 \times V_b}{0.63 \times V_b} = 0.33$	Verg. 4.7 NEN-EN 1991-1-4 Turbulentie intensiteit
$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 0.82 \times V_b^2$	Verg. 4.8 NEN-EN 1991-1-4 Extreme stuwdruk
$250 = 0.82 \times V_b^2 \rightarrow V_b = 16.66 \text{ m/s}$	Karakteristieke windsnelheid

Windsnelheid bebouwd gebied op 10m hoogte

$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.5}{0.05}\right)^{0.07} = 0.22$	Verg. 4.5 NEN-EN 1991-1-4 Terreinfactor
$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.22 \times \ln\left(\frac{7}{0.5}\right) = 0.59$	Verg. 4.4 NEN-EN 1991-1-4 Ruwheidsfactor
$V_m = C_r \times V_b = 0.59 \times V_b$	Verg. 4.3 NEN-EN 1991-1-4 Gemiddelde windsnelheid op hoogte
$\sigma_v = K_r \times V_b = 0.22 \times V_b$	Verg. 4.6 NEN-EN 1991-1-4 Standaardafwijking van de turbulentie
$L_v = \frac{\sigma_v}{V_m} = \frac{0.22 \times V_b}{0.59 \times V_b} = 0.38$	Verg. 4.7 NEN-EN 1991-1-4 Turbulentie intensiteit
$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 0.79 \times V_b^2$	Verg. 4.8 NEN-EN 1991-1-4 Extreme stuwdruk
$250 = 0.79 \times V_b^2 \rightarrow V_b = 17.76 \text{ m/s}$	Karakteristieke windsnelheid



Content: **Tentbook (according to EN 13782)**

Object: **10m stretchtent and alternatives**

Owner object: **Eurostretch B.V.**

Document code: **26.03.00007.2**

Author: **S. Banga**

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Date: 10.08.2026

Valid date: 10.08.2031

A handwritten signature in black ink, appearing to read 'S. Banga'.

Author: S. Banga

A handwritten signature in black ink, appearing to read 'R. Houtman'.

Authorized by: ir. R. Houtman

A. Introduction

Eurostretch B.V. has commissioned Tentech B.V. to perform a static analysis of a stretch tent. Eurostretch B.V. is a Dutch company that specializes in rental and selling of tent structures made of a stretchable membrane, so called Stretch Tents or Bedouin tents. The stretchable membrane enables a freedom of form as there is not a pre-described shape necessary. Depending on the location, the number of poles, length of the poles, and the placement of the poles, number and type of ties can be varied. This results in a custom canopy at each location.

The freedom of form is enabled by a stretchable fabric, as the desired form is stretched from a rectangular fabric. The drawback of this flexibility in shape is the difficulty of investigating all the different possibilities and generalizing them for a static analysis.

This report presents the static analysis of a 10m stretch tent series which consists of different sizes – **10x10, 10x15, 10x20, 10x12 and 9x12m**. The report shows the static analysis of the decisive sizes – 10x10m, 10x15m and 10x20m. The static analysis of tent 10x15m is valid for smaller sizes of 10x12m and 9x12m due to the application of the **scaling principle**. All models and variants are investigated in a “floating” set-up – membrane corners and sides are connected above ground, primarily to poles.

Scaling principle: A tent with a similar arrangement of poles and ties but smaller in size or having a more favourable shape is automatically verified when the calculation of the less favourable or bigger tent has been approved, provided that the spacing between the poles and ties of the scaled tent is similar or smaller than the original tent. While constructing smaller tent sizes of different shapes, it is significant to use similar (or more favourable) arrangements of poles and ties based on a ratio of distances (to ensure correct pole and tie spacing).

This document contains the data required for a tent book, according to EN 13782, for the 10m tent series of Eurostretch B.V., including:

- Ownership data;
- Drawings of the different variants of the tent, including dimensions, indications of elements and required anchoring/ballast;
- Permitted live load;
- Maximum wind speeds (according to EN 1991-1-4:2005);
- Structural analysis (according to EN 13782:2015);
- Material certificates (strength properties and fire properties).

Utrecht, S. Banga, 10.08.2026

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C. Codes and standards

- EN 1990 Eurocode, Basis of structural Design
- EN 1991 Eurocode 1, Part 1-4: General actions - wind actions.
- EN 1995 Eurocode 5, Design of wooden structures
- EN 13782 Temporary Structures – Tents - Safety
- EN 12195-2 Belts made from synthetic fibres
- ISO 1141 Fibre ropes – Polyester
- CTS French standard - Chapiteaux, Tentes, Structures

D. Summary

Manufacturer:	Eurostretch B.V. Sporstraat 25 7061 ZA Terborg Nederland info@eurostretch.com	
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This document is valid for five tent sizes: 10x10m, 10x15m, 10x20m, 10x12m and 9x12m. Although no separate calculation is made for 10x12m and 9x12m sizes, all elements, sizes and anchoring forces can be based on 10x15m model.

General information:				
Main dimensions:		10x10m	10x15m / 10x12m / 9x12m	10x20m
	Width:	10m	10m / 9m	10m
	Length:	10m	15m / 12m	20m
	Side height of membrane:	3m	3m	3m
	Max. height:	4m	4m	4m
	Main poles: *	4.0m Ø100 mm [Spruce,C18]	4.0m Ø100 mm [Spruce,C18]	4.0m Ø100 mm [Spruce, C18]
	Side poles: *	3.0m Ø 70 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]
	Corner poles: *	2.5m Ø 70 mm [Spruce, C18]	2.5m Ø 75 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
	Stormbelts:	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
	Guy ropes:	Ø 8mm polyester, breaking strength ≥ 900 daN **	Ø 8mm polyester, breaking strength ≥ 900 daN **	Ø 8mm polyester, breaking strength ≥ 900 daN **
	Long side – combined:	4 sections	4 sections	4 sections
	Long side - V-ropes:	3 sections	3 sections	3 sections
	Short side – combined:	-----	4 sections	4 sections
	Short side - V-ropes:	-----	3 sections	3 sections
	Belts:	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
	Fasteners:	Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN	Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN	Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN
	Side & corner pole base plate	25x25cm	25x25cm	25x25cm
	Main pole base plate	35x35cm	35x35cm	35x35cm

User load:	A maximum of 25 kg per main mast may be used for decorations, sound or lighting installations. The load must be applied centrally.
Snow load:	A snow load of 0.1 kN/m ² (4 cm) has been considered in accordance with the French CTS standard.
Wind load:	According to EN 13782 The following pressure values have been used: h < 5m: 0.50kN/m ² This value corresponds to wind force 8 on the Beaufort scale. Above this wind force (starting at wind force 9), the safety of the structure can no longer be guaranteed, and it must be evacuated.

* The average diameter, as a minimum required at the middle of the pole.

** The required breaking load (BL_{tot}) may be achieved by using multiple rope sections (n): BL_{rope} = BL_{tot} / n, see page 57

Wind speeds for default European terrain categories (not Country specific)

Out of service ⁽¹⁾					
	Coast	Flattened, open area	Rural area	Village	City
10 min. average wind speed ⁽²⁾	> 17.94 m/s > 64.57 km/h	> 18.89 m/s > 67.99 km/h	> 21.08 m/s > 75.88 km/h	> 24.99 m/s > 89.97 km/h	> 26.08 m/s > 93.89 km/h
Beaufort ⁽³⁾	> 7 BFT	> 7 BFT	> 8 BFT	> 9 BFT	> 9 BFT
Peak wind speed ⁽⁴⁾	> 113 km/h	> 113 km/h	> 113 km/h	> 113 km/h	> 113 km/h

(1) 'Out of service' means: above the given wind speed the structure is no longer guaranteed regarding strength and/or stability.

(2) 10min average wind speed at 10m height measured at the nearest weather stations.

(3) wind data in Beaufort (BFT) are indicative values.

(4) 3 second peak wind speed measured on site at 10m height.

Safety against sliding, overturning and uplifting:

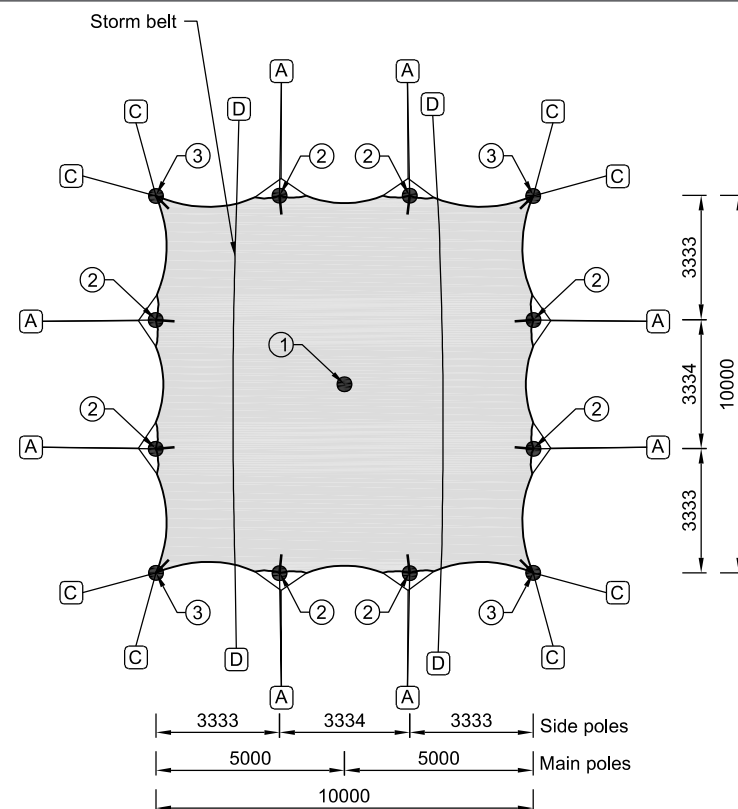
Anchor forces:	Assumptions: angle of 45 degrees The following design resistance of the anchor forces is required: (See H.8.3, page 67 for Anchor tests according to EN 13782)			
		10x10m	10x15m / 10x12m / 9x12m	10x20m
	Long side guy ropes:	6.05 kN (γ = 1.2)	6.24 kN (γ = 1.2)	6.92 kN (γ = 1.2)
	Short side guy ropes:		6.28 kN (γ = 1.2)	6.70 kN (γ = 1.2)
	Corner ties:	4.73 kN (γ = 1.2)	5.06 kN (γ = 1.2)	5.32 kN (γ = 1.2)
Required anchors:	Stormbelt:	6.74 kN (γ = 1.2)	7.21 kN (γ = 1.2)	7.12 kN (γ = 1.2)
	Based on dense, non-cohesive soil (e.g. sandy soils). In the case anchors Ø30mm x 1000mm (effective length) are being used:			
		10x10m	10x15m / 10x12m / 9x12m	10x20m

	Long side guy ropes:	2 anchors	2 anchors	2 anchors
	Short side guy ropes:		2 anchors	2 anchors
	Corner ties:	1 anchor	1 anchor	1 anchor
	Stormbelt:	2 anchors	2 anchors	2 anchors
Required ballast:	Ballast calculation is performed for 2 different situations:			
	- Concrete on concrete – concrete ballasts are placed on concrete - Concrete on wood or rubber – concrete ballasts are placed on wood, or an anti-slip mat is placed underneath the ballast			
	10x10m			
		Concrete on concrete	Concrete on wood or rubber	
	Long side guy ropes:	1280 kg	1140 kg	
	Corner ties:	1000 kg	890 kg	
	Stormbelt:	1430 kg	1270 kg	
	10x15m / 10x12m / 9x12m			
		Concrete on concrete	Concrete on wood or rubber	
	Long side guy ropes:	1320 kg	1180 kg	
	Short side guy ropes:	1330 kg	1180 kg	
	Corner ties:	1070 kg	960 kg	
	Stormbelt:	1530 kg	1360 kg	
	10x20m			
		Concrete on concrete	Concrete on wood or rubber	
Long side guy ropes:	1470 kg	1310 kg		
Short side guy ropes:	1420 kg	1260 kg		
Corner ties:	1130 kg	1000 kg		
Stormbelt:	1510 kg	1340 kg		

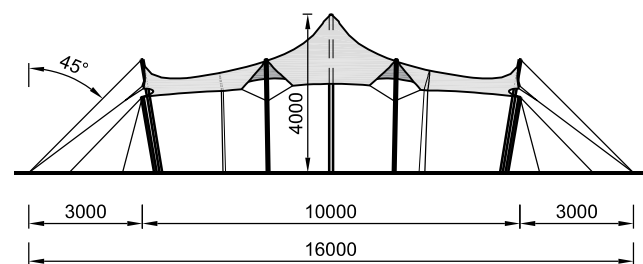
E. Drawings: main dimensions and anchoring

Drawings for the following sizes are presented in this chapter:

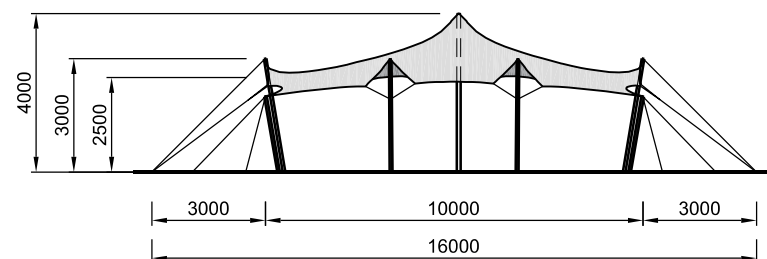
- 10x10m model
- 10x15m model
- 10x20m model
- 10x12m model (based on 10x15m model)
- 9x12m model (based on 10x15m model)



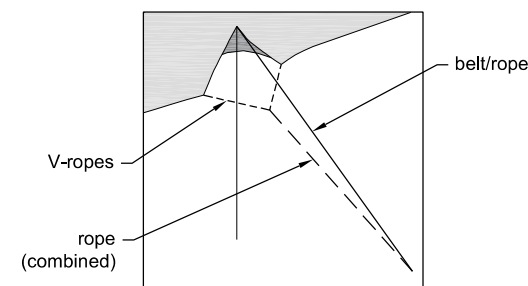
TOP VIEW
[1:200]



FRONT VIEW
[1:200]



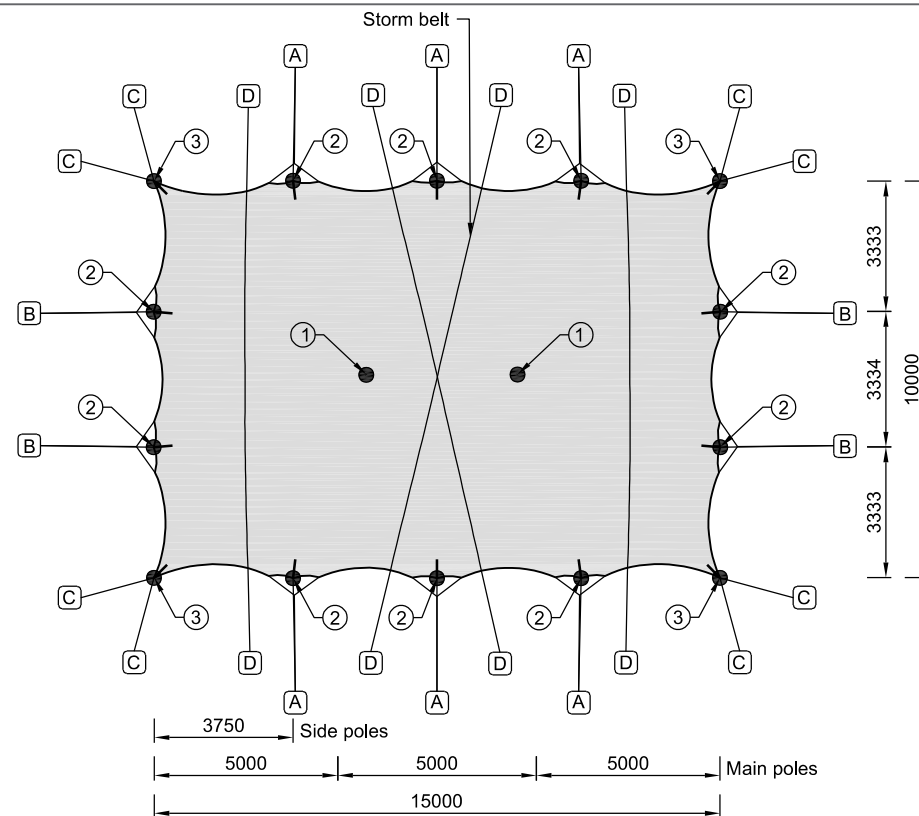
SIDE VIEW
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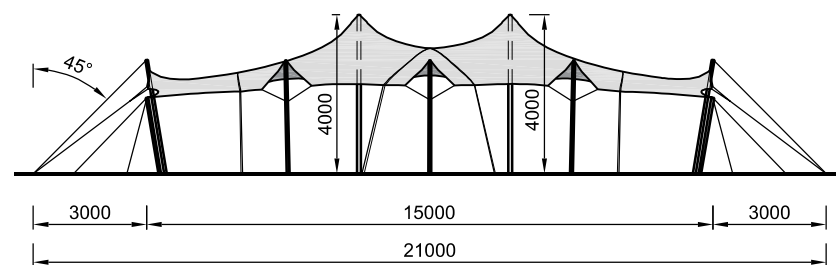
GUYING DIAGRAM
[1:100]

Safety against sliding, overturning and uplifting			
Based on dense, non-cohesive soil (e.g. sandy soils).			
	<u>Anchors</u> (Ø30x1000mm)	<u>Ballast</u> (concrete on concrete)	<u>Ballast</u> (concrete on wood or rubber)
(A)	Guying - long side	2x anchors	1280 kg
(B)	Guying - short side	-	-
(C)	Guying - corner	1x anchors	1000 kg
(D)	Storm belt (<i>per end</i>)	2x anchors	1430 kg
All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors.			
Poles			
Main poles 4.0m:			
① - Ø100mm [C18, Spruce]			
Side poles 3.0m:			
② - Ø70mm [C18, Spruce]			
Corner poles 2.5m:			
③ - Ø70mm [C18, Spruce]			
Guying			
Storm belt:	[PES] BL 3000 daN = LC 1000 daN		
Guying belt:	[PES] BL 3000 daN = LC 1000 daN		
Guying rope:	rope: Ø 8mm polyester, breaking strength ≥ 900 daN		
<u>Combined & v-ropes:</u>		<u>Rope</u>	
n - number of sections		n	
Guying - long side (combined)		4x	
Guying - long side (v-ropes)		3x	
Guying - short side (combined)		-	
Guying - short side (v-ropes)		-	
<u>Belts (alternative rope):</u>		<u>Belt</u>	<u>Rope</u>
n - number of sections		(number)	n
Guying - long side		1x	1x
Guying - short side		1x	1x
Guying - corner		1x	3x
Alternative configuration			
10 x 10 m	10 x 12 m		
10 x 15 m	9 x 12 m		
10 x 20 m			
* Basic configurations			

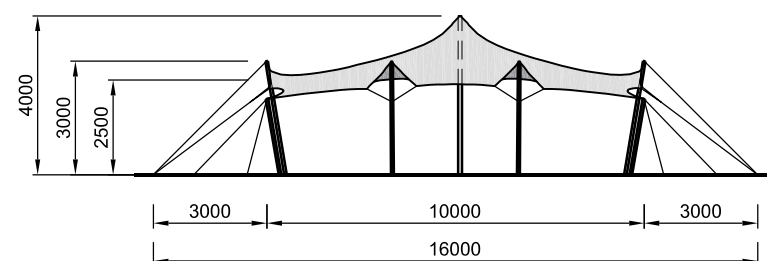
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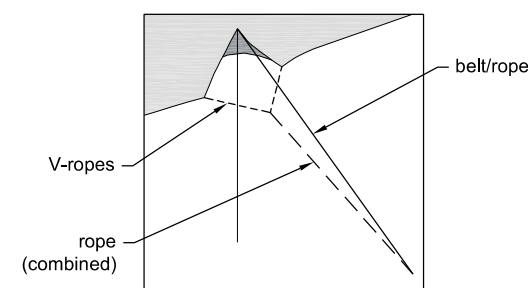
TOP VIEW
[1:200]



FRONT VIEW
[1:200]



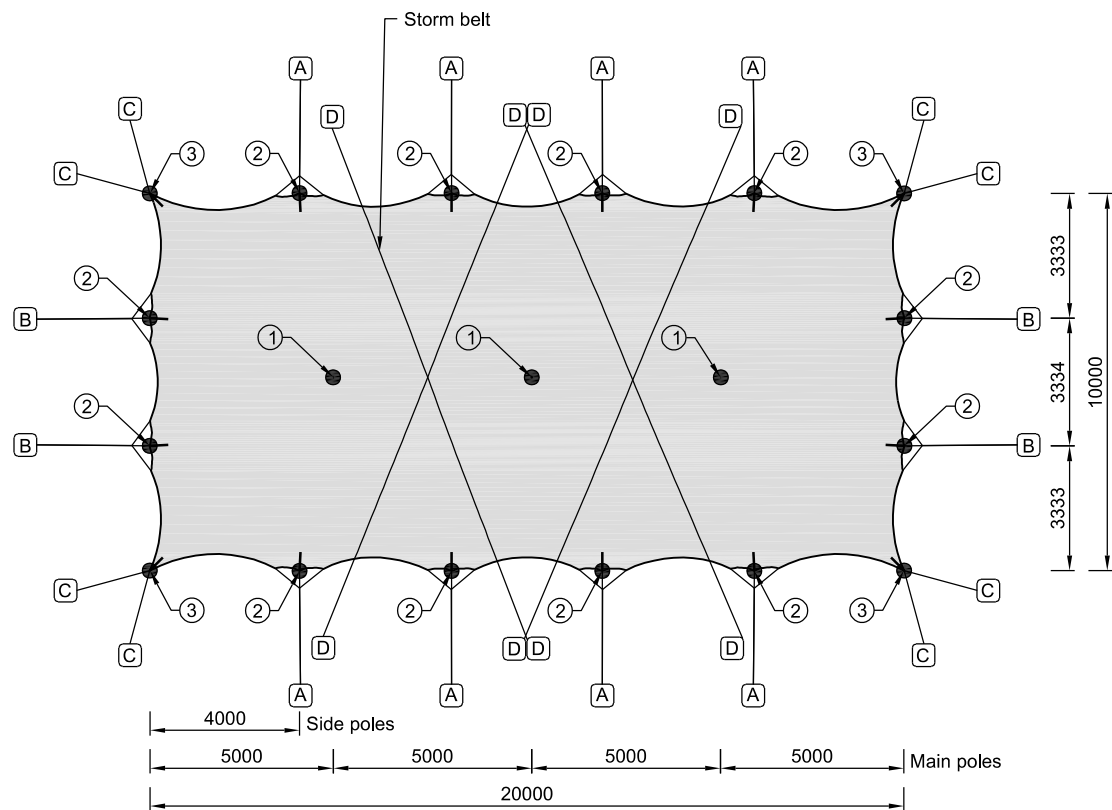
SIDE VIEW
[1:200]



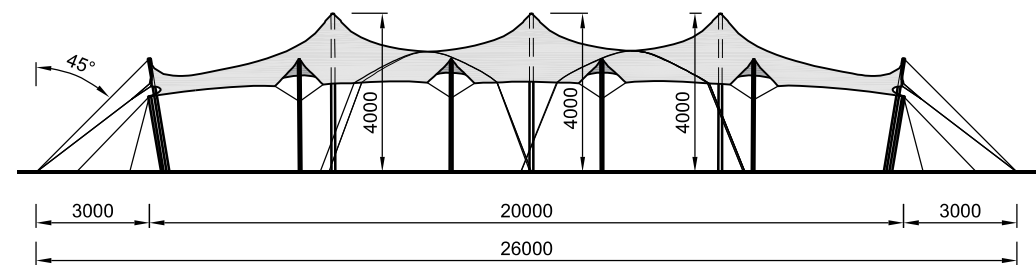
GUYING DIAGRAM
[1:100]

Safety against sliding, overturning and uplifting			
Based on dense, non-cohesive soil (e.g. sandy soils).			
	<u>Anchors</u> (Ø30x1000mm)	<u>Ballast</u> (concrete on concrete)	<u>Ballast</u> (concrete on wood or rubber)
(A)	Guying - long side	2x anchors	1320 kg
(B)	Guying - short side	2x anchors	1330 kg
(C)	Guying - corner	1x anchors	1070 kg
(D)	Storm belt (<i>per end</i>)	2x anchors	1530 kg
All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors.			
Poles			
Main poles 4.0m:			
① - Ø100mm [C18, Spruce]			
Side poles 3.0m:			
② - Ø75mm [C18, Spruce]			
Corner poles 2.5m:			
③ - Ø75mm [C18, Spruce]			
Guying			
Storm belt:		[PES] BL 3000 daN = LC 1000 daN	
Guying belt:		[PES] BL 3000 daN = LC 1000 daN	
Guying rope:		rope: Ø 8mm polyester, breaking strength ≥ 900 daN	
<u>Combined & v-ropes:</u>		<u>Rope</u>	
n - number of sections		n	
Guying - long side (combined)		4x	
Guying - long side (v-ropes)		3x	
Guying - short side (combined)		4x	
Guying - short side (v-ropes)		3x	
<u>Belts (alternative rope):</u>		<u>Belt</u>	<u>Rope</u>
n - number of sections		(number)	n
Guying - long side		1x	1x
Guying - short side		1x	1x
Guying - corner		1x	3x
Alternative configuration			
10 x 10 m	10 x 12 m		
10 x 15 m	9 x 12 m		
10 x 20 m			
* Basic configurations			

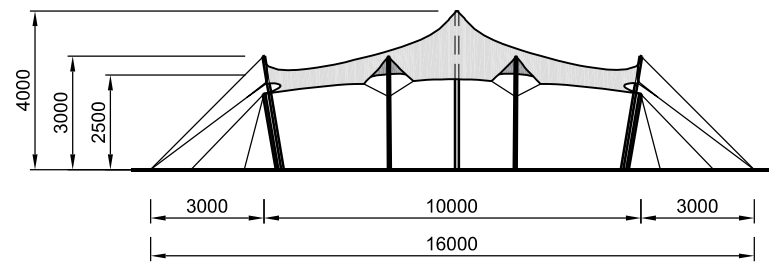
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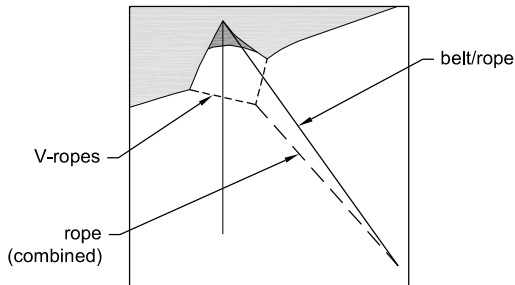
TOP VIEW
[1:200]



FRONT VIEW
[1:200]



SIDE VIEW
[1:200]



GUYING DIAGRAM
[1:100]

Safety against sliding, overturning and uplifting

Based on dense, non-cohesive soil (e.g. sandy soils).

	Anchors (Ø30x1000mm)	Ballast (concrete on concrete)	Ballast (concrete on wood or rubber)
A Guying - long side	2x anchors	1470 kg	1310 kg
B Guying - short side	2x anchors	1420 kg	1260 kg
C Guying - corner	1x anchors	1130 kg	1000 kg
D Storm belt (per end)	2x anchors	1510 kg	1340 kg

All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors.

Poles

Main poles 4.0m:
① - Ø100mm [C18, Spruce]
Side poles 3.0m:
② - Ø75mm [C18, Spruce]
Corner poles 2.5m:
③ - Ø70mm [C18, Spruce]

Guying

Storm belt: [PES] BL 3000 daN = LC 1000 daN
Guying belt: [PES] BL 3000 daN = LC 1000 daN
Guying rope: rope: Ø 8mm polyester, breaking strength ≥ 900 daN

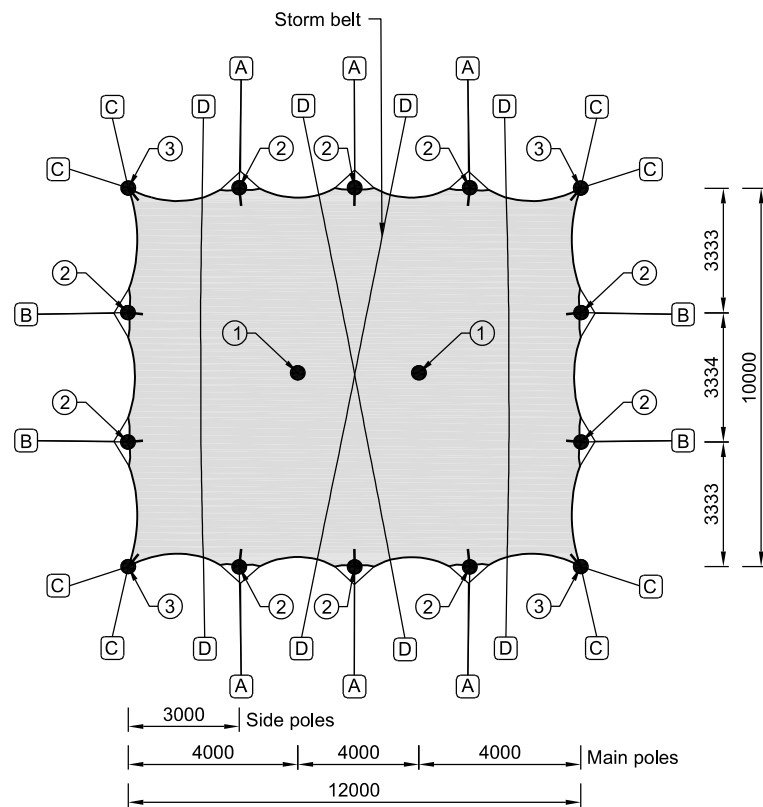
Combined & v-ropes:	Rope
n - number of sections	n
Guying - long side (combined)	4x
Guying - long side (v-ropes)	3x
Guying - short side (combined)	4x
Guying - short side (v-ropes)	3x

Belts (alternative rope):	Belt (number)	Rope
n - number of sections	n	n
Guying - long side	1x	1x
Guying - short side	1x	1x
Guying - corner	1x	3x

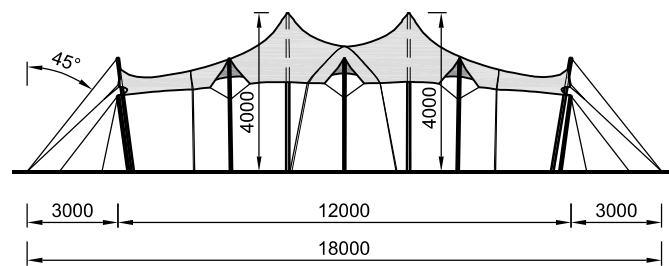
Alternative configuration

10 x 10 m	10 x 12 m
10 x 15 m	9 x 12 m
10 x 20 m	
* Basic configurations	

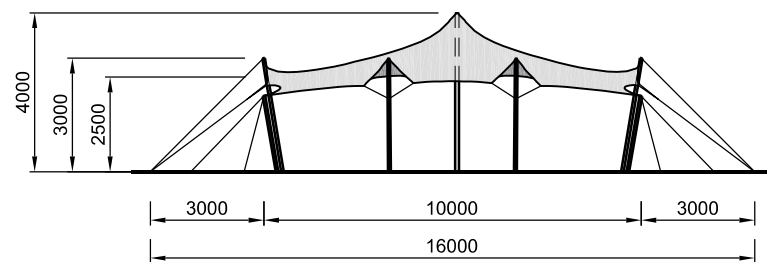
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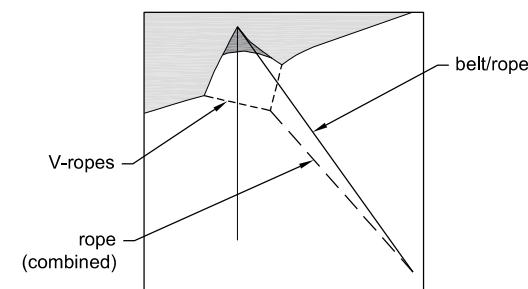
TOP VIEW
[1:200]



FRONT VIEW
[1:200]



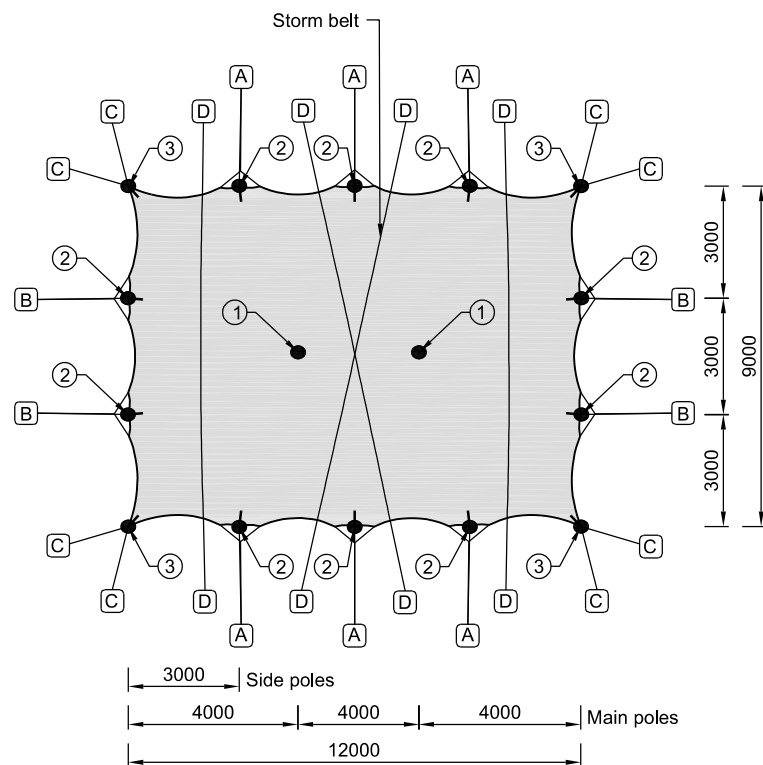
SIDE VIEW
[1:200]



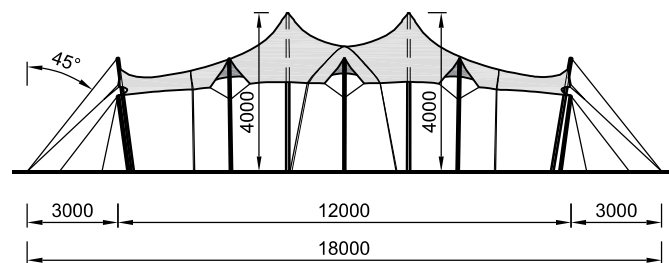
GUYING DIAGRAM
[1:100]

Safety against sliding, overturning and uplifting			
Based on dense, non-cohesive soil (e.g. sandy soils).			
	Anchors (Ø30x1000mm)	Ballast (concrete on concrete)	Ballast (concrete on wood or rubber)
A	Guying - long side	2x anchors	1320 kg
B	Guying - short side	2x anchors	1330 kg
C	Guying - corner	1x anchors	1070 kg
D	Storm belt (per end)	2x anchors	1530 kg
All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors.			
Poles			
Main poles 4.0m:			
① - Ø100mm [C18, Spruce]			
Side poles 3.0m:			
② - Ø75mm [C18, Spruce]			
Corner poles 2.5m:			
③ - Ø75mm [C18, Spruce]			
Guying			
Storm belt:	[PES] BL 3000 daN = LC 1000 daN		
Guying belt:	[PES] BL 3000 daN = LC 1000 daN		
Guying rope:	rope: Ø 8mm polyester, breaking strength ≥ 900 daN		
Combined & v-ropes:		Rope	
n - number of sections		n	
Guying - long side (combined)		4x	
Guying - long side (v-ropes)		3x	
Guying - short side (combined)		4x	
Guying - short side (v-ropes)		3x	
Belts (alternative rope):		Belt (number)	Rope n
n - number of sections			
Guying - long side		1x	1x
Guying - short side		1x	1x
Guying - corner		1x	3x
Alternative configuration			
10 x 10 m	10 x 12 m		
10 x 15 m	9 x 12 m		
10 x 20 m			
* Basic configurations			

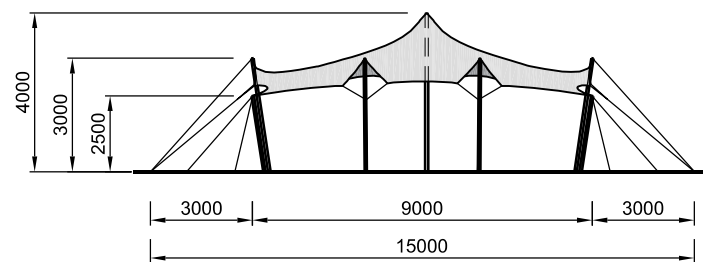
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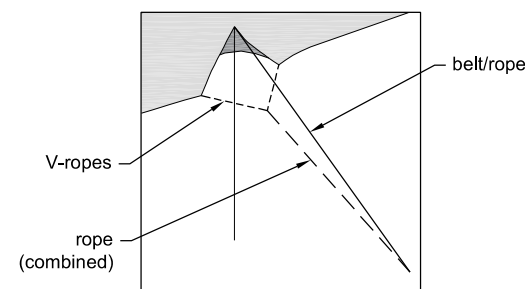
TOP VIEW
[1:200]



FRONT VIEW
[1:200]



SIDE VIEW
[1:200]



GUYING DIAGRAM
[1:100]

Safety against sliding, overturning and uplifting			
Based on dense, non-cohesive soil (e.g. sandy soils).			
	<u>Anchors</u> (Ø30x1000mm)	<u>Ballast</u> (concrete on concrete)	<u>Ballast</u> (concrete on wood or rubber)
A	Guying - long side	2x anchors	1320 kg
B	Guying - short side	2x anchors	1180 kg
C	Guying - corner	1x anchors	960 kg
D	Storm belt (<i>per end</i>)	2x anchors	1360 kg
All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors.			
Poles			
Main poles 4.0m:			
① - Ø100mm [C18, Spruce]			
Side poles 3.0m:			
② - Ø75mm [C18, Spruce]			
Corner poles 2.5m:			
③ - Ø75mm [C18, Spruce]			
Guying			
Storm belt:	[PES] BL 3000 daN = LC 1000 daN		
Guying belt:	[PES] BL 3000 daN = LC 1000 daN		
Guying rope:	rope: Ø 8mm polyester, breaking strength ≥ 900 daN		
<u>Combined & v-ropes:</u>		<u>Rope</u>	
n - number of sections		n	
Guying - long side (combined)		4x	
Guying - long side (v-ropes)		3x	
Guying - short side (combined)		4x	
Guying - short side (v-ropes)		3x	
<u>Belts (alternative rope):</u>		<u>Belt</u>	<u>Rope</u>
n - number of sections		(number)	n
Guying - long side		1x	1x
Guying - short side		1x	1x
Guying - corner		1x	3x
Alternative configuration			
10 x 10 m		10 x 12 m	
10 x 15 m		9 x 12 m	
10 x 20 m			
* Basic configurations			

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F. Important terms and conditions

This document applies to the built structure if the following principles and conditions are met:

- The used materials, parts and sections (membrane, poles, ties, anchoring) are in accordance with this document.
- The dimensions of the built structure match the dimensions stated in this document.
- Parts (poles, ties, anchors) may not be removed.
- Obstacles should be placed at least 0.5m from the membrane (measured perpendicular to the fabric); The fabric needs a certain freedom to deform in all directions to prevent damages caused by collision with objects located closely to the fabric (see also EN 13782, article 8.7).
- Above the maximum allowable wind speeds (see D) the structure should be evacuated and access for the public must be denied.
- Only decorations, music- and light installations of less than 25 kg per pole, can be attached to the structure. They must be attached centrally.
- A conventional load of 0.1 kN/m^2 is considered according to EN 13782, which corresponds with the required snow load (4cm) according to the French CTS.
- Anchoring is based on dense, non-cohesive soil. When soil differs, additional anchoring might be necessary, or anchor tests need to be performed.
- Poles must be stored dry. If they become wet, there needs to be a possibility for them to dry.
- All water accumulation must be removed as soon as possible.

G. Allowed wind speeds

The maximum wind speed is converted into a basic wind speed for a coastal area, flattened/open area, rural area, village and city according to EN 1991-1-4. Terrain roughness is taken according to the recommended general values for the different terrain categories for Europe. (not country specific) Illustrations of these terrain categories are presented on page 18.

Full wind load		
Wind pressure p_w	500 N/m ²	At 5m height
Corresponding peak wind pressure $p_{w,peak}$	605 N/m ²	At 10m height

Peak wind speed at 10m height

Equation:

$$605 = \frac{1}{2} \times \rho \times v^2 = \frac{1}{2} \times 1.25 \times v^2 \rightarrow v = 31.1 \text{ m/s} \rightarrow \pm 113 \text{ km/h}$$

Eq. 4.10 EN 1991-1-4
Basic wind pressure

Wind speed coastal area at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.003}{0.05}\right)^{0.07} = 0.156$$

Eq. 4.5 EN 1991-1-4
Terrain factor for coastal area

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.156 \times \ln\left(\frac{4.0}{0.003}\right) = 1.123$$

Eq. 4.4 EN 1991-1-4
Roughness factor
 $Z = 4.0 > Z_{min} = 1$

$$V_m = C_r \times V_b = 1.123 \times V_b$$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

$$\sigma_v = K_r \times V_b = 0.141 \times V_b$$

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.156 \times V_b}{1.158 \times V_b} = 0.139$$

Eq. 4.7 EN 1991-1-4
Turbulence intensity

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 1.554 \times V_b^2$$

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Equation:

$$500 = 1.554 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 17.94 \text{ m/s}$$

Characteristic wind speed

Wind speed flattened, open area at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.01}{0.05}\right)^{0.07} = 0.170$$

Eq. 4.5 EN 1991-1-4
Terrain factor for coastal area

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.170 \times \ln\left(\frac{4.0}{0.01}\right) = 1.017$$

Eq. 4.4 EN 1991-1-4
Roughness factor at 3.5m height
 $Z = 4.0 > Z_{min} = 1$

$$V_m = C_r \times V_b = 1.017 \times V_b$$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

$$\sigma_v = K_r \times V_b = 0.170 \times V_b$$

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.170 \times V_b}{1.017 \times V_b} = 0.167$$

Eq. 4.7 EN 1991-1-4
Turbulence intensity

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 1.402 \times V_b^2$$

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Equation:

$$500 = 1.402 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 18.89 \text{ m/s}$$

Characteristic wind speed

Wind speed rural area at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05} \right)^{0.07} = 0.19 \times \left(\frac{0.05}{0.05} \right)^{0.07} = 0.190$$

$$C_r = K_r \times \ln \left(\frac{z}{z_0} \right) = 0.190 \times \ln \left(\frac{4.0}{0.05} \right) = 0.833$$

$$V_m = C_r \times V_b = 0.833 \times V_b$$

$$\sigma_v = K_r \times V_b = 0.190 \times V_b$$

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.190 \times V_b}{0.833 \times V_b} = 0.228$$

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 1.125 \times V_b^2$$

Equation:

$$500 = 1.125 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 21.08 \text{ m/s}$$

Eq. 4.5 EN 1991-1-4
Terrain factor for unbuilt area

Eq. 4.4 EN 1991-1-4
Roughness factor at 4m height
 $Z = 4.0 > Z_{\min} = 2$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

Eq. 4.7 EN 1991-1-4
Turbulence intensity

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Characteristic wind speed

Wind speed village at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05} \right)^{0.07} = 0.19 \times \left(\frac{0.3}{0.05} \right)^{0.07} = 0.215$$

$$C_r = K_r \times \ln \left(\frac{z}{z_0} \right) = 0.215 \times \ln \left(\frac{5}{0.3} \right) = 0.606$$

$$V_m = C_r \times V_b = 0.606 \times V_b$$

$$\sigma_v = K_r \times V_b = 0.215 \times V_b$$

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.215 \times V_b}{0.606 \times V_b} = 0.355$$

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 0.801 \times V_b^2$$

Equation:

$$500 = 0.801 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 24.99 \text{ m/s}$$

Eq. 4.5 EN 1991-1-4
Terrain factor for unbuilt area

Eq. 4.4 EN 1991-1-4
Roughness factor at 7m height
 $Z = Z_{\min} = 5$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

Eq. 4.7 EN 1991-1-4
Turbulence intensity

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Characteristic wind speed

Wind speed city at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05} \right)^{0.07} = 0.19 \times \left(\frac{1}{0.05} \right)^{0.07} = 0.234$$

$$C_r = K_r \times \ln \left(\frac{z}{z_0} \right) = 0.234 \times \ln \left(\frac{10}{1} \right) = 0.540$$

$$V_m = C_r \times V_b = 0.540 \times V_b$$

$$\sigma_v = K_r \times V_b = 0.234 \times V_b$$

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.234 \times V_b}{0.540 \times V_b} = 0.434$$

$$Q_p = (1 + 7 \times L_v) \times \frac{1}{2} \times \rho \times V_m^2 = 0.735 \times V_b^2$$

Equation:

$$500 = 0.735 \times V_b^2 \rightarrow \text{solving gives} \rightarrow V_b = 26.08 \text{ m/s}$$

Eq. 4.5 EN 1991-1-4
Terrain factor for unbuilt area

Eq. 4.4 EN 1991-1-4
Roughness factor at 7m height
 $Z = Z_{\min} = 10$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

Eq. 4.7 EN 1991-1-4
Turbulence intensity

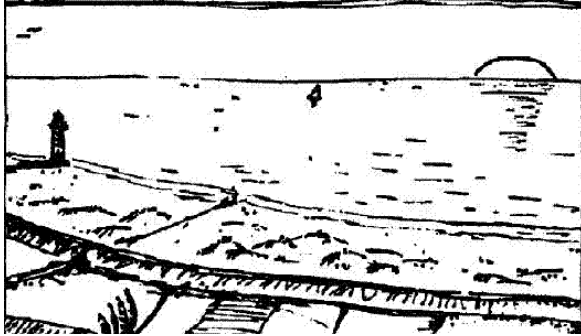
Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Characteristic wind speed

Terrain categories

Default European terrain categories (not Country specific)

0: Coastal area:



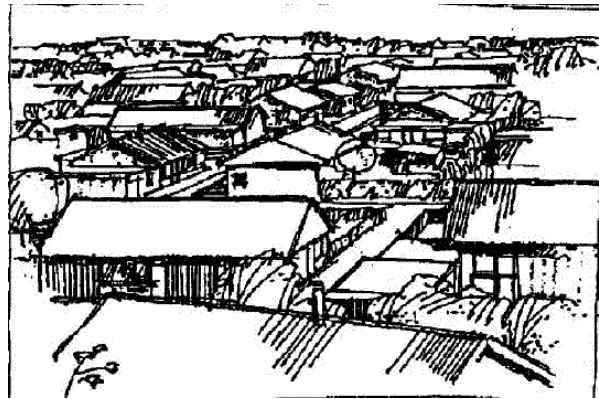
I: Flattened, open area:



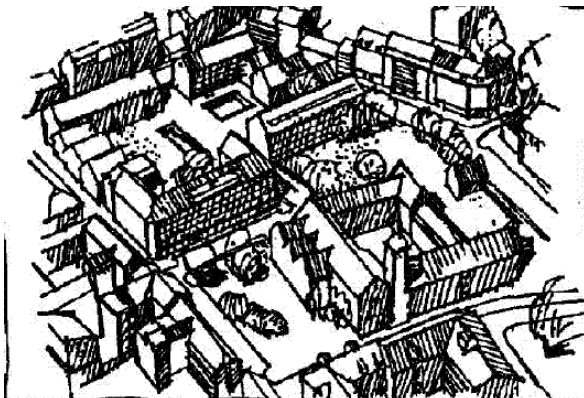
II: Rural area



III: Village



IV: City



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H.1. Project description

The principle of a stretchtent is based on a rectangular piece of stretchable membrane with a reinforcement along the boundary. The membrane is supported by poles, both at the edge and under the membrane. The side poles are stabilized by guy ropes and belts. The poles do not require a fixed position which ensures a freedom of shape.

Tentech B.V. has performed a static analysis of three tents – 10x10m, 10x15m and 10x20m always assumed to have stormbelts. Further chapters include material and cross-section description, calculation method, load cases and combinations, results, element checks, anchor and ballast calculations. These chapters present calculation results valid for Europe according to EN 13782.

As an exception, two additions to the static analysis are made (see I and J on pages 70 and 80 respectively):

1. For use only in Netherlands – element check for reduced wind speeds.
2. For use in France or countries following French CTS – element check for combined wind and snow loads

For use in other countries than Netherlands or France, the static analysis must be considered without its additions. When relevant information is mentioned for these specific cases either [NL] or [FR] immediately follows this information. This information is not considered while making the base calculations. Addition J on page 80 must always be considered for use in France.



Figure 1. 10x10m, 10x15m and 10x20m models

H.2. Materials

H.2.1 Fabric

Design tensile strength	f_d	f_{tk} / γ_m	art 8.6 EN 13782
Characteristic tensile strength (warp)	$f_{tk, \text{warp}}$		
Characteristic tensile strength (weft)	$f_{tk, \text{weft}}$		
Material factor – global, permanent load	γ_m	2.5	tbl 4. EN13782
Material factor – global, short duration load	γ_m	2.0	tbl 4. EN13782

Table 1. Used symbols, codes and standard for fabric materials

Material	Type	Weight	$f_{rd; \text{warp}; \text{perm}}$	$f_{rd; \text{weft}; \text{perm}}$	$f_{rd; \text{warp}; \text{short}}$	$f_{rd; \text{weft}; \text{short}}$
Sioen RP5639	-	$\approx 730 \text{ gr/m}^2$	9.6 kN/m	7.2 kN/m	12.0 kN/m	9.0 kN/m

Table 2. Used fabrics

H.2.2 Guy ropes

Design resistance	F_{rd}	R_m / γ_{m1}	art 10.2. EN13782
Characteristic tensile strength	R_m		art 10.2. EN13782
Material factor	$\gamma_{m1} < 12\text{mm}$	4.0	art 10.3. EN13782
	$\gamma_{m1} > 12\text{mm}$	3.3	art 10.3. EN13782

Table 3. Used symbols, codes and standard for rope materials

Material	Cross section	Breaking strength	R_m	F_{rd}
8mm Polyester Braid rope	$\varnothing 8 \text{ mm}$	$\geq 900 \text{ daN}$	9.0 kN	2.25 kN

Table 4. Used guy ropes

H.2.3 Belts

Design resistance	F_{rd}	R_m / γ_{m1}	art 10.2. EN13782
Characteristic breaking strength	R_m	$LC \times \gamma_{m2}$	art 10.2. EN13782
Lashing capacity	LC		Conform EN 12195-2
Material factor	γ_{m1}	2.0	art 10.2. EN13782
Material factor	γ_{m2}	3.0	EN1492-1

Table 5. Used symbols, codes and standard for belt materials

Material	LC	R_m	F_{rd}
Belt [PES] EN 12195-2, 50mm	1000 daN 10 kN	3000 daN 30 kN	15 kN

Table 6. Used belts

H.2.4 Wood

Material	γ_{m1}	1.3	tbl. 2.3. EN 1995-1-1
----------	---------------	-----	-----------------------

Table 7. Used material factors

Material	Weight	$E_{0.05}$	F_{c0k}	F_{mk}
Wood, Spruce, C18	320 kg/m ³	6.0 kN/m ²	18 N/mm ²	18 N/mm ²

Table 8. Used wooden materials

H.3. Cross sections

Profile	Material	$\varnothing$ mm	G kg/m	A mm ²	I_y mm ⁴	$W_{el,y}$ mm ³
$\varnothing \approx 100$ mm *	Wood C18	100	2.51	7854	49087839	98175
$\varnothing \approx 75$ mm *	Wood C18	75	1.41	4418	1553156	41417
$\varnothing \approx 70$ mm *	Wood C18	70	1.23	3848	1178588	33674

Table 9. Used cross sections

* the average diameter, as a minimum required at the middle of the pole.

H.4. Calculation method

H.4.1 Modelling

The analysis of the structure is performed with the software package EASY FCS supplied by TECHNET GmbH, Berlin. This software is specially developed for structures with large deformability, such as membrane structures. The performed analysis is a full non-linear second order analysis.

The membrane structure is modelled in 3D. The membrane is modelled as a cable net structure and is supported by poles. These centre poles will be stabilized by the tensioned membrane. The side poles are stabilized and tied down by tension belts or ropes, which are attached to ground anchors or ballast.

The membrane edges are reinforced. They consist of multiple layers of fabric about 10cm wide with additional patches where the lugs are placed. Storm belts are required constantly and placed in the valleys between the centre poles.

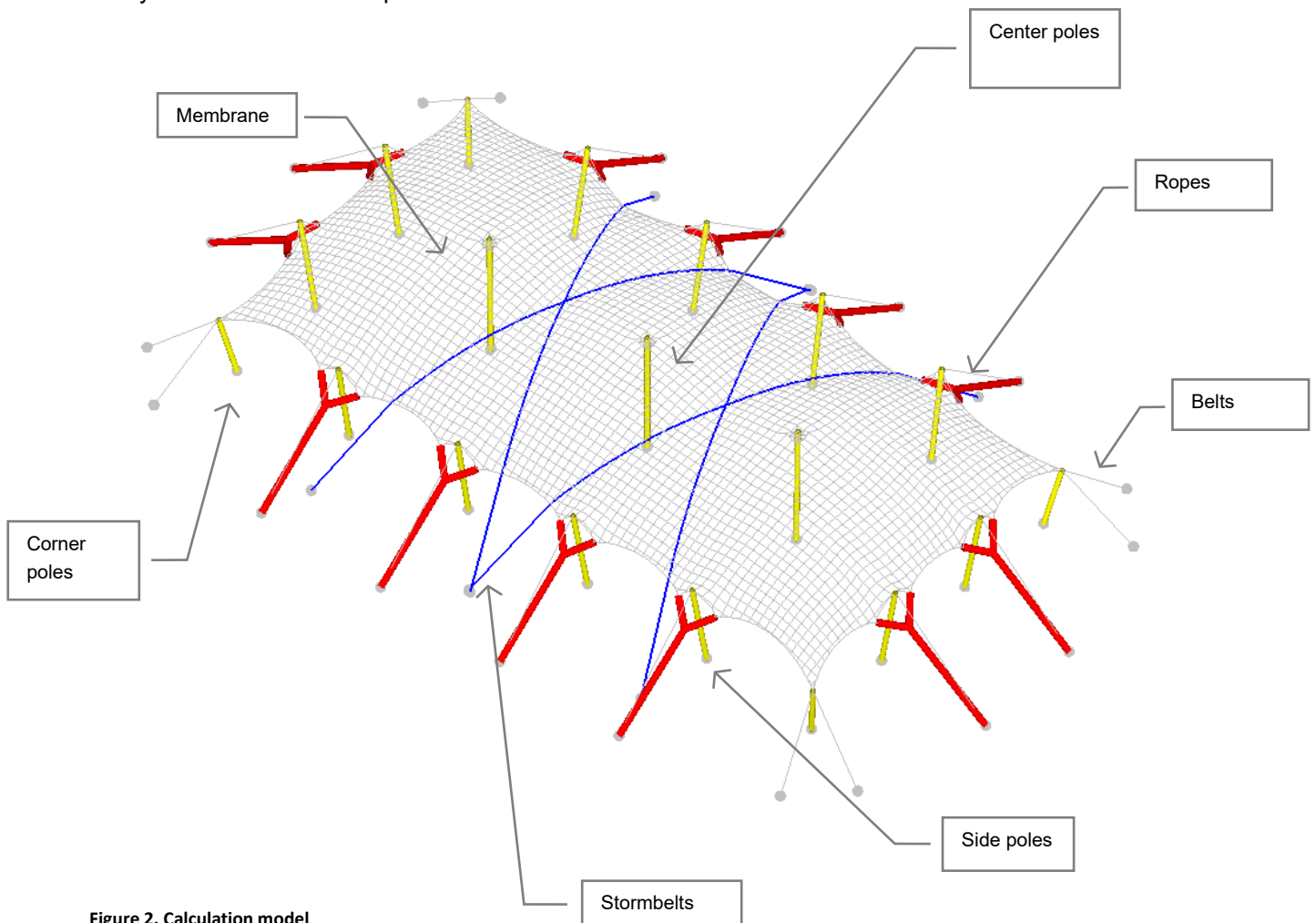


Figure 2. Calculation model

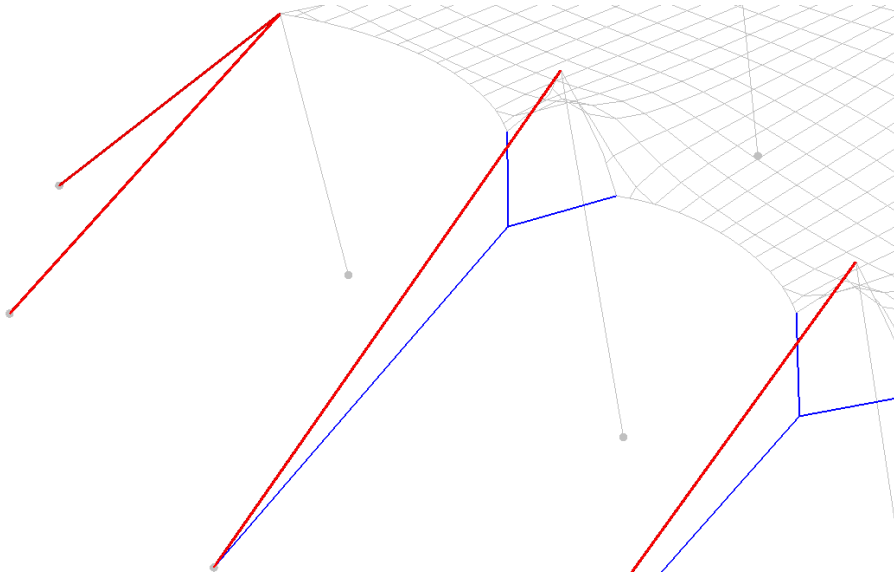


Figure 3. Principle of ties

Ties stabilize the tent and create the shape of the stretchable membrane. Guy ropes (see blue above) transfer the forces to the ground when wind suction is present. Belts (see red above) help transfer the pressure forces.

H.4.2 Structural behaviour of membrane structures

Since the fabric is a highly deformable material, it is only possible to calculate stresses and deformations with a non-linear method. FEM-software EASY is used to perform these calculations. Because of the non-linearity of the calculations the partial safety factors are not applied beforehand, since the deformations will be greater due to these safety factors, resulting in lower stresses in the fabric. See figure below.

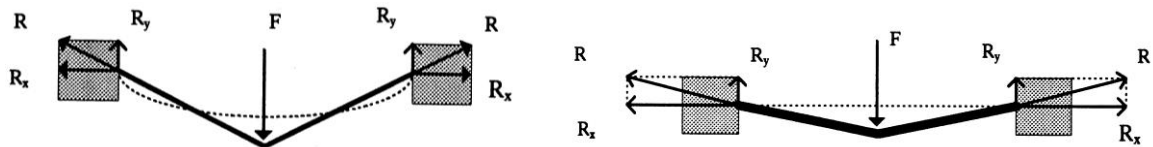


Figure 4. membrane behavior

As a membrane structure is a form-active structure, article 6.3 (4) b) of the EN-1990:2002 applies:

When the action effect increases less than the action, the partial factor γ_F should be applied to the action effect of the representative value of the action.

This means that no load factors are applied on the load beforehand, but afterwards.

H.4.3 Load combinations

H.4.3.1 Fundamental - Ultimate limit state

For determination of strength and to check elements and connections ultimate limit state is used.

	One variable load	Multiple variable loads
Unfavorable permanent load	$1.35 \times G + 1.5 \times Q$	$1.35 \times G + \sum 1.35 \times Q_i$
Favorable permanent load	$1.0 \times G + 1.5 \times Q$	$1.0 \times G + \sum 1.35 \times Q_i$

Table 10. Load combinations according to EN 13782

This means that the following load combinations will be checked/calculated:

1. 1.35 x Own weight + 1.5 x Wind load
2. 1.35 x Own weight + 1.5 x Snow load
3. 1.35 x Own weight + 1.35 x Wind load + 1.35 x Snow load [FR]

H.4.3.2 Safety against overturning, sliding and uplifting - Ultimate limit state

For determination and check of the needed ballast weight and/or anchors ultimate limit state is used

	One or multiple variable loads
Unfavorable permanent load	$1.1 \times G + 1.2 \times Q_{wind} + \sum 1.3 \times Q_i$
Favorable permanent load	$1.0 \times G + 1.2 \times Q_{wind} + \sum 1.3 \times Q_i$

Table 11. Load combinations according to EN 13782

This means the following load combinations will be checked/calculated:

1. 1.0 x Own weight + 1.2 x Wind load
2. 1.0 x Own weight + 1.2 x Wind load + 1.3 x Snow load [FR]

H.5. Load cases

H.5.1 Own weight

The own weight of the fabric is $730 \text{ g/m}^2 = 0.0073 \text{ kN/m}^2$ and is considered automatically within EASY software.

H.5.2 Pretension

The structure will be pretensioned with guy ropes and belts. With a ratchet it is possible to achieve 2-3 kN force in corner belts. Approximately 2kN force is achieved in side ropes. The tensile force in ropes and belts results in a membrane stress of approximately 0.50 kN/m^2 . Higher membrane forces can be present near connection points due to the stretchable nature of the fabric.

H.5.3 Conventional / snow load

Conventional load according to EN 13782: The stability shall be checked with a conventional vertical load of 0.1 kN/m^2 . This can be seen as a snow load of 0.1 kN/m^2 (4cm) according to the French CTS, where snow load must be combined with wind load. This condition is tackled in J. Addition 2 – element check for combined wind and snow load on page 80.

H.5.4 User load

A user defined load (for light, sound and/or decoration purposes) is 25 kg per pole and is added after the analysis while performing element checks.

H.5.5 Wind load

H.5.5.1 Wind pressure

Wind load according to EN 13782, 7.4.2.2:

$q_p(z_e)$ of **0.50 kN/m^2** if the height of the structure does not exceed 5m.

Wind load can be reduced for use in the Netherlands. A reduction factor $\alpha=0.7$ is applied, which leads to $q_p(z_e)$ of **0.25 kN/m^2 [NL]**, this is tackled in I. Addition 1 – element check for reduced wind speeds [NL]

H.5.5.2 Wind shape values (Cp-factors)

Two different wind situations are reviewed for the membrane:

1. The whole tent is subjected to wind suction (Cp values given in EN 13782)
2. The whole tent is subjected to wind pressure (Cp values given in EN 13782)

Wind suction

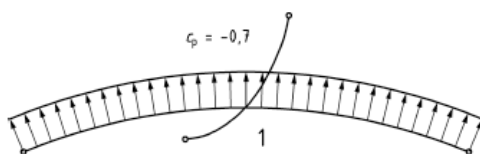


Figure 5. C_p values for rectangle structures EN 13782

P_w	$p_{w,rep}$	
500 N/m ²	$-0.7 \times 0.500 = -0.350$ kN/m ²	
350 N/m ²	$-0.7 \times 0.250 = -0.175$ kN/m ²	[NL]

Wind pressure

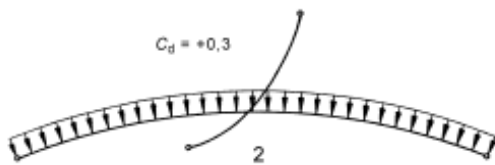


Figure 6. C_p values for rectangle structures EN 13782

P_w	$p_{w,rep}$
500 N/m ²	$0.3 \times 0.500 = 0.150$ kN/m ²

H.6. Calculation results

H.6.1 Listing of calculated load combinations

LC1 = Pretension

LC2 = Own weight

LC3 = Snow load

LC4 = Wind pressure

LC5 = Wind suction

LC6 = Wind suction reduced

The following load combinations are considered:

partial safety factors are added after the static analysis (see H.4.2).

	LC1	LC2	LC3	LC4	LC5	LC6 [NL]
CO1	1 x	1 x				
CO2	1 x	1 x	1 x			
CO3	1 x	1 x		1 x		
CO4	1 x	1 x			1 x	
CO5 [NL]	1 x	1 x				1 x
CO6 [FR]	1 x	1 x	1 x	1 x		

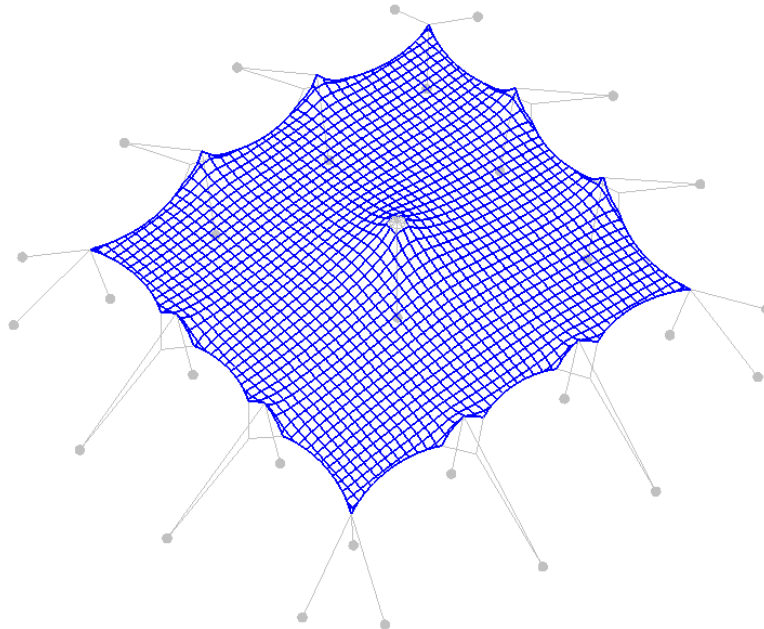
Table 12. Combinations (CO)

The results for load combinations CO5 and CO6 are mentioned in the upcoming chapter but considered in calculations only in chapters I and J on pages 70 and 80 respectively.

H.6.2 Overview: Global results of static analysis

H.6.2.1 Membrane

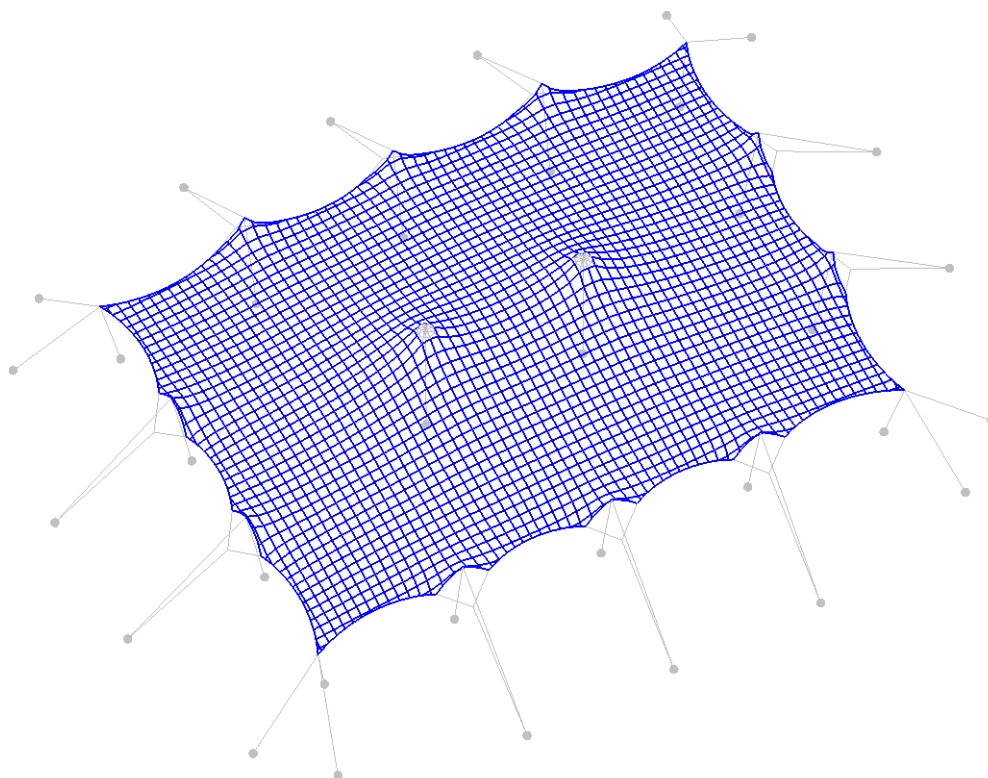
H.6.2.1.1. 10x10m model



	Load combination	F_{rep}	Pag.
Warp	CO1. Own weight + pretension	0.95 kN/m	119
	CO2. Own weight + pretension + snow	2.42 kN/m	123
	CO3. Own weight + pretension + wind pressure	3.09 kN/m	127
	Max CO4. Own weight + pretension + wind suction	3.39 kN/m	131
	CO5. Own weight + pretension + wind suction reduced [NL]	2.49 kN/m	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.55 kN/m	139
Weft	CO1. Own weight + pretension	0.91 kN/m	119
	CO2. Own weight + pretension + conventional / snow	2.11 kN/m	123
	Max CO3. Own weight + pretension + wind pressure	2.62 kN/m	127
	CO4. Own weight + pretension + wind suction	2.18 kN/m	131
	CO5. Own weight + pretension + wind suction reduced [NL]	1.99 kN/m	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.79 kN/m	139
Edge	CO1. Own weight + pretension	1.76 kN	119
	CO2. Own weight + pretension + conventional / snow	2.13 kN	123
	CO3. Own weight + pretension + wind pressure	2.44 kN	127
	Max CO4. Own weight + pretension + wind suction	3.41 kN	131
	CO5. Own weight + pretension + wind suction reduced [NL]	2.68 kN	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.85 kN	139

Table 13. Leading forces in membrane – 10x10m

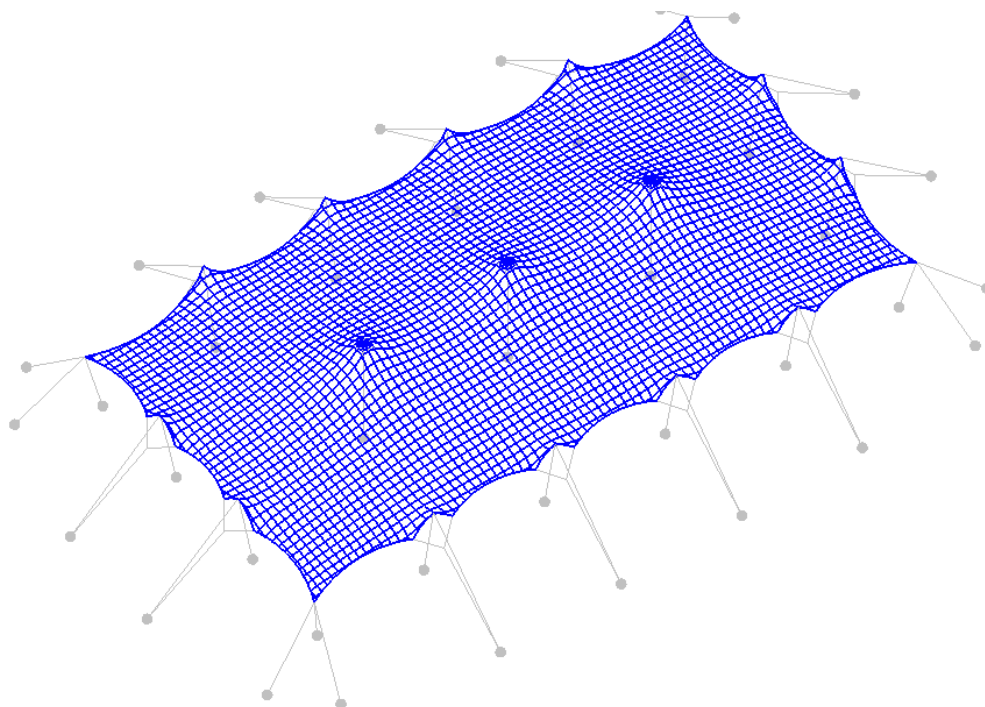
H.6.2.1.2. 10x15m model



	Load combination	F _{rep}	Pag.
Warp	CO1. Own weight + pretension	0.91 kN/m	143
	CO2. Own weight + pretension + snow	2.59 kN/m	147
	Max CO3. Own weight + pretension + wind pressure	3.35 kN/m	151
	CO4. Own weight + pretension + wind suction	3.13 kN/m	155
	CO5. Own weight + pretension + wind suction reduced [NL]	2.51 kN/m	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.97 kN/m	163
Weft	CO1. Own weight + pretension	0.89 kN/m	143
	CO2. Own weight + pretension + conventional / snow	2.18 kN/m	147
	Max CO3. Own weight + pretension + wind pressure	2.93 kN/m	151
	CO4. Own weight + pretension + wind suction	2.27 kN/m	155
	CO5. Own weight + pretension + wind suction reduced [NL]	2.48 kN/m	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.24 kN/m	163
Edge	CO1. Own weight + pretension	1.90 kN	143
	CO2. Own weight + pretension + conventional / snow	2.35 kN	147
	CO3. Own weight + pretension + wind pressure	2.74 kN	151
	Max CO4. Own weight + pretension + wind suction	3.55 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	3.17 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.19 kN	163

Table 14. Leading forces in membrane – 10x15m

H.6.2.1.3. 10x20m model

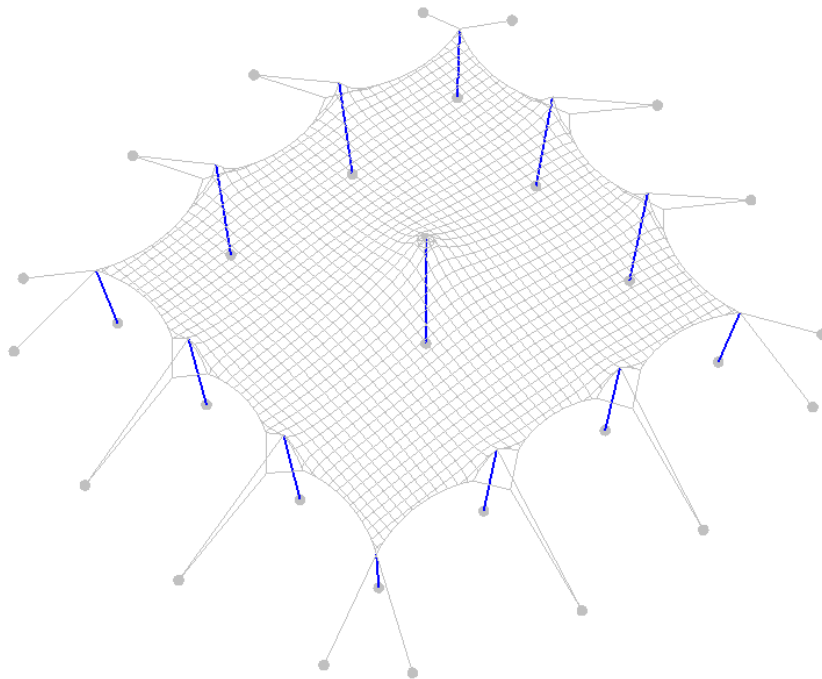


	Load combination	F_{rep}	Pag.
Warp	CO1. Own weight + pretension	0.95 kN/m	167
	CO2. Own weight + pretension + snow	2.63 kN/m	171
	CO3. Own weight + pretension + wind pressure	3.41 kN/m	175
	Max CO4. Own weight + pretension + wind suction	4.45 kN/m	179
	CO5. Own weight + pretension + wind suction reduced [NL]	2.53 kN/m	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	5.01 kN/m	187
Weft	CO1. Own weight + pretension	0.88 kN/m	167
	CO2. Own weight + pretension + conventional / snow	2.16 kN/m	171
	Max CO3. Own weight + pretension + wind pressure	2.94 kN/m	175
	CO4. Own weight + pretension + wind suction	2.89 kN/m	179
	CO5. Own weight + pretension + wind suction reduced [NL]	2.68 kN/m	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.20 kN/m	187
Edge	CO1. Own weight + pretension	1.80 kN	167
	CO2. Own weight + pretension + conventional / snow	2.26 kN	171
	CO3. Own weight + pretension + wind pressure	2.67 kN	175
	Max CO4. Own weight + pretension + wind suction	4.43 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	3.43 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.16 kN	187

Table 15. Leading forces in membrane – 10x20m

H.6.2.2 Poles

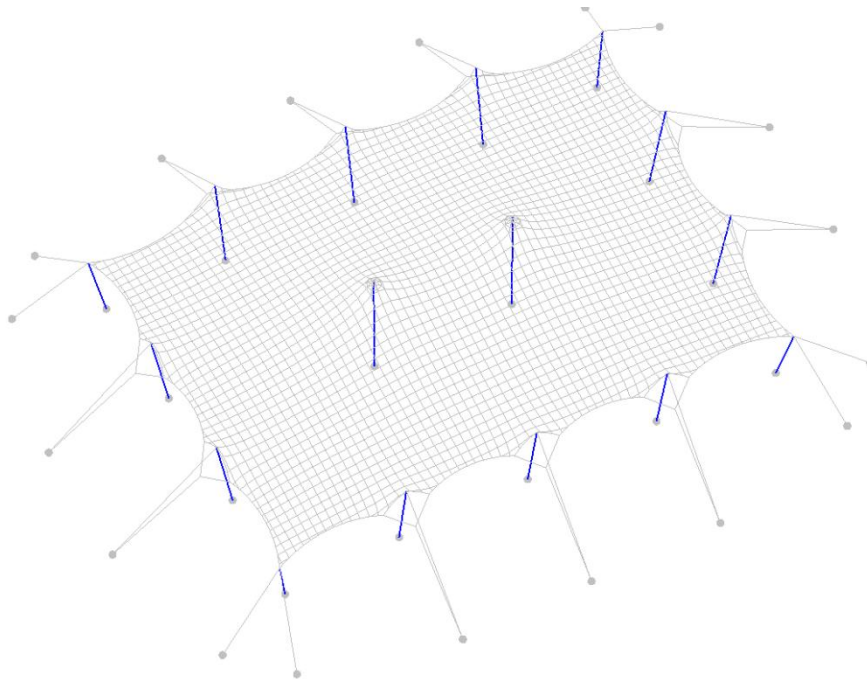
H.6.2.2.1. 10x10m model



	Load combination	F_{rep}	Pag.
Main poles	CO1. Own weight + pretension	-2.12 kN	119
	CO2. Own weight + pretension + snow	-5.41 kN	123
	Max CO3. Own weight + pretension + wind pressure	-6.95 kN	127
	CO4. Own weight + pretension + wind suction	-0.05 kN	131
	CO5. Own weight + pretension + wind suction reduced [NL]	-0.08 kN	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	-10.32 kN	139
Side poles	CO1. Own weight + pretension	-1.30 kN	119
	CO2. Own weight + pretension + conventional / snow	-2.43 kN	123
	Max CO3. Own weight + pretension + wind pressure	-3.13 kN	127
	CO4. Own weight + pretension + wind suction	-2.02 kN	131
	CO5. Own weight + pretension + wind suction reduced [NL]	-0.83 kN	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	-4.33 kN	139
Corner poles	CO1. Own weight + pretension	-2.64 kN	119
	CO2. Own weight + pretension + conventional / snow	-3.38 kN	123
	CO3. Own weight + pretension + wind pressure	-3.95 kN	127
	Max CO4. Own weight + pretension + wind suction	-4.52 kN	131
	CO5. Own weight + pretension + wind suction reduced [NL]	-3.43 kN	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	-4.75 kN	139

Table 16. Leading forces in poles – 10x10m model

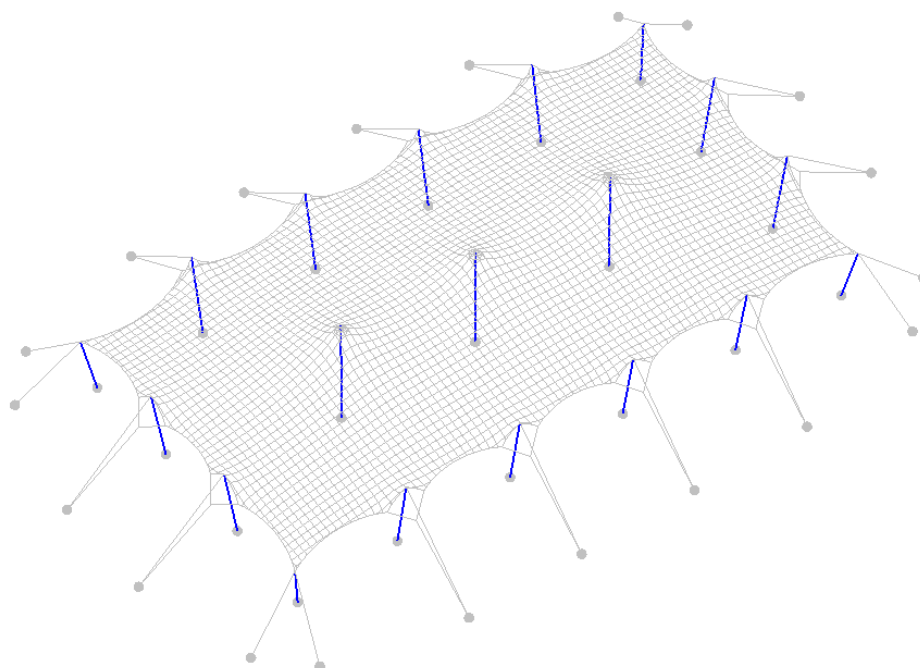
H.6.2.2.2. 10x15m model



	Load combination	F _{rep}	Pag.
Main poles	CO1. Own weight + pretension	-1.74 kN	143
	CO2. Own weight + pretension + snow	-4.96 kN	147
	Max CO3. Own weight + pretension + wind pressure	-6.49 kN	151
	CO4. Own weight + pretension + wind suction	0.06 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	-0.02 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	-9.78 kN	163
Side poles	CO1. Own weight + pretension	-1.37 kN	143
	CO2. Own weight + pretension + conventional / snow	-2.74 kN	147
	Max CO3. Own weight + pretension + wind pressure	-3.60 kN	151
	CO4. Own weight + pretension + wind suction	-2.47 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	-0.84 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	-4.99 kN	163
Corner poles	CO1. Own weight + pretension	-2.86 kN	143
	CO2. Own weight + pretension + conventional / snow	-3.81 kN	147
	CO3. Own weight + pretension + wind pressure	-4.56 kN	151
	Max CO4. Own weight + pretension + wind suction	-4.77 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	-3.87 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	-5.50 kN	163

Table 17. Leading forces in poles – 10x15m model

H.6.2.2.3. 10x20m model

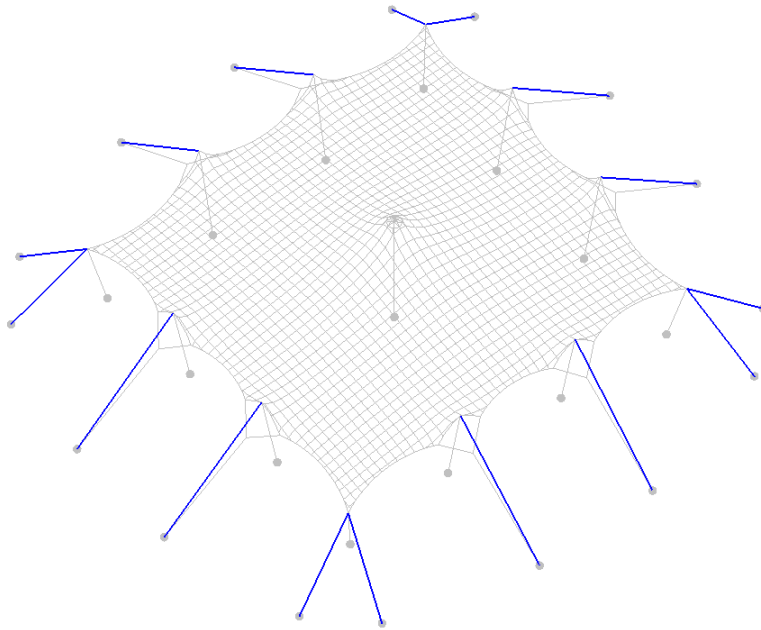


	Load combination	F _{rep}	Pag.
Main poles	CO1. Own weight + pretension	-1.84 kN	167
	CO2. Own weight + pretension + snow	-5.15 kN	171
	Max CO3. Own weight + pretension + wind pressure	-6.75 kN	175
	CO4. Own weight + pretension + wind suction	-0.06 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	0.03 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	-10.07 kN	187
Side poles	CO1. Own weight + pretension	-1.35 kN	167
	CO2. Own weight + pretension + conventional / snow	-2.69 kN	171
	Max CO3. Own weight + pretension + wind pressure	-3.55 kN	175
	CO4. Own weight + pretension + wind suction	-2.21 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	-0.80 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	-4.88 kN	187
Corner poles	CO1. Own weight + pretension	-2.73 kN	167
	CO2. Own weight + pretension + conventional / snow	-3.67 kN	171
	Max CO3. Own weight + pretension + wind pressure	-4.43 kN	175
	CO4. Own weight + pretension + wind suction	-4.31 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	-3.63 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	-5.40 kN	187

Table 18. Leading forces in poles – 10x20m model

H.6.2.3 Belts

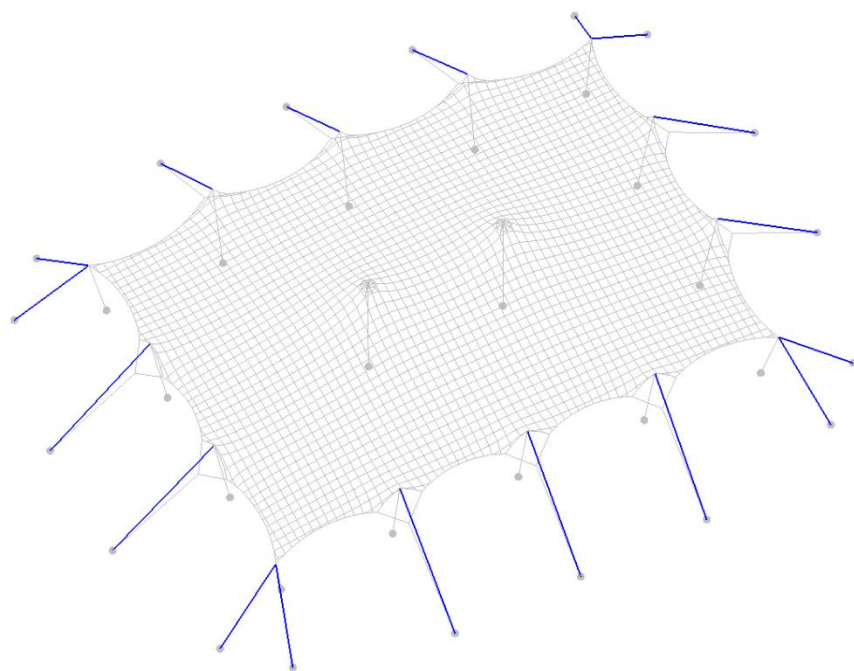
H.6.2.3.1. 10x10m model



	Load combination	F_{rep}	Pag.
Corner belts	CO1. Own weight + pretension	2.10 kN	119
	CO2. Own weight + pretension + snow	2.50 kN	123
	CO3. Own weight + pretension + wind pressure	2.86 kN	127
	Max CO4. Own weight + pretension + wind suction	3.94 kN	131
	CO5. Own weight + pretension + wind suction reduced [NL]	3.19 kN	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.33 kN	139
Belts – side	CO1. Own weight + pretension	0.12 kN	119
	CO2. Own weight + pretension + conventional / snow	0.47 kN	123
	Max CO3. Own weight + pretension + wind pressure	0.70 kN	127
	CO4. Own weight + pretension + wind suction	0.35 kN	131
	CO5. Own weight + pretension + wind suction reduced [NL]	0.16 kN	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	1.13 kN	139

Table 19. Leading forces in belts – 10x10m model

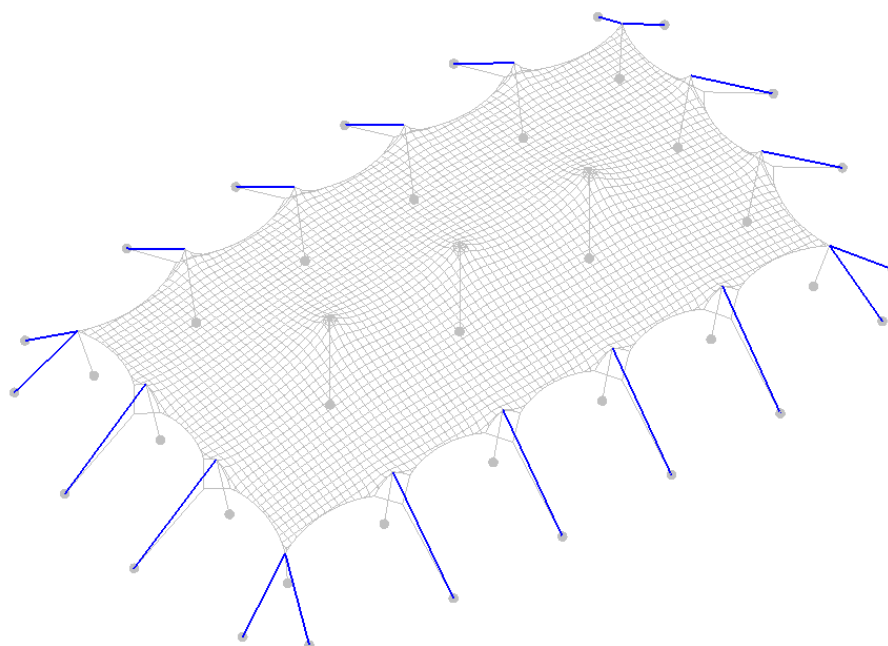
H.6.2.3.2. 10x15m model



	Load combination	F _{rep}	Pag.
Corner belts	CO1. Own weight + pretension	2.31 kN	143
	CO2. Own weight + pretension + snow	2.85 kN	147
	CO3. Own weight + pretension + wind pressure	3.28 kN	151
	Max CO4. Own weight + pretension + wind suction	4.22 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	3.51 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.81 kN	163
Belts – long side	CO1. Own weight + pretension	0.14 kN	143
	CO2. Own weight + pretension + conventional / snow	0.51 kN	147
	Max CO3. Own weight + pretension + wind pressure	0.81 kN	151
	CO4. Own weight + pretension + wind suction	0.50 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	0.18 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	1.30 kN	163
Belts – short side	CO1. Own weight + pretension	0.12 kN	143
	CO2. Own weight + pretension + conventional / snow	0.56 kN	147
	Max CO3. Own weight + pretension + wind pressure	0.84 kN	151
	CO4. Own weight + pretension + wind suction	0.22 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	0.18 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	1.37 kN	163

Table 20. Leading forces in belts – 10x15m model

H.6.2.3.3. 10x20m model

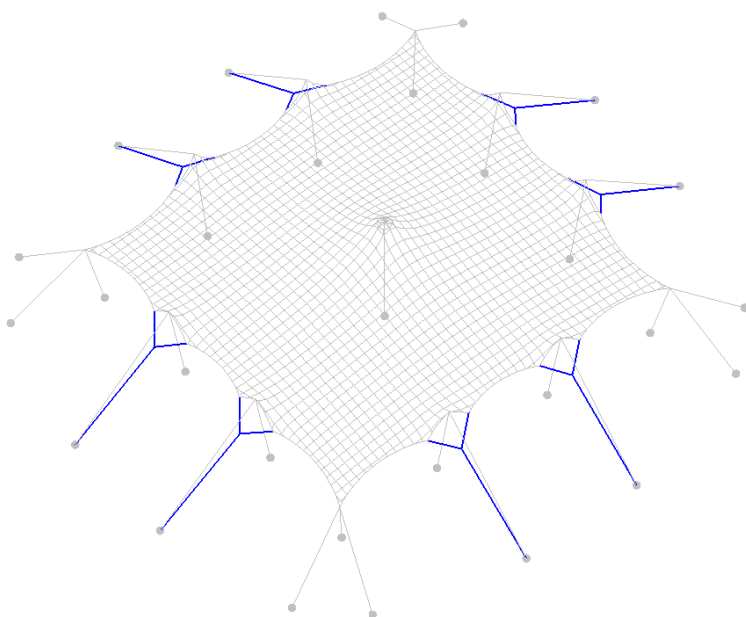


	Load combination	F _{rep}	Pag.
Corner belts	CO1. Own weight + pretension	2.33 kN	167
	CO2. Own weight + pretension + snow	2.95 kN	171
	CO3. Own weight + pretension + wind pressure	3.44 kN	175
	Max CO4. Own weight + pretension + wind suction	4.43 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	3.44 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.01 kN	187
Belts – long side	CO1. Own weight + pretension	0.22 kN	167
	CO2. Own weight + pretension + conventional / snow	0.70 kN	171
	Max CO3. Own weight + pretension + wind pressure	1.01 kN	175
	CO4. Own weight + pretension + wind suction	0.57 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	0.26 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	1.50 kN	187
Belts – short side	CO1. Own weight + pretension	0.13 kN	167
	CO2. Own weight + pretension + conventional / snow	0.55 kN	171
	Max CO3. Own weight + pretension + wind pressure	0.85 kN	175
	CO4. Own weight + pretension + wind suction	0.13 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	0.13 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	1.35 kN	187

Table 21. Leading forces in belts – 10x20m model

H.6.2.4 Guy ropes

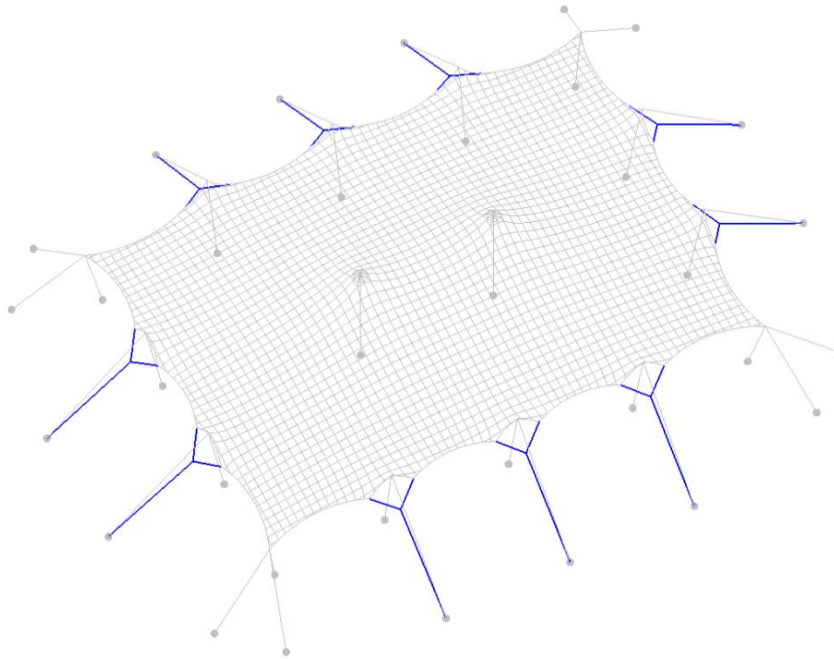
H.6.2.4.1. 10x10m model



	Load combination	F _{rep}	Pag.
Combined rope – side	CO1. Own weight + pretension	1.83 kN	119
	CO2. Own weight + pretension + snow	2.12 kN	123
	CO3. Own weight + pretension + wind pressure	2.41 kN	127
	Max CO4. Own weight + pretension + wind suction	4.83 kN	131
	CO5. Own weight + pretension + wind suction reduced [NL]	3.64 kN	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.76 kN	139
V-ropes –side	CO1. Own weight + pretension	1.40 kN	119
	CO2. Own weight + pretension + conventional / snow	1.63 kN	123
	CO3. Own weight + pretension + wind pressure	1.84 kN	127
	Max CO4. Own weight + pretension + wind suction	3.51 kN	131
	CO5. Own weight + pretension + wind suction reduced [NL]	2.72 kN	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.10 kN	139

Table 22. Leading forces in ropes – 10x10m model

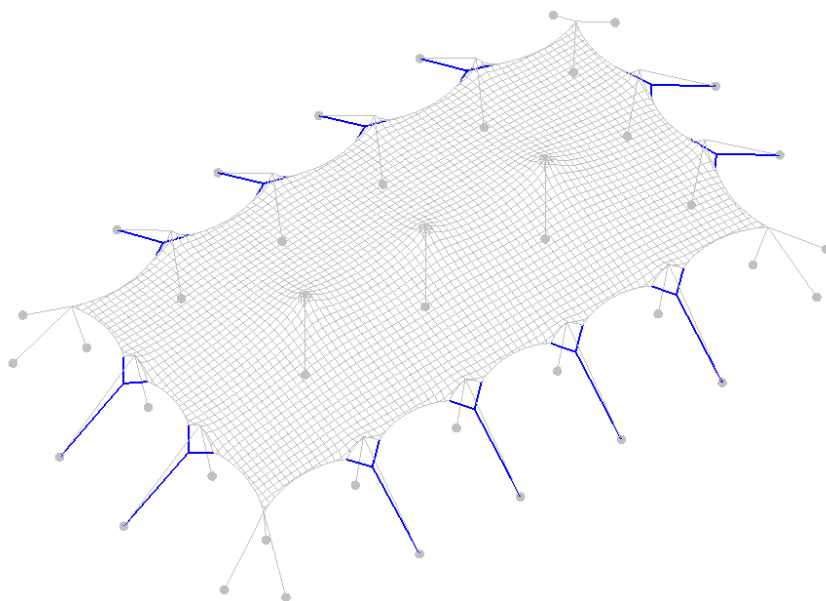
H.6.2.4.2. 10x15m model



	Load combination	F _{rep}	Pag.
Combined rope – long side	CO1. Own weight + pretension	2.11 kN	143
	CO2. Own weight + pretension + snow	2.59 kN	147
	CO3. Own weight + pretension + wind pressure	3.04 kN	151
	Max CO4. Own weight + pretension + wind suction	4.70 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	4.66 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.50 kN	163
V-ropes – long side	CO1. Own weight + pretension	1.53 kN	143
	CO2. Own weight + pretension + conventional / snow	1.83 kN	147
	CO3. Own weight + pretension + wind pressure	2.11 kN	151
	Max CO4. Own weight + pretension + wind suction	3.38 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	3.30 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.39 kN	163
Combined rope – short side	CO1. Own weight + pretension	1.92 kN	143
	CO2. Own weight + pretension + conventional / snow	2.38 kN	147
	CO3. Own weight + pretension + wind pressure	2.77 kN	151
	Max CO4. Own weight + pretension + wind suction	5.01 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	4.04 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.21 kN	163
V-ropes – short side	CO1. Own weight + pretension	1.45 kN	143
	CO2. Own weight + pretension + conventional / snow	1.80 kN	147
	CO3. Own weight + pretension + wind pressure	2.09 kN	151
	Max CO4. Own weight + pretension + wind suction	3.64 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	2.94 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.41 kN	163

Table 23. Leading forces in ropes – 10x15m model

H.6.2.4.3. 10x20m model

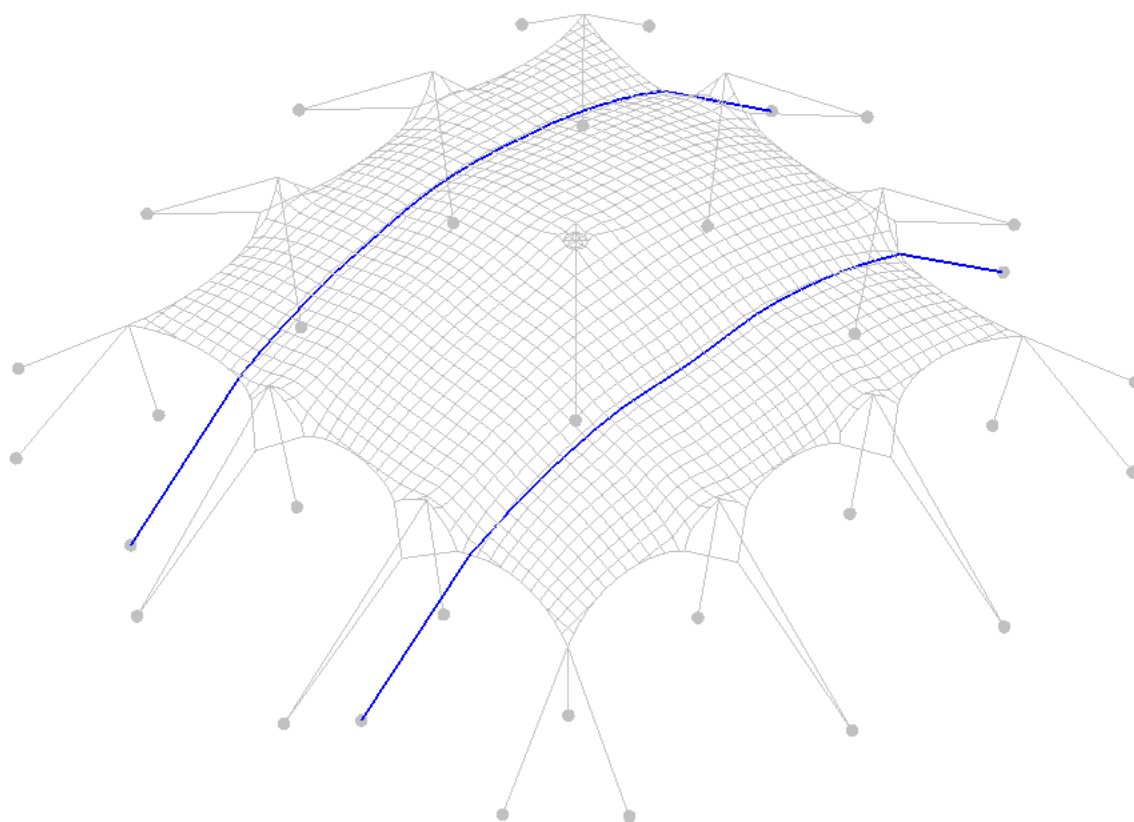


	Load combination	F_{rep}	Pag.
Combined rope – long side	CO1. Own weight + pretension	2.16 kN	167
	CO2. Own weight + pretension + snow	2.48 kN	171
	CO3. Own weight + pretension + wind pressure	2.91 kN	175
	Max CO4. Own weight + pretension + wind suction	5.43 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	5.46 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.34 kN	187
V-ropes – long side	CO1. Own weight + pretension	1.50 kN	167
	CO2. Own weight + pretension + conventional / snow	1.76 kN	171
	CO3. Own weight + pretension + wind pressure	2.05 kN	175
	Max CO4. Own weight + pretension + wind suction	3.91 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	3.68 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.28 kN	187
Combined rope – short side	CO1. Own weight + pretension	1.87 kN	167
	CO2. Own weight + pretension + conventional / snow	2.42 kN	171
	CO3. Own weight + pretension + wind pressure	2.89 kN	175
	Max CO4. Own weight + pretension + wind suction	5.46 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	3.52 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.39 kN	187
V-ropes – short side	CO1. Own weight + pretension	1.37 kN	167
	CO2. Own weight + pretension + conventional / snow	1.80 kN	171
	CO3. Own weight + pretension + wind pressure	2.14 kN	175
	Max CO4. Own weight + pretension + wind suction	3.83 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	2.54 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	2.51 kN	187

Table 24. Leading forces in ropes – 10x20m model

H.6.2.5 Stormbelts

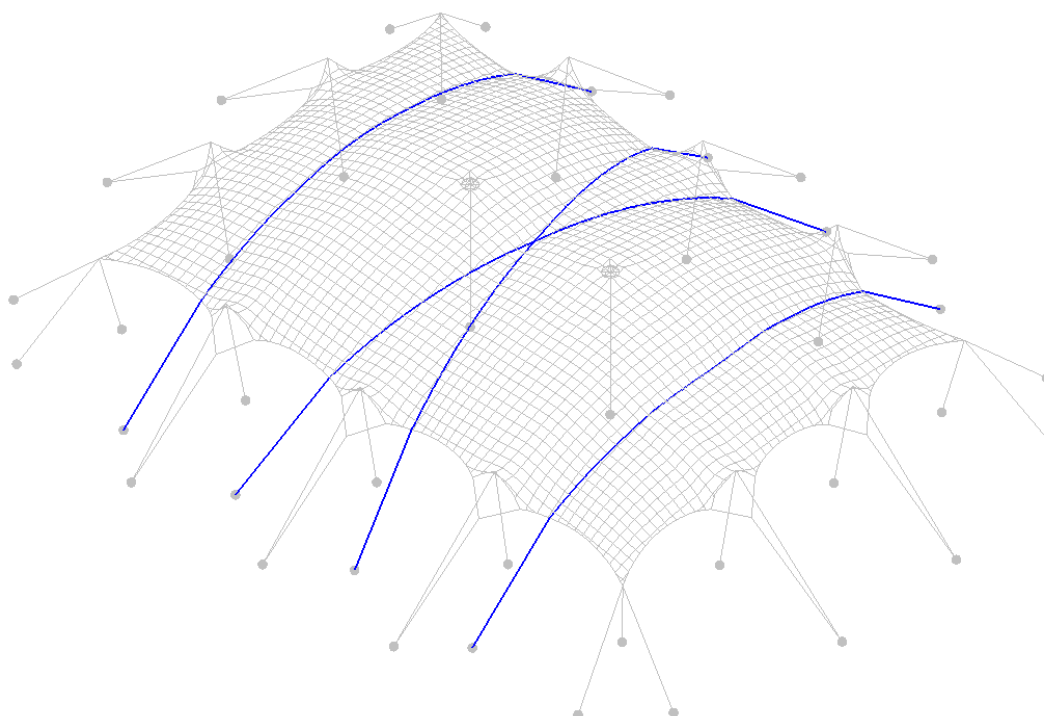
H.6.2.5.1. 10x10m model



	Load combination	F_{rep}	Pag.
Stormbelts	CO1. Own weight + pretension	0.00 kN	119
	CO2. Own weight + pretension + snow	0.00 kN	123
	CO3. Own weight + pretension + wind pressure	0.00 kN	127
	Max CO4. Own weight + pretension + wind suction	5.62 kN	131
	CO5. Own weight + pretension + wind suction reduced [NL]	0.00 kN	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	0.00 kN	139

Table 25. Leading forces in stormbelts – 10x10m model

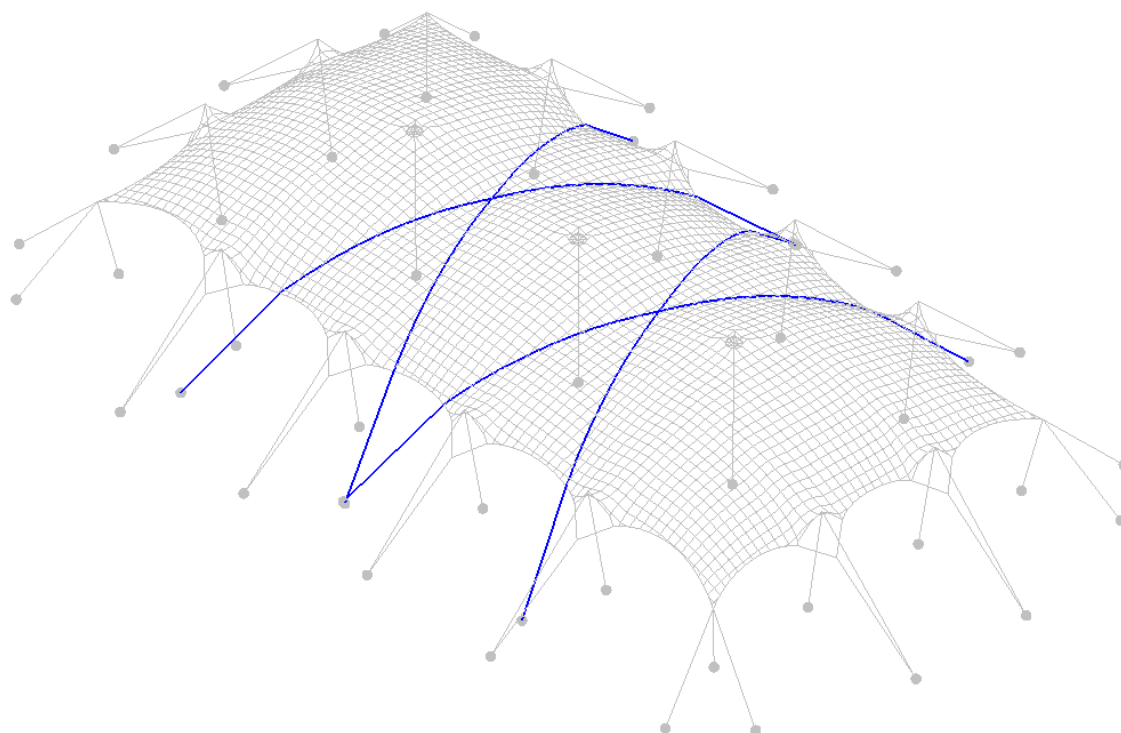
H.6.2.5.2. 10x15m model



	Load combination	F_{rep}	Pag.
Stormbelts	CO1. Own weight + pretension	0.00 kN	143
	CO2. Own weight + pretension + snow	0.00 kN	147
	CO3. Own weight + pretension + wind pressure	0.00 kN	151
	Max CO4. Own weight + pretension + wind suction	6.01 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	0.00 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	0.00 kN	163

Table 26. Leading forces in stormbelts – 10x15m model

H.6.2.5.3. 10x20m model

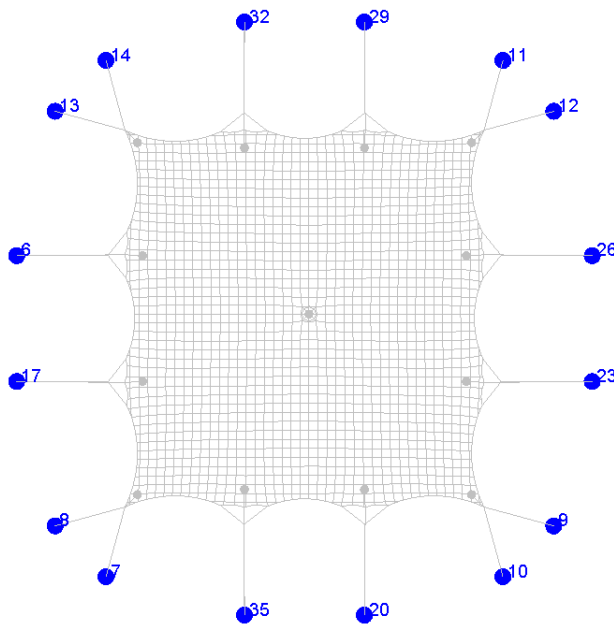


	Load combination	F_{rep}	Pag.
Stormbelts	CO1. Own weight + pretension	0.00 kN	167
	CO2. Own weight + pretension + snow	0.00 kN	171
	CO3. Own weight + pretension + wind pressure	0.00 kN	175
	Max CO4. Own weight + pretension + wind suction	5.93 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	0.00 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	0.00 kN	187

Table 27. Leading forces in stormbelts – 10x20m model

H.6.2.6 Reaction forces

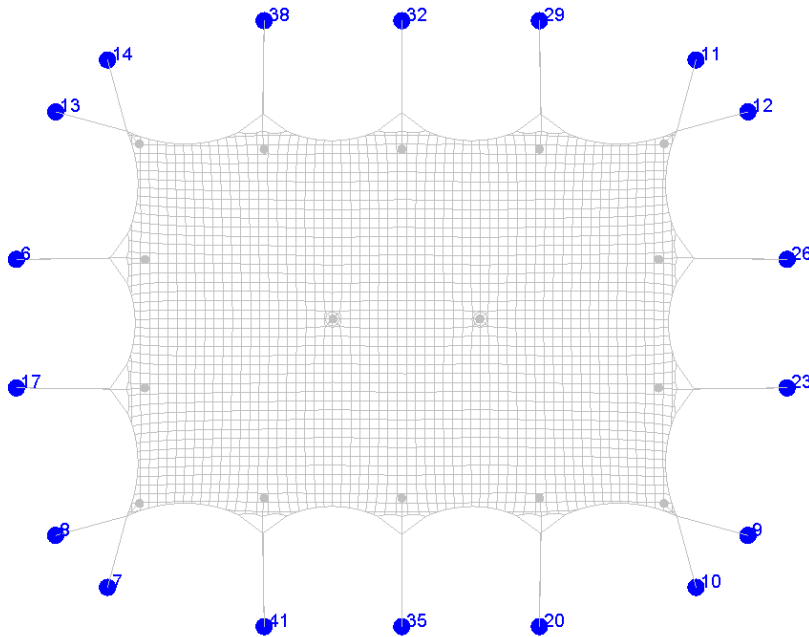
H.6.2.6.1. 10x10m model



	Load combination	F _{rep}	Pag.
Side	CO1. Own weight + pretension	1.94 kN	119
	CO2. Own weight + pretension + snow	2.59 kN	123
	CO3. Own weight + pretension + wind pressure	3.10 kN	127
	Max CO4. Own weight + pretension + wind suction	5.04 kN	131
	CO5. Own weight + pretension + wind suction reduced [NL]	3.80 kN	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.94 kN	139
Corner	CO1. Own weight + pretension	2.10 kN	119
	CO2. Own weight + pretension + conventional / snow	2.50 kN	123
	CO3. Own weight + pretension + wind pressure	2.86 kN	127
	Max CO4. Own weight + pretension + wind suction	3.94 kN	131
	CO5. Own weight + pretension + wind suction reduced [NL]	3.19 kN	135
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.36 kN	139

Table 28. Leading reaction forces – 10x10m model

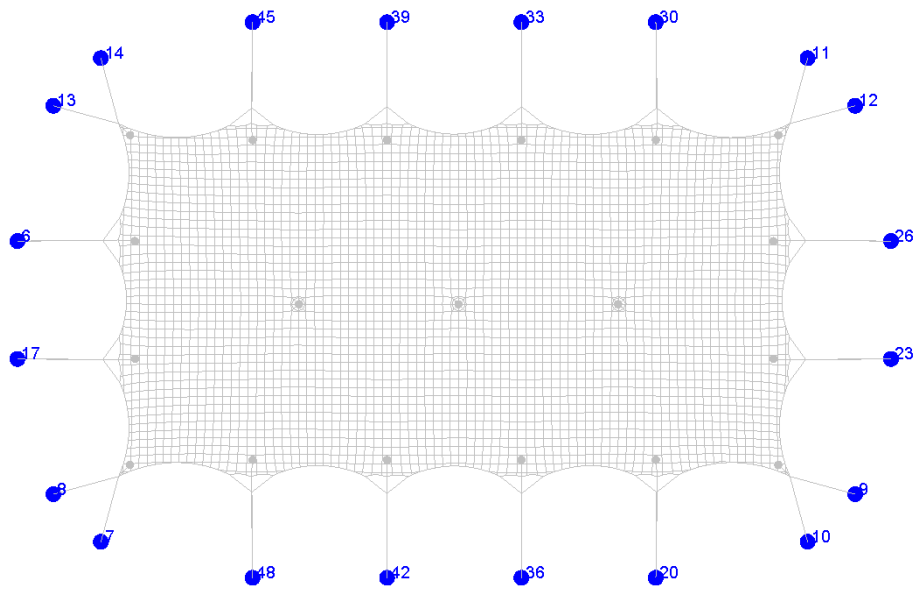
H.6.2.6.2. 10x15m model



	Load combination	F _{rep}	Pag.
Long side	CO1. Own weight + pretension	2.22 kN	143
	CO2. Own weight + pretension + snow	3.08 kN	147
	CO3. Own weight + pretension + wind pressure	3.83 kN	151
	Max CO4. Own weight + pretension + wind suction	5.20 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	4.82 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.83 kN	163
Short side	CO1. Own weight + pretension	2.04 kN	143
	CO2. Own weight + pretension + conventional / snow	2.92 kN	147
	CO3. Own weight + pretension + wind pressure	3.60 kN	151
	Max CO4. Own weight + pretension + wind suction	5.23 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	4.22 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.62 kN	163
Corner	CO1. Own weight + pretension	2.31 kN	143
	CO2. Own weight + pretension + conventional / snow	2.85 kN	147
	CO3. Own weight + pretension + wind pressure	3.28 kN	151
	Max CO4. Own weight + pretension + wind suction	4.22 kN	155
	CO5. Own weight + pretension + wind suction reduced [NL]	3.51 kN	159
	CO6. Own weight + pretension + wind pressure + snow [FR]	3.85 kN	163

Table 29. Leading reaction forces – 10x15m model

H.6.2.6.3. 10x20m model



	Load combination	F _{rep}	Pag.
Long side	CO1. Own weight + pretension	2.31 kN	167
	CO2. Own weight + pretension + snow	3.16 kN	171
	CO3. Own weight + pretension + wind pressure	3.87 kN	175
	Max CO4. Own weight + pretension + wind suction	5.77 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	5.71 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.82 kN	187
Short side	CO1. Own weight + pretension	1.99 kN	167
	CO2. Own weight + pretension + conventional / snow	2.95 kN	171
	CO3. Own weight + pretension + wind pressure	3.72 kN	175
	Max CO4. Own weight + pretension + wind suction	5.58 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	3.65 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.81 kN	187
Corner	CO1. Own weight + pretension	2.33 kN	167
	CO2. Own weight + pretension + conventional / snow	2.95 kN	171
	CO3. Own weight + pretension + wind pressure	3.44 kN	175
	Max CO4. Own weight + pretension + wind suction	4.43 kN	179
	CO5. Own weight + pretension + wind suction reduced [NL]	3.44 kN	183
	CO6. Own weight + pretension + wind pressure + snow [FR]	4.05 kN	187

Table 30. Leading reaction forces – 10x20m model

H.6.3 Summary of forces

H.6.3.1 10x10m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.95	2.42	3.09	3.39	2.49	4.55
Weft	0.91	2.11	2.62	2.18	1.99	3.79
Edge	1.76	2.13	2.44	3.41	2.68	2.85
Main pole	-2.12	-5.41	-6.95	0.05	-0.08	-10.32
Side pole	-1.30	-2.43	-3.13	-2.02	-0.83	-4.33
Corner pole	-2.64	-3.38	-3.95	-4.52	-3.43	-4.75
Corner belt	2.10	2.50	2.86	3.94	3.19	3.33
Belts - long side	0.12	0.47	0.70	0.35	0.16	1.13
Stormbelts	0.00	0.00	0.00	5.62	0.00	0.00
Combined rope - long side	1.83	2.12	2.41	4.83	3.64	2.76
V-ropes - long side	1.40	1.63	1.84	3.51	2.72	2.10
Reaction - long side	1.94	2.59	3.10	5.04	3.80	3.94
Reaction - corner	2.10	2.50	2.86	3.94	3.19	3.36

Table 31. Summary of forces for 10x10m model

H.6.3.2 10x15m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.91	2.59	3.35	3.13	2.51	4.97
Weft	0.89	2.18	2.93	2.27	2.48	4.24
Edge	1.90	2.35	2.74	3.55	3.17	3.19
Main pole	-1.74	-4.96	-6.49	0.06	-0.02	-9.78
Side pole	-1.37	-2.74	-3.60	-2.47	-0.84	-4.99
Corner pole	-2.86	-3.81	-4.56	-4.77	-3.87	-5.50
Corner belt	2.31	2.85	3.28	4.22	3.51	3.81
Belts - long side	0.14	0.51	0.81	0.50	0.18	1.30
Belts - short side	0.12	0.56	0.84	0.22	0.18	1.37
Stormbelts	0.00	0.00	0.00	6.01	0.00	0.00
Combined rope - long side	2.11	2.59	3.04	4.70	4.66	3.50
V-ropes - long side	1.53	1.83	2.11	3.38	3.30	2.39
Combined rope - short side	1.92	2.38	2.77	5.01	4.04	3.21
V-ropes short side	1.45	1.80	2.09	3.64	2.94	2.41
Reaction - long side	2.22	3.08	3.83	5.20	4.82	4.83
Reaction - short side	2.04	2.92	3.60	5.23	4.22	4.62
Reaction - corner	2.31	2.85	3.28	4.22	3.51	3.85

Table 32. Summary of forces for 10x15m model

H.6.3.3 10x20m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.95	2.63	3.41	4.45	2.53	5.01
Weft	0.88	2.16	2.94	2.89	2.68	4.20
Edge	1.80	2.26	2.67	4.43	3.43	3.16
Main pole	-1.84	-5.15	-6.75	0.06	0.03	-10.07
Side pole	-1.35	-2.69	-3.55	-2.21	-0.80	-4.88
Corner pole	-2.73	-3.67	-4.43	-4.31	-3.63	-5.40
Corner belt	2.33	2.95	3.44	4.43	3.44	4.01
Belts - long side	0.22	0.70	1.01	0.57	0.26	1.50
Belts - short side	0.13	0.55	0.85	0.13	0.13	1.35
Stormbelts	0.00	0.00	0.00	5.93	0.00	0.00
Combined rope - long side	2.16	2.48	2.91	5.43	5.46	3.34
V-ropes - long side	1.50	1.76	2.05	3.91	3.68	2.28
Combined rope - short side	1.87	2.42	2.89	5.46	3.52	3.39
V-ropes short side	1.37	1.80	2.14	3.83	2.54	2.51
Reaction - long side	2.31	3.16	3.87	5.77	5.71	4.82
Reaction - short side	1.99	2.95	3.72	5.58	3.65	4.81
Reaction - corner	2.33	2.95	3.44	4.43	3.44	4.05

Table 33. Summary of forces for 10x20m model

H.7. Element check

H.7.1 Membrane

	Load combination	Element	Representative stress	Design value stress	Pag.
10x10m	CO1. Own weight + pretension	Long term warp	0.95 kN/m	1.28 kN/m ($\gamma = 1.35$)	51
10x10m	CO1. Own weight + pretension	Long term weft	0.91 kN/m	1.23 kN/m ($\gamma = 1.35$)	51
10x20m	CO4. Own weight + pretension + wind suction	Short term warp	4.45 kN/m	6.68 kN/m ($\gamma = 1.5$)	52
10x20m	CO3. Own weight + pretension + wind pressure	Short term weft	2.94 kN/m	4.41 kN/m ($\gamma = 1.5$)	52

Sioen RP5639 fabric is being used.

UC.1a	Long term warp	$S_{Ed} / S_{rd} < 1$	$1.28 / 9.6 = 0.13 < 1$	OK
UC.1b	Long term weft	$S_{Ed} / S_{rd} < 1$	$1.23 / 7.2 = 0.17 < 1$	OK
UC.1c	Short term warp	$S_{Ed} / S_{rd} < 1$	$6.68 / 12.0 = 0.56 < 1$	OK
UC.1d	Short term weft	$S_{Ed} / S_{rd} < 1$	$4.41 / 9.0 = 0.49 < 1$	OK

For capacity of membrane see H.2.1, page 24

H.7.2 Main poles

	Load combinations	Element	Representative force	Design value force	Pag.
10x10m	CO3. Own weight + pretension + wind pressure	Main pole 4.0m	-6.95 kN	-10.43 kN ($\gamma = 1.5$)	51

User load of max. 25 kg is applied, loaded centrically.

4.0m pole $N_{ed} = -6.95 \times 1.50 + (-0.25 \times 1.35) = -10.76 \text{ kN}$

H.7.2.1 4.0m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 100 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 4.00m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.2	4.0m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.37 / (12.46 \times 0.120) = 0.92 \leq 1$	OK
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See Annex G.1 Main pole 4.0m – Wood, on page 191 for the elaborate check

H.7.3 Side poles

H.7.3.1 10x10m model

	Load combinations	Element	Representative force	Design value force	Pag.
10x10m	CO3. Own weight + pretension + wind pressure	Side pole 3.0m	-3.13 kN	-4.70 kN ($\gamma = 1.5$)	51

User load of max. 25 kg is applied, loaded centrically.

$$3\text{m pole} \quad N_{ed} = -3.13 \times 1.50 + (-0.25 \times 1.35) = -5.03 \text{ kN}$$

H.7.3.1.1. 3m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 70 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 3.0m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.3	3m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.31 / (12.46 \times 0.105) = 1.00 \leq 1$	OK
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See Annex G.2 Side pole 3m – Wood, Spruce C18, on page 192 for the elaborate check

H.7.3.2 10x15m and 10x20m model

	Load combinations	Element	Representative force	Design value force	Pag.
10x15m	CO3. Own weight + pretension + wind pressure	Side pole 3.0m	-3.60 kN	-5.40 kN ($\gamma = 1.5$)	51

User load of max. 25 kg is applied, loaded centrically.

$$3\text{m pole} \quad N_{ed} = -3.60 \times 1.50 + (-0.25 \times 1.35) = -5.74 \text{ kN}$$

H.7.3.2.1. 3m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 75 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 3.0m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.3	3m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.30 / (12.46 \times 0.120) = 0.87 \leq 1$	OK
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See Annex G.2 Side pole 3m – Wood, Spruce C18, on page 192 for the elaborate check

H.7.4 Corner poles

H.7.4.1 10x10 and 10x20m model

	Load combinations	Element	Representative force	Design value force	Pag.
10x10m	CO4. Own weight + pretension + wind suction	Corner pole 2.5m	-4.52 kN	-6.78 kN ($\gamma = 1.5$)	51

User load of max. 25 kg is applied, loaded centrally.

2.5m corner pole $N_{ed} = -4.52 \times 1.50 + (-0.25 \times 1.35) = -7.12 \text{ kN}$

H.7.4.1.1. 2.5m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 70 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 2.5m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.4	2.5m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.85 / (12.46 \times 0.149) = 1.00 \leq 1$	OK
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See Annex G.3 Corner pole 2.5m - Wood, C18, on page 194 for the elaborate check

H.7.4.2 10x15m model

	Load combinations	Element	Representative force	Design value force	Pag.
10x15m	CO4. Own weight + pretension + wind suction	Corner pole 2.5m	-4.77 kN	-7.16 kN ($\gamma = 1.5$)	51

User load of max. 25 kg is applied, loaded centrally.

2.5m corner pole $N_{ed} = -4.77 \times 1.50 + (-0.25 \times 1.35) = -7.49 \text{ kN}$

H.7.4.2.1. 2.5m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 75 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 2.5m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.4	2.5m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.70 / (12.46 \times 0.170) = 0.80 \leq 1$	OK
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See Annex G.3 Corner pole 2.5m - Wood, C18, on page 194 for the elaborate check

H.7.5 Guy ropes

	Load combination	Element	Representative force	Design value force	Pag.
10x10m	CO4. Own weight + pretension + wind suction	side - combined	4.83 kN	7.25 kN ($\gamma = 1.5$)	51
10x15m	CO4. Own weight + pretension + wind suction	long side - combined	4.70 kN	7.05 kN ($\gamma = 1.5$)	51
10x20m	CO4. Own weight + pretension + wind suction	long side - combined	5.43 kN	8.15 kN ($\gamma = 1.5$)	52
10x10m	CO4. Own weight + pretension + wind suction	side – V-ropes	3.51 kN	5.27 kN ($\gamma = 1.5$)	51
10x15m	CO4. Own weight + pretension + wind suction	long side – V-ropes	3.38 kN	5.07 kN ($\gamma = 1.5$)	51
10x20m	CO4. Own weight + pretension + wind suction	long side – V-ropes	3.91 kN	5.87 kN ($\gamma = 1.5$)	52
10x15m	CO4. Own weight + pretension + wind suction	short side - combined	5.01 kN	7.52 kN ($\gamma = 1.5$)	51
10x20m	CO4. Own weight + pretension + wind suction	short side - combined	5.46 kN	8.19 kN ($\gamma = 1.5$)	52
10x15m	CO4. Own weight + pretension + wind suction	short side – V-ropes	3.64 kN	5.46 kN ($\gamma = 1.5$)	51
10x20m	CO4. Own weight + pretension + wind suction	short side – V-ropes	3.83 kN	5.75 kN ($\gamma = 1.5$)	52

Synthetic rope Ø8mm, BL 900 daN. Is being used $\rightarrow F_{rd,8mm} = 2.25$ kN.

In a case of using individual guy ropes made of multiple sections, the ropes need to be tied back, creating multiple rope sections that together will take the load.

10x10m model

UC.5a	side - combined	1 individual rope made of <u>4 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$7.25 / (1 \times 4 \times 2.25) = 0.81 < 1$	OK
UC.5b	side – V-ropes	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$5.27 / (1 \times 3 \times 2.25) = 0.78 < 1$	OK

For capacity of ropes see H.2.2, page 24

10x15m model

UC.5e	long side - combined	1 individual rope made of <u>4 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$7.05 / (1 \times 4 \times 2.25) = 0.78 < 1$	OK
UC.5f	long side – V-ropes	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$5.07 / (1 \times 3 \times 2.25) = 0.75 < 1$	OK

UC.5g	short side - combined	1 individual rope made of <u>4 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$7.52 / (1 \times 4 \times 2.25) = 0.84 < 1$	OK
UC.5h	short side – V-ropes	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$5.46 / (1 \times 3 \times 2.25) = 0.81 < 1$	OK

For capacity of ropes see H.2.2, page 24

10x20m model

UC.5e	long side - combined	1 individual rope made of <u>4 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$8.15 / (1 \times 4 \times 2.25) = 0.91 < 1$	OK
UC.5f	long side – V-ropes	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$5.87 / (1 \times 3 \times 2.25) = 0.87 < 1$	OK
UC.5g	short side - combined	1 individual rope made of <u>4 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$8.19 / (1 \times 4 \times 2.25) = 0.91 < 1$	OK
UC.5h	short side – V-ropes	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$5.75 / (1 \times 3 \times 2.25) = 0.85 < 1$	OK

For capacity of ropes see H.2.2, page 24

It is also possible to use alternative **ropes** with a higher breaking strength (BL_{rope}) and less rope sections (n), as long as $BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}$

Required breaking loads for ropes:

10x10m model

Side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2900 \text{ kg}$

Side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2108 \text{ kg}$

10x15m model

Long side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2820 \text{ kg}$

Long side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2028 \text{ kg}$

Short side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 3008 \text{ kg}$

Short side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2184 \text{ kg}$

10x20m model

Long side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 3260 \text{ kg}$

Long side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2348 \text{ kg}$

Short side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 3276 \text{ kg}$

Short side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2300 \text{ kg}$

H.7.6 Belts

	Load combination	Element	Representative force	Design value force	Pag.
10x20m	CO4. Own weight + pretension + wind suction	Corner belt	4.43 kN	6.65 kN ($\gamma = 1.5$)	52
10x20m	CO3. Own weight + pretension + wind pressure	Belt – long side	1.01 kN	1.52 kN ($\gamma = 1.5$)	52
10x20m	CO3. Own weight + pretension + wind pressure	Belt – short side	0.85 kN	1.28 kN ($\gamma = 1.5$)	52

H.7.6.1 Belt capacity check

PES belt with a breaking strength of 3000kg is being used.

UC.6a	Corner belt	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$6.65 / (1 \times 15) = 0.44 < 1$	OK
UC.6b	Belt – long side	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$1.52 / (1 \times 15) = 0.10 < 1$	OK
UC.6c	Belt – short side	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$1.28 / (1 \times 15) = 0.09 < 1$	OK

For capacity of belts see H.2.3, page 24

H.7.6.2 Alternative – rope capacity check

It is possible to use either a rope or a belt. Synthetic rope Ø8mm, BL 900 daN. Is being used $\rightarrow F_{rd,8mm} = 2.25$ kN.

In a case of using individual guy ropes, the ropes need to be tied back, creating multiple rope sections that together will take the load.

UC.6d	Corner rope	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$6.65 / (1 \times 3 \times 2.25) = 0.99 < 1$	OK
UC.6e	Rope – long side	1 individual rope made of <u>1 section</u> Ø8mm	$F_d / F_{rd} < 1$	$1.52 / (1 \times 1 \times 2.25) = 0.68 < 1$	OK
UC.6f	Rope – short side	1 individual rope made of <u>1 section</u> Ø8mm	$F_d / F_{rd} < 1$	$1.28 / (1 \times 1 \times 2.25) = 0.57 < 1$	OK

For capacity of ropes see H.2.2, page 24

It is also possible to use alternative **ropes** with a higher breaking strength (BL_{rope}) and less rope sections (n), as long as $BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}$

Required breaking loads for ropes:

Corner rope: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2660$ kg

Rope – long side: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 608$ kg

Rope – long side: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 512$ kg

H.7.7 Stormbelts

	Load combination	Element	Representative force	Design value force	Pag.
10x15m	CO4. Own weight + pretension + wind suction	Stormbelt	6.01 kN	9.02 kN ($\gamma = 1.5$)	51

PES belt with a breaking strength of 3000kg is being used.

UC.7	Storm belt	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$9.02 / (1 \times 15) = 0.60 < 1.0$	OK
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For capacity of belts see H.2.3, page 24

H.7.8 Fasteners

Force acting on fasteners is equal to the force in belts or ropes. See these values in chapters H.7.6 and H.7.5.

H.7.8.1 Corner belt connection

H.7.8.1.1. M16 Eye bolt, DIN580

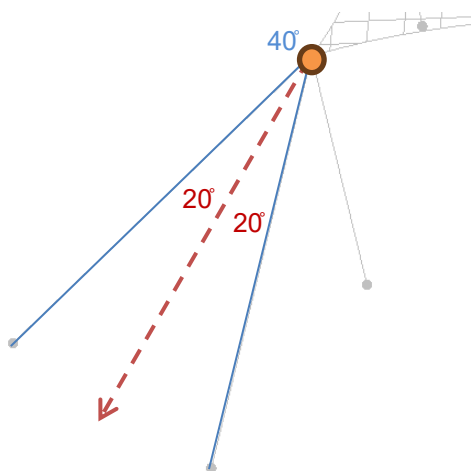
For $\leq 45^\circ$:

M16, WLL 700kg $\rightarrow$ BL = 6 x 7 kN = 42 kN $\rightarrow$ Frd = BL / γ_m = 42 / 2 = 21 kN

$F_d = 2 \times F_{\text{corner belt}} \times \cos 20^\circ = 2 \times 6.65 \times \cos 20^\circ = 12.65 \text{ kN}$

UC.8a	Corner eye bolt	$F_d / F_{rd} < 1$	$12.65 / 21 = 0.60 < 1.0$	OK
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For capacity of eye bolts see K.3 Capacity of Eye bolts on page 94.



H.7.8.1.2. Corner belt connection

A similar connection can be used if the capacity of it is at least 12.7 kN.

Orange Pin D-sluiting, WLL 1.0 t $\rightarrow$ BL = 6 x 10 kN = 60 kN $\rightarrow$ Frd = BL / γ_m = 60 kN / 2 = 30 kN

UC.8b	D-sluiting	$F_d / F_{rd} < 1$	$12.65 / 30 = 0.42 < 1.0$	OK
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For capacity of D-sluiting see K.4 Capacity of Orange Pin 1 Ton D-sluiting, DIN EN 13889 on page 94.

H.7.8.1.3. Rope ratchet – for alternative corner connection with ropes

Ratchet, BL = 3.5 kN $\rightarrow$ Frd = BL / γ_m = 3.5 kN / 1.5 = 2.33 kN

$F_d = 6.65 \text{ kN}$ (Instead of corner belts corner ropes are used.)

UC.8c	Corner ropes (alternative)	$F_d / F_{rd} < 1$	$6.65 / 3 \times 2.33 = 0.95 < 1$	OK
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For capacity of rope ratchet see K.6 Rope ratchet, page 95

* Number of cross sections making a rope. Ratchet is applied only on one cross-section.

H.7.8.2 Side belt connection

H.7.8.2.1. M12 Eye bolt, DIN580

For $\leq 45^\circ$:

M12, WLL 340kg $\rightarrow$ BL = 6 x 3.4 kN = 20.4 kN $\rightarrow$ Frd = BL / γ_m = 20.4 / 2 = 10.2 kN

F_d=1.52 kN

UC.8d	Corner eye bolt	$F_d / F_{rd} < 1$	$1.52 / 10.2 = 0.15 < 1.0$	OK
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For capacity of eye bolts see K.3 Capacity of Eye bolts on page 94.

H.7.8.2.2. Carabine, DIN5299C

Carabine 180x14, WLL 580kg $\rightarrow$ BL = 2 x 5.8 kN = 11.6 kN $\rightarrow$ Frd = BL / γ_m = 11.6 kN / 2 = 5.8 kN

F_d=1.52 kN

UC.8e	$F_d / F_{rd} < 1$	$1.52 / 5.80 = 0.26 < 1$	OK
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For capacity of carabines see K.5 Carabine, DIN 5299C, page 95

H.7.8.3 Side rope connection

H.7.8.3.1. Carabine, DIN5299C

Carabine 180x14, WLL 580kg $\rightarrow$ BL = 2 x 5.8 kN = 11.6 kN $\rightarrow$ Frd = BL / γ_m = 11.6 kN / 2 = 5.8 kN

F_d=5.87 kN

UC.8f	$F_d / F_{rd} < 1$	$5.87 / 5.80 = 1.01 \sim 1$	Acceptable
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For capacity of carabines see K.5 Carabine, DIN 5299C, page 95

H.7.8.3.2. Rope ratchet

Ratchet, BL = 3.5 kN $\rightarrow$ Frd = BL / γ_m = 3.5 kN / 1.5 = 2.33 kN

10x10m	UC.8g	Side – combined rope	$F_d / F_{rd} < 1$	$7.25 / 4 \times 2.33 = 0.78 < 1$	OK
10x15m	UC.8h	Long side – combined rope	$F_d / F_{rd} < 1$	$7.05 / 4 \times 2.33 = 0.76 < 1$	OK
10x20m	UC.8i	Long side – combined rope	$F_d / F_{rd} < 1$	$8.15 / 4 \times 2.33 = 0.87 < 1$	OK
10x15m	UC.8j	Short side – combined rope	$F_d / F_{rd} < 1$	$7.52 / 4 \times 2.33 = 0.81 < 1$	OK
10x20m	UC.8k	Short side – combined rope	$F_d / F_{rd} < 1$	$8.19 / 4 \times 2.33 = 0.88 < 1$	OK

For capacity of rope ratchet see K.6 Rope ratchet, page 95

* Number of cross sections making a rope. Ratchet is applied only on one cross-section.

H.7.8.4 Base plate

H.7.8.4.1. Side pole & corner pole

See H.6.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

- Soil (base plate size 25x25 cm, $p_{\text{soil}} = 125 \text{ kN/m}^2$)

UC.8l	Soil	$\frac{F_d}{A} / p \leq 1$	$7.49 / (0.25 \cdot 0.25) / 125 = 0.96 < 1$	OK
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H.7.8.4.2. Main pole

See H.6.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

- Soil (base plate size 35x35 cm, $p_{\text{soil}} = 175 \text{ kN/m}^2$)

UC.8m	Soil	$\frac{F_d}{A} / p \leq 1$	$10.76 / (0.35 \cdot 0.35) / 175 = 0.50 < 1$	OK
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H.8. Safety against overturning, sliding and uplifting

The calculations in this chapter provide a guideline for dense cohesionless soil, in case no anchor tests have been performed. Anchor tests on location can show that a different number of anchors or different size of anchors have sufficient capacity for the specific soil conditions. (see paragraph H.8.3)

H.8.1 Anchor capacity

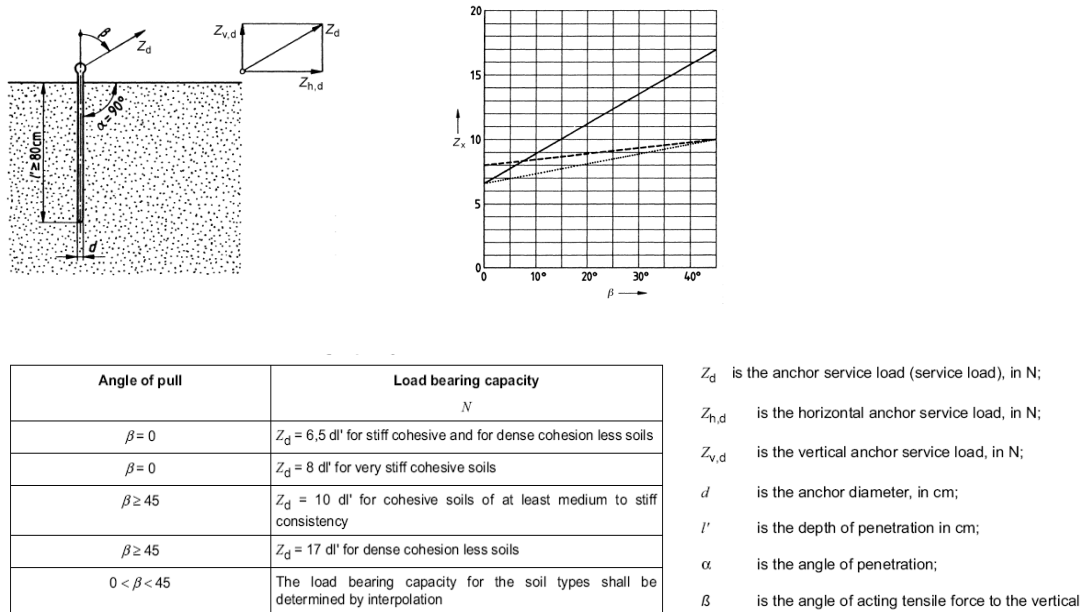


figure 7. Taken from EN 13782: Figures 4 & 5, table 5

Anchors of $\varnothing 30\text{mm}$ are used, considering an **effective length of at least 1000 mm**. If multiple anchors are placed at the same location, the anchors need to be at least $5 \times \varnothing$ apart to use the full capacity of the anchors.

Anchor capacity for anchors based in dense cohesionless soil (sandy soil)

Ø30 x 1000 mm		
Angle	β	≥ 45
Effective length anchor	l'	100 cm
Diameter anchor	d	3.0 cm
Anchor capacity*	Z_d	5.10 kN

H.8.2 Required anchor pins

	Load combination	Element	Representative force	Design value force	Pag.
10x10m	CO4. Own weight + pretension + wind suction	Side anchor point	5.04 kN	6.05 kN ($\gamma = 1.2$)	51
10x15m	CO4. Own weight + pretension + wind suction	Long side anchor point	5.20 kN	6.24 kN ($\gamma = 1.2$)	51
10x20m	CO4. Own weight + pretension + wind suction	Long side anchor point	5.77 kN	6.92 kN ($\gamma = 1.2$)	52
10x15m	CO4. Own weight + pretension + wind suction	Short side anchor point	5.23 kN	6.28 kN ($\gamma = 1.2$)	51
10x20m	CO4. Own weight + pretension + wind suction	Short side anchor point	5.58 kN	6.70 kN ($\gamma = 1.2$)	52
10x10m	CO4. Own weight + pretension + wind suction	Corner anchor point	3.94 kN	4.73 kN ($\gamma = 1.2$)	51
10x15m	CO4. Own weight + pretension + wind suction	Corner anchor point	4.22 kN	5.06 kN ($\gamma = 1.2$)	51
10x20m	CO4. Own weight + pretension + wind suction	Corner anchor point	4.43 kN	5.32 kN ($\gamma = 1.2$)	52
10x10m	CO4. Own weight + pretension + wind suction	Stormbelt	5.62 kN	6.74 kN ($\gamma = 1.2$)	51
10x15m	CO4. Own weight + pretension + wind suction	Stormbelt	6.01 kN	7.21 kN ($\gamma = 1.2$)	51
10x20m	CO4. Own weight + pretension + wind suction	Stormbelt	5.93 kN	7.12 kN ($\gamma = 1.2$)	52

Required anchor amount for dense cohesionless soil – **10x10m model**

Side guy ropes	2x	Ø30 x 1000 mm	$F_d / F_{rd} = 6.05 / (2 \times 5.10) = 0.59 \leq 1$	OK
Corner ties	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.73 / (1 \times 5.10) = 0.93 < 1$	OK
Stormbelt	2x	Ø30 x 1000 mm	$F_d / F_{rd} = 6.74 / (2 \times 5.10) = 0.66 < 1$	OK

Required anchor amount for dense cohesionless soil – **10x15m model**

Long side guy ropes	2x	Ø30 x 1000 mm	$F_d / F_{rd} = 6.24 / (2 \times 5.10) = 0.61 < 1$	OK
Short side guy ropes	2x	Ø30 x 1000 mm	$F_d / F_{rd} = 6.28 / (2 \times 5.10) = 0.62 < 1$	OK
Corner ties	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 5.06 / (1 \times 5.10) = 0.99 < 1$	OK
Stormbelts	2x	Ø30 x 1000 mm	$F_d / F_{rd} = 7.21 / (2 \times 5.10) = 0.71 < 1$	OK

Required anchor amount for dense cohesionless soil – **10x20m model**

Long side guy ropes	2x	Ø30 x 1000 mm	$F_d / F_{rd} = 6.92 / (2 \times 5.10) = 0.68 < 1$	OK
Short side guy ropes	2x	Ø30 x 1000 mm	$F_d / F_{rd} = 6.70 / (2 \times 5.10) = 0.66 < 1$	OK
Corner ties	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 5.32 / (1 \times 5.10) = 1.04 \sim 1$	Acceptable
Stormbelt	2x	Ø30 x 1000 mm	$F_d / F_{rd} = 7.12 / (2 \times 5.10) = 0.70 < 1$	OK

H.8.3 Anchor tests according to BS-EN 13782

It is advised to perform anchor test on location when there is a reason to doubt the “pull-out force” of the anchors, which could be when ground conditions differ from dense, non-cohesive soil.

Anchor tests should be carried out according to the following procedure:

Three anchors spread throughout the terrain should be put perpendicular into the ground. The anchors should be pulled out with the aid of a spring balance in the direction of the force acting on the anchor. The lowest of the three measured values should be used.

A safety factor of $v = 1.6$ regarding ultimate limit load is to apply for the lowest test value to determine the load bearing capacity in subsequent calculation. The load bearing capacity determined in this manner shall not result in anchor movement which would result in stresses, deformations or instability inadmissible for the structure.

If the foundation conditions are comparable, test loadings carried out in another location may be adduced for substantiation purposes.

For example:

Force in belts: $F_{rep} = 5.20 \text{ kN}$

$F_{sd, belt} = 1.2 \times F_{rep} = 1.2 \times 5.20 = 6.24 \text{ kN}$

The partial safety factor $\gamma = 1.6$ is applied on the ultimate limit load:

$Z_{u,d, test} > 1.6 \times F_{sd} = 1.6 \times 6.24 = 9.98 \text{ kN}$

If for example the anchor test point has a minimal anchor capacity of 5.5 kN (550 kg), then 2 anchors are needed: $2 \times 5.5 = 11 \text{ kN} > Z_{u,d, test}$

H.8.4 Required ballast

The safety against sliding shall be calculated from:

$$\sum \mu * \gamma * N_k \geq \sum \gamma * H_k \quad (\text{eq. 5. NEN-EN 13782})$$

where

μ the coefficient of friction according to table 3

γ The safety factor according to Table 2, $\gamma = 1.2$ (unfavorable wind load)

N the vertical load components (characteristic value), corresponds to the required ballast

H the horizontal load components (characteristic value).

The safety against lifting shall be calculated from:

$$\sum \gamma * N_{Rk, stb} \geq \sum \gamma * N_{Ek, dst} \quad (\text{eq. 7. NEN-EN 13782})$$

where

$N_{Rk, stb}$ are the vertical stabilizing load components (characteristic value);

$N_{Ek, dst}$ are the vertical lifting destabilizing load components (characteristic value).

For load calculations against sliding and lifting, F_{rep} is split into the horizontal (F_h) and vertical (F_v) forces. If the separate reaction force factors are not known:

$$F_h = F_{rep} \times \cos \alpha \quad \text{and} \quad F_v = F_{rep} \times \sin \alpha$$

Where α is equal to the angle between F_{rep} and the ground plane.

It follows from the above equation:

$$\text{Required load (against sliding):} \quad B_h = (\gamma F_h) / \mu$$

$$\text{Required load (against lifting):} \quad B_v = \gamma F_v$$

$$B_{tot} = B_h + B_v$$

B_{tot} = total required load

B_h = required load due to horizontal forces

B_v = required load due to vertical forces

10x10m model

Location	F_{rep} ($\gamma = 1.2$)	F_H	F_V	α	Ballast for	Ballast for
					$\mu = 0.5$	$\mu = 0.6$
					(concrete on concrete)	(concrete on rubber or wood)
	(kN)	(kN)	(kN)	°	(KG)	(KG)
Long side guy ropes	6.05	4.28	4.28	45	1280	1140
Corner ties	4.73	3.34	3.34	45	1000	890
Stormbelts	6.74	4.77	4.77	45	1430	1270

10x15m model

Location	F_{rep} ($\gamma = 1.2$)	F_H	F_V	α	Ballast for	Ballast for
					$\mu = 0.5$	$\mu = 0.6$
					(concrete on concrete)	(concrete on rubber or wood)
	(kN)	(kN)	(kN)	°	(KG)	(KG)
Long side guy ropes	6.24	4.41	4.41	45	1320	1180
Short side guy ropes	6.28	4.44	4.44	45	1330	1180
Corner ties	5.06	3.58	3.58	45	1070	960
Stormbelts	7.21	5.10	5.10	45	1530	1360

10x20m model

Location	F_{rep} ($\gamma = 1.2$)	F_H	F_V	α	Ballast for	Ballast for
					$\mu = 0.5$	$\mu = 0.6$
					(concrete on concrete)	(concrete on rubber or wood)
	(kN)	(kN)	(kN)	°	(KG)	(KG)
Long side guy ropes	6.92	4.90	4.90	45	1470	1310
Short side guy ropes	6.70	4.73	4.73	45	1420	1260
Corner ties	5.32	3.76	3.76	45	1130	1000
Stormbelts	7.12	5.03	5.03	45	1510	1340

I. Addition 1 – element check for reduced wind speeds [NL]

For stretchtent use in the Netherlands, a reduction of maximal allowable wind speeds is specified. As a result, the stretchtent is calculated for two wind scenarios – full wind and reduced wind. Maximal wind speeds are specified and if the structure exceeds them, the tent must be evacuated and access must be denied. For these wind speeds no stormbelts are necessary.

A reduction factor $\alpha=0.7$ is applied, which leads to $q_p(z_e)$ of **0.25 kN/m² [NL]**. **No stormbelts are required.**

I.1. Summary of results – low wind in Netherlands

This document is valid for five tent sizes: 10x10m, 10x15m, 10x20m, 10x12m and 9x12m. Although no separate calculation is made for 10x12m and 9x12m sizes, all elements, sizes and anchoring forces can be based on 10x15m model.

Changes that come from lower wind and no stormbelts are highlighted in orange.

General information:				
Main dimensions:		10x10m	10x15m / 10x12m / 9x12m	10x20m
Width:		10m	10m / 9m	10m
Length:		10m	15m / 12m	20m
Side height of membrane:		3m	3m	3m
Max. height:		4m	4m	4m
Main poles: *		4.0m Ø100 mm [Spruce,C18]	4.0m Ø100 mm [Spruce,C18]	4.0m Ø100 mm [Spruce, C18]
Side poles: *		3.0m Ø 70 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]
Corner poles: *		2.5m Ø 70 mm [Spruce, C18]	2.5m Ø 75 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
Guy ropes:		Ø 8mm polyester, breaking strength ≥ 900 daN **	Ø 8mm polyester, breaking strength ≥ 900 daN **	Ø 8mm polyester, breaking strength ≥ 900 daN **
Long side – combined:		3 sections	3 sections	4 sections
Long side - V-ropes:		2 sections	3 sections	3 sections
Short side – combined:		-----	3 sections	3 sections
Short side - V-ropes:		-----	2 sections	2 sections
Belts:		50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
Fasteners:		Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN	Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN	Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN

	Side & corner pole base plate	25x25cm	25x25cm	25x25cm
	Main pole base plate	35x35cm	35x35cm	35x35cm
User load:	A maximum of 25 kg per main mast may be used for decorations, sound or lighting installations. The load must be applied centrally.			
Snow load:	A snow load of 0.1 kN/m ² (4 cm) has been considered in accordance with the French CTS standard.			
Wind load:	According to EN 13782 The following pressure values have been used: h < 5m: 0.25kN/m ²			

* The average diameter, as a minimum required at the middle of the pole.

** The required breaking load (BL_{tot}) may be achieved by using multiple rope sections (n): BL_{rope} = BL_{tot} / n, see page 74

Safety against sliding, overturning and uplifting:

Anchor forces:	Assumptions: angle of 45 degrees The following design resistance of the anchor forces is required: (See H.8.3, page 67 for Anchor tests according to EN 13782)			
		10x10m	10x15m / 10x12m / 9x12m	10x20m
	Long side guy ropes:	4.56 kN (γ = 1.2)	5.78 kN (γ = 1.2)	6.85 kN (γ = 1.2)
	Short side guy ropes:	5.06 kN (γ = 1.2)	4.38 kN (γ = 1.2)	4.13 kN (γ = 1.2)
Required anchors:	Corner ties:	3.83 kN (γ = 1.2)	4.21 kN (γ = 1.2)	4.13 kN (γ = 1.2)
	Based on dense, non-cohesive soil (e.g. sandy soils). In the case anchors Ø30mm x 1000mm (effective length) are being used:			
		10x10m	10x15m / 10x12m / 9x12m	10x20m
	Long side guy ropes:	1 anchor	2 anchors	2 anchors
Required ballast:	Short side guy ropes:	1 anchor	1 anchor	1 anchor
	Corner ties:	1 anchor	1 anchor	1 anchor
	Ballast calculation is performed for 2 different situations: - Concrete on concrete – concrete ballasts are placed on concrete - Concrete on wood or rubber – concrete ballasts are placed on wood, or an anti-slip mat is placed underneath the ballast			
	10x10m			
		Concrete on concrete	Concrete on wood or rubber	
	Long side guy ropes:	970 kg	860 kg	
	Corner ties:	810 kg	720 kg	
	10x15m / 10x12m / 9x12m			

		Concrete on concrete	Concrete on wood or rubber
Long side guy ropes:		1230 kg	1090 kg
Short side guy ropes:		1070 kg	960 kg
Corner ties:		980 kg	790 kg
10x20m			
		Concrete on concrete	Concrete on wood or rubber
Long side guy ropes:		1450 kg	1290 kg
Short side guy ropes:		950 kg	840 kg
Corner ties:		880 kg	780 kg

I.2. Summary of forces

Further chapters include the calculation only for highlighted elements in the CO5 [NL] column if they show higher values than calculated in the previous chapters or if recalculating them show a clear difference in results. Other element checks can be looked up in H.7.

I.2.1 10x10m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.95	2.42	3.09	3.39	2.49	4.55
Weft	0.91	2.11	2.62	2.18	1.99	3.79
Edge	1.76	2.13	2.44	3.41	2.68	2.85
Main pole	-2.12	-5.41	-6.95	0.05	-0.08	-10.32
Side pole	-1.30	-2.43	-3.13	-2.02	-0.83	-4.33
Corner pole	-2.64	-3.38	-3.95	-4.52	-3.43	-4.75
Corner belt	2.10	2.50	2.86	3.94	3.19	3.33
Belts - side	0.12	0.47	0.70	0.35	0.16	1.13
Stormbelts	0.00	0.00	0.00	5.62	0.00	0.00
Combined rope - side	1.83	2.12	2.41	4.83	3.64	2.76
V-ropes - side	1.40	1.63	1.84	3.51	2.72	2.10
Reaction - side	1.94	2.59	3.10	5.04	3.80	3.94
Reaction - corner	2.10	2.50	2.86	3.94	3.19	3.36

Table 34. Summary of forces for 10x10m model

I.2.2 10x15m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.91	2.59	3.35	3.13	2.51	4.97
Weft	0.89	2.18	2.93	2.27	2.48	4.24
Edge	1.90	2.35	2.74	3.55	3.17	3.19
Main pole	-1.74	-4.96	-6.49	0.06	-0.02	-9.78
Side pole	-1.37	-2.74	-3.60	-2.47	-0.84	-4.99
Corner pole	-2.86	-3.81	-4.56	-4.77	-3.87	-5.50
Corner belt	2.31	2.85	3.28	4.22	3.51	3.81
Belts - long side	0.14	0.51	0.81	0.50	0.18	1.30
Belts - short side	0.12	0.56	0.84	0.22	0.18	1.37
Stormbelts	0.00	0.00	0.00	6.01	0.00	0.00
Combined rope - long side	2.11	2.59	3.04	4.70	4.66	3.50
V-ropes - long side	1.53	1.83	2.11	3.38	3.30	2.39
Combined rope - short side	1.92	2.38	2.77	5.01	4.04	3.21
V-ropes short side	1.45	1.80	2.09	3.64	2.94	2.41
Reaction - long side	2.22	3.08	3.83	5.20	4.82	4.83
Reaction - short side	2.04	2.92	3.60	5.23	4.22	4.62
Reaction - corner	2.31	2.85	3.28	4.22	3.51	3.85

Table 35. Summary of forces for 10x15m model

I.2.3 10x20m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.95	2.63	3.41	4.45	2.53	5.01
Weft	0.88	2.16	2.94	2.89	2.68	4.20
Edge	1.80	2.26	2.67	4.43	3.43	3.16
Main pole	-1.84	-5.15	-6.75	0.06	0.03	-10.07
Side pole	-1.35	-2.69	-3.55	-2.21	-0.80	-4.88
Corner pole	-2.73	-3.67	-4.43	-4.31	-3.63	-5.40
Corner belt	2.33	2.95	3.44	4.43	3.44	4.01
Belts - long side	0.22	0.70	1.01	0.57	0.26	1.50
Belts - short side	0.13	0.55	0.85	0.13	0.13	1.35
Stormbelts	0.00	0.00	0.00	5.93	0.00	0.00
Combined rope - long side	2.16	2.48	2.91	5.43	5.46	3.34
V-ropes - long side	1.50	1.76	2.05	3.91	3.68	2.28
Combined rope - short side	1.87	2.42	2.89	5.46	3.52	3.39
V-ropes short side	1.37	1.80	2.14	3.83	2.54	2.51
Reaction - long side	2.31	3.16	3.87	5.77	5.71	4.82
Reaction - short side	1.99	2.95	3.72	5.58	3.65	4.81
Reaction - corner	2.33	2.95	3.44	4.43	3.44	4.05

Table 36. Summary of forces for 10x20m model

I.3. Element check - low wind in Netherlands

I.3.1 Guy ropes

	Load combination	Element	Representative force	Design value force	Pag.
10x10m	CO5. Own weight + pretension + wind suction reduced	side - combined	3.64 kN	5.46 kN ($\gamma = 1.5$)	72
10x15m	CO5. Own weight + pretension + wind suction reduced	long side - combined	4.66 kN	6.99 kN ($\gamma = 1.5$)	73
10x20m	CO5. Own weight + pretension + wind suction reduced	long side - combined	5.46 kN	8.19 kN ($\gamma = 1.5$)	73
10x10m	CO5. Own weight + pretension + wind suction reduced	side – V-ropes	2.72 kN	4.08 kN ($\gamma = 1.5$)	72
10x15m	CO5. Own weight + pretension + wind suction reduced	long side – V-ropes	3.30 kN	4.95 kN ($\gamma = 1.5$)	73
10x20m	CO5. Own weight + pretension + wind suction reduced	long side – V-ropes	3.68 kN	5.52 kN ($\gamma = 1.5$)	73
10x15m	CO5. Own weight + pretension + wind suction reduced	short side - combined	4.04 kN	6.06 kN ($\gamma = 1.5$)	73
10x20m	CO5. Own weight + pretension + wind suction reduced	short side - combined	3.52 kN	5.28 kN ($\gamma = 1.5$)	73
10x15m	CO5. Own weight + pretension + wind suction reduced	short side – V-ropes	2.94 kN	4.41 kN ($\gamma = 1.5$)	73
10x20m	CO5. Own weight + pretension + wind suction reduced	short side – V-ropes	2.54 kN	3.76 kN ($\gamma = 1.5$)	73

Synthetic rope Ø8mm, BL 900 daN. Is being used → $F_{rd,8mm} = 2.25$ kN.

In a case of using individual guy ropes made of multiple sections, the ropes need to be tied back, creating multiple rope sections that together will take the load.

10x10m model

UC.5a	side - combined	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$5.46 / (1 \times 3 \times 2.25) = 0.81 < 1$	OK
UC.5b	side – V-ropes	1 individual rope made of <u>2 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$4.08 / (1 \times 2 \times 2.25) = 0.91 < 1$	OK

For capacity of ropes see H.2.2, page 24

10x15m model

UC.5e	long side - combined	1 individual rope made of 3 sections Ø8mm	$F_d / F_{rd} < 1$	$6.99 / (1 \times 3 \times 2.25) = 1.04 \sim 1$	OK
UC.5f	long side – V-ropes	1 individual rope made of 3 sections Ø8mm	$F_d / F_{rd} < 1$	$4.95 / (1 \times 3 \times 2.25) = 0.73 < 1$	OK
UC.5g	short side - combined	1 individual rope made of 3 sections Ø8mm	$F_d / F_{rd} < 1$	$6.06 / (1 \times 3 \times 2.25) = 0.90 < 1$	OK
UC.5h	short side – V-ropes	1 individual rope made of 2 sections Ø8mm	$F_d / F_{rd} < 1$	$4.41 / (1 \times 2 \times 2.25) = 0.98 < 1$	OK

For capacity of ropes see H.2.2, page 24

10x20m model

UC.5e	long side - combined	1 individual rope made of 4 sections Ø8mm	$F_d / F_{rd} < 1$	$8.19 / (1 \times 4 \times 2.25) = 0.91 < 1$	OK
UC.5f	long side – V-ropes	1 individual rope made of 3 sections Ø8mm	$F_d / F_{rd} < 1$	$5.52 / (1 \times 3 \times 2.25) = 0.82 < 1$	OK
UC.5g	short side - combined	1 individual rope made of 3 sections Ø8mm	$F_d / F_{rd} < 1$	$5.28 / (1 \times 3 \times 2.25) = 0.78 < 1$	OK
UC.5h	short side – V-ropes	1 individual rope made of 2 sections Ø8mm	$F_d / F_{rd} < 1$	$3.76 / (1 \times 2 \times 2.25) = 0.84 < 1$	OK

For capacity of ropes see H.2.2, page 24

It is also possible to use alternative **ropes** with a higher breaking strength (BL_{rope}) and less rope sections (n), as long as $BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}$

Required breaking loads for ropes:

10x10m model

Side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2184 \text{ kg}$

Side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1632 \text{ kg}$

10x15m model

Long side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2796 \text{ kg}$

Long side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1900 \text{ kg}$

Short side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2424 \text{ kg}$

Short side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1764 \text{ kg}$

10x20m model

Long side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 3276 \text{ kg}$

Long side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2208 \text{ kg}$

Short side - combined: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2112 \text{ kg}$

Short side – V-ropes: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 1504 \text{ kg}$

I.3.2 Fasteners

Force acting on fasteners is equal to the force in belts or ropes. See these values in chapter I.3.1.

I.3.2.1 Side rope connection

I.3.2.1.1. Carabine, DIN5299C

Carabine 180x14, WLL 580kg $\rightarrow$ BL = 2 x 5.8 kN = 11.6 kN $\rightarrow$ Frd = BL / γ_m = 11.6 kN / 2 = 5.8 kN
F_d=5.52 kN

UC.3a	F_d / F_{rd} < 1	5.52 / 5.8 = 0.95 ≤ 1	OK
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For capacity of carabines see K.5 Carabine, DIN 5299C, page 95

I.3.2.1.2. Rope ratchet

Ratchet, BL = 3.5 kN $\rightarrow$ Frd = BL / γ_m = 3.5 kN / 1.5 = 2.33 kN

10x10m	UC.3b	Side – combined rope	F_d / F_{rd} < 1	5.46 / 3*x2.33= 0.78 < 1	OK
10x15m	UC.3c	Long side – combined rope	F_d / F_{rd} < 1	6.99 / 3*x2.33= 1.00 < 1	OK
10x20m	UC.3d	Long side – combined rope	F_d / F_{rd} < 1	8.19 / 4*x2.33= 0.88 < 1	OK
10x15m	UC.3e	Short side – combined rope	F_d / F_{rd} < 1	6.06 / 3*x2.33= 0.87 < 1	OK
10x20m	UC.3f	Short side – combined rope	F_d / F_{rd} < 1	5.28 / 3*x2.33= 0.76 < 1	OK

For capacity of rope ratchet see K.6 Rope ratchet, page 95

* Number of cross sections making a rope. Ratchet is applied only on one cross-section.

I.4. Safety against overturning, sliding and uplifting - low wind in Netherlands

The calculation in this chapter provides a guideline for dense cohesion less soil, in case no anchor tests have been performed. Anchor tests on location can show that a different number of anchors or different size of anchors have sufficient capacity for the specific soil conditions. (see paragraph H.8.3)

Assumed anchor capacity $F_{rd}=5.1$ kN (see H.8.1)

I.4.1 Required anchor pins

	Load combination	Element	Representative force	Design value force	Pag.
10x10m	CO5. Own weight + pretension + wind suction reduced	Side anchor point	3.80 kN	4.56 kN ($\gamma = 1.2$)	72
10x15m	CO5. Own weight + pretension + wind suction reduced	Long side anchor point	4.82 kN	5.78 kN ($\gamma = 1.2$)	73
10x20m	CO5. Own weight + pretension + wind suction reduced	Long side anchor point	5.71 kN	6.85 kN ($\gamma = 1.2$)	73
10x15m	CO5. Own weight + pretension + wind suction reduced	Short side anchor point	4.22 kN	5.06 kN ($\gamma = 1.2$)	73
10x20m	CO3. Own weight + pretension + wind pressure	Short side anchor point	3.72 kN	4.46 kN ($\gamma = 1.2$)	73
10x10m	CO5. Own weight + pretension + wind suction reduced	Corner anchor point	3.19 kN	3.83 kN ($\gamma = 1.2$)	72
10x15m	CO5. Own weight + pretension + wind suction reduced	Corner anchor point	3.51 kN	4.21 kN ($\gamma = 1.2$)	73
10x20m	CO5. Own weight + pretension + wind suction reduced	Corner anchor point	3.44 kN	4.13 kN ($\gamma = 1.2$)	73

Required anchor amount for dense cohesionless soil – **10x10m model**

Side guy ropes	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.56 / (1 \times 5.10) = 0.89 \leq 1$	OK
Corner ties	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 3.83 / (1 \times 5.10) = 0.75 < 1$	OK

Required anchor amount for dense cohesionless soil – **10x15m model**

Long side guy ropes	2x	Ø30 x 1000 mm	$F_d / F_{rd} = 5.78 / (2 \times 5.10) = 0.57 < 1$	OK
Short side guy ropes	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 5.06 / (1 \times 5.10) = 0.99 < 1$	OK
Corner ties	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.21 / (1 \times 5.10) = 0.83 < 1$	OK

Required anchor amount for dense cohesionless soil – **10x20m model**

Long side guy ropes	2x	Ø30 x 1000 mm	$F_d / F_{rd} = 6.85 / (2 \times 5.10) = 0.67 < 1$	OK
Short side guy ropes	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.46 / (1 \times 5.10) = 0.87 < 1$	OK
Corner ties	1x	Ø30 x 1000 mm	$F_d / F_{rd} = 4.13 / (1 \times 5.10) = 0.81 < 1$	OK

I.4.2 Required ballast

For calculation method see H.8.4.

10x10m model

Location	F_{rep} ($\gamma = 1.2$)	F_H	F_V	α	Ballast for	Ballast for
					$\mu = 0.5$	$\mu = 0.6$
					(concrete on concrete)	(concrete on rubber or wood)
	(kN)	(kN)	(kN)	°	(KG)	(KG)
Long side guy ropes	4.56	3.22	3.22	45	970	860
Corner ties	3.83	2.71	2.71	45	810	720

10x15m model

Location	F_{rep} ($\gamma = 1.2$)	F_H	F_V	α	Ballast for	Ballast for
					$\mu = 0.5$	$\mu = 0.6$
					(concrete on concrete)	(concrete on rubber or wood)
	(kN)	(kN)	(kN)	°	(KG)	(KG)
Long side guy ropes	5.78	4.09	4.09	45	1230	1090
Short side guy ropes	5.06	3.58	3.58	45	1070	960
Corner ties	4.21	2.98	2.98	45	890	790

10x20m model

Location	F_{rep} ($\gamma = 1.2$)	F_H	F_V	α	Ballast for	Ballast for
					$\mu = 0.5$	$\mu = 0.6$
					(concrete on concrete)	(concrete on rubber or wood)
	(kN)	(kN)	(kN)	°	(KG)	(KG)
Long side guy ropes	6.85	4.85	4.85	45	1450	1290
Short side guy ropes	4.46	3.10	3.10	45	930	830
Corner ties	4.13	2.92	2.92	45	880	780

J. Addition 2 – element check for combined wind and snow load [FR]

For stretchtent use in France, the static analysis must include a load combination including snow and wind loads, corresponding to article 7 of CTS. This addition represents changes in element checks and anchor and ballast determination if this load combination is considered. Stormbelts are always required.

J.1. Summary of results – snow+wind in France

This document is valid for five tent sizes: 10x10m, 10x15m, 10x20m, 10x12m and 9x12m. Although no separate calculation is made for 10x12m and 9x12m sizes, all elements, sizes and anchoring forces can be based on 10x15m model.

Changes that come from snow and wind combination are highlighted in orange.

General information:				
Main dimensions:		10x10m	10x15m / 10x12m / 9x12m	10x20m
Width:		10m	10m / 9m	10m
Length:		10m	15m / 12m	20m
Side height of membrane:		3m	3m	3m
Max. height:		4m	4m	4m
Main poles: *		4.0m Ø105 mm [Spruce,C18]	4.0m Ø105 mm [Spruce,C18]	4.0m Ø105 mm [Spruce, C18]
Side poles: *		3.0m Ø 75 mm [Spruce, C18]	3.0m Ø 80 mm [Spruce, C18]	3.0m Ø 80 mm [Spruce, C18]
Corner poles: *		2.5m Ø 75 mm [Spruce, C18]	2.5m Ø 75 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
Stormbelts:		50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
Guy ropes:		Ø 8mm polyester, breaking strength ≥ 900 daN **	Ø 8mm polyester, breaking strength ≥ 900 daN **	Ø 8mm polyester, breaking strength ≥ 900 daN **
Long side – combined:		4 sections	4 sections	4 sections
Long side - V-ropes:		3 sections	3 sections	3 sections
Short side – combined:		-----	4 sections	4 sections
Short side - V-ropes:		-----	3 sections	3 sections
Belts:		50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
Fasteners:		Carabine 180x14 M16 & M8 Eye bolt Ratchet, BL = 3.5 kN	Carabine 180x14 M16 & M8 Eye bolt Ratchet, BL = 3.5 kN	Carabine 180x14 M16 & M8 Eye bolt Ratchet, BL = 3.5 kN

	Side & corner pole base plate	25x25cm	25x25cm	25x25cm
	Main pole base plate	35x35cm	35x35cm	35x35cm
User load:	A maximum of 25 kg per main mast may be used for decorations, sound or lighting installations. The load must be applied centrally.			
Snow load:	A snow load of 0.1 kN/m ² (4 cm) has been considered in accordance with the French CTS standard.			
Wind load:	According to EN 13782 The following pressure values have been used: h < 5m: 0.50kN/m ² This value corresponds to wind force 8 on the Beaufort scale. Above this wind force (starting at wind force 9), the safety of the structure can no longer be guaranteed, and it must be evacuated.			

* The average diameter, as a minimum required at the middle of the pole.

** The required breaking load (BL_{tot}) may be achieved by using multiple rope sections (n): BL_{rope} = BL_{tot} / n, see page 57

Safety against sliding, overturning and uplifting:

Anchor forces:	Assumptions: angle of 45 degrees The following design resistance of the anchor forces is required: (See H.8.3, page 67 for Anchor tests according to EN 13782)			
		10x10m	10x15m / 10x12m / 9x12m	10x20m
	Long side guy ropes:	6.05 kN (γ = 1.2)	6.24 kN (γ = 1.2)	6.92 kN (γ = 1.2)
	Short side guy ropes:		6.28 kN (γ = 1.2)	6.70 kN (γ = 1.2)
	Corner ties:	4.73 kN (γ = 1.2)	5.06 kN (γ = 1.2)	5.32 kN (γ = 1.2)
Required anchors:	Stormbelt:	6.74 kN (γ = 1.2)	7.21 kN (γ = 1.2)	7.12 kN (γ = 1.2)
	Based on dense, non-cohesive soil (e.g. sandy soils). In the case anchors Ø30mm x 1000mm (effective length) are being used:			
		10x10m	10x15m / 10x12m / 9x12m	10x20m
	Long side guy ropes:	2 anchors	2 anchors	2 anchors
	Short side guy ropes:		2 anchors	2 anchors
Required ballast:	Corner ties:	1 anchor	1 anchor	1 anchor
	Stormbelt:	2 anchors	2 anchors	2 anchors
	Ballast calculation is performed for 2 different situations:			
	- Concrete on concrete – concrete ballasts are placed on concrete - Concrete on wood or rubber – concrete ballasts are placed on wood, or an anti-slip mat is placed underneath the ballast			
	10x10m			
		Concrete on concrete	Concrete on wood or rubber	

Long side guy ropes:	1280 kg	1140 kg
Corner ties:	1000 kg	890 kg
Stormbelt:	1430 kg	1270 kg
10x15m / 10x12m / 9x12m		
	Concrete on concrete	Concrete on wood or rubber
Long side guy ropes:	1320 kg	1180 kg
Short side guy ropes:	1330 kg	1180 kg
Corner ties:	1070 kg	960 kg
Stormbelt:	1530 kg	1360 kg
10x20m		
	Concrete on concrete	Concrete on wood or rubber
Long side guy ropes:	1470 kg	1310 kg
Short side guy ropes:	1420 kg	1260 kg
Corner ties:	1130 kg	1000 kg
Stormbelt:	1510 kg	1340 kg

J.2. Cross sections

Profile	Material	Ø mm	G kg/m	A mm ²	I _y mm ⁴	W _{el,y} mm ³
Ø ≈ 105 mm *	Wood C18	105	2.77	8659	5966602	113649
Ø ≈ 80 mm *	Wood C18	80	1.61	5026	2010619	50265
Ø ≈ 75 mm *	Wood C18	75	1.41	4418	1553156	41417

Table 37. Used cross sections

* The average diameter, as a minimum required at the middle of the pole.

J.3. Summary of forces

Further chapters include the calculation only for highlighted elements in the CO6 [FR] column if they show higher values than calculated in the previous chapters or if recalculating them show a clear difference in results. Other element checks can be looked up in H.7.

J.3.1 10x10m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.95	2.42	3.09	3.39	2.49	4.55
Weft	0.91	2.11	2.62	2.18	1.99	3.79
Edge	1.76	2.13	2.44	3.41	2.68	2.85
Main pole	-2.12	-5.41	-6.95	0.05	-0.08	-10.32
Side pole	-1.30	-2.43	-3.13	-2.02	-0.83	-4.33
Corner pole	-2.64	-3.38	-3.95	-4.52	-3.43	-4.75
Corner belt	2.10	2.50	2.86	3.94	3.19	3.33

Belts - side	0.12	0.47	0.70	0.35	0.16	1.13
Stormbelts	0.00	0.00	0.00	5.62	0.00	0.00
Combined rope - side	1.83	2.12	2.41	4.83	3.64	2.76
V-ropes - side	1.40	1.63	1.84	3.51	2.72	2.10
Reaction - side	1.94	2.59	3.10	5.04	3.80	3.94
Reaction - corner	2.10	2.50	2.86	3.94	3.19	3.36

Table 38. Summary of forces for 10x10m model

J.3.2 10x15m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.91	2.59	3.35	3.13	2.51	4.97
Weft	0.89	2.18	2.93	2.27	2.48	4.24
Edge	1.90	2.35	2.74	3.55	3.17	3.19
Main pole	-1.74	-4.96	-6.49	0.06	-0.02	-9.78
Side pole	-1.37	-2.74	-3.60	-2.47	-0.84	-4.99
Corner pole	-2.86	-3.81	-4.56	-4.77	-3.87	-5.50
Corner belt	2.31	2.85	3.28	4.22	3.51	3.81
Belts - long side	0.14	0.51	0.81	0.50	0.18	1.30
Belts - short side	0.12	0.56	0.84	0.22	0.18	1.37
Stormbelts	0.00	0.00	0.00	6.01	0.00	0.00
Combined rope - long side	2.11	2.59	3.04	4.70	4.66	3.50
V-ropes - long side	1.53	1.83	2.11	3.38	3.30	2.39
Combined rope - short side	1.92	2.38	2.77	5.01	4.04	3.21
V-ropes short side	1.45	1.80	2.09	3.64	2.94	2.41
Reaction - long side	2.22	3.08	3.83	5.20	4.82	4.83
Reaction - short side	2.04	2.92	3.60	5.23	4.22	4.62
Reaction - corner	2.31	2.85	3.28	4.22	3.51	3.85

Table 39. Summary of forces for 10x15m model

J.3.3 10x20m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.95	2.63	3.41	4.45	2.53	5.01
Weft	0.88	2.16	2.94	2.89	2.68	4.20
Edge	1.80	2.26	2.67	4.43	3.43	3.16
Main pole	-1.84	-5.15	-6.75	0.06	0.03	-10.07
Side pole	-1.35	-2.69	-3.55	-2.21	-0.80	-4.88
Corner pole	-2.73	-3.67	-4.43	-4.31	-3.63	-5.40
Corner belt	2.33	2.95	3.44	4.43	3.44	4.01
Belts - long side	0.22	0.70	1.01	0.57	0.26	1.50
Belts - short side	0.13	0.55	0.85	0.13	0.13	1.35
Stormbelts	0.00	0.00	0.00	5.93	0.00	0.00
Combined rope - long side	2.16	2.48	2.91	5.43	5.46	3.34
V-ropes - long side	1.50	1.76	2.05	3.91	3.68	2.28
Combined rope - short side	1.87	2.42	2.89	5.46	3.52	3.39
V-ropes short side	1.37	1.80	2.14	3.83	2.54	2.51
Reaction - long side	2.31	3.16	3.87	5.77	5.71	4.82

Reaction - short side	1.99	2.95	3.72	5.58	3.65	4.81
Reaction - corner	2.33	2.95	3.44	4.43	3.44	4.05

Table 40. Summary of forces for 10x20m model

J.4. Element check – snow + wind in France

J.4.1 Membrane

	Load combination	Element	Representative stress	Design value stress	Pag.
10x10m	CO1. Own weight + pretension	Long term warp	0.95 kN/m	1.28 kN/m ($\gamma = 1.35$)	82
10x10m	CO1. Own weight + pretension	Long term weft	0.91 kN/m	1.23 kN/m ($\gamma = 1.35$)	82
10x20m	CO6. Own weight + pretension + wind pressure + snow	Short term warp	5.01 kN/m	6.76 kN/m ($\gamma = 1.35$)	83
10x15m	CO6. Own weight + pretension + wind pressure + snow	Short term weft	4.24 kN/m	5.72 kN/m ($\gamma = 1.35$)	83

Sioen RP5639 fabric is being used.

UC.1a	Long term warp	$S_{Ed} / S_{rd} < 1$	$1.28 / 9.6 = 0.13 < 1$	OK
UC.1b	Long term weft	$S_{Ed} / S_{rd} < 1$	$1.23 / 7.2 = 0.17 < 1$	OK
UC.1c	Short term warp	$S_{Ed} / S_{rd} < 1$	$6.76 / 12.0 = 0.56 < 1$	OK
UC.1d	Short term weft	$S_{Ed} / S_{rd} < 1$	$5.72 / 9.0 = 0.64 < 1$	OK

For capacity of membrane see H.2.1, page 24

J.4.2 Main poles

	Load combinations	Element	Representative force	Design value force	Pag.
10x10m	CO6. Own weight + pretension + wind pressure + snow	Main pole 4.0m	-10.32 kN	-13.93 kN ($\gamma = 1.35$)	82

User load of max. 25 kg is applied, loaded centrically.

4.0m pole $N_{ed} = -10.32 \times 1.35 + (-0.25 \times 1.35) = -14.27 \text{ kN}$

J.4.2.1 4.0m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 105 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 4.00m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.2	4.0m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.65 / (12.46 \times 0.132) = 1.01 \sim 1$	Acceptable
------	------	--	------------

See Annex G.4 Main pole 4.0m – Wood, Spruce C18 – for snow and wind combination [FR], on page 196 for the elaborate check

J.4.3 Side poles

J.4.3.1 10x10m model

	Load combinations	Element	Representative force	Design value force	Pag.
10x10m	CO6. Own weight + pretension + wind pressure + snow	Side pole 3.0m	-4.33 kN	-5.85 kN ($\gamma = 1.5$)	83

User load of max. 25 kg is applied, loaded centrically.

$$3\text{m pole} \quad N_{ed} = -4.33 \times 1.35 + (-0.25 \times 1.35) = -6.18 \text{ kN}$$

J.4.3.1.1. 3m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 75 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 3.0m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.3	3m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.40 / (12.46 \times 0.120) = 0.94 \leq 1$	OK
------	----	--	----

See Annex G.5 Side pole 3.0m – Wood, Spruce C18 – for snow and wind combination [FR], on page 197 for the elaborate check

J.4.3.2 10x15m and 10x20m model

	Load combinations	Element	Representative force	Design value force	Pag.
10x15m	CO6. Own weight + pretension + wind pressure + snow	Side pole 3.0m	-4.99 kN	-6.74 kN ($\gamma = 1.5$)	83

User load of max. 25 kg is applied, loaded centrically.

$$3\text{m pole} \quad N_{ed} = -4.99 \times 1.35 + (-0.25 \times 1.35) = -7.07 \text{ kN}$$

J.4.3.2.1. 3m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 80 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 3.0m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.3	3m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.41 / (12.46 \times 0.136) = 0.83 \leq 1$	OK
------	----	--	----

See Annex G.5 Side pole 3.0m – Wood, Spruce C18 – for snow and wind combination [FR], on page 197 for the elaborate check

J.4.4 Corner poles

	Load combinations	Element	Representative force	Design value force	Pag.
10x15m	CO6. Own weight + pretension + wind pressure + snow	Corner pole 2.5m	-5.50 kN	-7.43 kN ($\gamma = 1.35$)	83

User load of max. 25 kg is applied, loaded centrically.

2.5m corner pole $N_{ed} = -5.50 \times 1.35 + (-0.25 \times 1.35) = -7.76 \text{ kN}$

J.4.4.1 2.5m pole - Wood, Spruce C18

Profile	=	Pole, $\varnothing \approx 75 \text{ mm}$ average diameter, as a minimum required at the middle of the pole
Length	=	max. 2.5m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.4	2.5m	$\sigma_{c,0,d} / (f_{c,0,d} \times k_{cy}) = 1.76 / (12.46 \times 0.170) = 0.80 < 1$	OK
------	------	---	----

See Annex G.6 Corner pole 2.5m – Wood, Spruce C18 – for snow and wind combination [FR], on page 199 for the elaborate check

J.4.5 Belts

	Load combination	Element	Representative force	Design value force	Pag.
10x20m	CO4. Own weight + pretension + wind suction	Corner belt	4.43 kN	6.65 kN ($\gamma = 1.5$)	83
10x20m	CO6. Own weight + pretension + wind pressure + snow	Belt – long side	1.50 kN	2.03 kN ($\gamma = 1.35$)	83
10x15m	CO6. Own weight + pretension + wind pressure + snow	Belt – short side	1.37 kN	1.85 kN ($\gamma = 1.35$)	83

J.4.5.1 Belt capacity check

PES belt with a breaking strength of 3000kg is being used.

UC.5a	Corner belt	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$6.65 / (1 \times 15) = 0.44 < 1$	OK
UC.5b	Belt – long side	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$2.03 / (1 \times 15) = 0.14 < 1$	OK
UC.5c	Belt – short side	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	$1.85 / (1 \times 15) = 0.12 < 1$	OK

For capacity of belts see H.2.3, page 24

J.4.5.2 Alternative – rope capacity check

It is possible to use either a rope or a belt. Synthetic rope Ø8mm, BL 900 daN. Is being used → $F_{rd,8mm} = 2.25$ kN.

In a case of using individual guy ropes, the ropes need to be tied back, creating multiple rope sections that together will take the load.

UC.5d	Corner rope	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	$6.65 / (1 \times 3 \times 2.25) = 0.99 < 1$	OK
UC.5e	Rope – long side	1 individual rope made of <u>1 section</u> Ø8mm	$F_d / F_{rd} < 1$	$2.13 / (1 \times 1 \times 2.25) = 0.95 < 1$	OK
UC.5f	Rope – short side	1 individual rope made of <u>1 section</u> Ø8mm	$F_d / F_{rd} < 1$	$1.85 / (1 \times 1 \times 2.25) = 0.82 < 1$	OK

For capacity of ropes see H.2.2, page 24

It is also possible to use alternative **ropes** with a higher breaking strength (BL_{rope}) and less rope sections (n), as long as $BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}$

Required breaking loads for ropes:

Corner rope: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 2660$ kg

Rope – long side: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 852$ kg

Rope – long side: $BL_{rope} \times n_{sections} \times n_{ropes} \geq 740$ kg

J.4.6 Fasteners

Force acting on fasteners is equal to the force in belts or ropes. See these values in chapters H.7.6 and H.7.5.

J.4.6.1 Side belt connection

J.4.6.1.1. M12 Eye bolt, DIN580

For $\leq 45^\circ$:

M12, WLL 340kg $\rightarrow$ BL = 6 x 3.4 kN = 20.4 kN $\rightarrow$ Frd = BL / γ_m = 20.4 / 2 = 10.2 kN

F_d=2.13 kN

UC.6a	Corner eye bolt	$F_d / F_{rd} < 1$	$2.13 / 10.2 = 0.21 < 1.0$	OK
-------	-----------------	--------------------	----------------------------	----

For capacity of eye bolts see K.3 Capacity of Eye bolts on page 94.

J.4.6.1.2. Carabine, DIN5299C

Carabine 180x14, WLL 580kg $\rightarrow$ BL = 2 x 5.8 kN = 11.6 kN $\rightarrow$ Frd = BL / γ_m = 11.6 kN / 2 = 5.8 kN

F_d=2.13 kN

UC.6b	$F_d / F_{rd} < 1$	$2.13 / 5.80 = 0.78 < 1$	OK
-------	--------------------	--------------------------	----

For capacity of carabines see K.5 Carabine, DIN 5299C, page 95

J.4.6.2 Base plate

J.4.6.2.1. Side pole & corner pole

See J.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

- Soil (base plate size 25x25 cm, $p_{soil} = 125 \text{ kN/m}^2$)

UC.6c	Soil	$\frac{F_d}{A} / p \leq 1$	$7.76 / (0.25 \cdot 0.25) / 125 = 0.99 < 1$	OK
-------	------	----------------------------	---	----

J.4.6.2.2. Main pole

See J.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

- Soil (base plate size 35x35 cm, $p_{soil} = 175 \text{ kN/m}^2$)

UC.6d	Soil	$\frac{F_d}{A} / p \leq 1$	$14.27 / (0.35 \cdot 0.35) / 175 = 0.63 < 1$	OK
-------	------	----------------------------	--	----

K. Material certificates

K.1. Fabric specifications

SIOEN FABRICS

Z.I. du Blanc Ballot, Avenue Urbino 6, B-7700 Mouscron - Tel +32(0)56 85 68 80 - Fax +32(0)56 34 61 31
sioenfabrics@sioen.be - <http://www.sioen.com>

Technical Data Sheet

RP5639

FLEXITENT SINGLE M2 (fiche provisoire) - Width 153 cm

Total weight / Poids total	730 [+/-10%] g/m ²	ISO 3801/5 1977
Weight of fabric [PES] / Poids tissu [PES]	280 [+/-10] g/m ²	ISO 3801/5 1977
Weight of coating [PU/PVC] / Poids enduction [PU/PVC]	450 [+/-10%] g/m ²	ISO 3801/5 1977
Breaking strength warp Force de rupture chaîne	1200 N/5cm	ISO 13934 1999
Breaking strength weft Force de rupture trame	900 N/5cm	ISO 13934 1999
Breaking extension warp Allongement à la rupture chaîne	110 %	ISO 13934 1999
Breaking extension weft Allongement à la rupture trame	110 %	ISO 13934 1999
Tear strenght warp Résistance à la déchirure chaîne	100 N	ISO 4674A1 1977
Tear strenght weft Résistance à la déchirure trame	180 N	ISO 4674A1 1977
Adhesion warp / Adhérence chaîne	23 N/5cm	D41 1015
Adhesion weft / Adhérence trame	23 N/5cm	D41 1015
Waterproofness before washing Imperméabilité avant lavage	OK:2000 G M D mm	ISO 811 1981
Waterproofness after washing Imperméabilité après lavage	OK:2000 G M D mm	ISO 811 1981
Flammability / Inflammabilité	OK:M2 -	NF P 92-507 (2004)



May be cleaned with mild, PH neutral, non-ionic detergents. Certainly avoid solvent containing cleaning agents (alcohols, polar solvents, etc.). Testing is always advised before using cleaning agents.
After surface cleaning, please let air dry.

All our technical characteristics are indicative
Rev. 10/2012

K.2. Fabric fire certificate



INSTITUT TECHNOLOGIQUE

INSTITUT TECHNOLOGIQUE

FCBA

Laboratoire d'essais feu

10 rue Galilée - 77420 Champs-sur-Marne

Laboratoire agréé par le Ministère de l'Intérieur et de l'Aménagement du Territoire

JO du 10 février 2007

PROCES-VERBAL DE CLASSEMENT DE REACTION AU FEU

D'UN MATERIAU PREVU A L'ARTICLE 5

DE L'ARRETE DU 21 NOVEMBRE 2002

N° CM - 21 - P - 096

et 1 annexe de 4 pages.

Valable 5 ans à compter du mardi 13 juillet 2021

Matériau présenté par : SIOEN FABRICS SA

Avenue Urbino 6

7700 MOUSCRON

Marque commerciale : F5639

Description sommaire:

Type de textile enduit tricot polyester enduit de PVC ignifugé

Composition support textile/ fabric 100% polyester (280 g/m²) + enduction/ coating PVC ignifugé (450 g/m²)

Masse surfacique 730 g/m²

Epaisseur 0,75 mm

Coloris tous coloris/ all colors

Nature de l'essai : Essai au brûleur électrique
Essai de persistance de flamme
Essai de gouttes

Classement : M2

Observations : /

Durabilité du classement (l'annexe 22) : Non limitée a priori

Usage : Revêtement de murs tendus, voilages, rideaux,... **Conditions d'entretien :** selon préconisations du client

Compte tenu des critères résultant des essais annexés décrits dans le(s) rapport(s) d'essais N° : FE 21-0216-1182

Ce procès-verbal atteste uniquement des caractéristiques de l'échantillon soumis aux essais et ne préjuge pas des caractéristiques de produits similaires. Il ne constitue donc pas une certification de produits au sens de l'article L.115-27 du code de la consommation et de la loi du 3 juin 1994.

NOTA : Seules sont autorisées les reproductions intégrales et par photocopies du présent procès-verbal de classement ou de l'ensemble du procès-verbal de classement et rapport(s) d'essais annexé(s). Ces conclusions ne portent que sur les performances de résistance / réaction au feu de l'élément objet du présent procès verbal de classement. Elles ne préjugent, en aucun cas, des autres performances liées à son incorporation à un ouvrage"

"Non valable pour toute application couverte par l'article AM 18 (Arrêté du 6 mars 2006) concernant les sièges rembourrés"

Champs-sur-Marne, le vendredi 23 juillet 2021

Technicien d'essais du Laboratoire Feu



Bilel HAMILA

Responsable Technique du laboratoire Feu
du Pôle Ameublement



Clotilde ETIEMVRE

Siège social
10 rue Galilée
77420 Champs-sur-Marne
Tél + 33 (0)1 72 84 97 84
www.fcba.fr
Siret 775 680 903 00132
APE 7219 Z
Code TVA CEE : FR 14 775 680 903



Institut technologique FCBA : Forêt, Cellulose, Bois - construction, Ameublement

Laboratory approved by the Ministry for the Interior
JO of February 10, 2007
OFFICIAL REPORT OF CLASSIFICATION OF FIRE PERFORMANCE
Of a MATERIAL ENVISAGED IN ARTICLE 5
DECREE OF November 21, 2002

N° **CM - 21 - P - 096**

Page 1

et 1 annexe de 4 pages.

Valid 5 years as from mardi 13 juillet 2021

Material presents by : **SIOEN FABRICS SA**

Avenue Urbino 6
7700 MOUSCRON

Trade mark: **F5639**

Summary description

Kind of coating fabric **tricot polyester enduit de PVC ignifugé**
Composition **support textile/ fabric 100% polyester (280 g/m²) + enduction/ coating PVC ignifugé (450 g/m²)**
Mass per unit area **730 g/m²**
Thickness **0,75 mm**
Color **tous coloris/ all colors**

Nature of the test : **Electrical burner test**
Flame persistence
Dripping test

Classification : **M2**

Observations :

Durability of the classification (Article 5 de l'annexe 2) Not limited

Usual term **Walls hung covering, sheers, curtains,...**

Cleaning conditions **Recommendation of Customer**

Taking into account the criteria resulting from the tests annexed described in the test report N°FE 21-0216-1182

This official report attests only characteristics of the sample submitted for testing and does not prejudice similar characteristics of products. It thus does not constitute a certification of products within the meaning of the L.115-27 article of the code of the consumption and the law of June 3rd, 1994.
NOTE: The integral reproductions and by photocopy of this official report of classification are only authorized or the official report unit of classification and report of annexed test. These conclusions relate only to the resistance / reaction to fire performance of the element that is the subject of the present classification report. They do not prejudice, in any case, other performances related to its incorporation to a work *

"Not legally acceptable for any use under the pretence of the article AM 18 (Decree of March 6th, 2006) for upholstered seat"

Champs-sur-Marne, le vendredi 23 juillet 2021

Technicien d'essais du Laboratoire Feu

Responsable Technique du laboratoire Feu
du Pôle Ameublement

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Bilal HAMILA



Only the french version is valid
Institut technologique FCBA : Forêt, Closely, Bois - construction, Ameublement



Clotilde ETIEMVRE

K.5. Carabine, DIN 5299C

Karta Techniczna Nr 10GM/KAR

1. Nazwa dostawcy: „GÓRALMET” M. i J. Góral Sp. J.

Adres dostawcy: ul. Krakowska 68, 32-860 Czchów, tel/fax 014 6635260

2. Nazwa wyrobu: Karabińczyk do lin 5299C

3. Opis i przeznaczenie: Zatrzaśniki stalowe, służące jako elementy łączące w zakończeniach lin, łańcuchów gospodarczych, oraz do innych zastosowań ogólnych. Do stosowania w zakresie parametrów technicznych podanych w punkcie 5.

4. Karabińczyki do lin nie mogą być użyte w zastosowaniach posiadających inne - uregulowane prawem czy normami wymagania bezpieczeństwa, np. do podnoszenia ładunków, jako środek do ochrony indywidualnej przed upadkiem z wysokości, lub osprzęt alpinistyczny. Nie przeciążać ponad wartość WLL (kG) podaną w kolumnie 13.

5. Parametry techniczne wyrobu:

LP.	Rozmiar [mm]	Wymiary nominalne [mm]										Masa [kg/szt]	WLL Obciążenie robocze [kG]	BF Siła niszcząca [min. kG]	Materiał	Powłoki
		A [mm]	C min [mm]	D [mm]	E min [mm]	Ø [mm]	L [mm]	---	---	---	---					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
1	4	19,5	4,5	6	4,5	3,9	40	---	---	---	0,008	70	140	C10 ~ C15	Ocynk elektrolityczny	
2	5	24	5,5	7,5	5,5	4,8	50	---	---	---	0,015	100	200			
3	6	29	8	8,5	6	5,5	60	---	---	---	0,023	120	240			
4	7	34	9	10,5	8,5	6,7	70	---	---	---	0,040	180	360			
5	8	39	10	11,5	9	7,4	80	---	---	---	0,056	230	460			
6	9	44	11	13,5	10	8,4	90	---	---	---	0,083	250	500			
7	10	48	11,5	14,5	11	9,4	100	---	---	---	0,115	350	700			
8	11	57	17	17,5	13	10,7	120	---	---	---	0,179	400	800			
9	12	67	22	20	15	11,5	140	---	---	---	0,246	450	900			
10	13	80	26	21	17	12,6	160	---	---	---	0,343	530	1060			
11	14	87	28	24	20	13,6	180	---	---	---	0,443	580	1160			
12	15	98	32	24	20	14,7	200	---	---	---	0,563	700	1400			

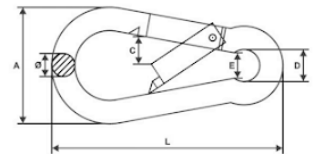
6. Informacje dodatkowe:

➤ Współczynnik bezpieczeństwa - 2

➤ BF = 2,0 x WLL

➤ Cechowanie: symbol producenta – GM lub GMG; kod rozmiaru

➤ Informacje dodatkowe – etykieta: nazwa wyrobu; rozmiar; ilość szt.; numer dostawy; dane producenta i adres strony internetowej.



K.6. Rope ratchet



3/8" Rope Ratchet

Carolina North Manufacturing
10022

\$10.89 ~~\$12.89~~

Maximum load limit 250 lbs. Outer casings are made from #6-33% glass filled nylon with interior mechanisms of die cast zinc. Perfect for large jobs:

- heavy loads for transport
- hoisting items
- straightening items
- securing objects under tension
- product weight 0.16 lbs

Product is compatible with our 3/8" S-hooks and 3/8" solid braided rope.

Bulk purchasing options available.

ADD TO CART

BUY IT NOW

Share:    

Collections: Bulk Rope Ratchets
Type: Bulk Ratchet

Annex

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Annex A. Software input (load combination) – 10x10m model

Annex A.1. CO1. Own weight + pretension

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT) ORDERED BY LOADGROUPS							
LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.7115	0.009063
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0291	0.000371
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4864	97.285966
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4836	96.722595
SUM				0.0000	0.0000	-1.7107	
EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADGROUPS							
LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
SUM				0.0000	0.0000	0.0000	
EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADMODES							
	LOADMODE			SUM_X	SUM_Y	SUM_Z	
SUM	AREA-LOADS			0.0000	0.0000	0.0000	
EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS							
				SUM_X	SUM_Y	SUM_Z	
				0.0000	0.0000	-1.7107	

Inner Energy= 1.574062

StatikD: Sum of external loads: 0.000000 0.000000 -1.710614
 StatikD: Sum of support forces: 0.000004 0.000011 1.710763
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\C01_OW+VO
 StatikD: Time of calculation: 0.24s
 StatikD: Version 2025.38 done.

Annex A.2. CO2. Own weight + pretension + conventional / snow

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.7115	0.009063
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0291	0.000371
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4864	97.285966
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4836	96.722595
SUM				0.0000	0.0000	-1.7107	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1000	1.00	0.0000	0.0000	-9.4121	94.12
SUM				0.0000	0.0000	-9.4121	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-9.4121
SUM	AREA-LOADS	0.0000	0.0000	-9.4121

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0000	0.0000	-11.1227

Inner Energy= 2.607481

StatikD: Sum of external loads: 0.000000 0.000000 -11.122765
 StatikD: Sum of support forces: 0.000003 0.000012 11.124678
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\CO2_OW+VO+SNOW
 StatikD: Time of calculation: 0.27s
 StatikD: Version 2025.38 done.

Annex A.3. CO3. Own weight + pretension + wind pressure

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.7115	0.009063
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0291	0.000371
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4864	97.285966
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4836	96.722595
SUM				0.0000	0.0000	-1.7107	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	-0.1500	1.00	-0.0000	0.0000	-14.1181	98.07
SUM				-0.0000	0.0000	-14.1181	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	-0.0000	0.0000	-14.1181
SUM	AREA-LOADS	-0.0000	0.0000	-14.1181

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0000	0.0000	-15.8288

Inner Energy= 3.515106

StatikD: Sum of external loads: -0.000001 0.000001 -15.828783
 StatikD: Sum of support forces: 0.000007 0.000009 15.829106
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\CO3_OW+V0+WD
 StatikD: Time of calculation: 0.28s
 StatikD: Version 2025.38 done.

Annex A.4. CO4. Own weight + pretension + wind suction

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.7115	0.009063
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0385	0.000491
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4864	97.285966
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4836	96.722595
SUM				0.0000	0.0000	-1.7200	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	0.3500	1.00	-0.0013	-0.0041	32.9139	98.05
SUM				-0.0013	-0.0041	32.9139	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	-0.0013	-0.0041	32.9139
SUM	AREA-LOADS	-0.0013	-0.0041	32.9139

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0013	-0.0041	31.1939

Inner Energy= 7.087801

StatikD: Sum of external loads: -0.001328 -0.004069 31.193895
 StatikD: Sum of support forces: 0.001270 0.004033 -31.194731
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\CO4_OW+VO+WU
 StatikD: Time of calculation: 0.99s
 StatikD: Version 2025.38 done.

Annex A.5. CO5. Own weight + pretension + wind suction reduced [NL]

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.7115	0.009063
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0291	0.000371
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4864	97.285966
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4836	96.722595
SUM				0.0000	0.0000	-1.7107	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	0.1750	1.00	0.0000	-0.0000	16.4711	98.07
SUM				0.0000	-0.0000	16.4711	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	0.0000	-0.0000	16.4711
SUM	AREA-LOADS	0.0000	-0.0000	16.4711

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0000	-0.0000	14.7605

Inner Energy= 4.086116

StatikD: Sum of external loads: -0.000003 -0.000004 14.760512
 StatikD: Sum of support forces: 0.000000 0.000005 -14.760568
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\CO5_OW+VO+WU_red
 StatikD: Time of calculation: 1.98s
 StatikD: Version 2025.38 done.

Annex A.6. CO6. Own weight + pretension + wind pressure + snow [FR]

Annex A.6.1. For element check

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT) ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.7115	0.009063
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0291	0.000371
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4864	97.285966
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4836	96.722595
SUM				0.0000	0.0000	-1.7107	

EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1000	1.00	0.0000	0.0000	-9.4121	94.12
1	WIND	-0.1500	1.00	-0.0000	0.0000	-14.1181	98.07
SUM				-0.0000	0.0000	-23.5302	

EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-9.4121
SUM	WIND	-0.0000	0.0000	-14.1181
SUM	AREA-LOADS	-0.0000	0.0000	-23.5302

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0000	0.0000	-25.2409

Inner Energy= 5.265614

StatikD: Sum of external loads: -0.000001 0.000001 -25.240880
 StatikD: Sum of support forces: 0.000009 0.000008 25.240553
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\CO6_OW+VO+WD+SNOW
 StatikD: Time of calculation: 0.30s
 StatikD: Version 2025.38 done.

Annex A.6.2. For anchor and ballast calculation

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.7115	0.009063
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0291	0.000371
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4864	97.285966
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.4836	96.722595
SUM				0.0000	0.0000	-1.7107	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1083	1.00	0.0000	0.0000	-10.1961	94.12
1	WIND	-0.1500	1.00	-0.0000	0.0000	-14.1181	98.07
SUM				-0.0000	0.0000	-24.3142	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-10.1961
SUM	WIND	-0.0000	0.0000	-14.1181
SUM	AREA-LOADS	-0.0000	0.0000	-24.3142

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0000	0.0000	-26.0249

Inner Energy= 5.420759

StatikD: Sum of external loads: -0.000001 0.000001 -26.024883
 StatikD: Sum of support forces: 0.000009 0.000009 26.026117
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x10\C06a_OW+VO+WD+SNOW_anchors
 StatikD: Time of calculation: 0.24s
 StatikD: Version 2025.38 done.

Annex B. Software input (load combination) – 10x15m model

Annex B.1. CO1. Own weight + pretension

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT) ORDERED BY LOADGROUPS							
LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.9460	0.012051
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0393	0.000500
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7673	153.469253
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7628	152.565787
SUM				0.0000	0.0000	-2.5155	
EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADGROUPS							
LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
SUM				0.0000	0.0000	0.0000	
EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADMODES							
	LOADMODE			SUM_X	SUM_Y	SUM_Z	
SUM	AREA-LOADS			0.0000	0.0000	0.0000	
EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS							
				SUM_X	SUM_Y	SUM_Z	
				0.0000	0.0000	-2.5155	

Inner Energy= 2.345706

StatikD: Sum of external loads: 0.000000 0.000000 -2.515503
 StatikD: Sum of support forces: -0.002219 -0.001612 2.515072
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\CO1_OW+VO
 StatikD: Time of calculation: 0.68s
 StatikD: Version 2025.38 done.

Annex B.2. CO2. Own weight + pretension + conventional / snow

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.9460	0.012051
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0393	0.000500
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7673	153.469253
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7628	152.565787
SUM				0.0000	0.0000	-2.5155	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1000	1.00	0.0000	0.0000	-14.8990	148.99
SUM				0.0000	0.0000	-14.8990	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-14.8990
SUM	AREA-LOADS	0.0000	0.0000	-14.8990

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0000	0.0000	-17.4145

Inner Energy= 4.245182

StatikD: Sum of external loads: 0.000000 0.000000 -17.414544
 StatikD: Sum of support forces: -0.002619 0.001603 17.415259
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\CO2_OW+VO+SNOW
 StatikD: Time of calculation: 1.13s
 StatikD: Version 2025.38 done.

Annex B.3. CO3. Own weight + pretension + wind pressure

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.9460	0.012051
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0393	0.000500
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7673	153.469253
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7628	152.565787
SUM				0.0000	0.0000	-2.5155	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	-0.1500	1.00	0.0042	0.0084	-22.3486	154.64
SUM				0.0042	0.0084	-22.3486	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	0.0042	0.0084	-22.3486
SUM	AREA-LOADS	0.0042	0.0084	-22.3486

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0042	0.0084	-24.8641

Inner Energy= 5.886284

StatikD: Sum of external loads: 0.004168 0.008447 -24.864059
 StatikD: Sum of support forces: -0.005342 -0.006802 24.865402
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\CO3_OW+VO+WD
 StatikD: Time of calculation: 1.23s
 StatikD: Version 2025.38 done.

Annex B.4. CO4. Own weight + pretension + wind suction

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.9460	0.012051
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0586	0.000747
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7673	153.469253
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7628	152.565787
SUM				0.0000	0.0000	-2.5349	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	0.3500	1.00	-0.0091	-0.0203	52.0521	154.58
SUM				-0.0091	-0.0203	52.0521	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	-0.0091	-0.0203	52.0521
SUM	AREA-LOADS	-0.0091	-0.0203	52.0521

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0091	-0.0203	49.5172

Inner Energy= 11.014835

StatikD: Sum of external loads: -0.009041 -0.020279 49.517238
 StatikD: Sum of support forces: 0.009277 0.019985 -49.516573
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\CO4_OW+VO+WU
 StatikD: Time of calculation: 2.25s
 StatikD: Version 2025.38 done.

Annex B.5. CO5. Own weight + pretension + wind suction reduced [NL]

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.9460	0.012051
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0393	0.000500
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7673	153.469253
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7628	152.565787
SUM				0.0000	0.0000	-2.5155	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	0.1750	1.00	-0.0049	-0.0098	26.0733	154.64
SUM				-0.0049	-0.0098	26.0733	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	-0.0049	-0.0098	26.0733
SUM	AREA-LOADS	-0.0049	-0.0098	26.0733

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0049	-0.0098	23.5579

Inner Energy= 7.521792

StatikD: Sum of external loads: -0.004882 -0.009859 23.557848
 StatikD: Sum of support forces: 0.006997 0.018128 -23.596153
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\CO5_OW+VO+WU_red
 StatikD: Time of calculation: 2.12s
 StatikD: Version 2025.38 done.

Annex B.6. CO6. Own weight + pretension + wind pressure + snow [FR]

Annex B.6.1. For element check

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.9460	0.012051
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0393	0.000500
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7673	153.469253
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7628	152.565787
SUM				0.0000	0.0000	-2.5155	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1000	1.00	0.0000	0.0000	-14.8990	148.99
1	WIND	-0.1500	1.00	0.0042	0.0084	-22.3486	154.64
SUM				0.0042	0.0084	-37.2476	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-14.8990
SUM	WIND	0.0042	0.0084	-22.3486
SUM	AREA-LOADS	0.0042	0.0084	-37.2476

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0042	0.0084	-39.7631

Inner Energy= 8.975232

StatikD: Sum of external loads: 0.004168 0.008447 -39.763095
 StatikD: Sum of support forces: -0.006145 -0.008567 39.762601
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\CO6_OW+VO+WD+SNOW
 StatikD: Time of calculation: 1.43s
 StatikD: Version 2025.38 done.

Annex B.6.2. For anchor and ballast calculation

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.9460	0.012051
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0393	0.000500
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7673	153.469253
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.7628	152.565787
SUM				0.0000	0.0000	-2.5155	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1083	1.00	0.0000	0.0000	-16.1401	148.99
1	WIND	-0.1500	1.00	0.0042	0.0084	-22.3486	154.64
SUM				0.0042	0.0084	-38.4887	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-16.1401
SUM	WIND	0.0042	0.0084	-22.3486
SUM	AREA-LOADS	0.0042	0.0084	-38.4887

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0042	0.0084	-41.0042

Inner Energy= 9.248682

StatikD: Sum of external loads: 0.004168 0.008447 -41.004176
 StatikD: Sum of support forces: -0.005850 -0.008439 41.004555
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x15\C06a_OW+VO+WD+SNOW_anchors
 StatikD: Time of calculation: 1.46s
 StatikD: Version 2025.38 done.

Annex C. Software input (load combination) – 10x20m model

Annex C.1. CO1. Own weight + pretension

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-1.1806	0.015039
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0494	0.000629
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9720	194.398401
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9663	193.267579
SUM				0.0000	0.0000	-3.1683	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
SUM				0.0000	0.0000	0.0000	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

LOADMODE		SUM_X	SUM_Y	SUM_Z
SUM	AREA-LOADS	0.0000	0.0000	0.0000

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0000	0.0000	-3.1683

Inner Energy= 2.667523

StatikD: Sum of external loads: 0.000000 0.000000 -3.168216
 StatikD: Sum of support forces: -0.001067 -0.000072 3.167956
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20\CO1_OW+VO
 StatikD: Time of calculation: 0.93s
 StatikD: Version 2025.38 done.

Annex C.2. CO2. Own weight + pretension + conventional / snow

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-1.1806	0.015039
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0494	0.000629
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9720	194.398401
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9663	193.267579
SUM				0.0000	0.0000	-3.1683	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1000	1.00	0.0000	0.0000	-18.8287	188.29
SUM				0.0000	0.0000	-18.8287	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-18.8287
SUM	AREA-LOADS	0.0000	0.0000	-18.8287

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0000	0.0000	-21.9970

Inner Energy= 4.932091

StatikD: Sum of external loads: 0.000000 0.000000 -21.996996
 StatikD: Sum of support forces: -0.003667 -0.001001 21.997637
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20\CO2_OW+VO+SNOW
 StatikD: Time of calculation: 1.35s
 StatikD: Version 2025.38 done.

Annex C.3. CO3. Own weight + pretension + wind pressure

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT) ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-1.1806	0.015039
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0494	0.000629
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9720	194.398401
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9663	193.267579
SUM				0.0000	0.0000	-3.1683	

EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	-0.1500	1.00	0.0004	0.0148	-28.2431	195.79
SUM				0.0004	0.0148	-28.2431	

EXTERNAL LOADS (AREA-DEPENDENT) ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	0.0004	0.0148	-28.2431
SUM	AREA-LOADS	0.0004	0.0148	-28.2431

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0004	0.0148	-31.4113

Inner Energy= 6.986727

StatikD: Sum of external loads: 0.000460 0.014788 -31.411354
 StatikD: Sum of support forces: -0.002743 -0.011967 31.411551
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20\CO3_OW+VO+WD
 StatikD: Time of calculation: 1.54s
 StatikD: Version 2025.38 done.

Annex C.4. CO4. Own weight + pretension + wind suction

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-1.1806	0.015039
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0688	0.000877
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9720	194.398401
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9663	193.267579
SUM				0.0000	0.0000	-3.1877	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	0.3500	1.00	-0.0032	-0.0323	65.8053	195.75
SUM				-0.0032	-0.0323	65.8053	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	-0.0032	-0.0323	65.8053
SUM	AREA-LOADS	-0.0032	-0.0323	65.8053

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0032	-0.0323	62.6176

Inner Energy= 14.312429

StatikD: Sum of external loads: -0.003195 -0.032267 62.617612
 StatikD: Sum of support forces: -0.001255 0.032404 -62.605597
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20\CO4_OW+VO+WU
 StatikD: Time of calculation: 2.71s
 StatikD: Version 2025.38 done.

Annex C.5. CO5. Own weight + pretension + wind suction reduced [NL]

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-1.1806	0.015039
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0494	0.000629
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9720	194.398401
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9663	193.267579
SUM				0.0000	0.0000	-3.1683	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	WIND	0.1750	1.00	-0.0005	-0.0172	32.9503	195.79
SUM				-0.0005	-0.0172	32.9503	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	WIND	-0.0005	-0.0172	32.9503
SUM	AREA-LOADS	-0.0005	-0.0172	32.9503

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
-0.0005	-0.0172	29.7820

Inner Energy= 9.377589

StatikD: Sum of external loads: -0.000487 -0.017266 29.782032
 StatikD: Sum of support forces: -0.001501 0.018529 -29.791202
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20 - final\CO5_OW+VO+WU_red
 StatikD: Time of calculation: 3.07s
 StatikD: Version 2025.38 done.

Annex C.6. CO6. Own weight + pretension + wind pressure + snow [FR]

Annex C.6.1. For element check

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-1.1806	0.015039
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0494	0.000629
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9720	194.398401
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9663	193.267579
SUM				0.0000	0.0000	-3.1683	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1000	1.00	0.0000	0.0000	-18.8287	188.29
1	WIND	-0.1500	1.00	0.0004	0.0148	-28.2431	195.79
SUM				0.0004	0.0148	-47.0718	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-18.8287
SUM	WIND	0.0004	0.0148	-28.2431
SUM	AREA-LOADS	0.0004	0.0148	-47.0718

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0004	0.0148	-50.2401

Inner Energy= 10.819541

StatikD: Sum of external loads: 0.000460 0.014788 -50.240093
 StatikD: Sum of support forces: -0.004591 -0.012770 50.240338
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20\CO6_OW+VO+WD+SNOW
 StatikD: Time of calculation: 1.56s
 StatikD: Version 2025.38 done.

Annex C.6.2. For anchor and ballast calculation

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-1.1806	0.015039
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0494	0.000629
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9720	194.398401
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.9663	193.267579
SUM				0.0000	0.0000	-3.1683	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1083	1.00	0.0000	0.0000	-20.3972	188.29
1	WIND	-0.1500	1.00	0.0004	0.0148	-28.2431	195.79
SUM				0.0004	0.0148	-48.6403	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-20.3972
SUM	WIND	0.0004	0.0148	-28.2431
SUM	AREA-LOADS	0.0004	0.0148	-48.6403

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0004	0.0148	-51.8085

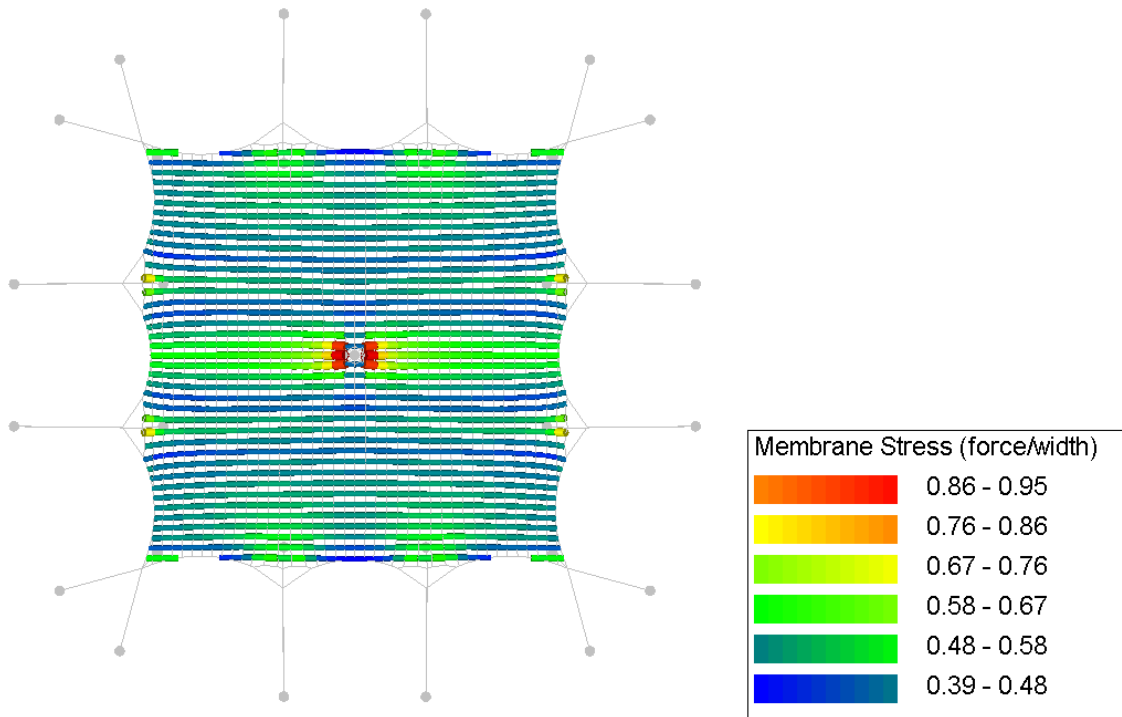
Inner Energy= 11.160164

StatikD: Sum of external loads: 0.000460 0.014788 -51.808511
 StatikD: Sum of support forces: -0.005047 -0.013400 51.808345
 StatikD: P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek2\10x20\C06a_OW+VO+WD+SNOW_anchors
 StatikD: Time of calculation: 1.55s
 StatikD: Version 2025.38 done.

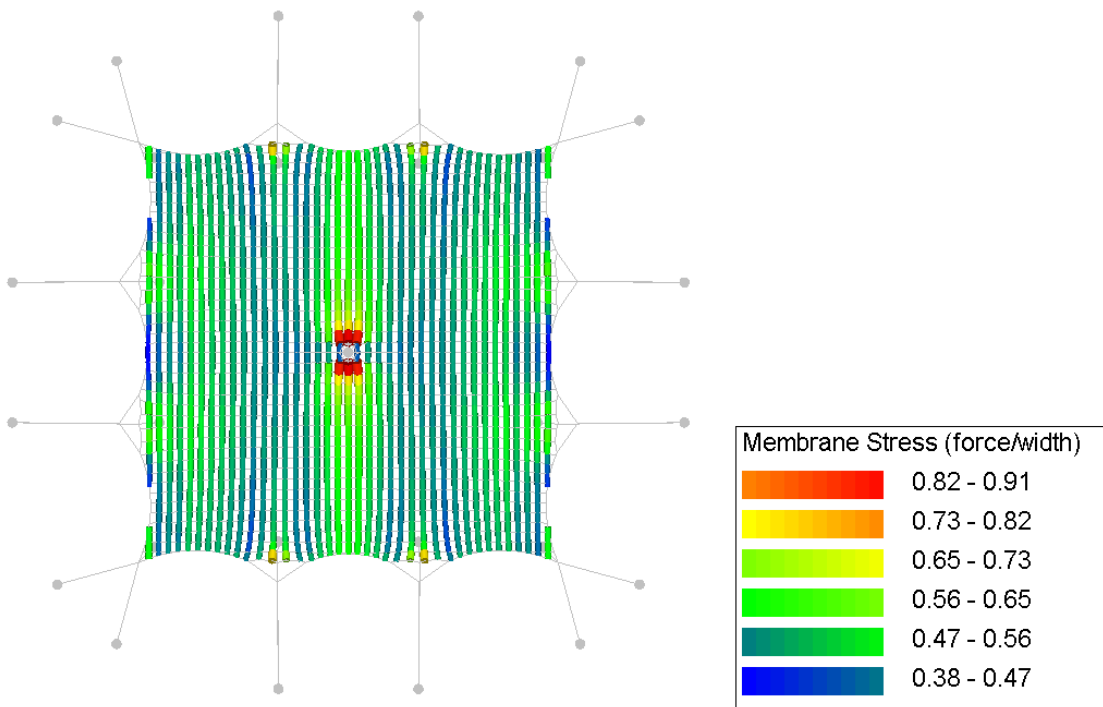
Annex D. Results per load combination – 10x10m model

Annex D.1. CO1. Own weight + pretension

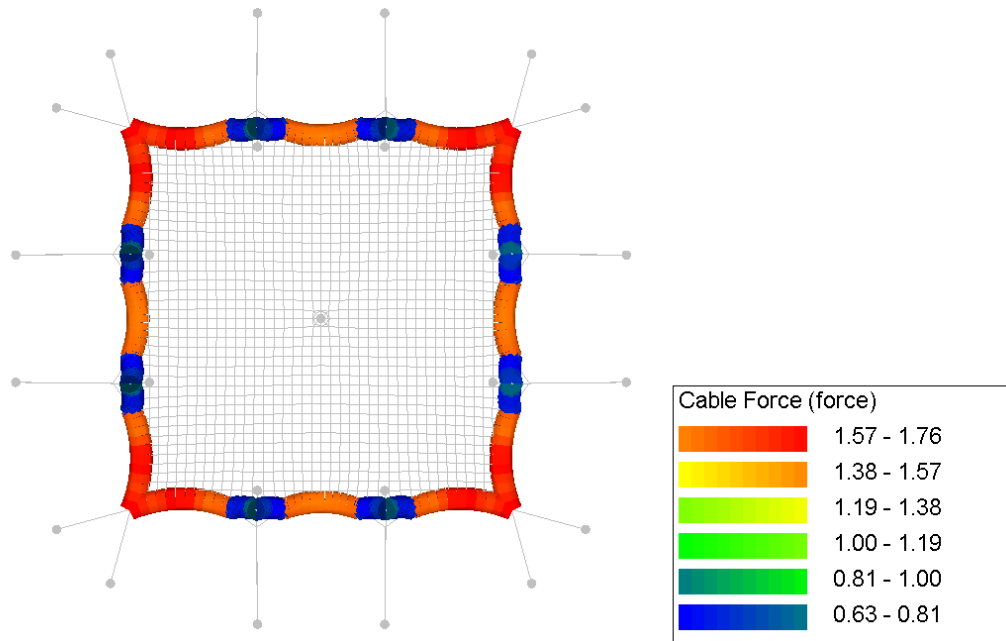
Annex D.1.1. Membrane stress (warp)



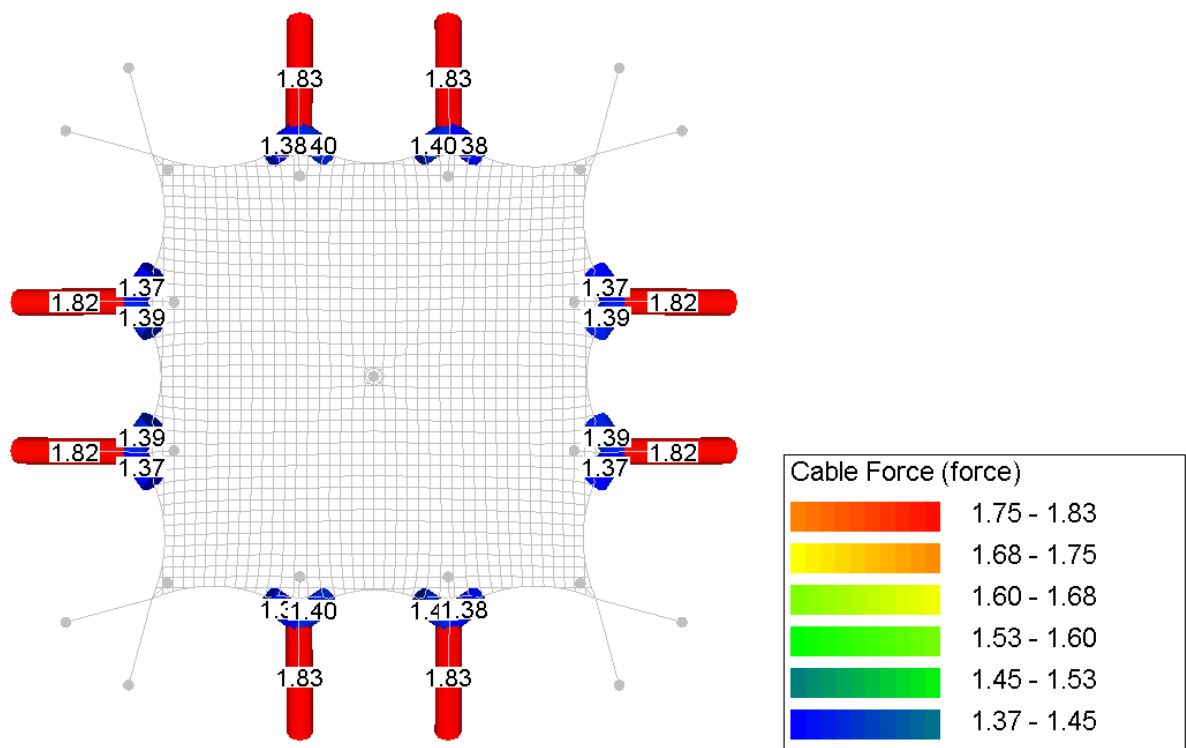
Annex D.1.2. Membrane stress (weft)



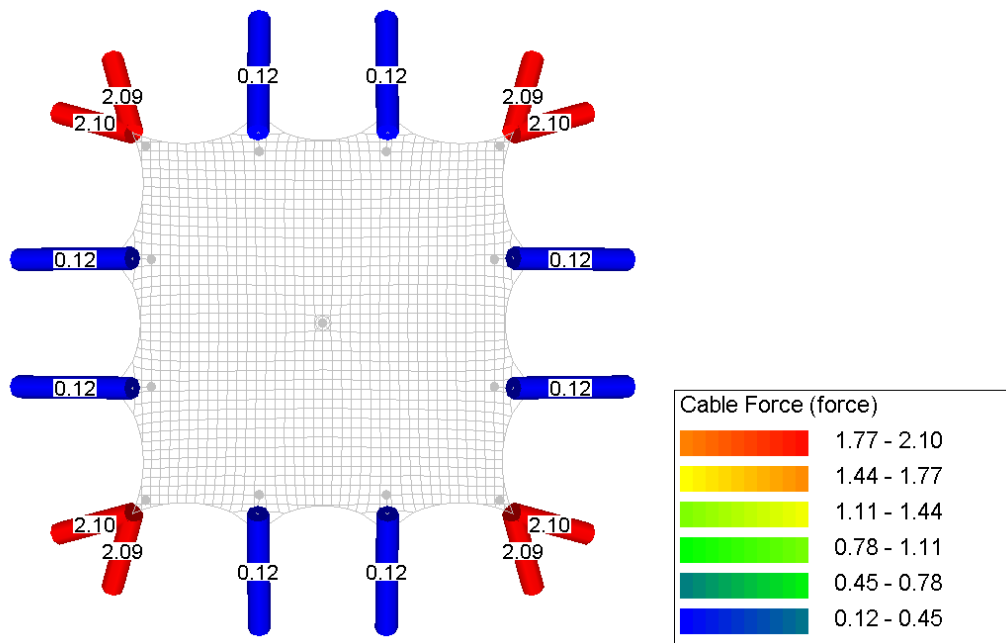
Annex D.1.3. Circumference rope forces



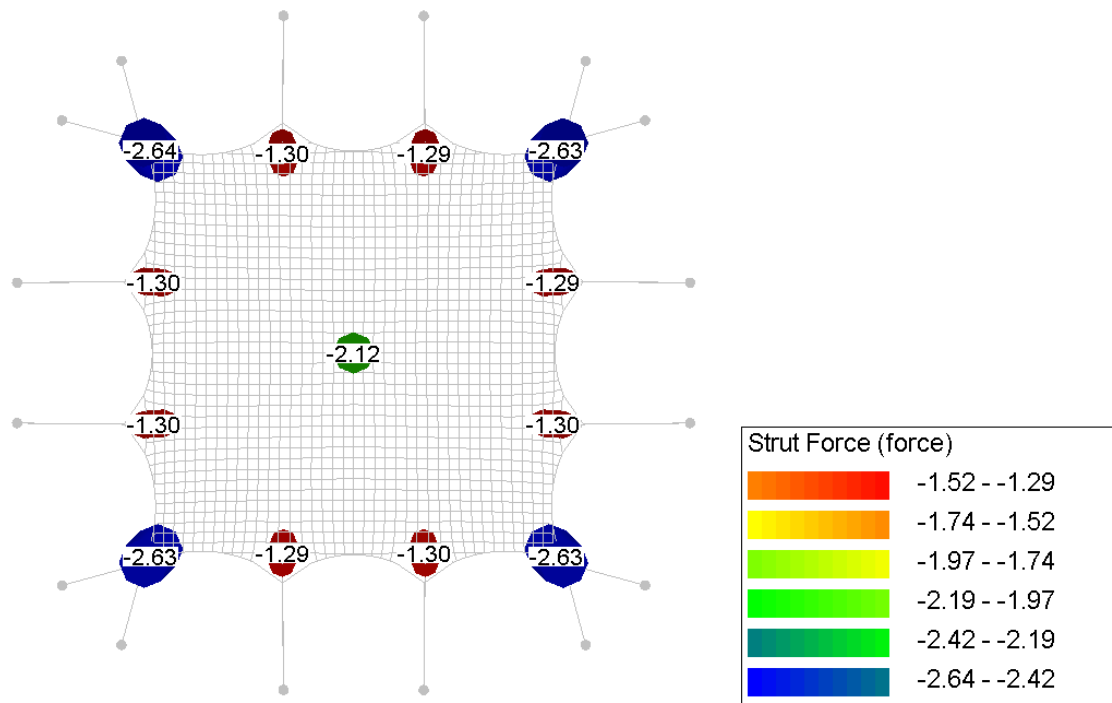
Annex D.1.4. Rope forces



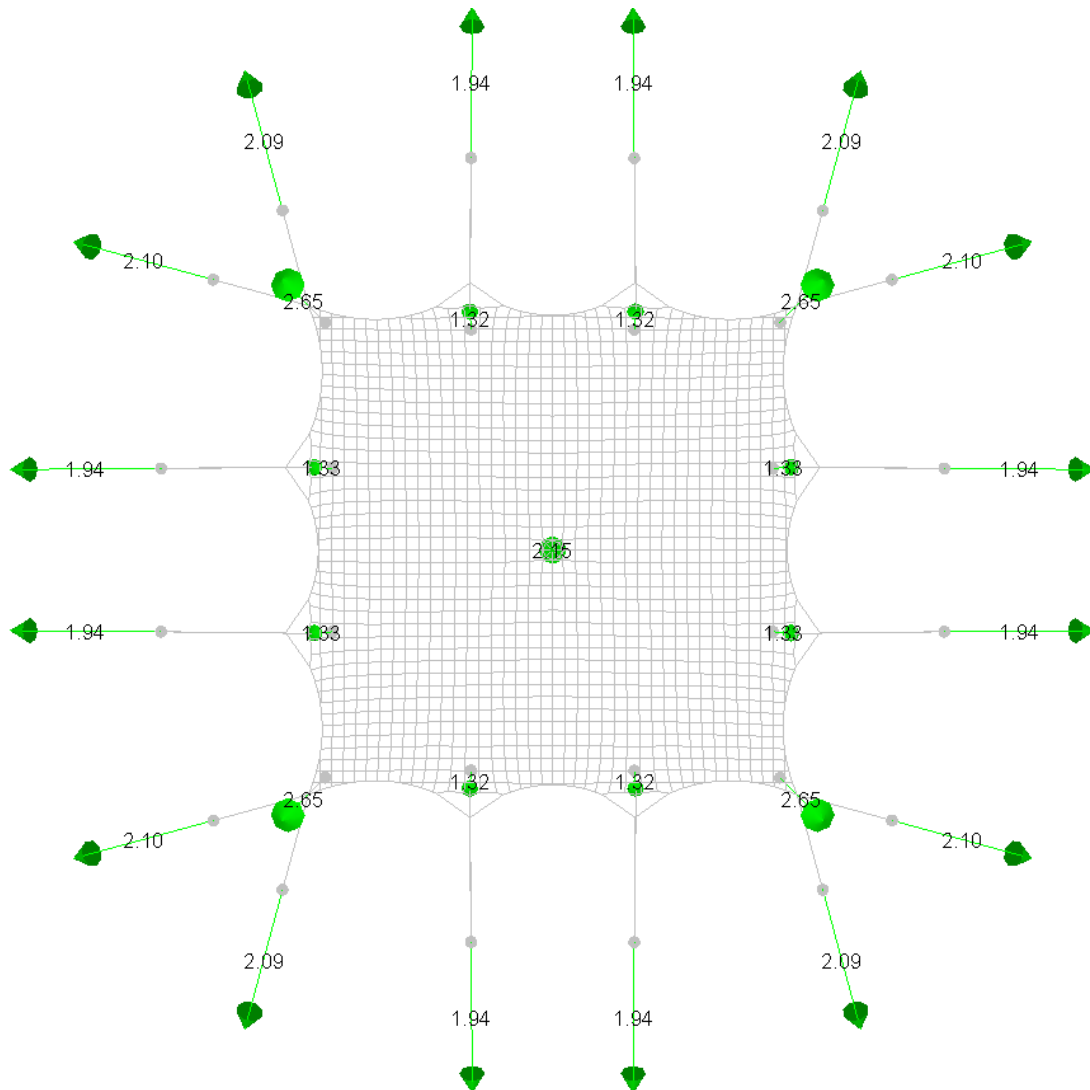
Annex D.1.5. Belt forces



Annex D.1.6. Pole forces

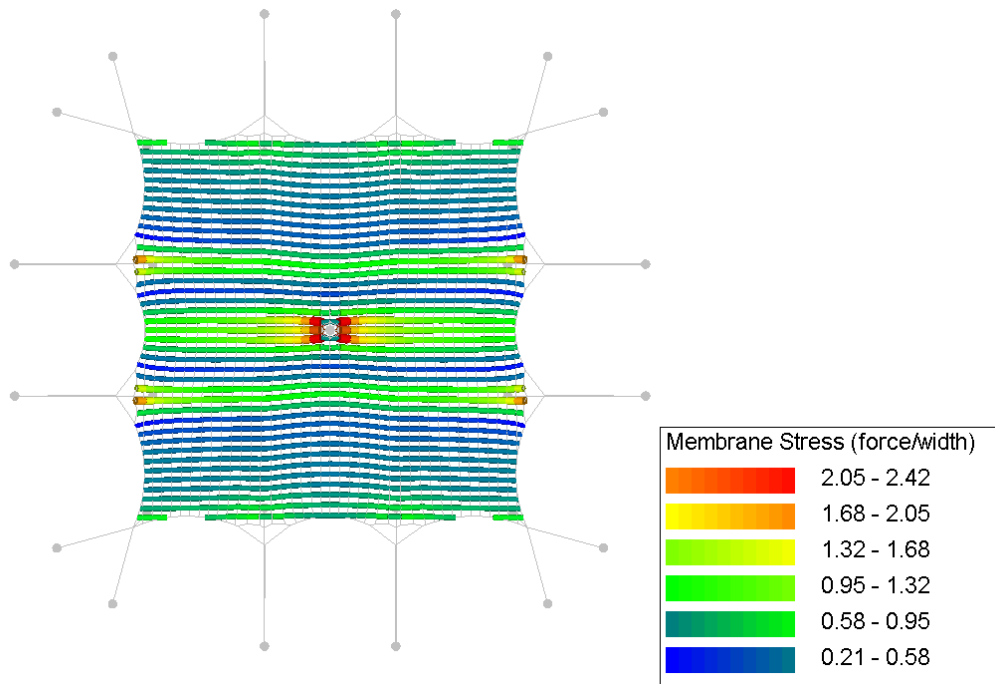


Annex D.1.7. Reaction forces

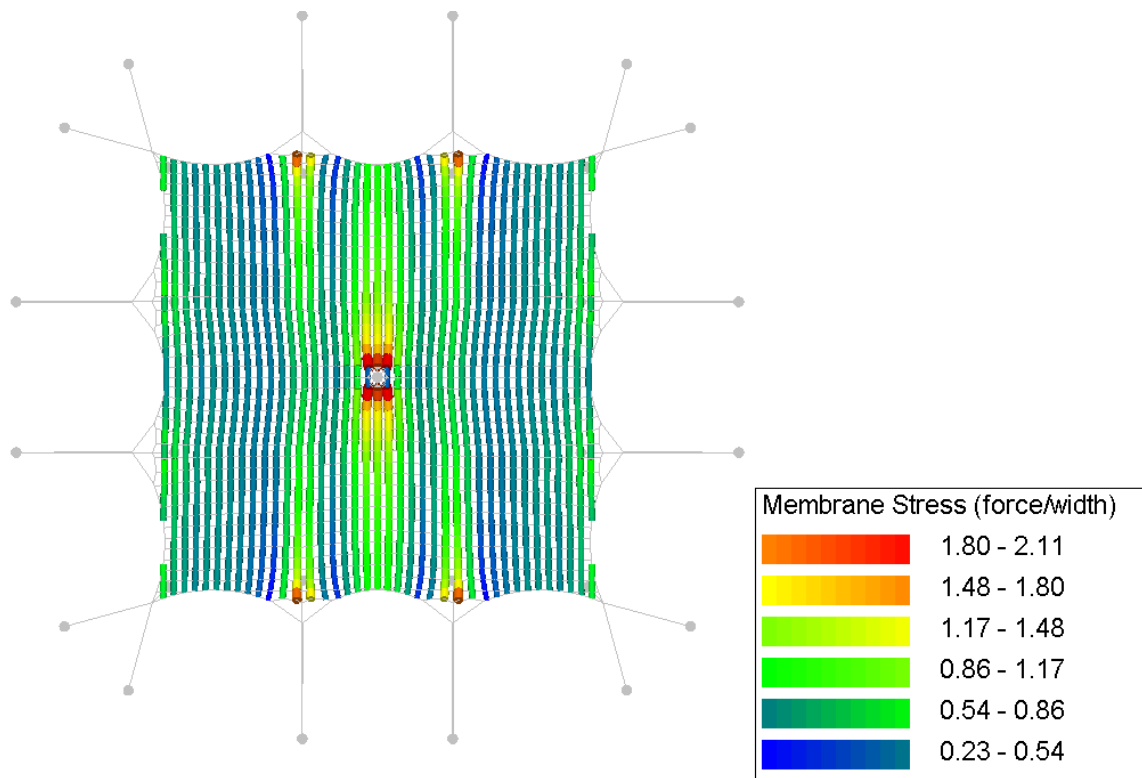


Annex D.2. CO2. Own weight + pretension + snow

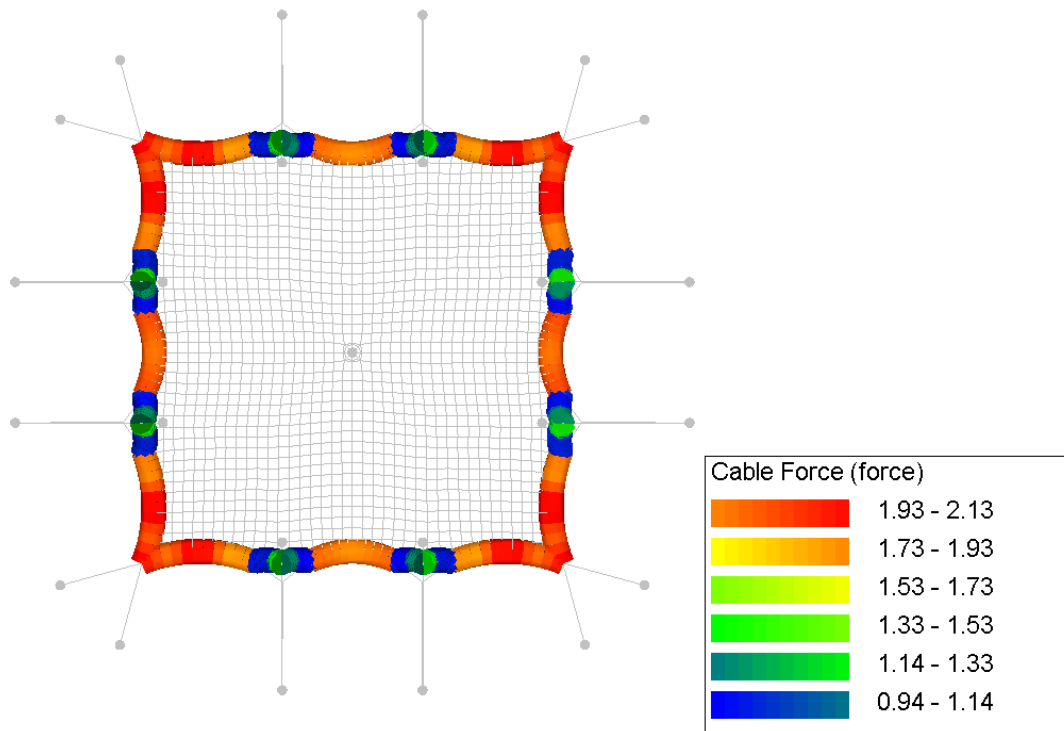
Annex D.2.1. Membrane stress (warp)



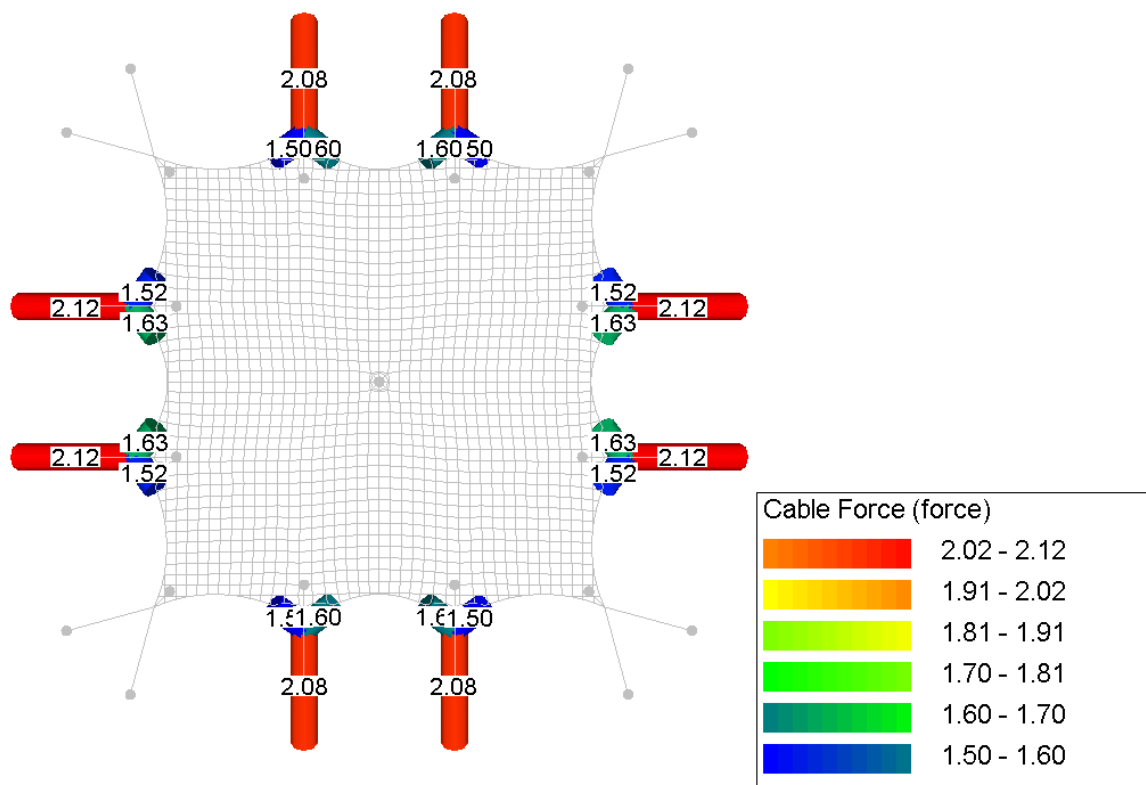
Annex D.2.2. Membrane stress (weft)



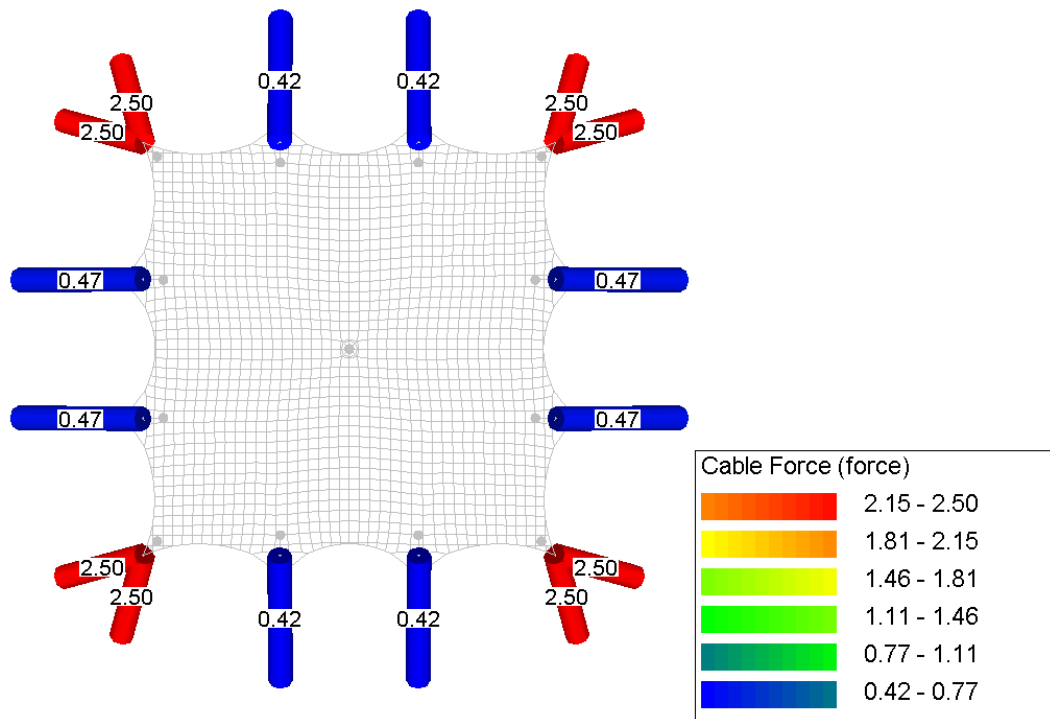
Annex D.2.3. Circumference rope forces



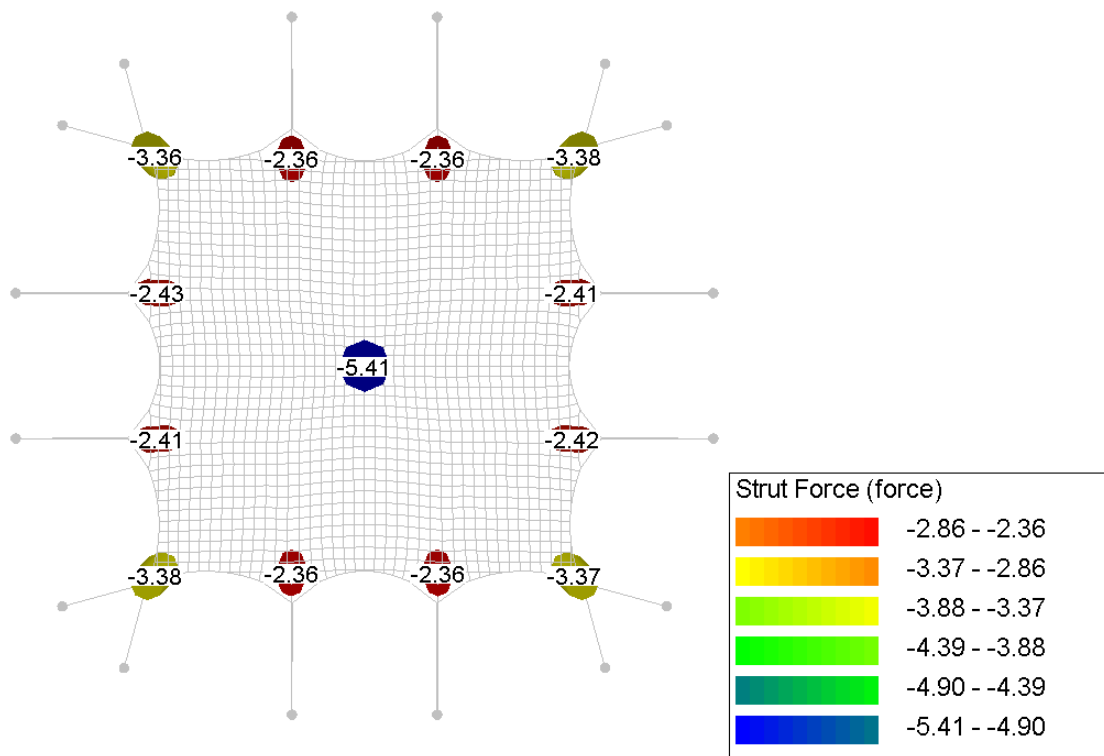
Annex D.2.4. Rope forces



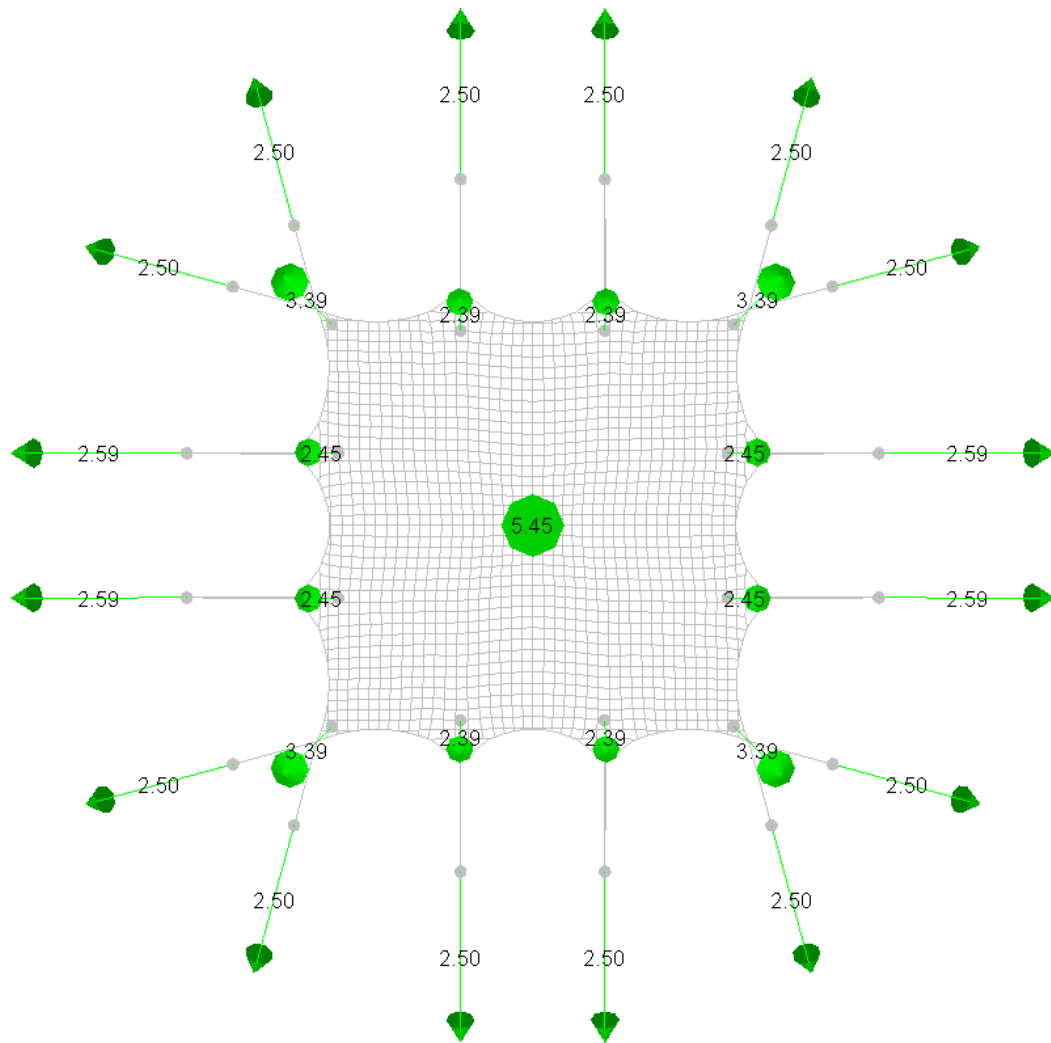
Annex D.2.5. Belt forces



Annex D.2.6. Pole forces

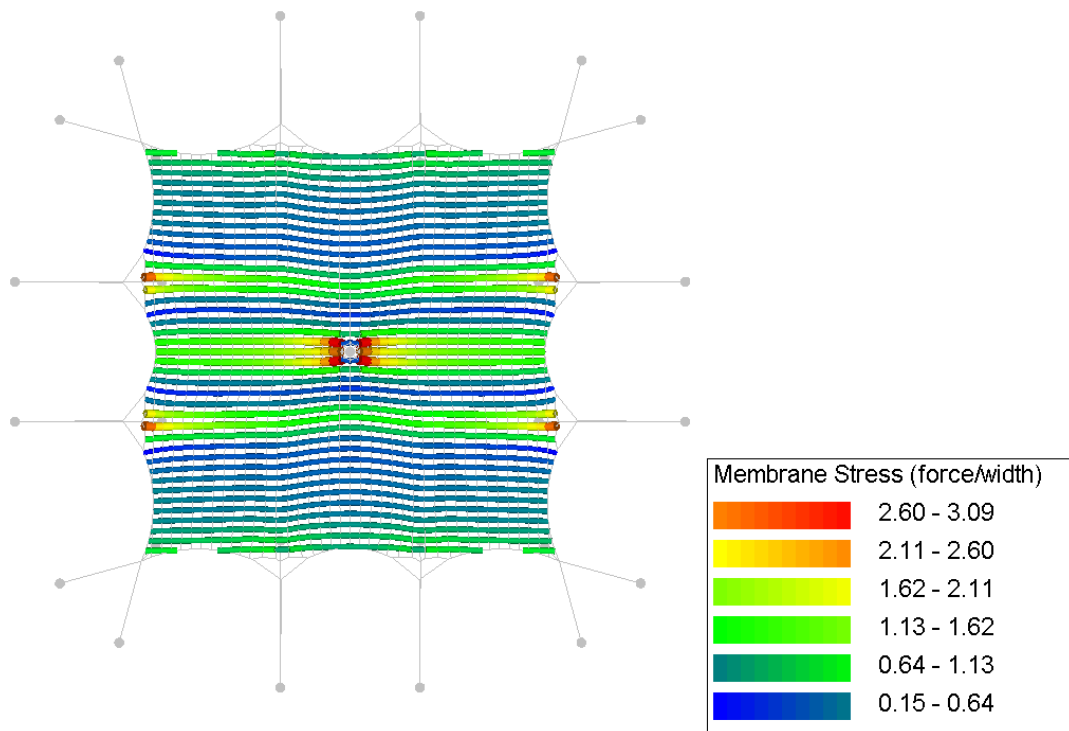


Annex D.2.7. Anchor forces

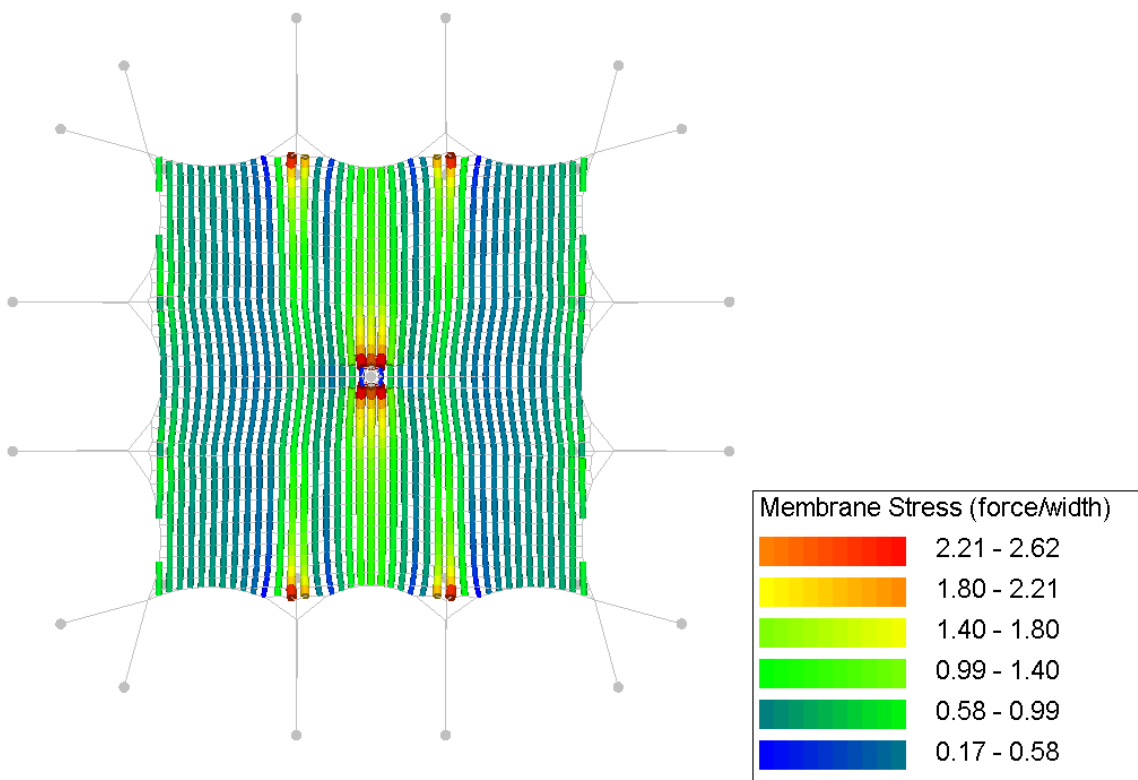


Annex D.3. CO3. Own weight + pretension + wind pressure

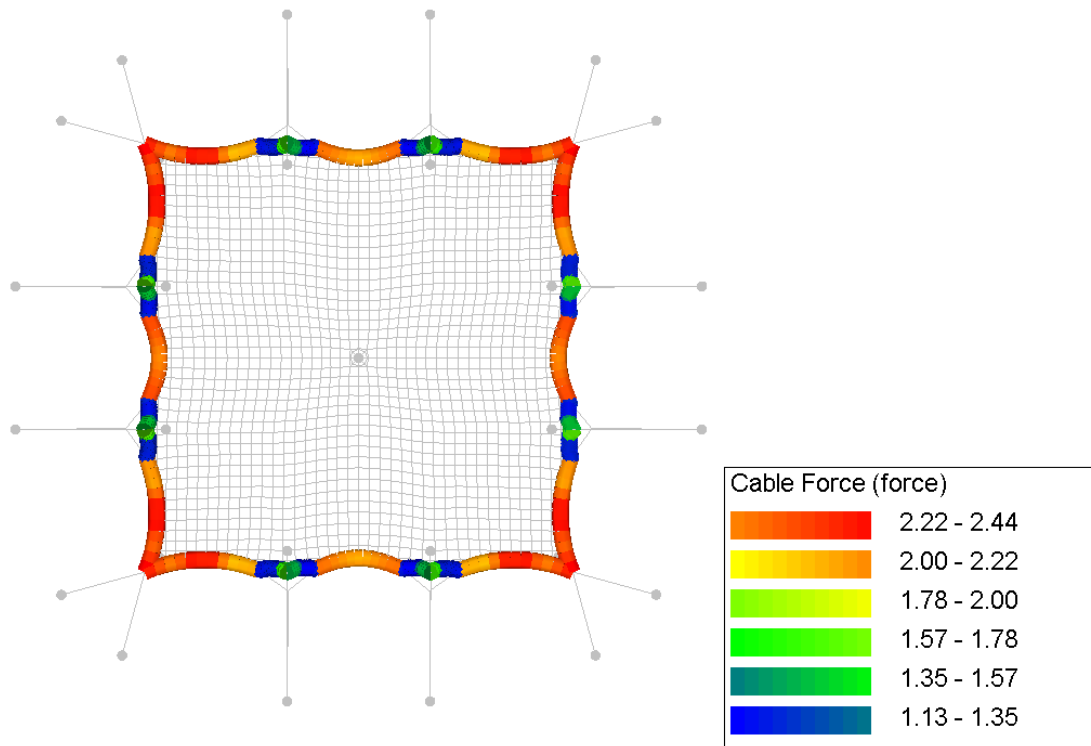
Annex D.3.1. Membrane stress (warp)



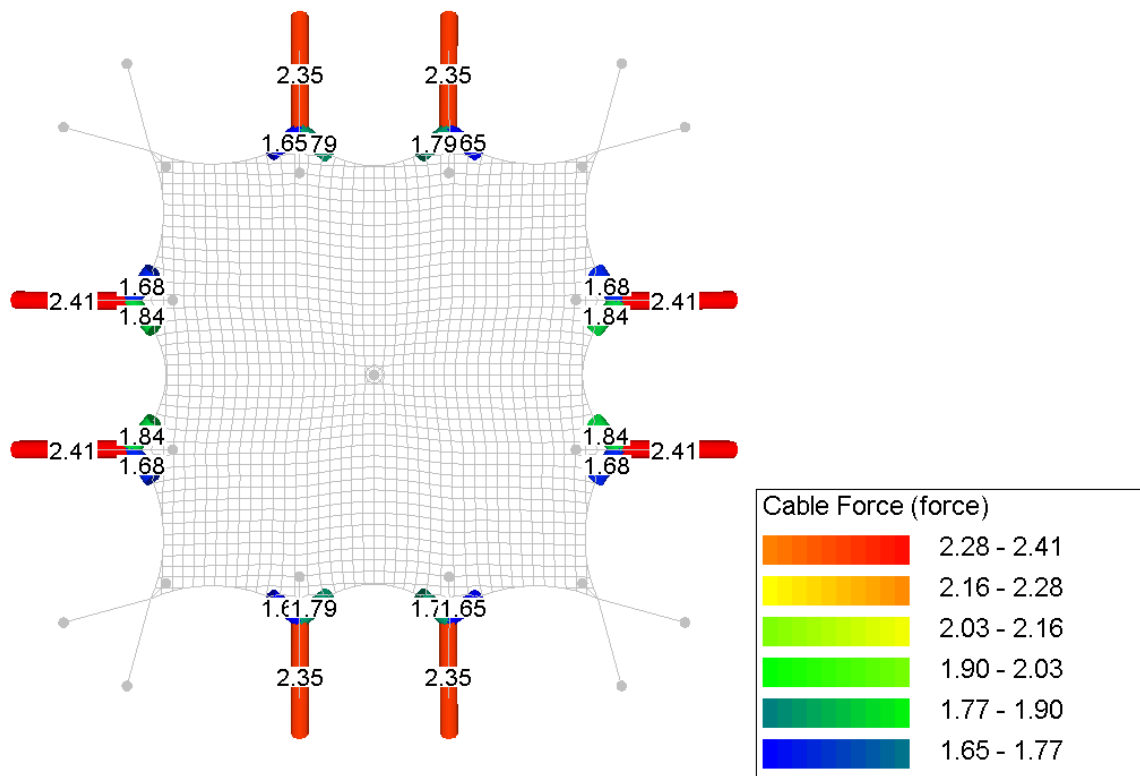
Annex D.3.2. Membrane stress (weft)



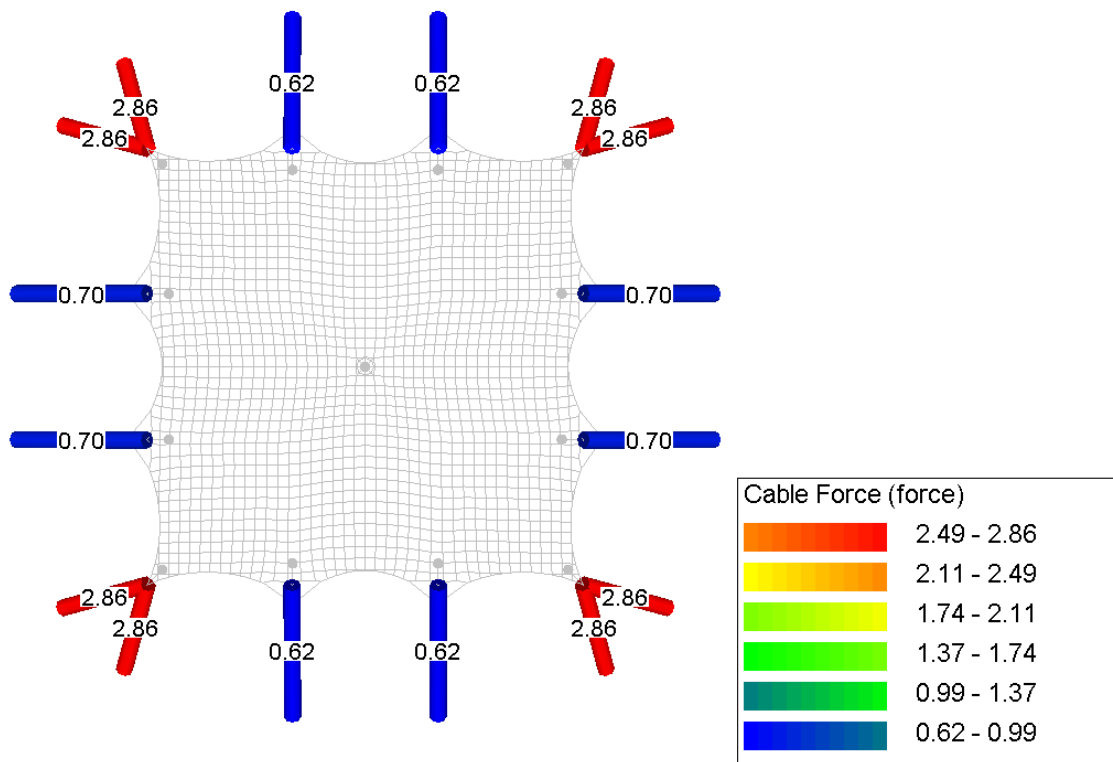
Annex D.3.3. Circumference rope forces



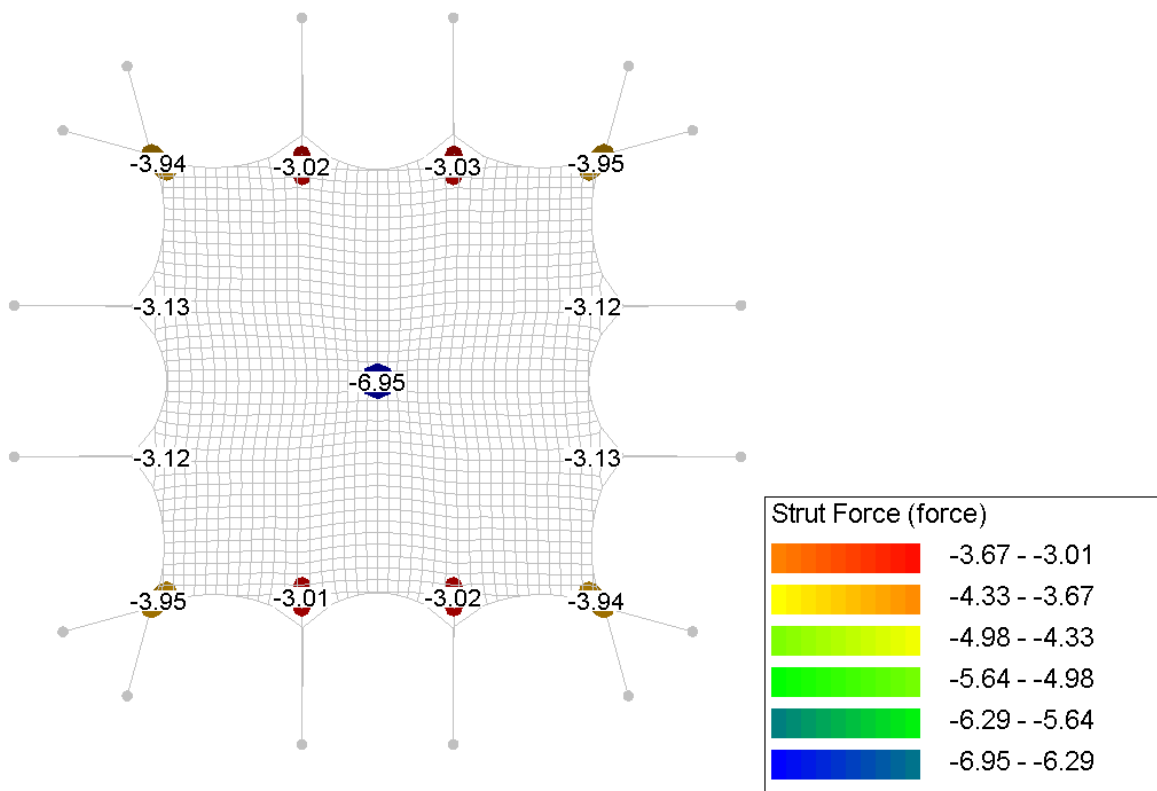
Annex D.3.4. Rope forces



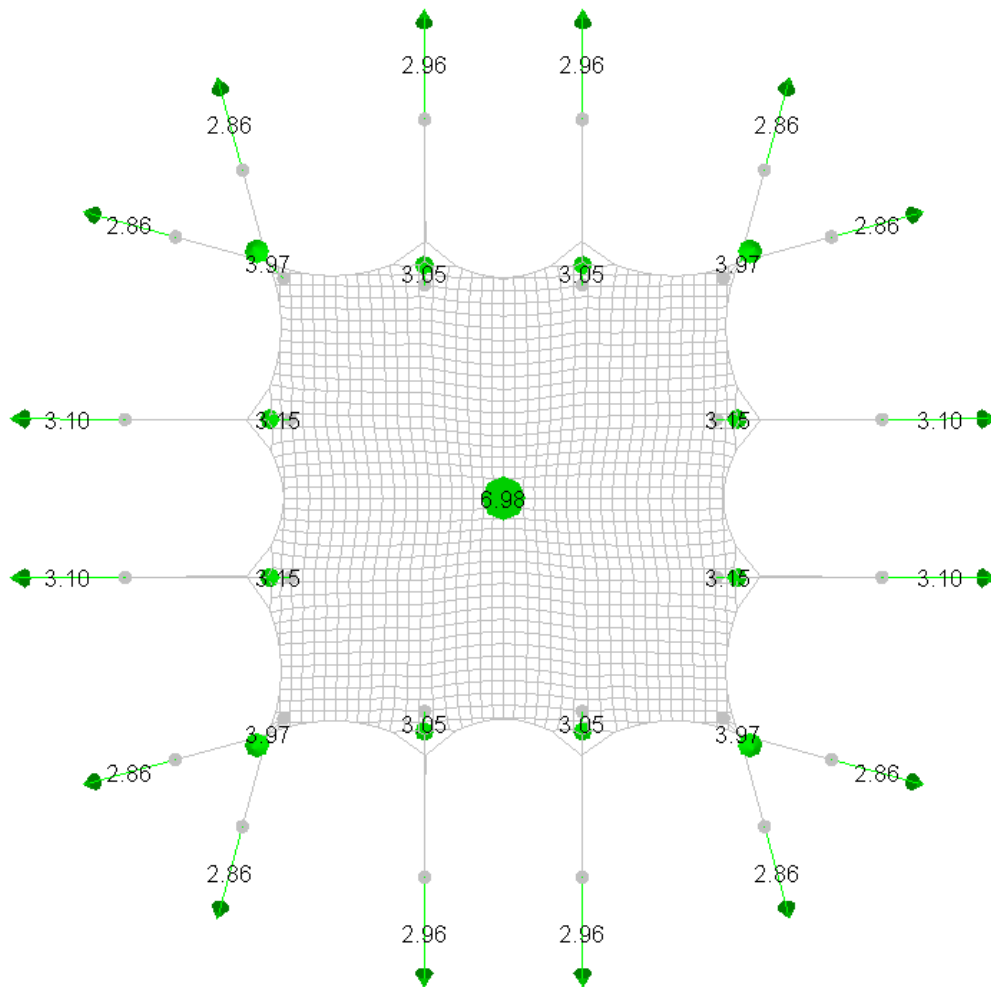
Annex D.3.5. Belt forces



Annex D.3.6. Poles forces

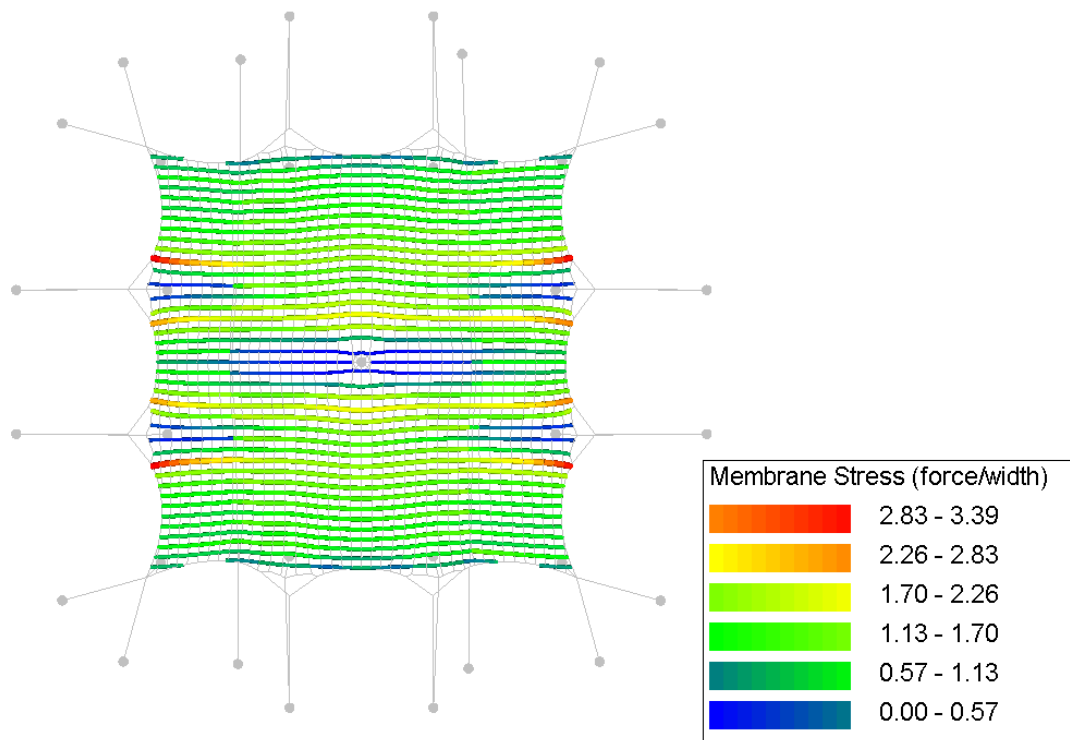


Annex D.3.7. Reaction forces

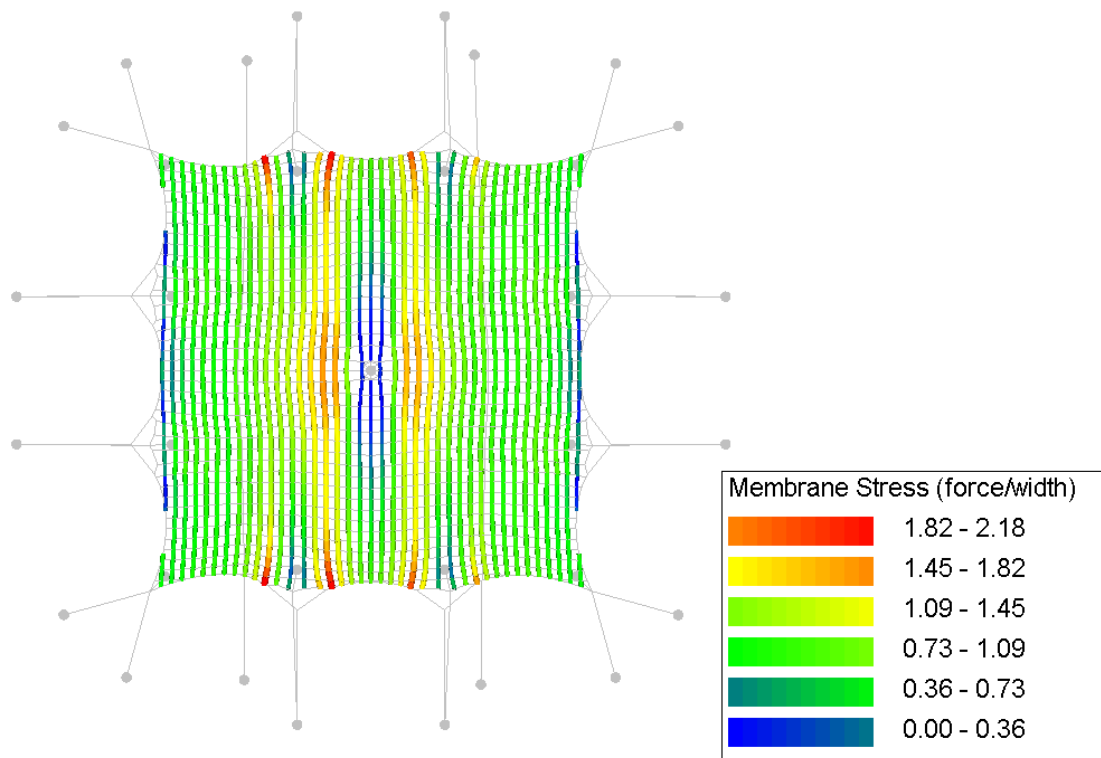


Annex D.4. CO4. Own weight + pretension + wind suction

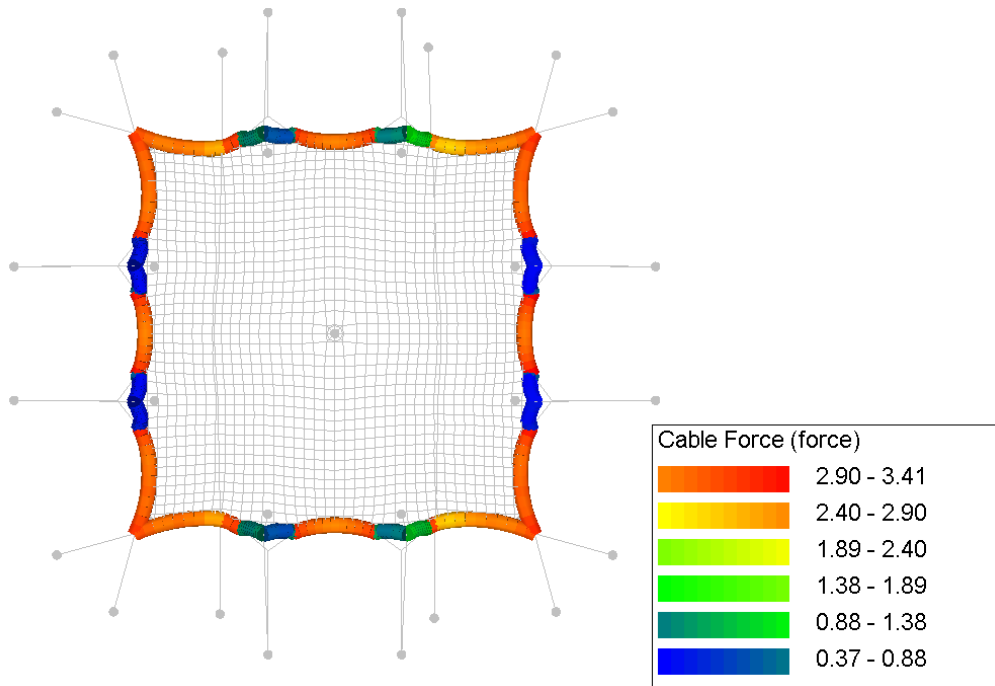
Annex D.4.1. Membrane stress (warp)



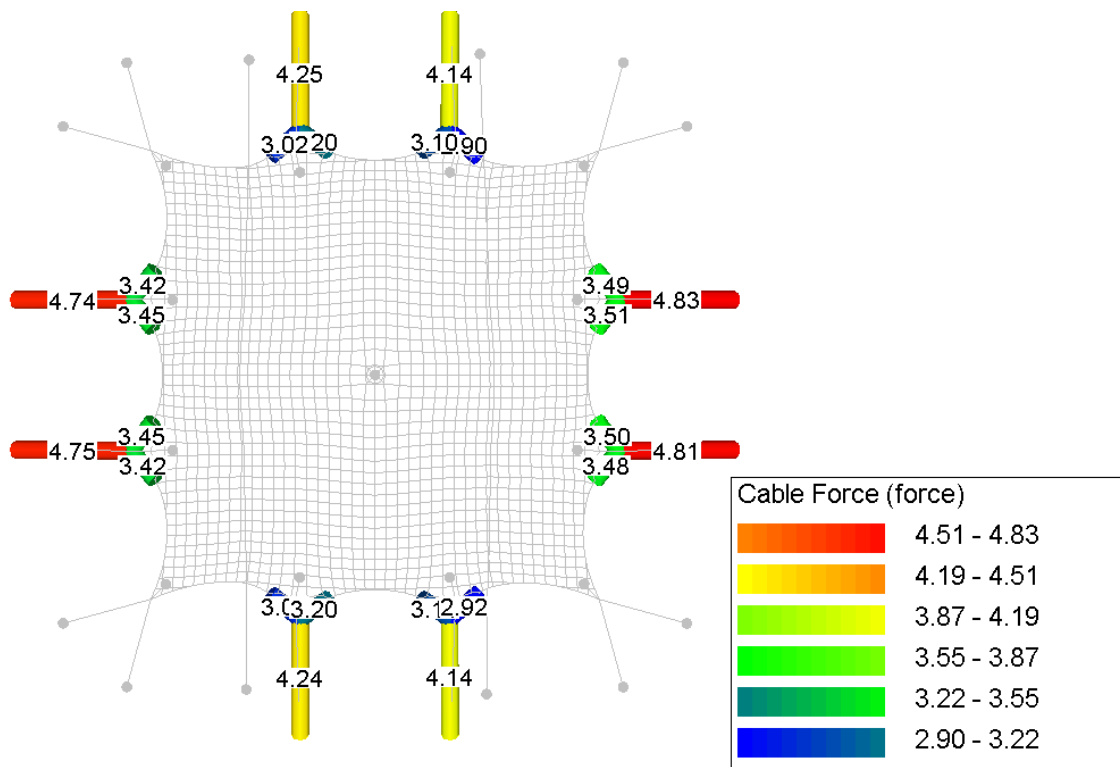
Annex D.4.2. Membrane stress (weft)



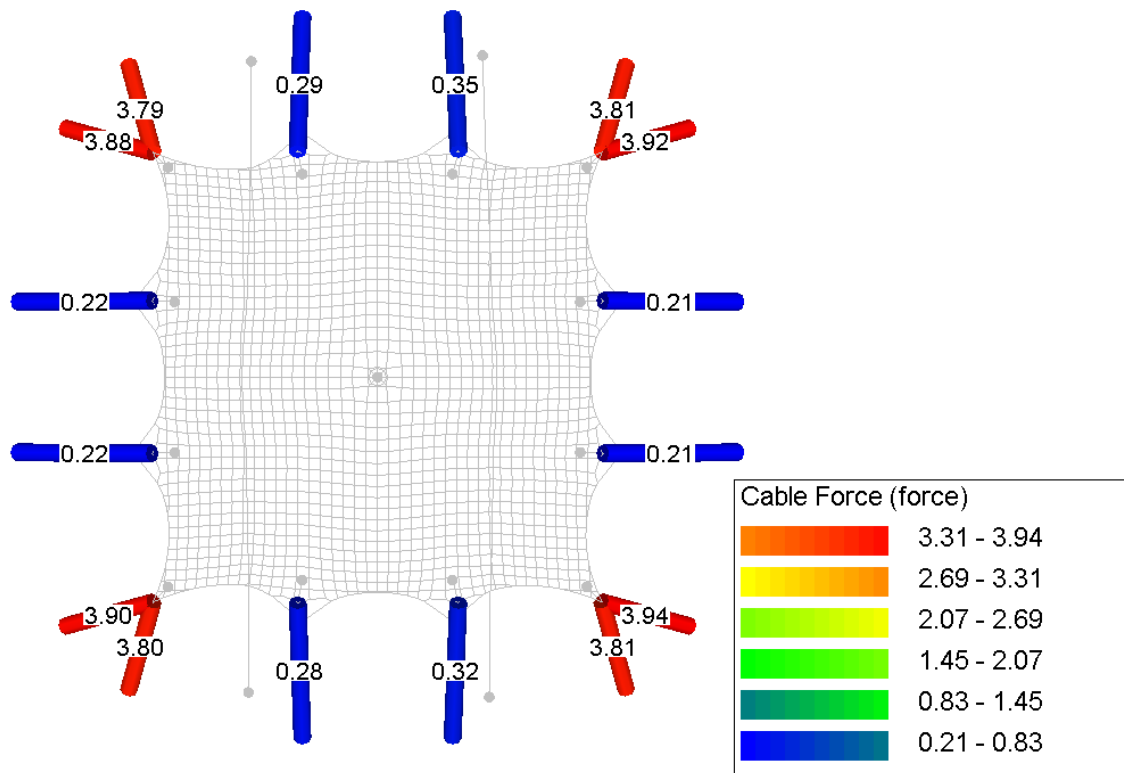
Annex D.4.3. Circumference rope forces



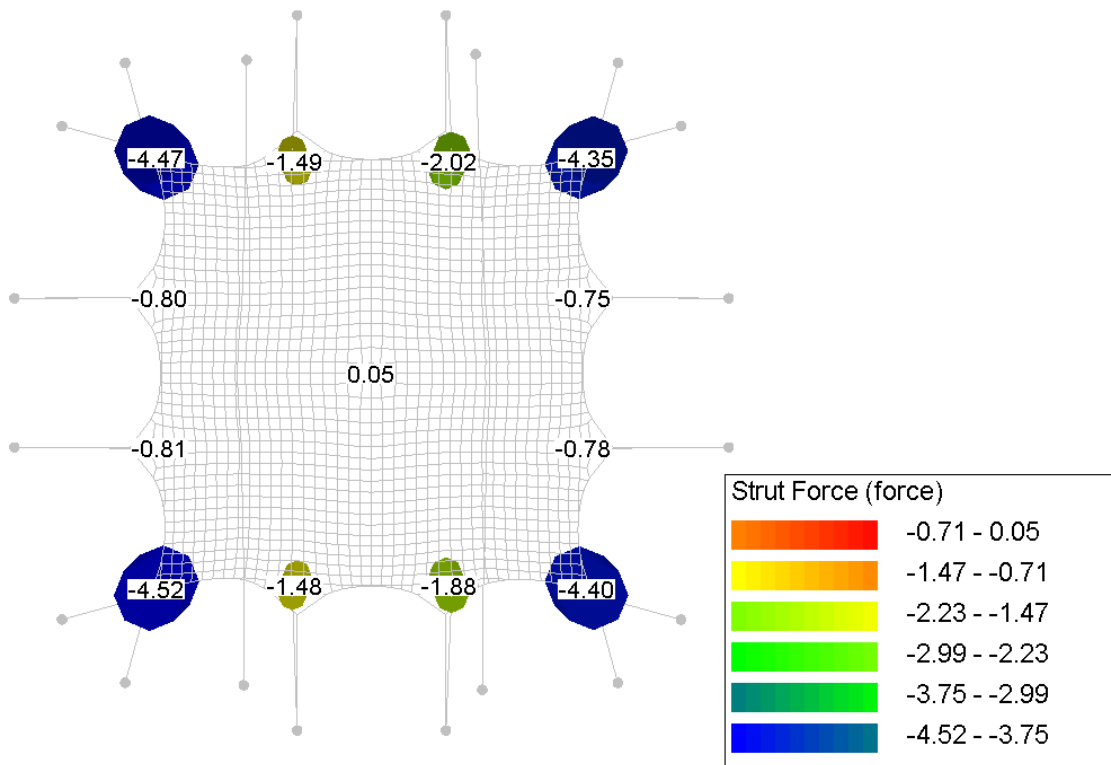
Annex D.4.4. Guy rope forces



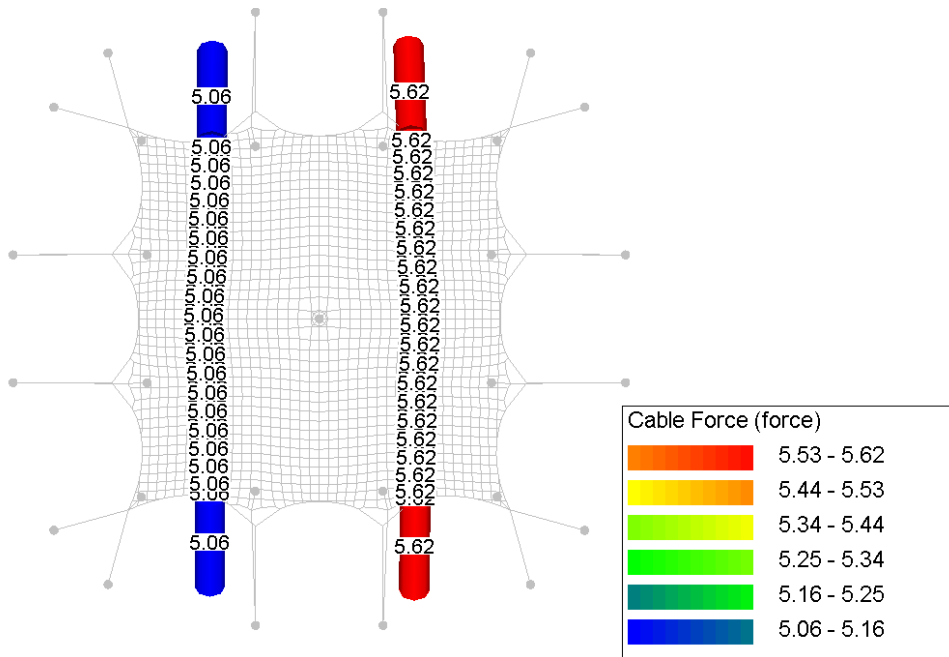
Annex D.4.5. Belt forces



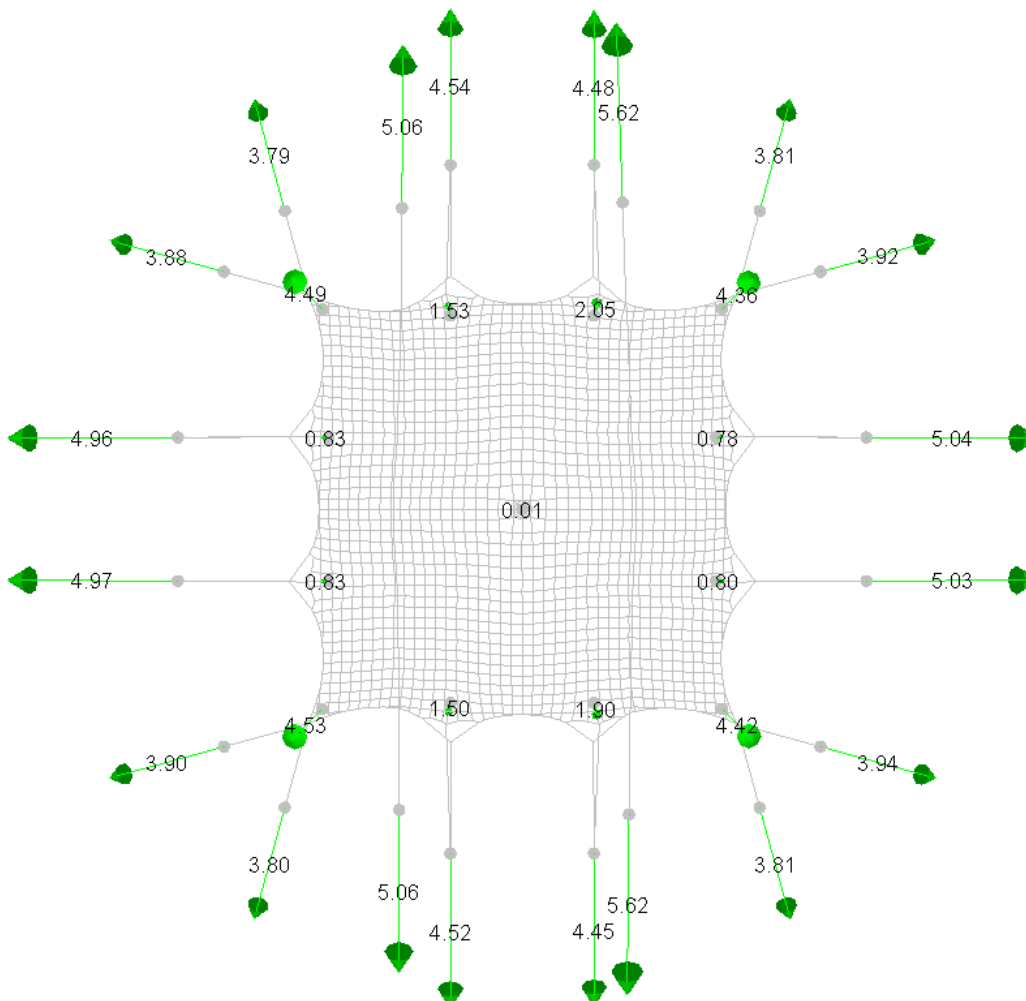
Annex D.4.6. Poles forces



Annex D.4.7. Storm belt forces

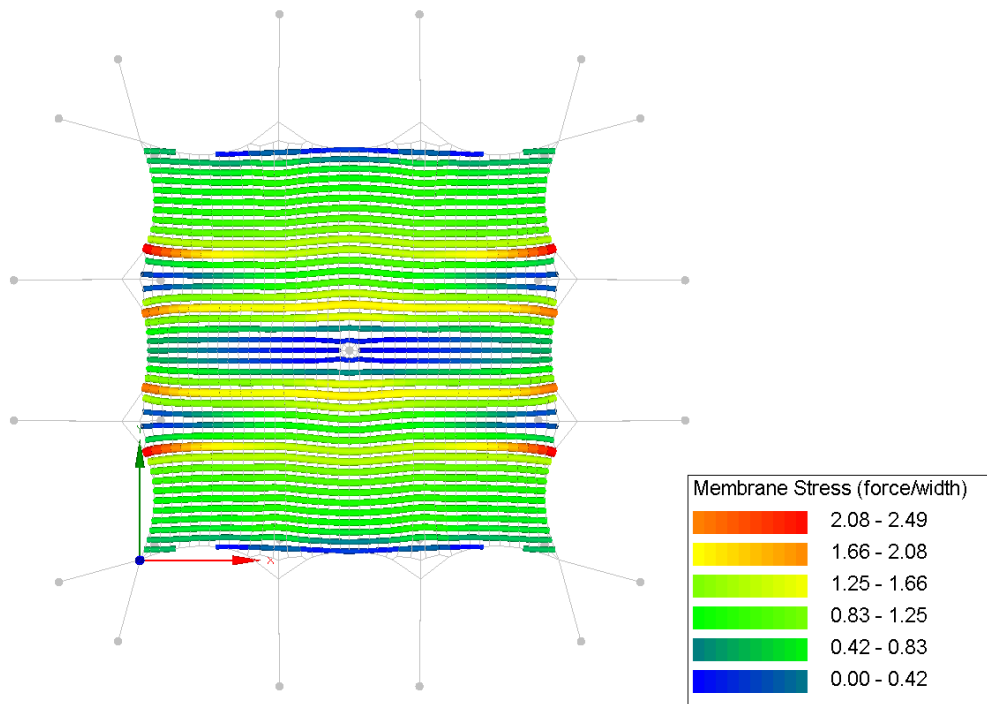


Annex D.4.8. Reaction forces

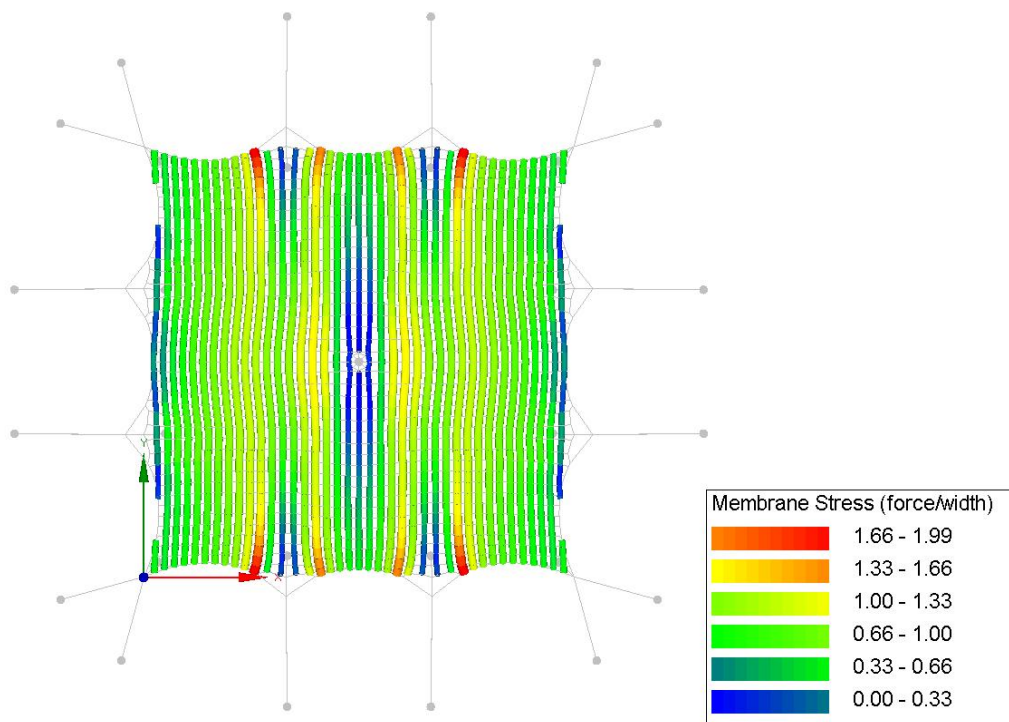


Annex D.5. CO5. Own weight + pretension + wind suction reduced [NL]

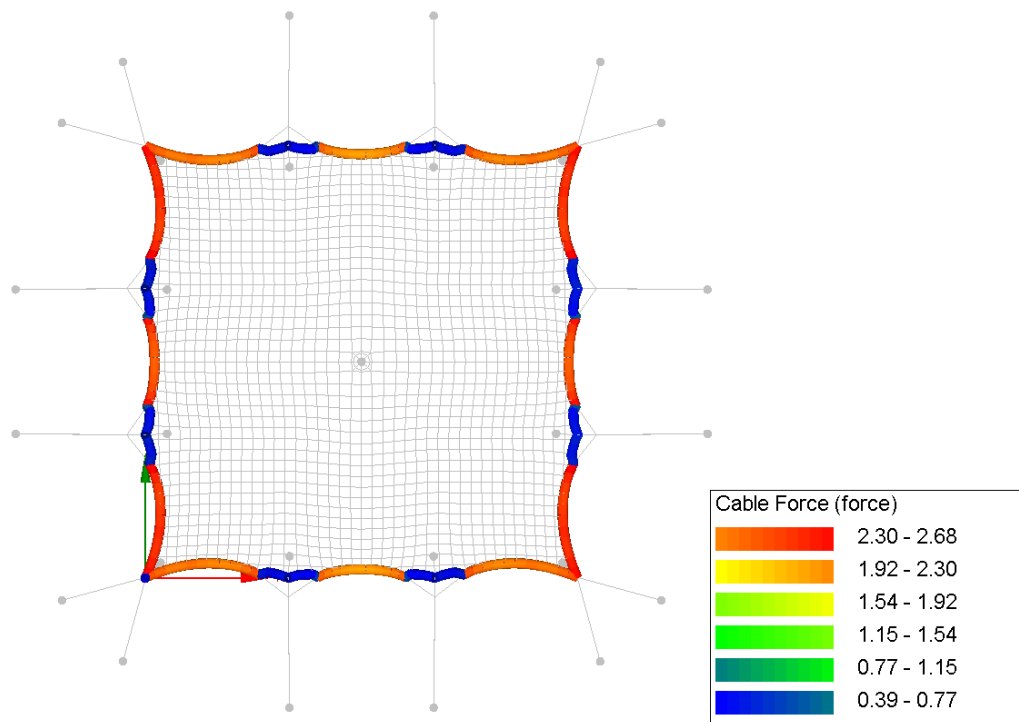
Annex D.5.1. Membrane stress (warp)



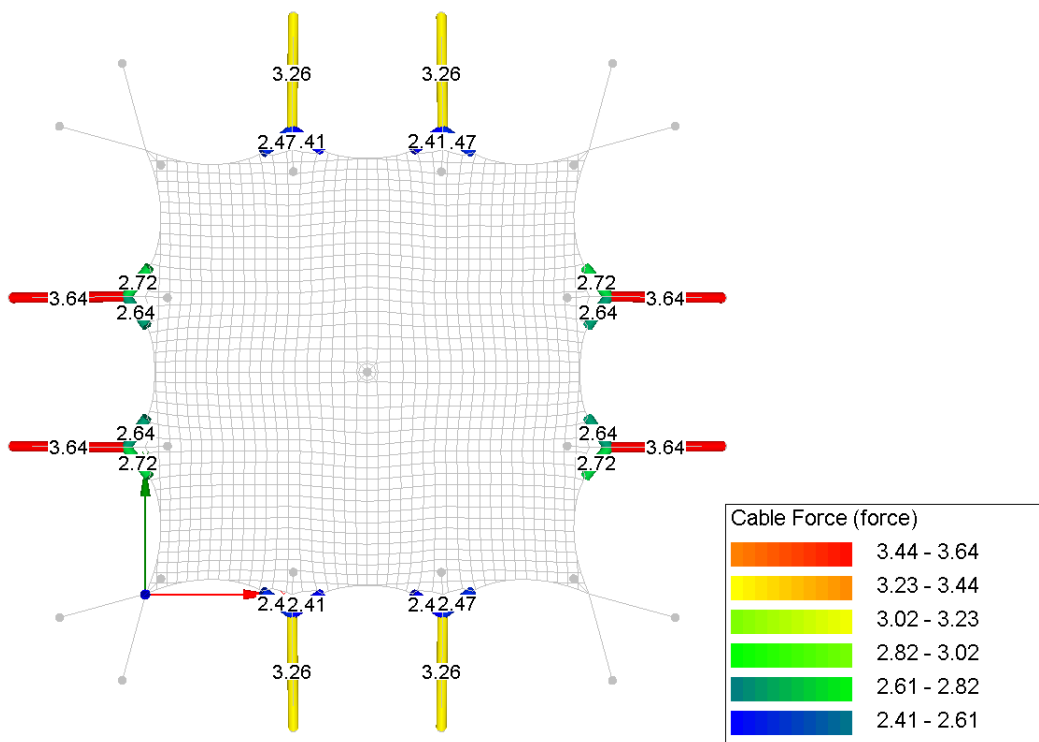
Annex D.5.2. Membrane stress (weft)



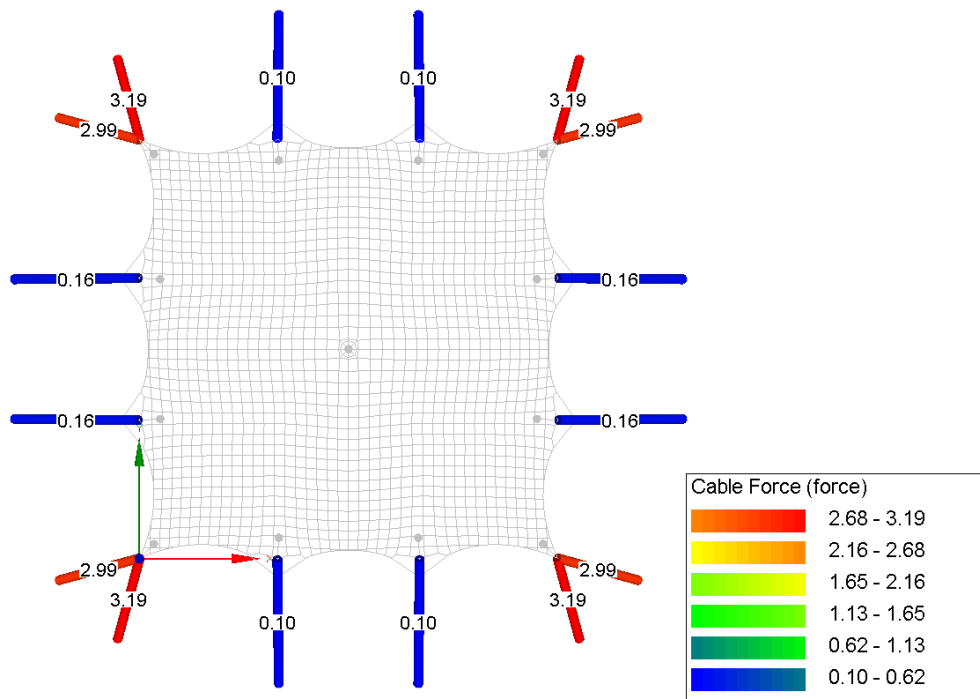
Annex D.5.3. Circumference rope forces



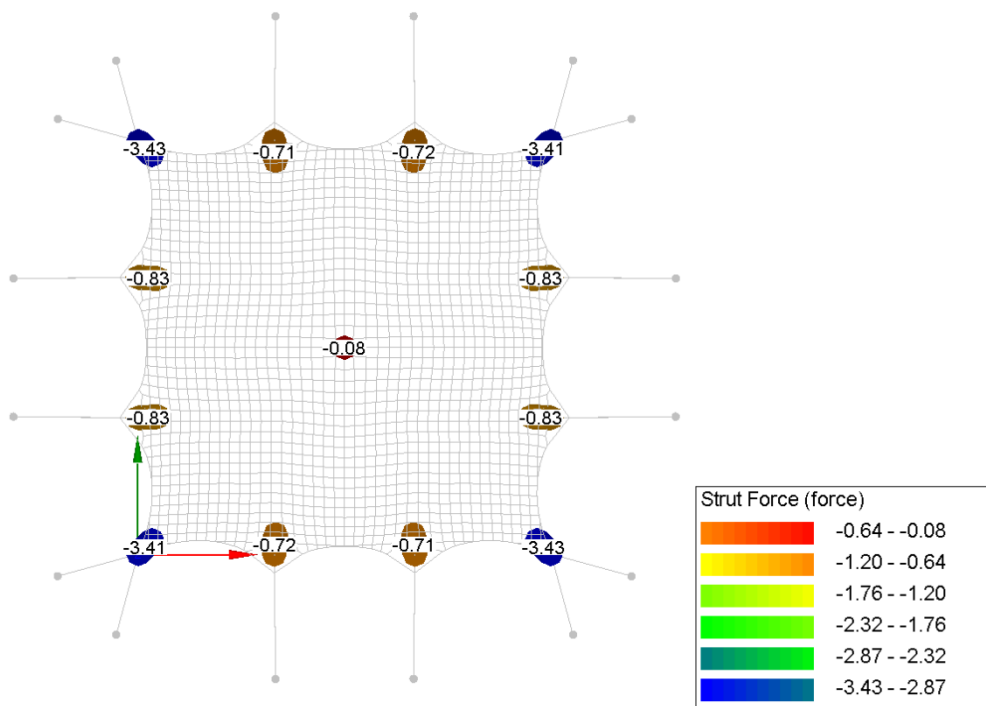
Annex D.5.4. Rope forces



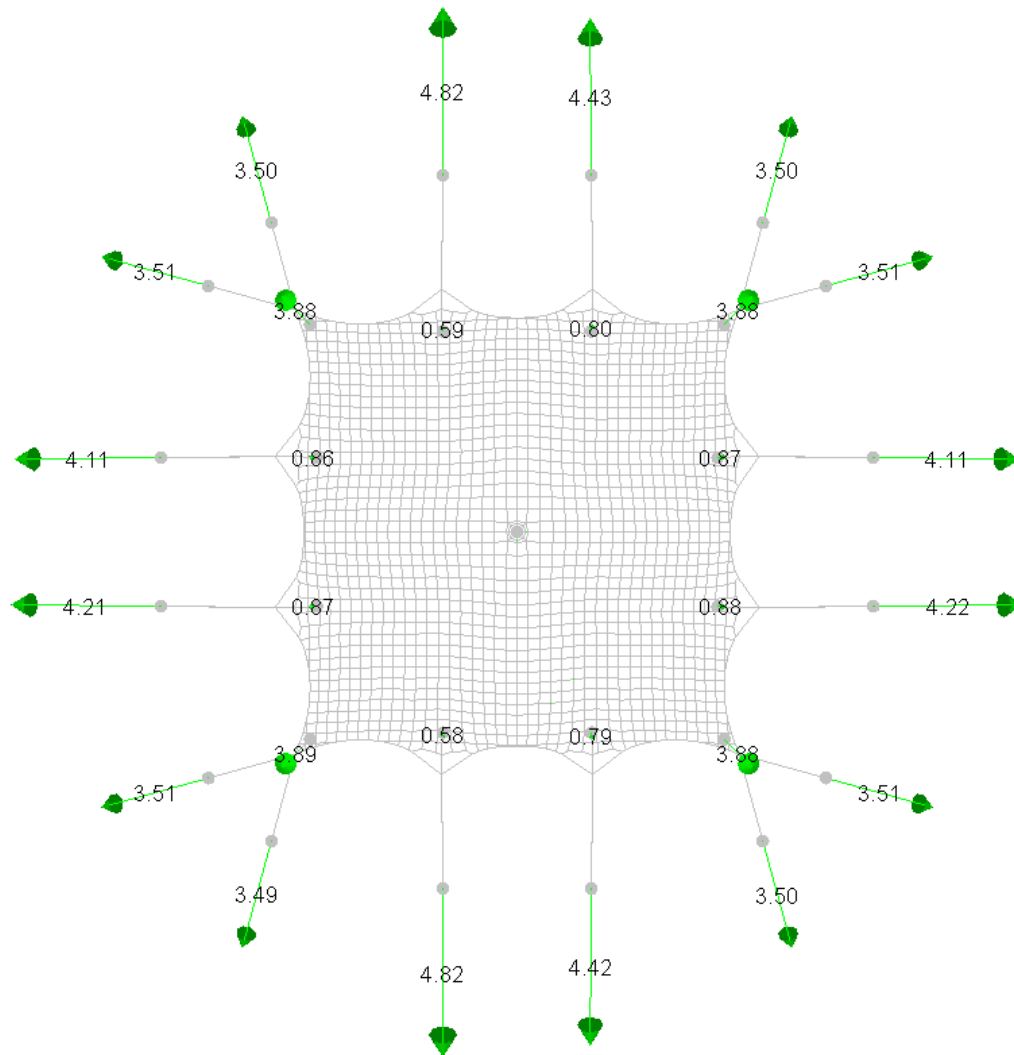
Annex D.5.5. Belt forces



Annex D.5.6. Poles forces

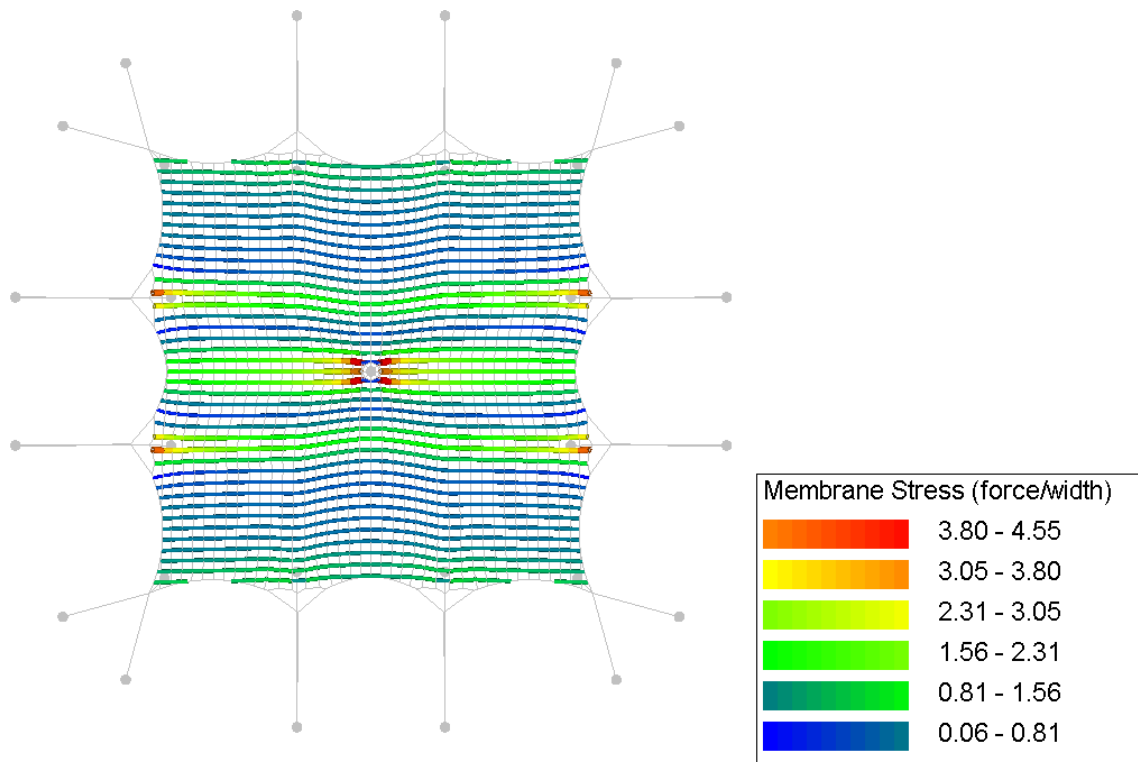


Annex D.5.7. Reaction forces

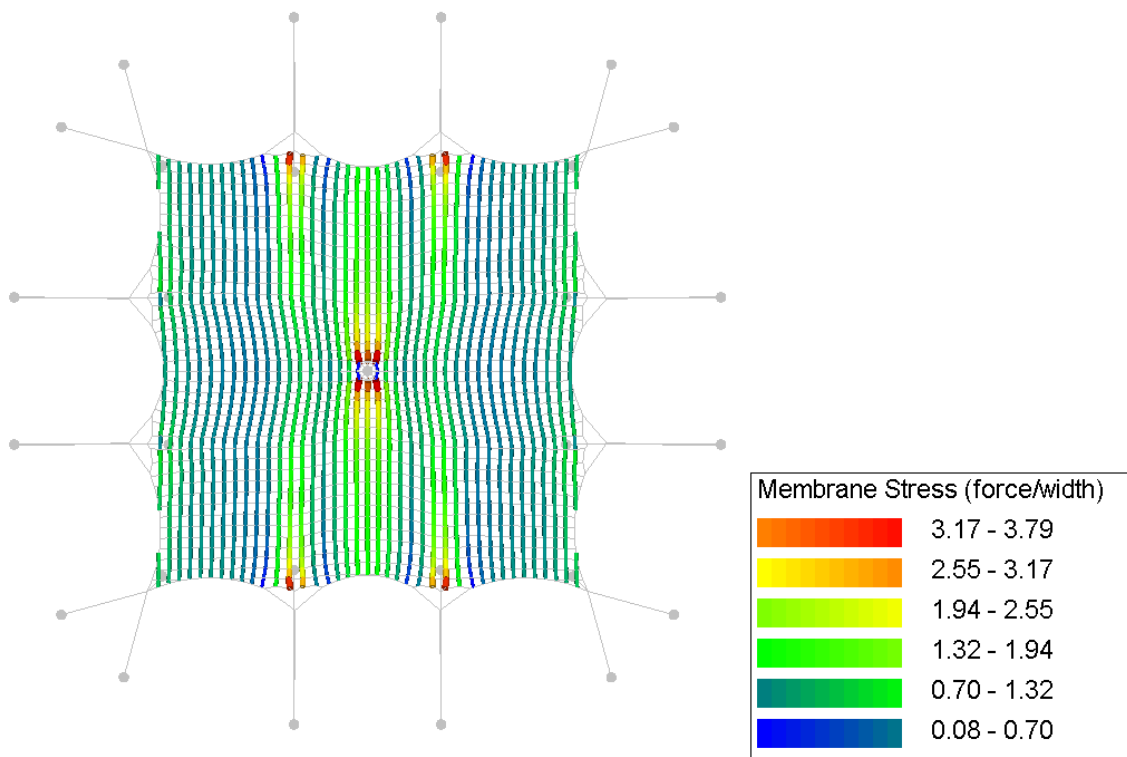


Annex D.6. CO6. Own weight + pretension + wind pressure + snow [FR]

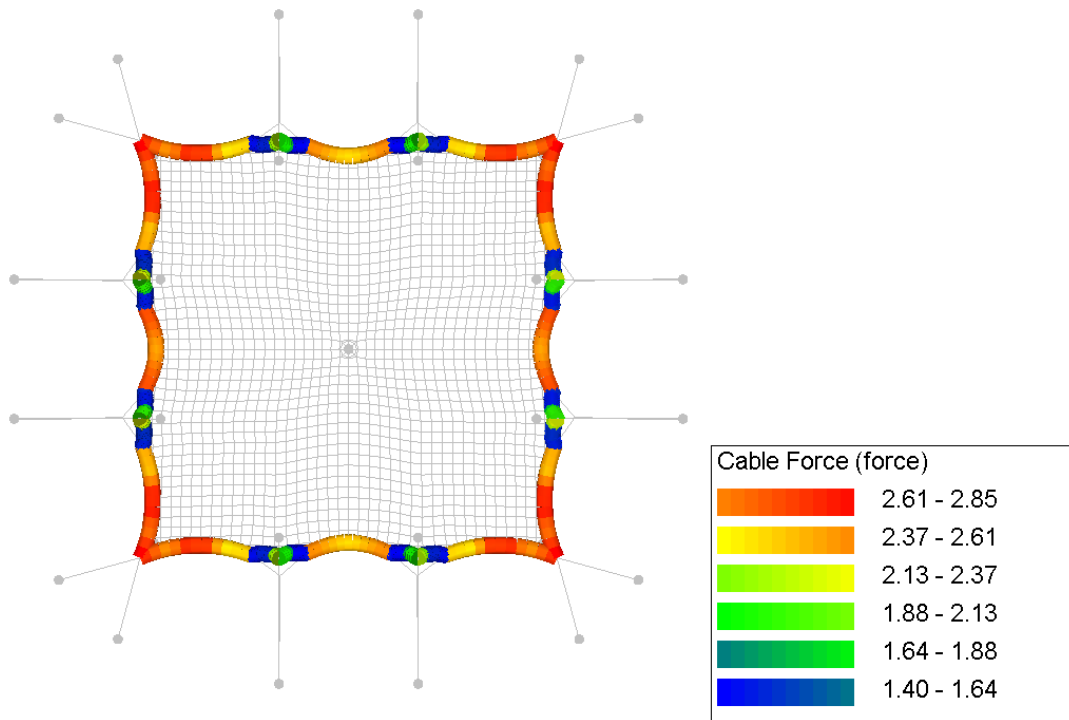
Annex D.6.1. Membrane stress (warp)



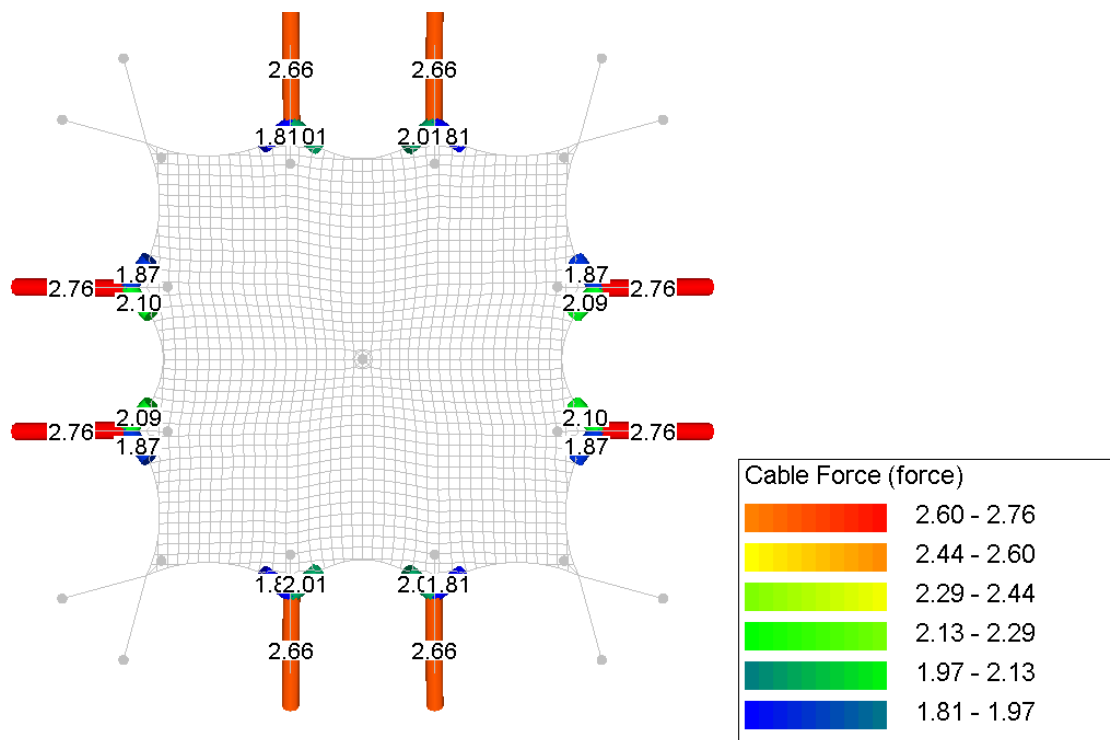
Annex D.6.2. Membrane stress (weft)



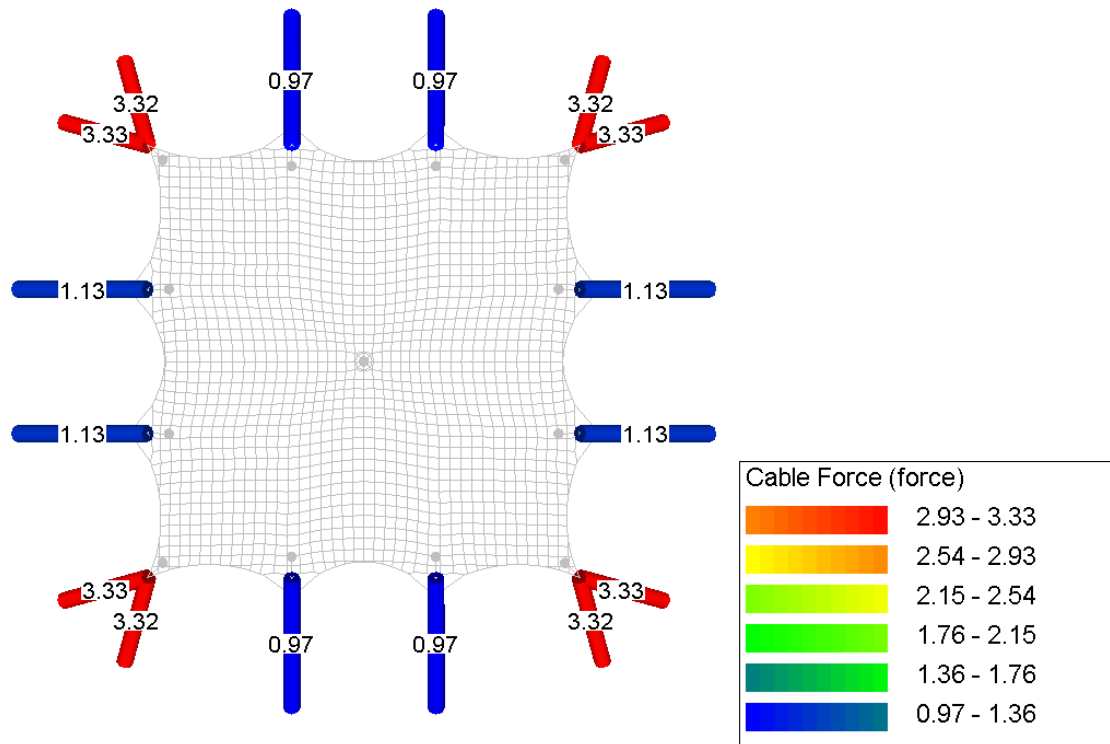
Annex D.6.3. Circumference rope forces



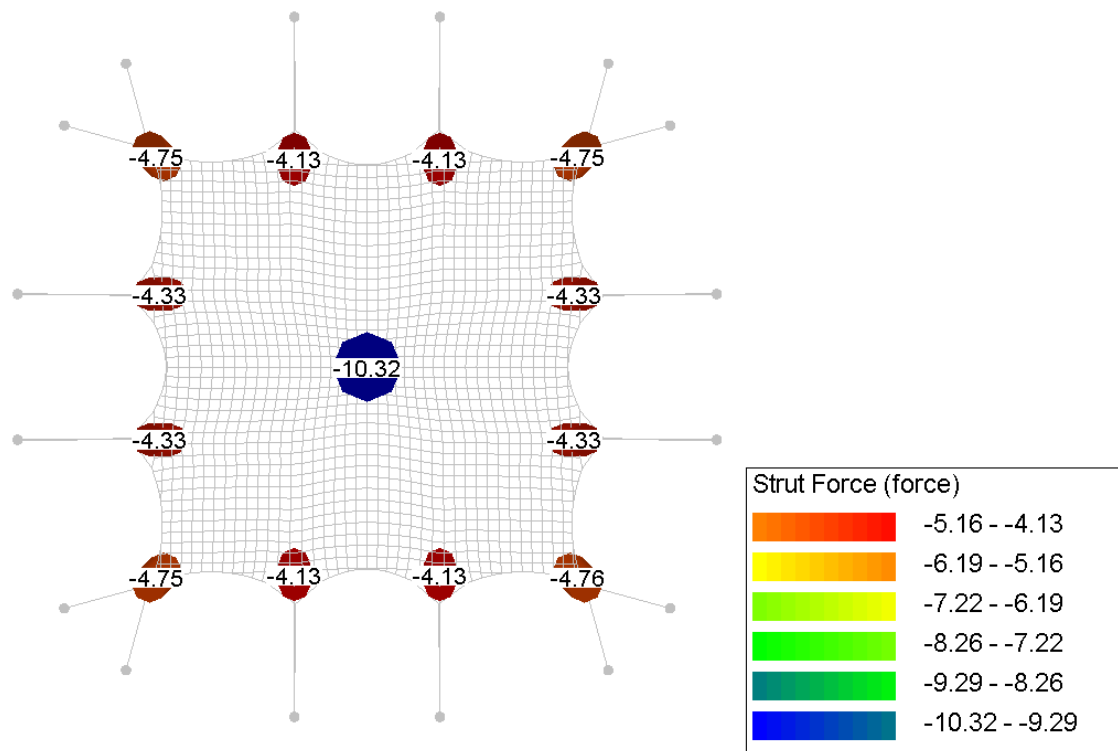
Annex D.6.4. Guy rope forces



Annex D.6.5. Belt forces

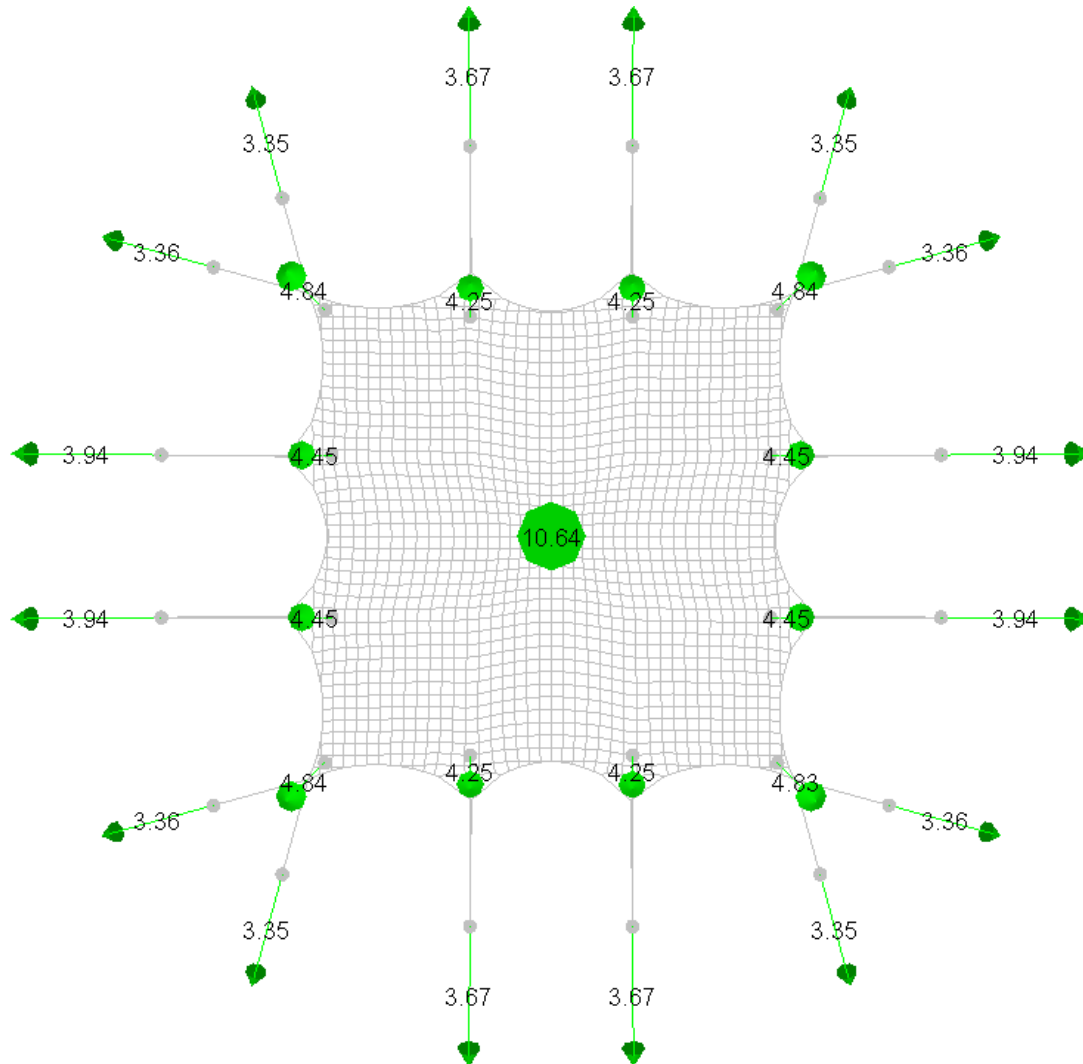


Annex D.6.6. Poles forces



Annex D.6.7. Reaction forces

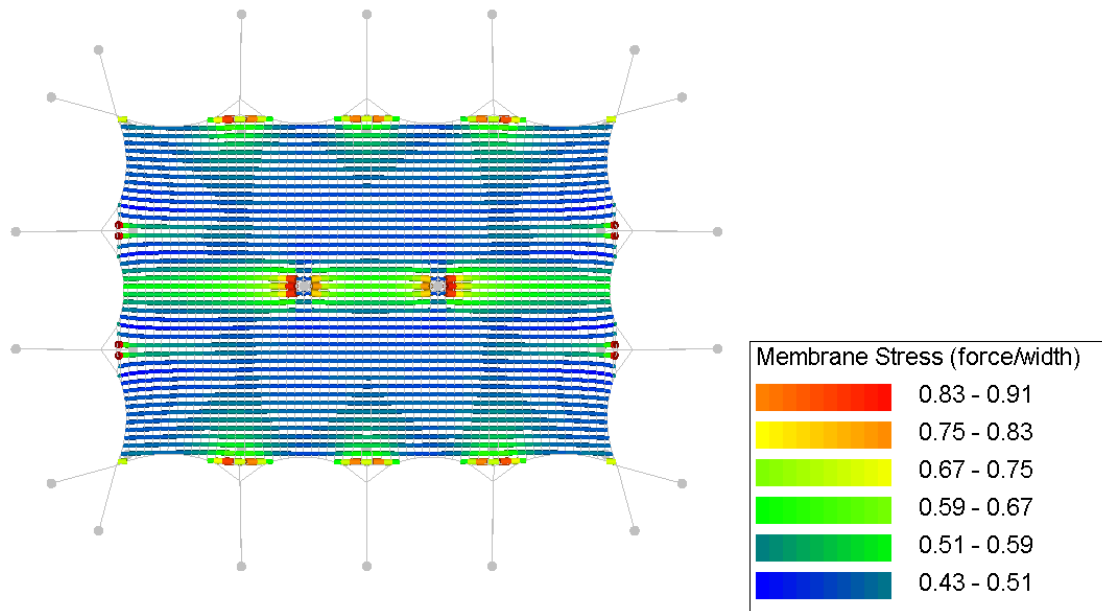
Increased snow load of $0.1 \times 1.5/1.35 = 0.1083 \text{ kN/m}^2$ is applied for ballast and anchor calculation



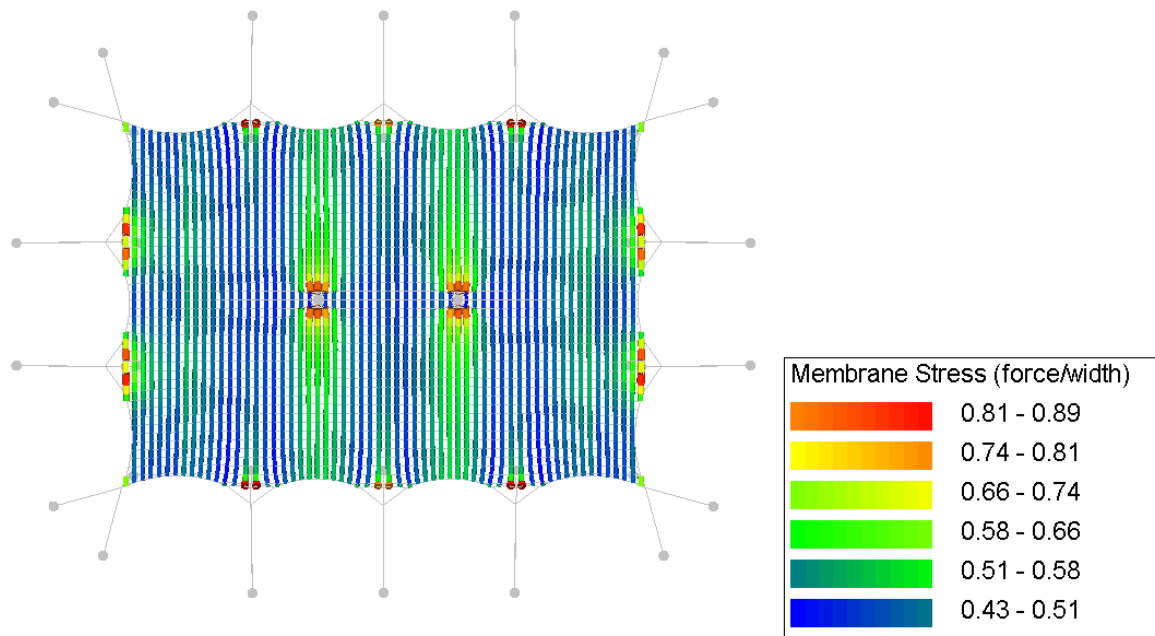
Annex E. Results per load combination – 10x15m model

Annex E.1. CO1. Own weight + pretension

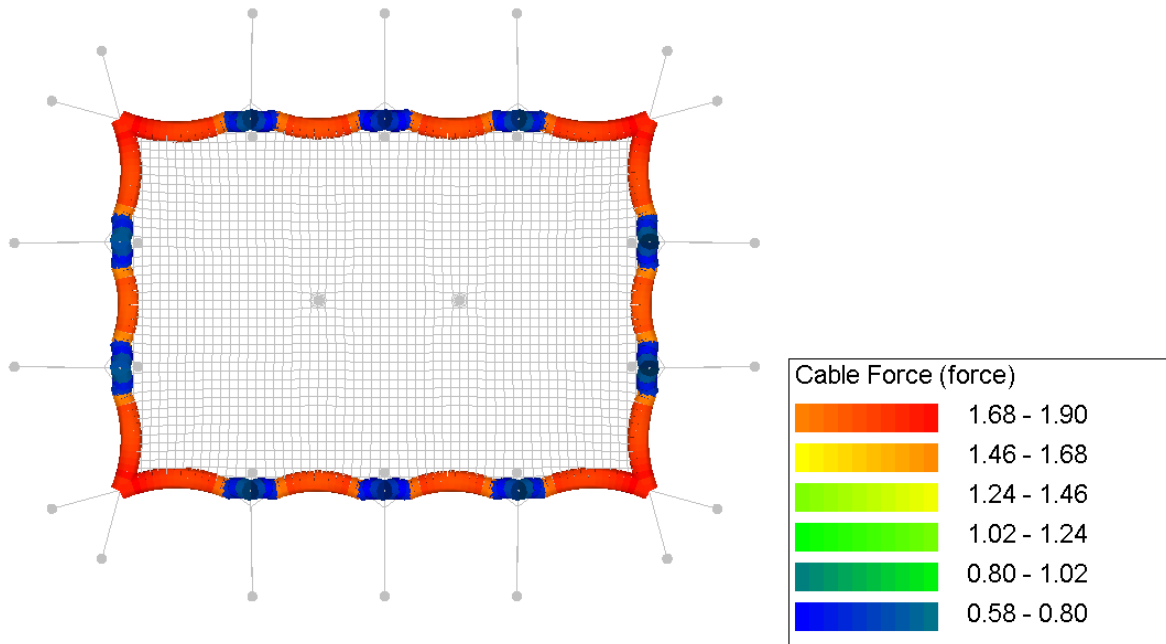
Annex E.1.1. Membrane stress (warp)



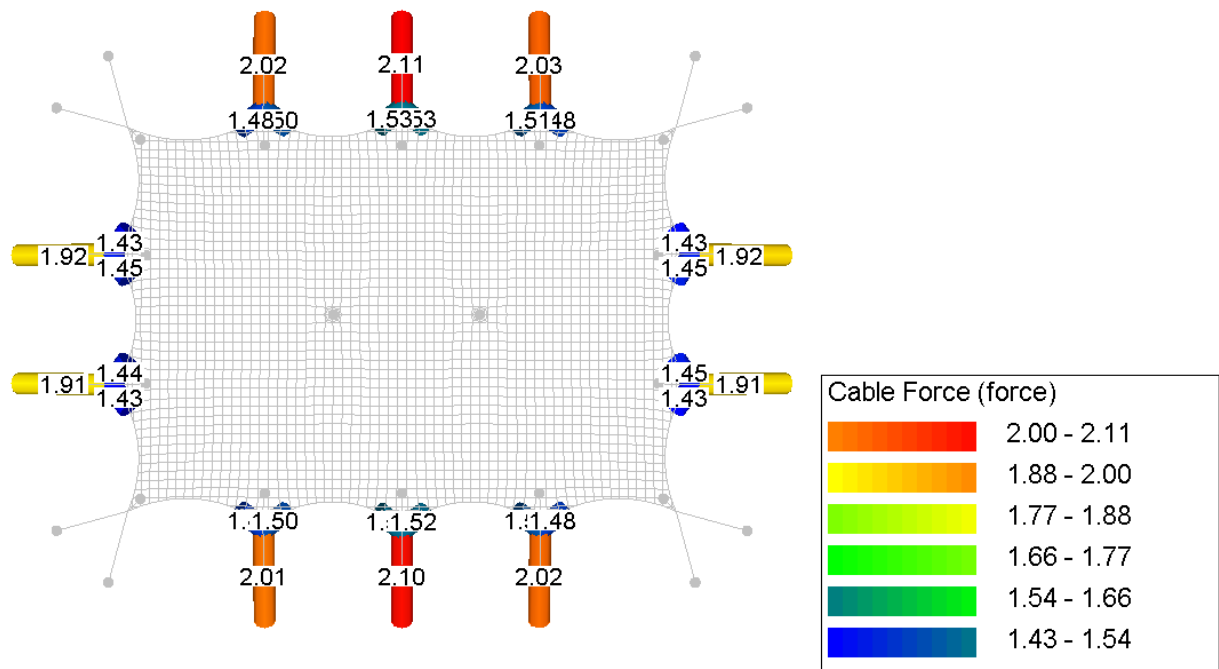
Annex E.1.2. Membrane stress (weft)



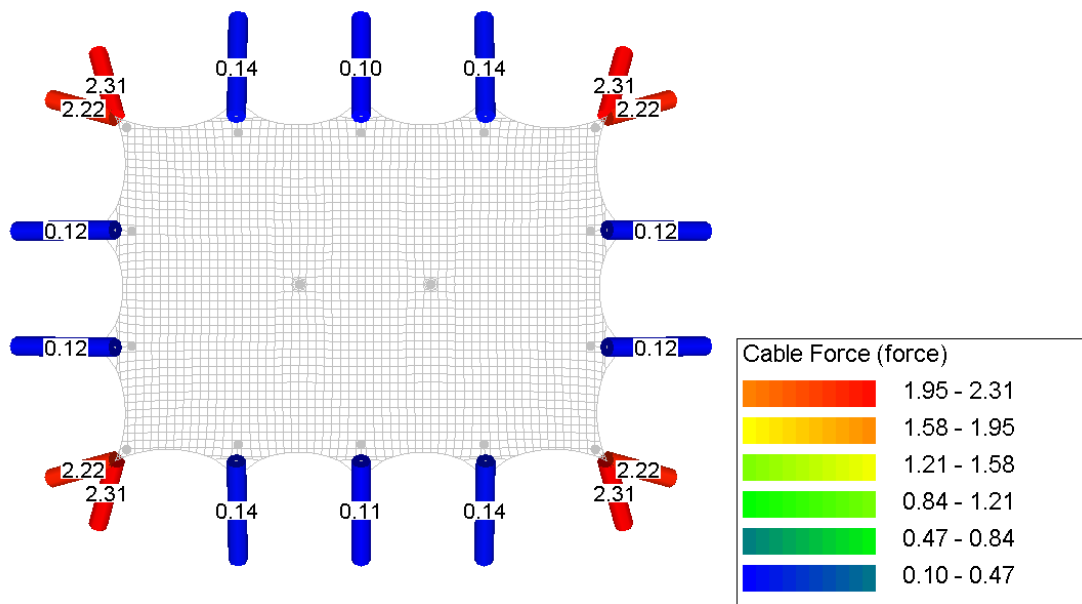
Annex E.1.3. Circumference rope forces



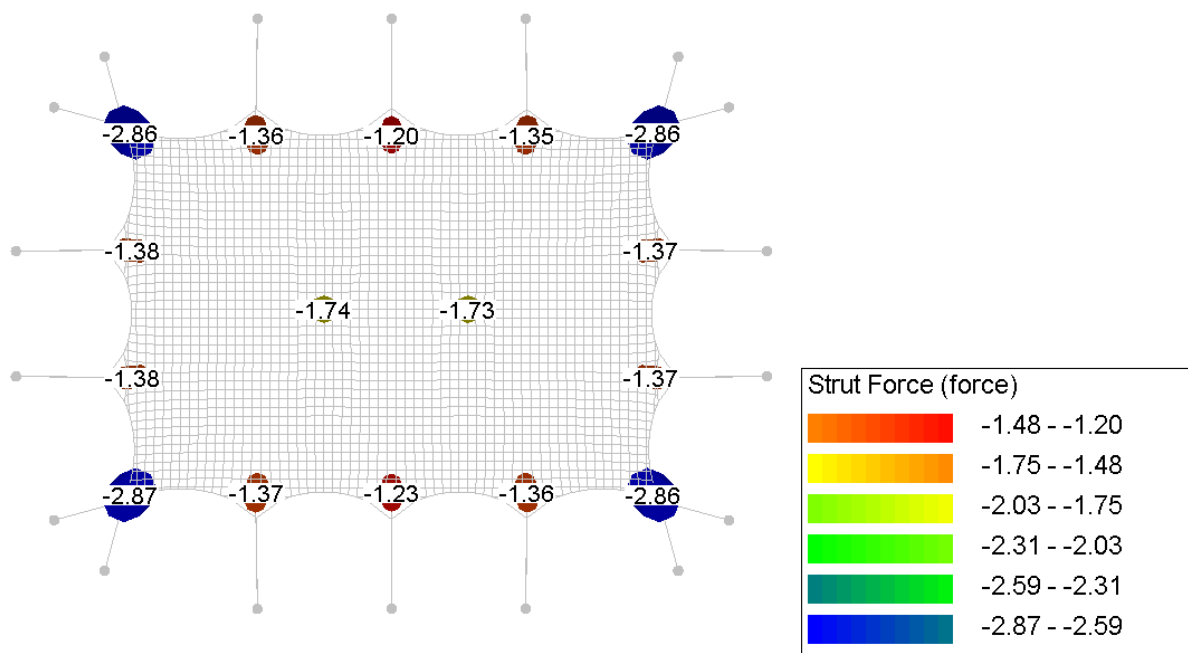
Annex E.1.4. Rope forces



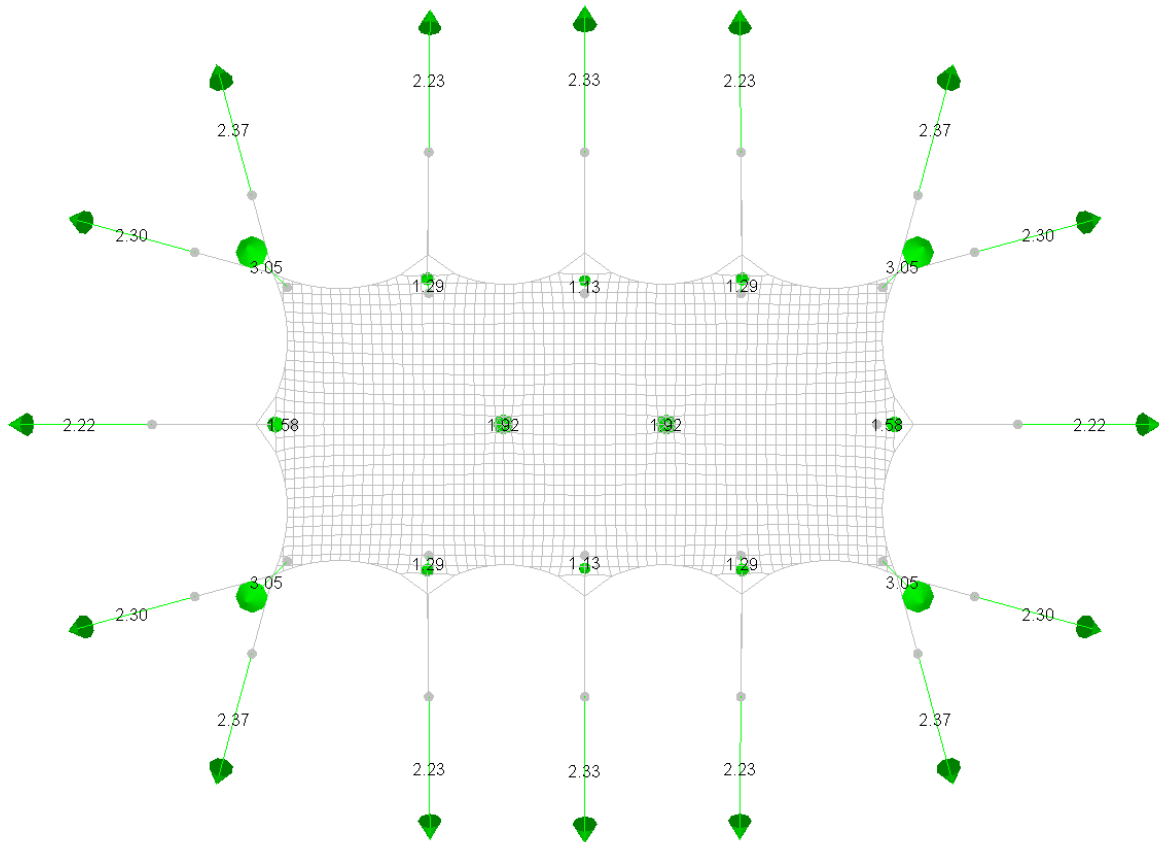
Annex E.1.5. Belt forces



Annex E.1.6. Pole forces

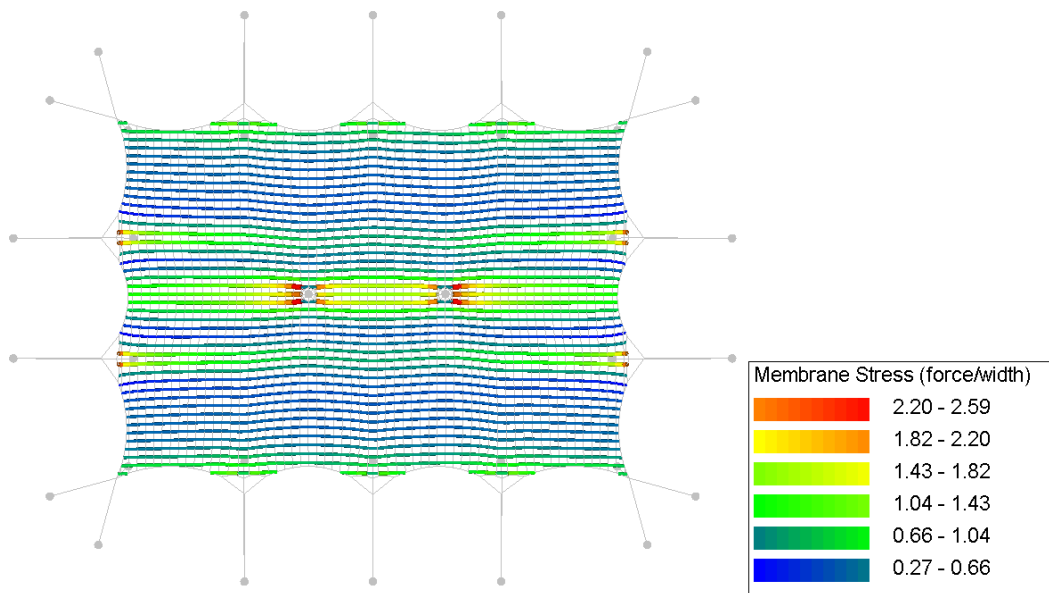


Annex E.1.7. Reaction forces

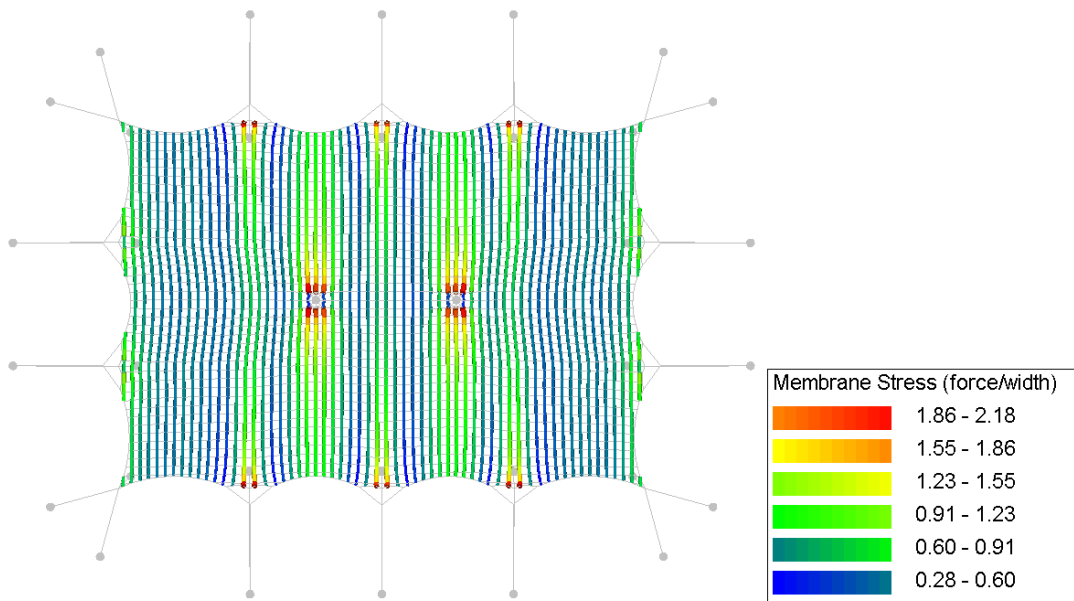


Annex E.2. CO2. Own weight + pretension + snow

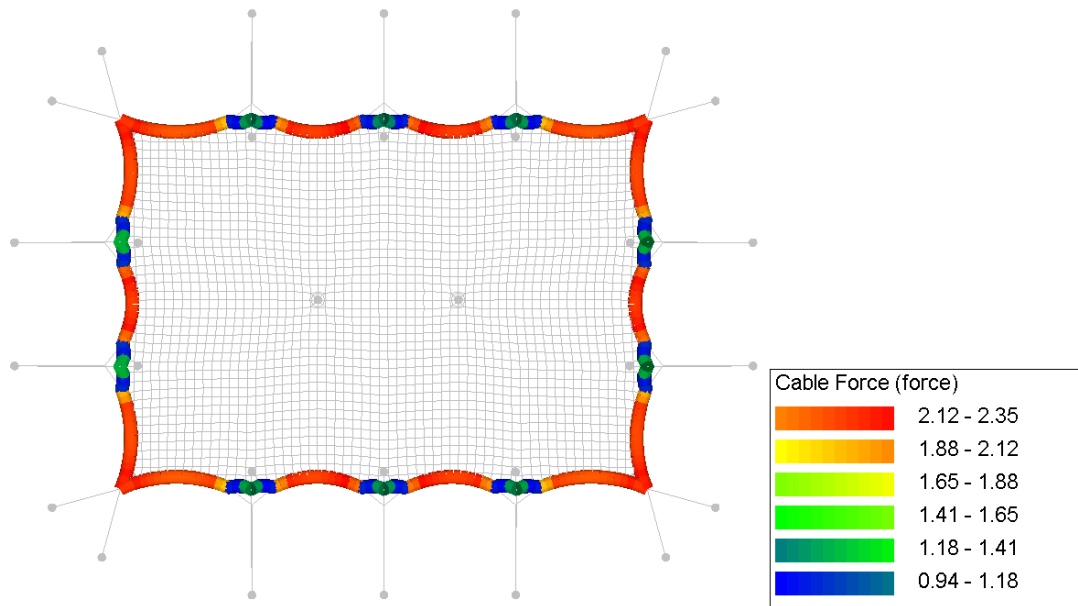
Annex E.2.1. Membrane stress (warp)



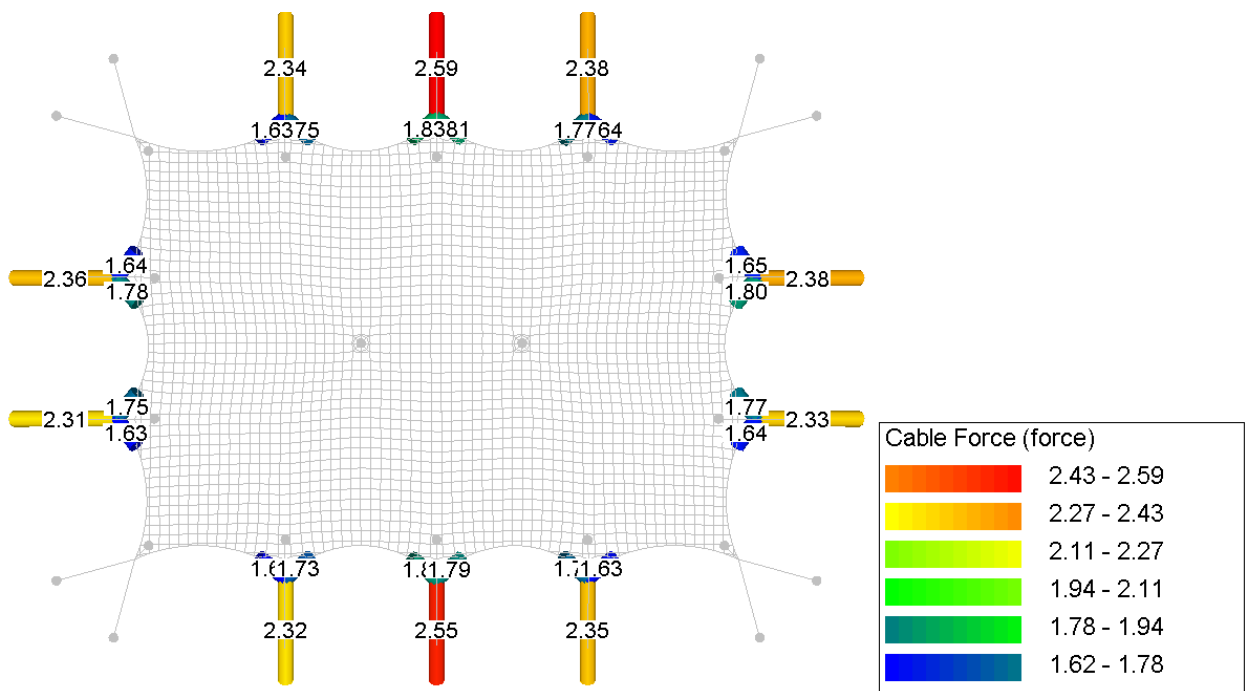
Annex E.2.2. Membrane stress (weft)



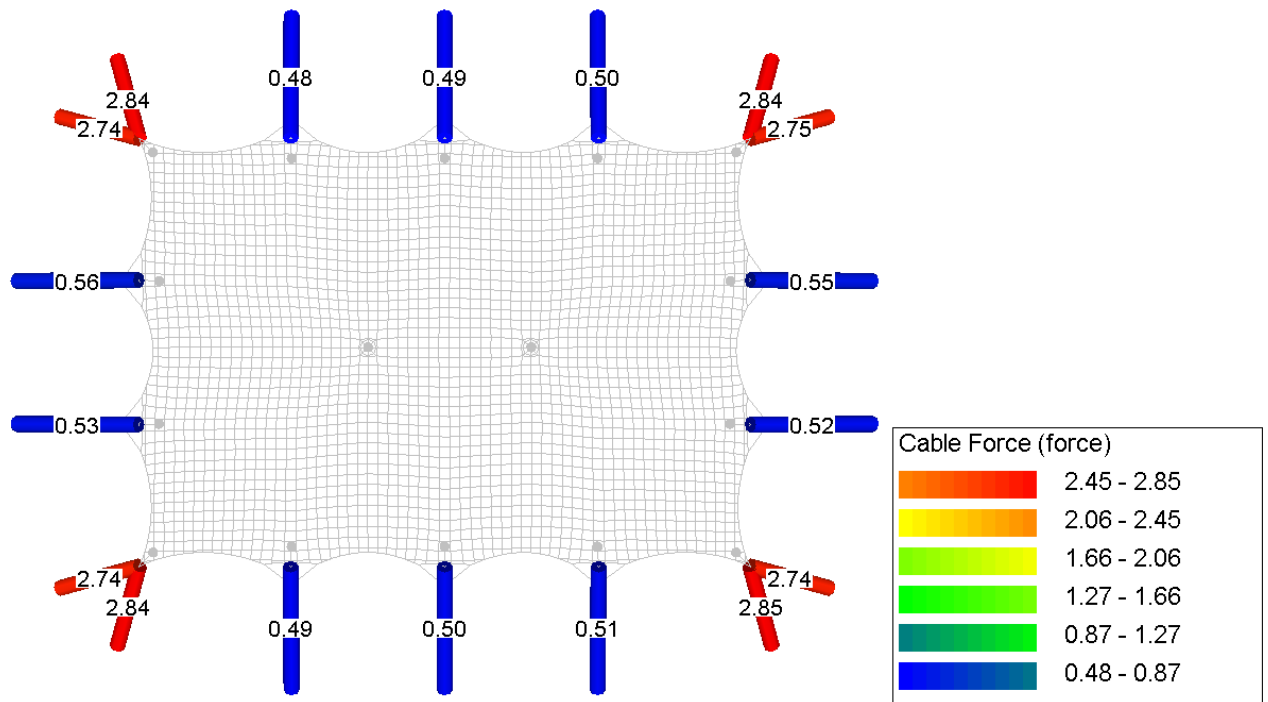
Annex E.2.3. Circumference rope forces



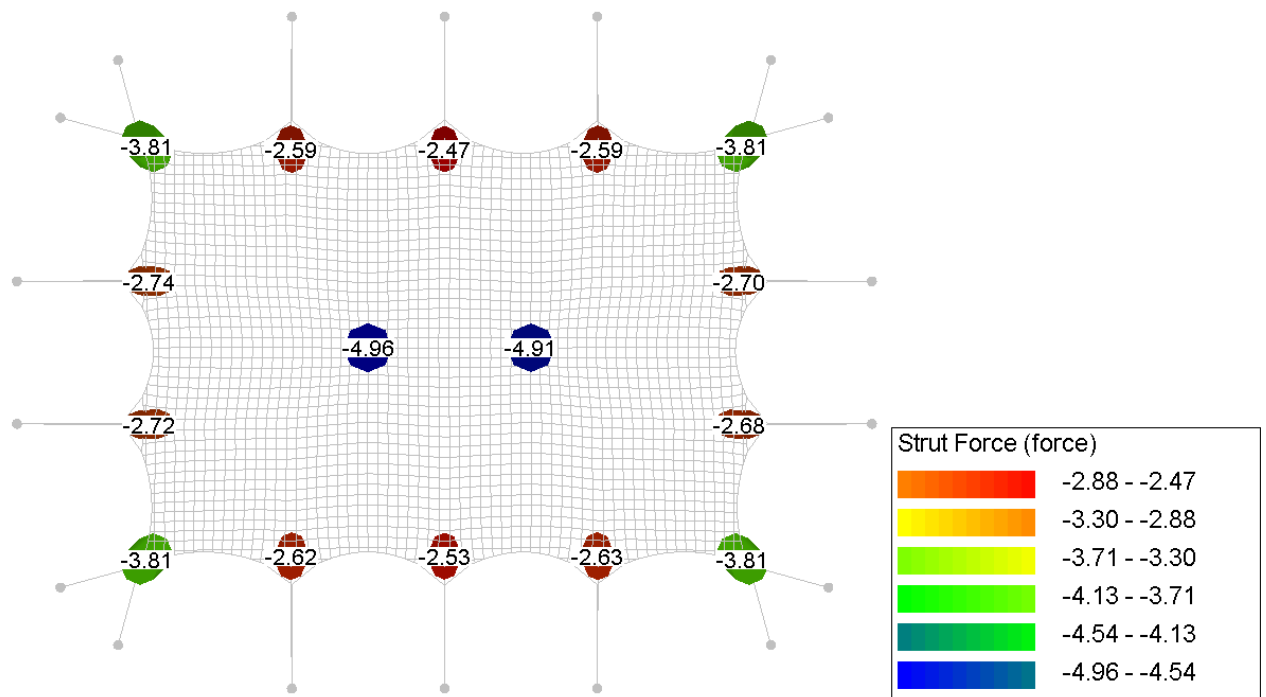
Annex E.2.4. Rope forces



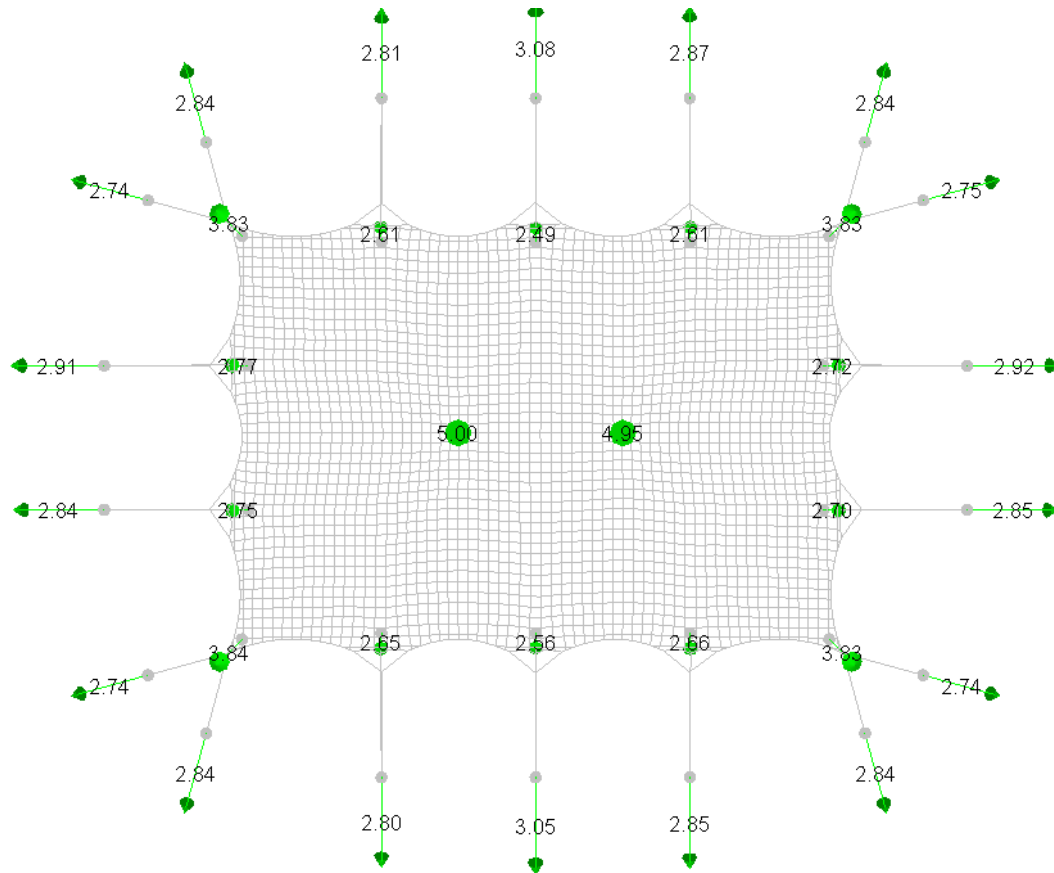
Annex E.2.5. Belt forces



Annex E.2.6. Pole forces

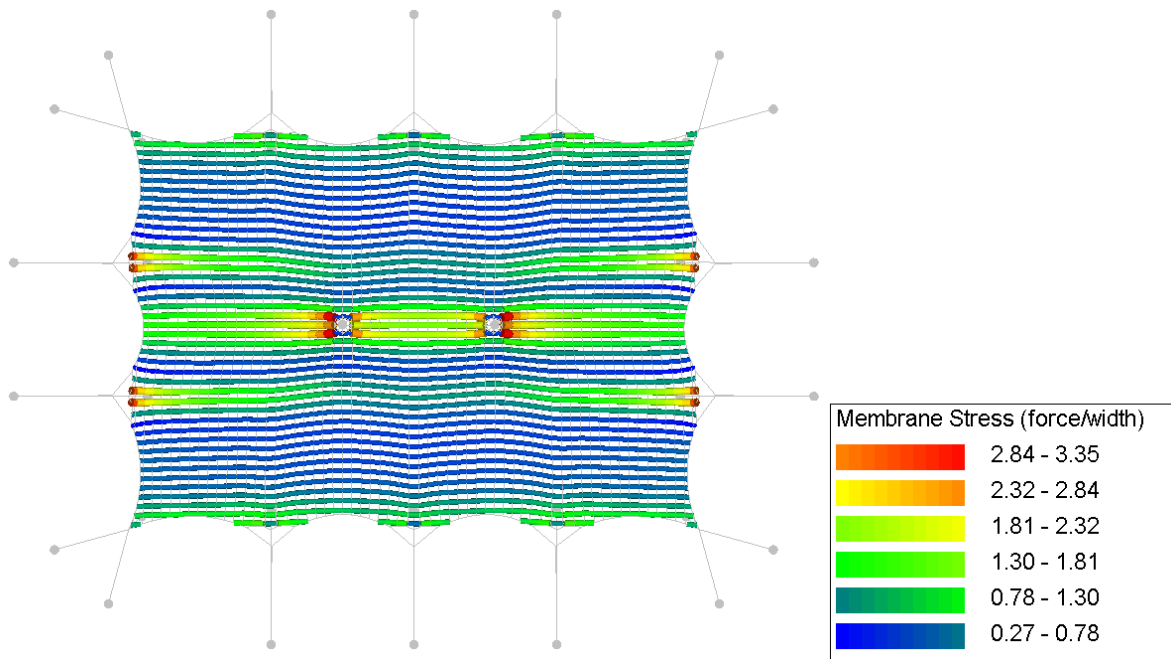


Annex E.2.7. Anchor forces

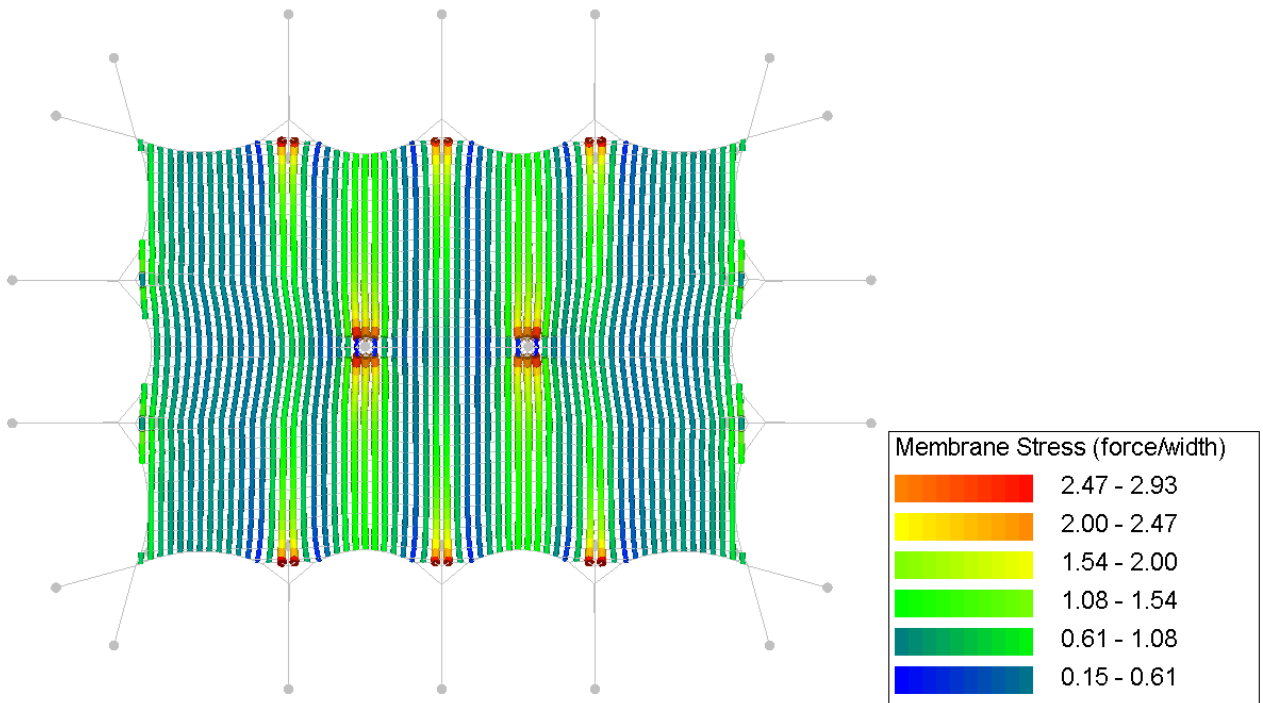


Annex E.3. CO3. Own weight + pretension + wind pressure

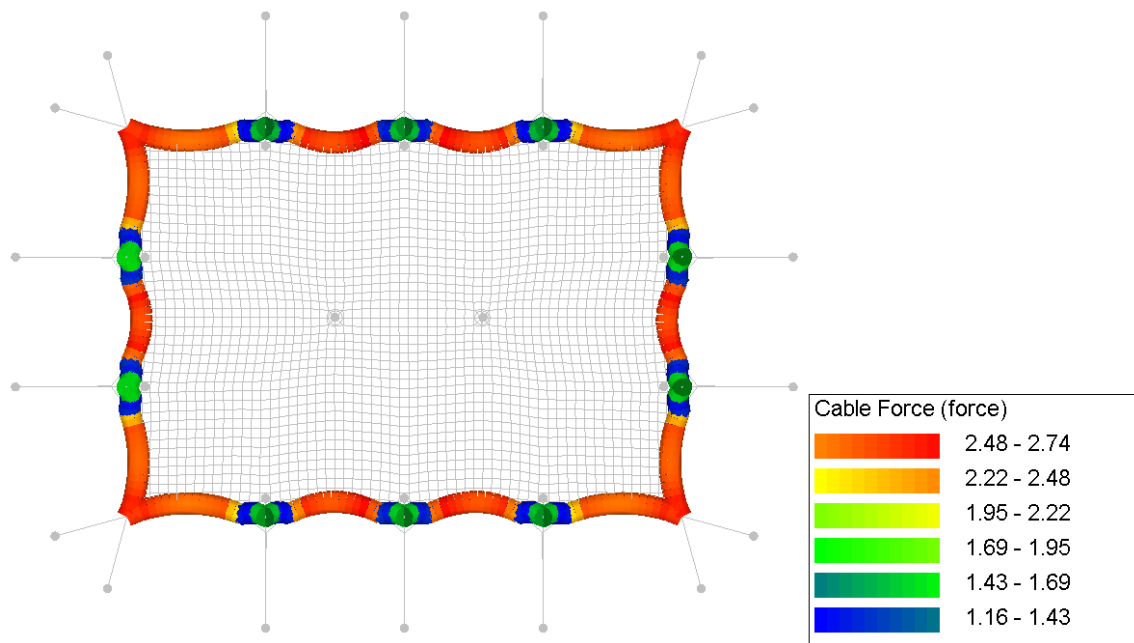
Annex E.3.1. Membrane stress (warp)



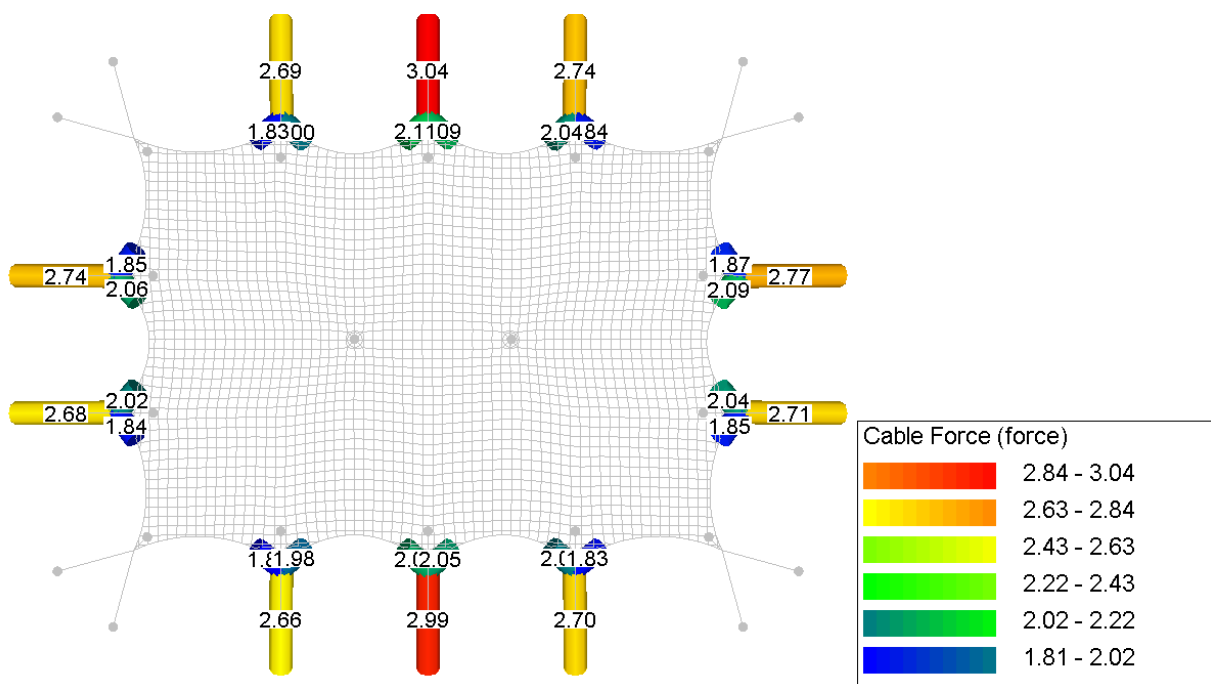
Annex E.3.2. Membrane stress (weft)



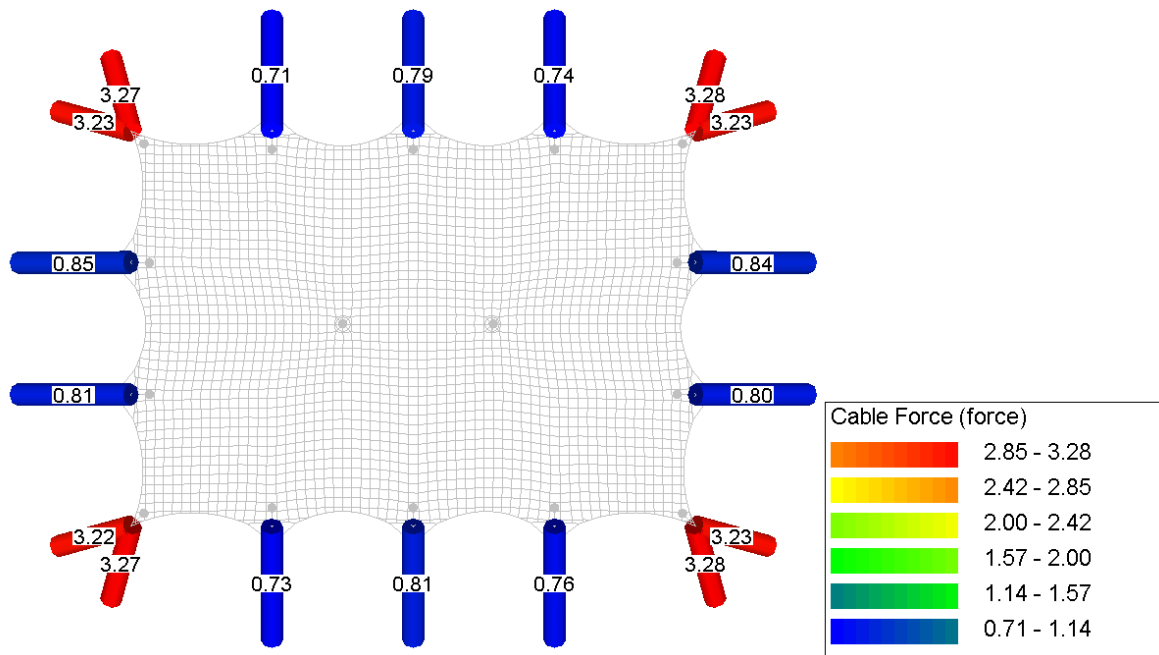
Annex E.3.3. Circumference rope forces



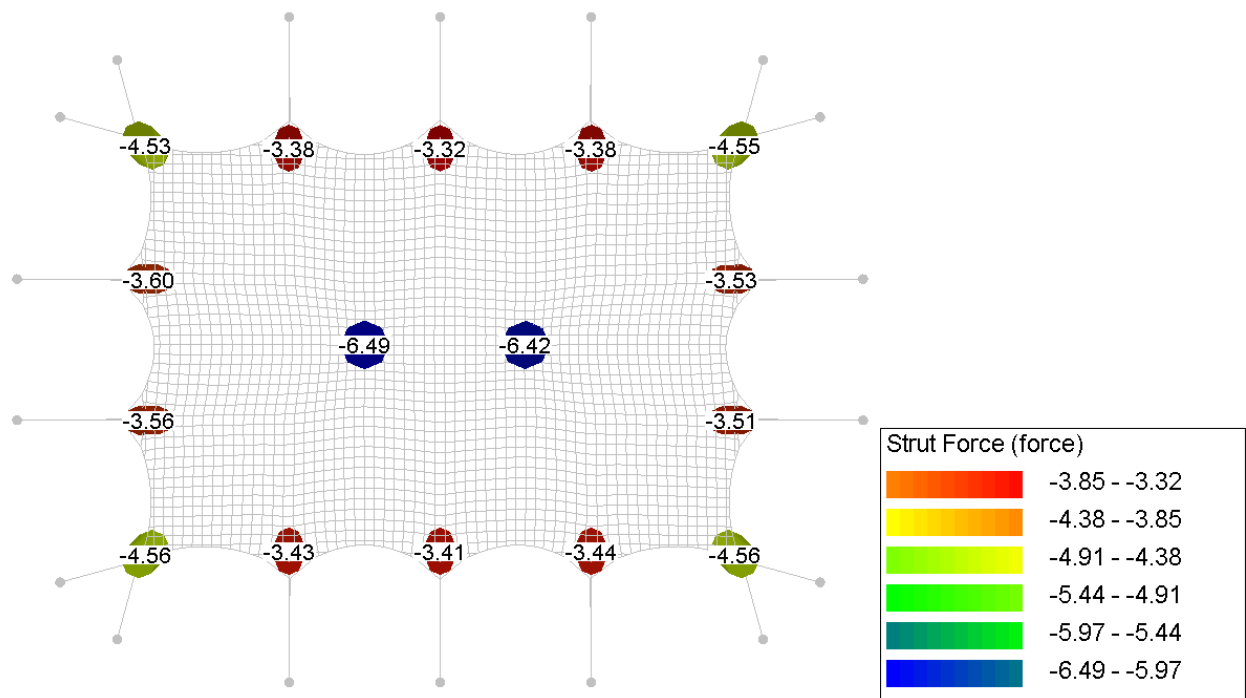
Annex E.3.4. Rope forces



Annex E.3.5. Belt forces

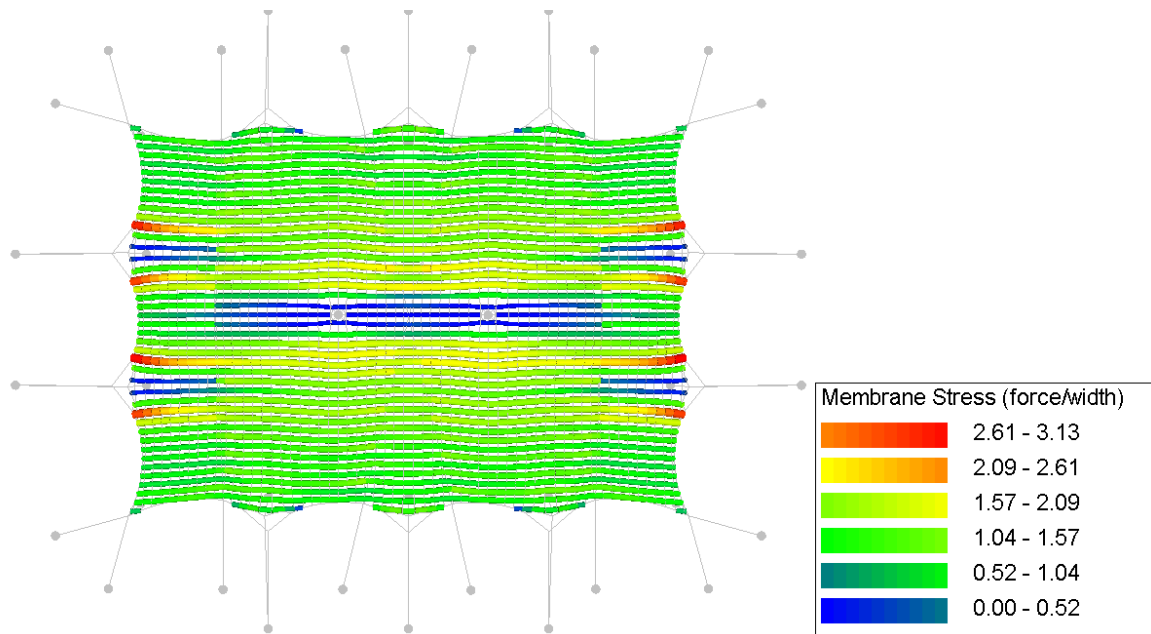


Annex E.3.6. Poles forces

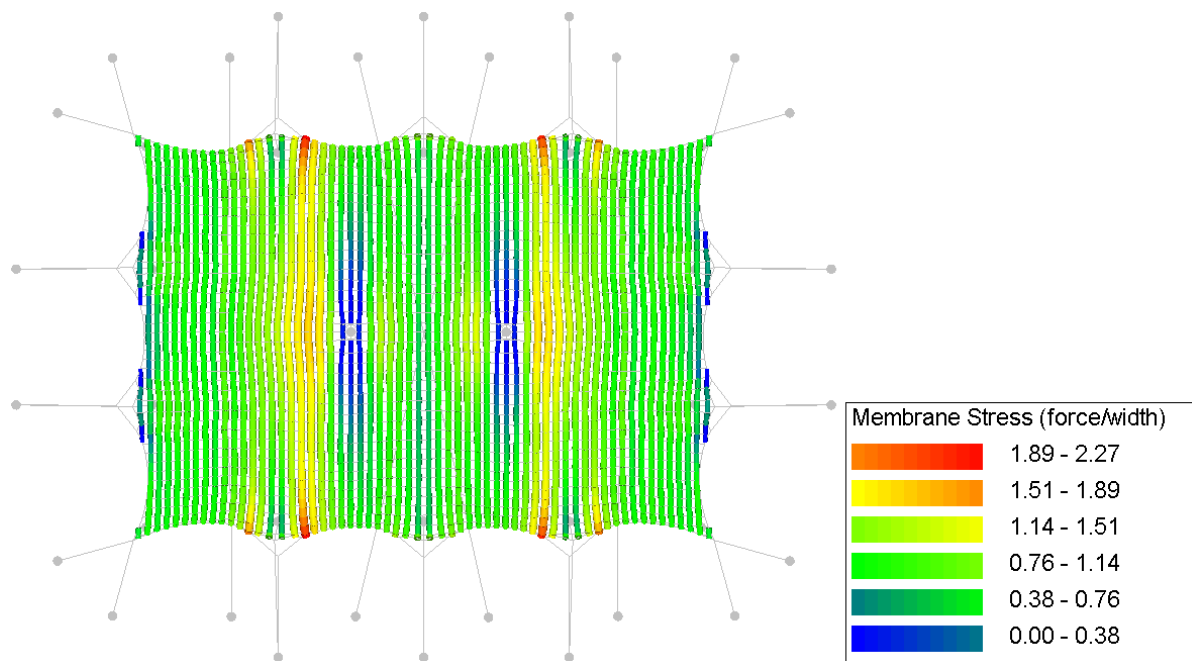


Annex E.4. CO4. Own weight + pretension + wind suction

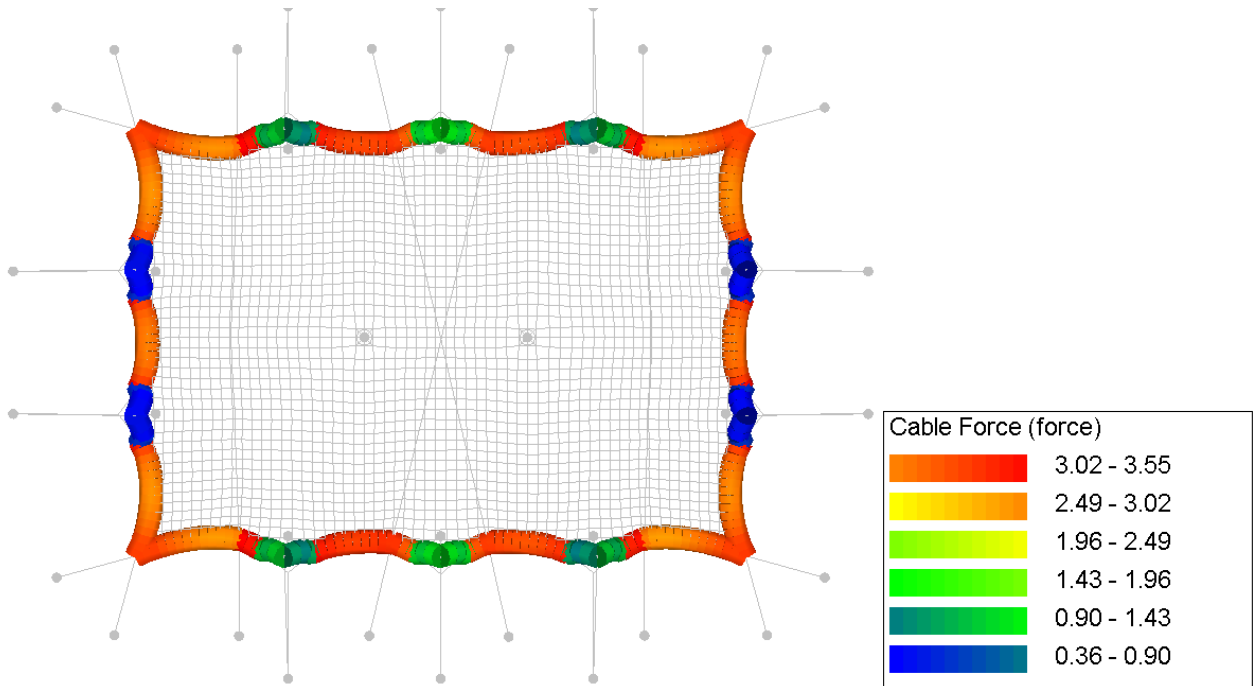
Annex E.4.1. Membrane stress (warp)



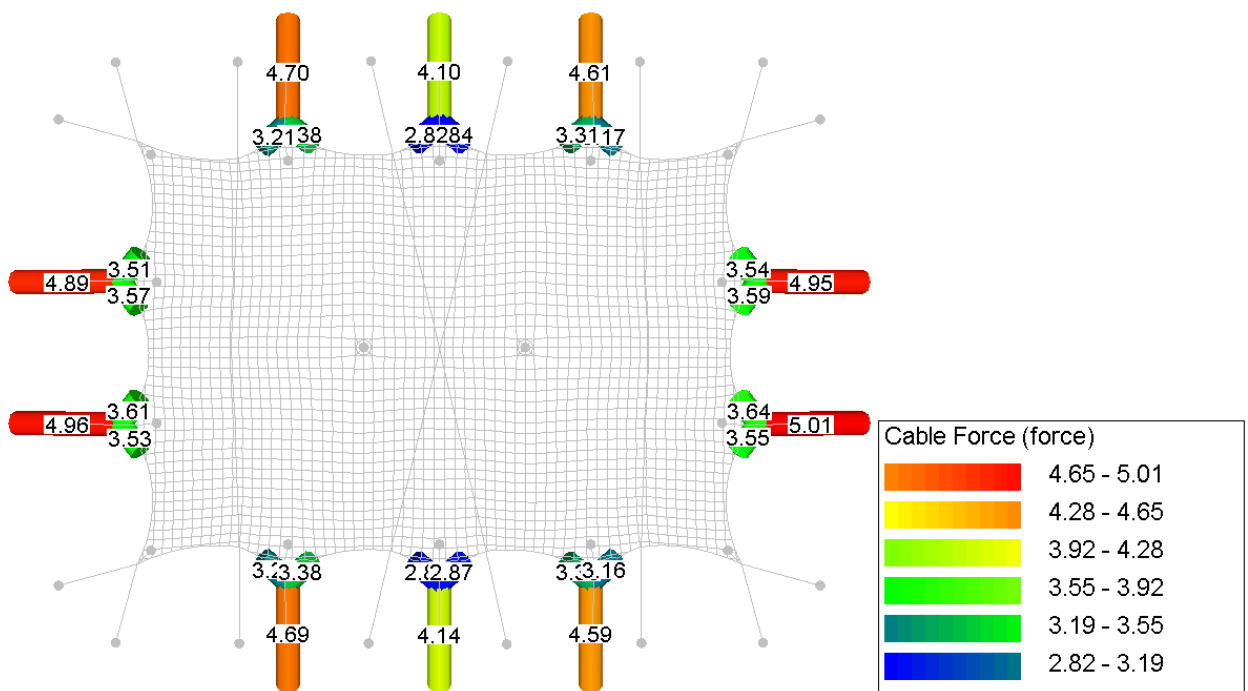
Annex E.4.2. Membrane stress (weft)



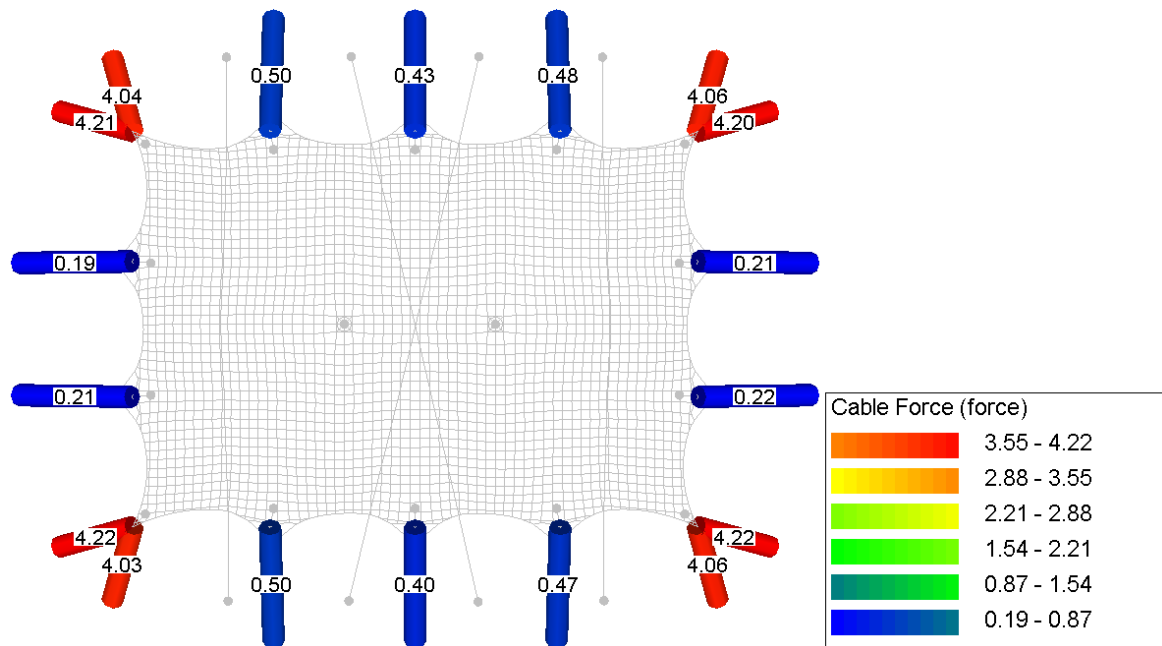
Annex E.4.3. Circumference rope forces



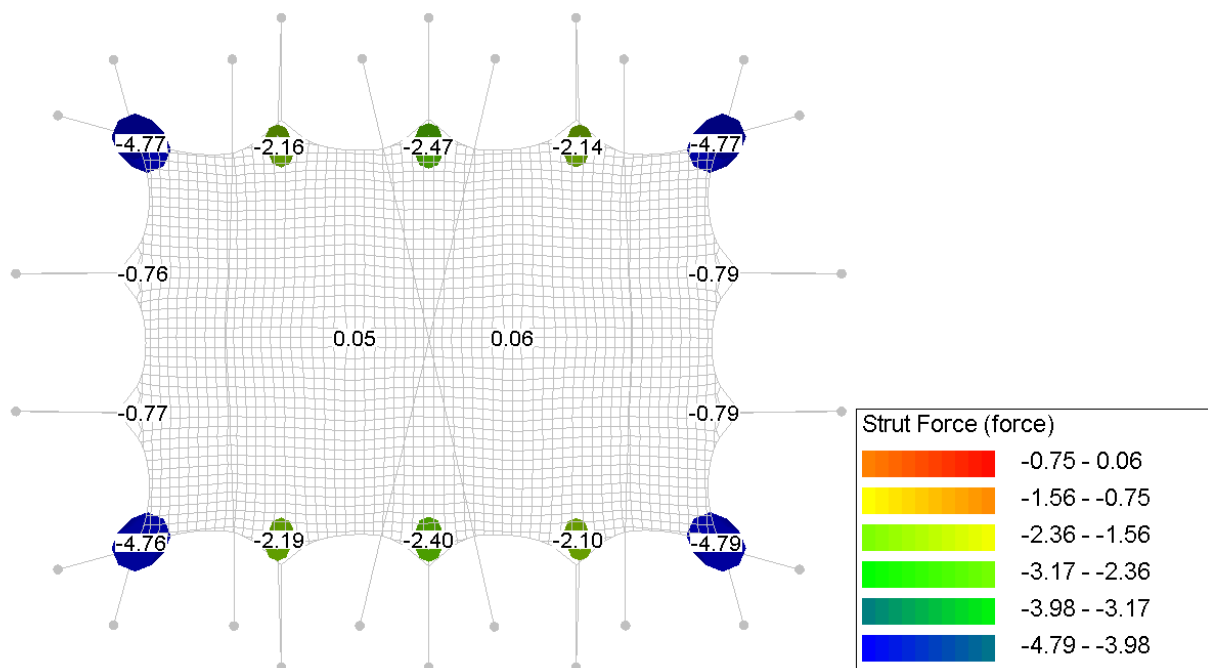
Annex E.4.4. Guy rope forces



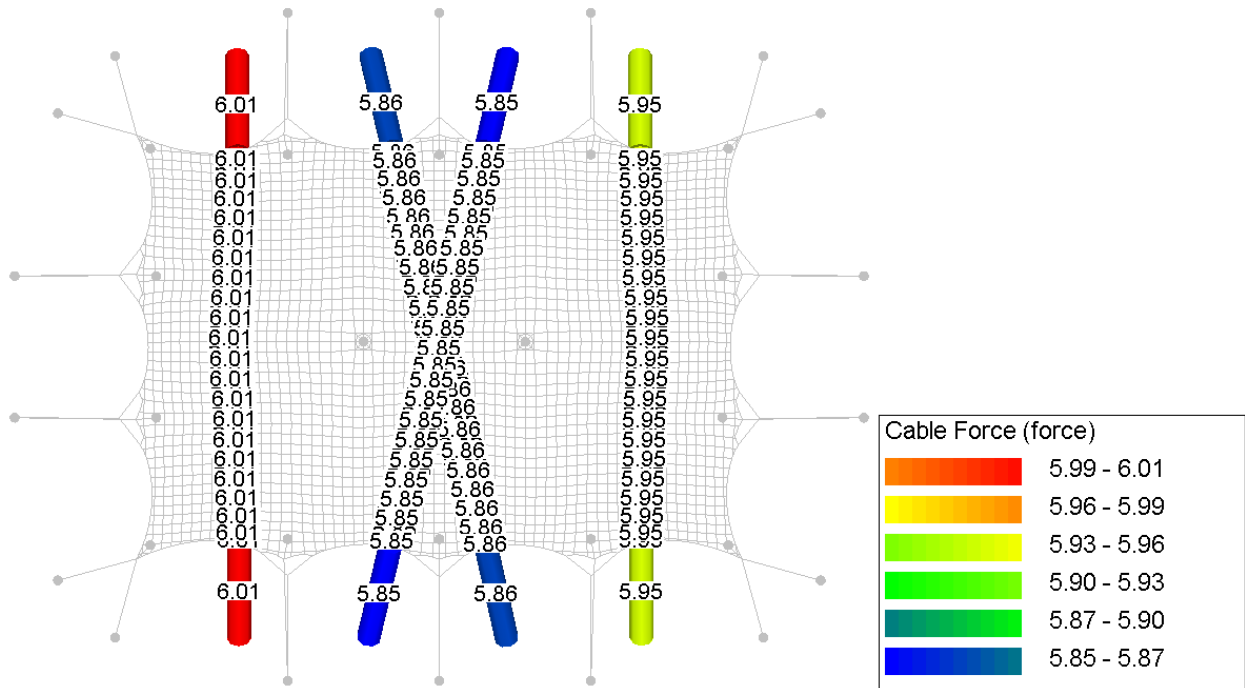
Annex E.4.5. Belt forces



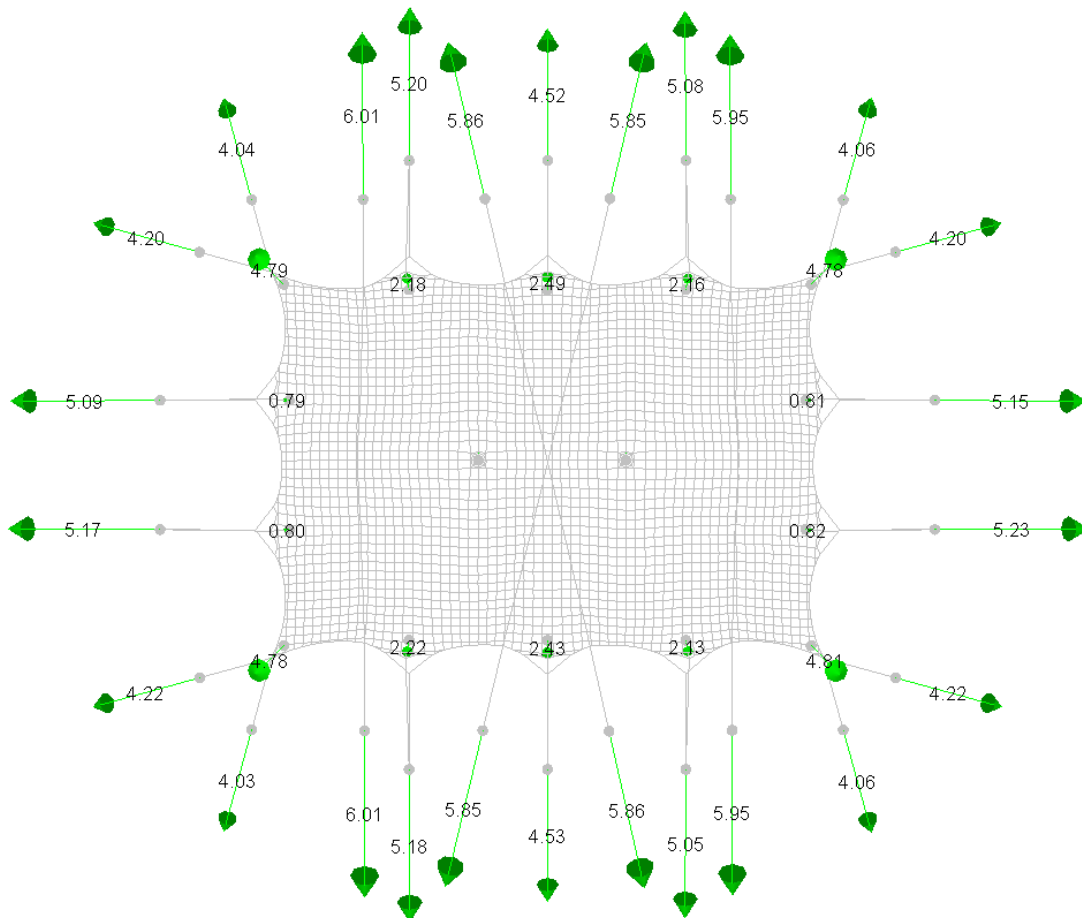
Annex E.4.6. Poles forces



Annex E.4.7. Storm belt forces

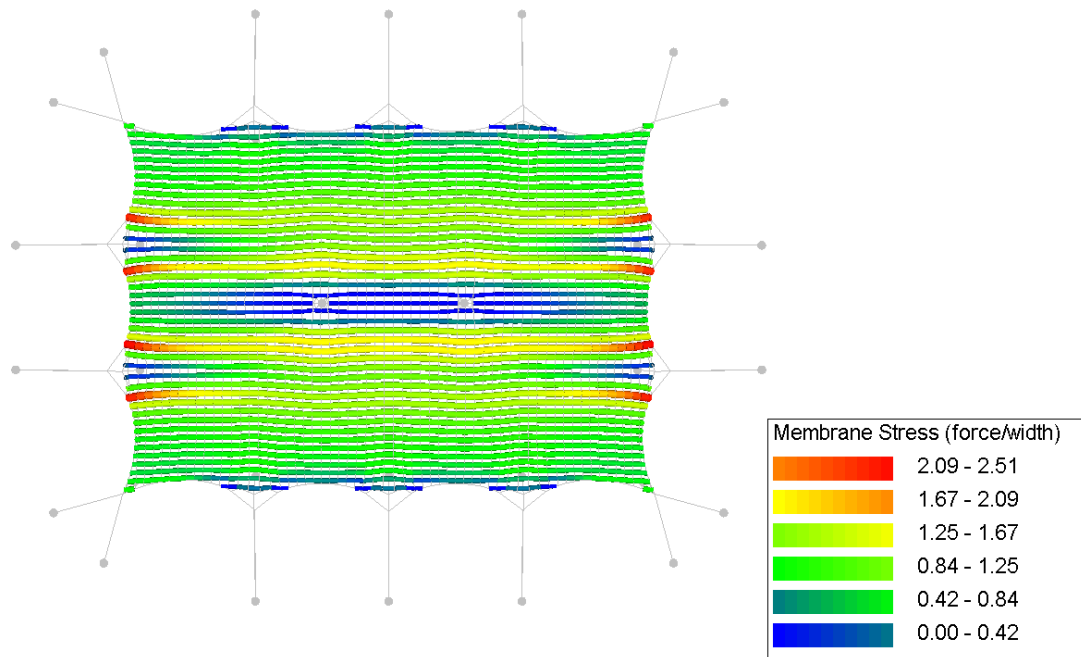


Annex E.4.8. Reaction forces

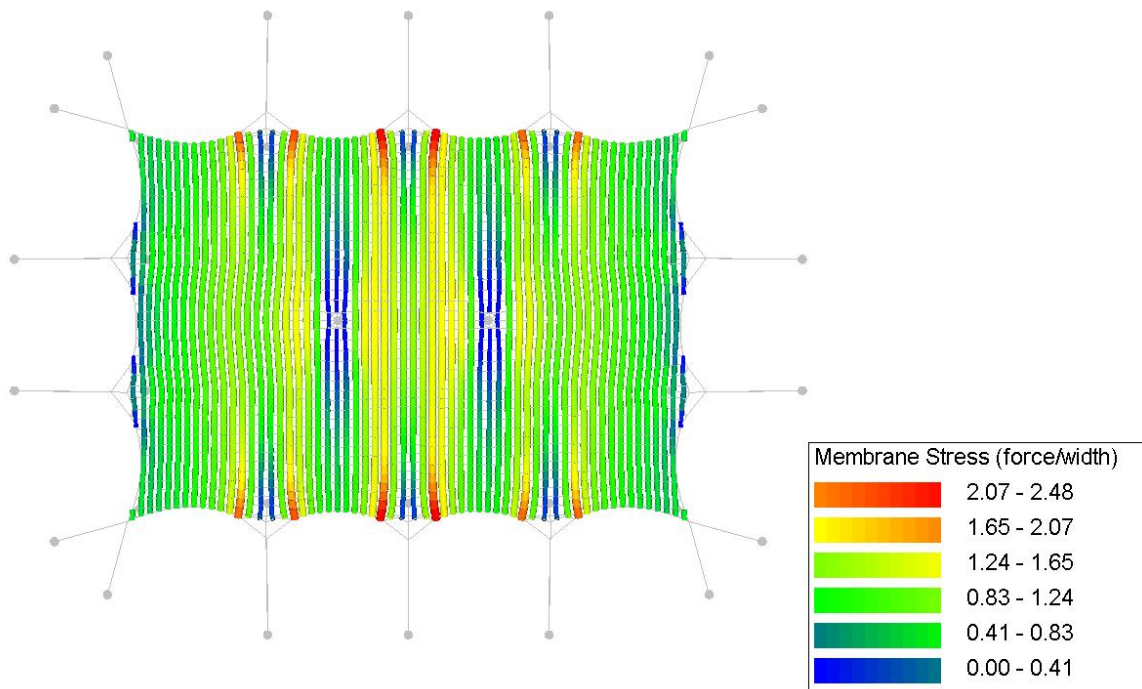


Annex E.5. CO5. Own weight + pretension + wind suction reduced [NL]

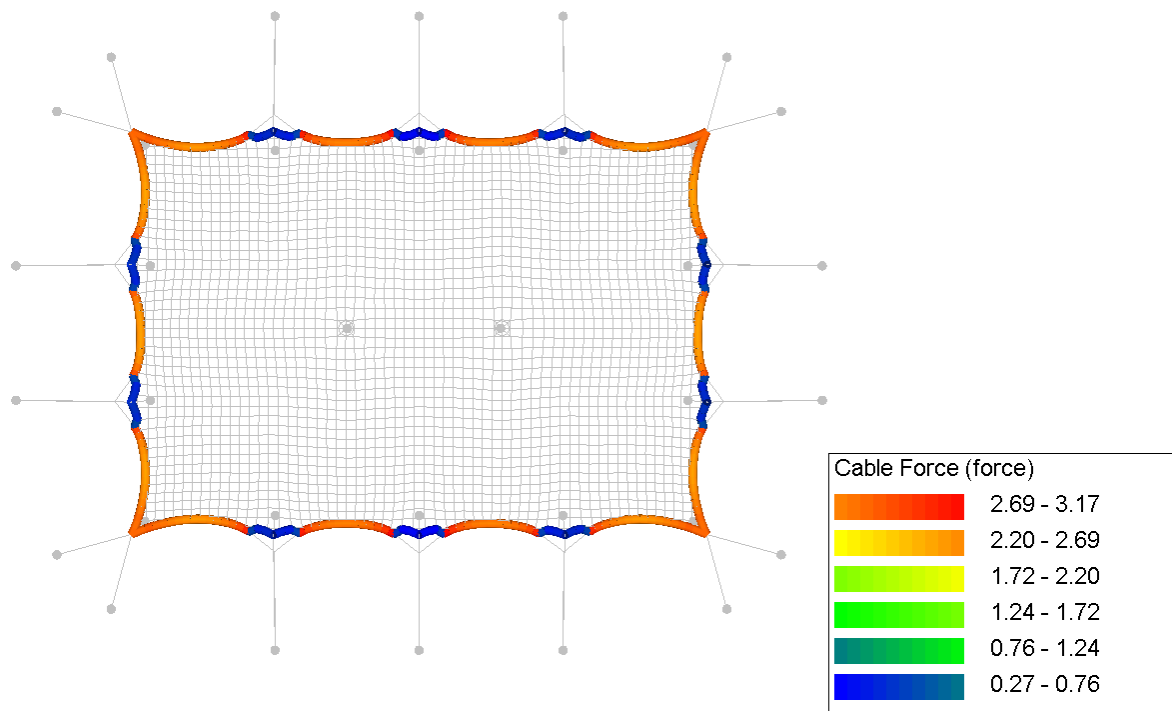
Annex E.5.1. Membrane stress (warp)



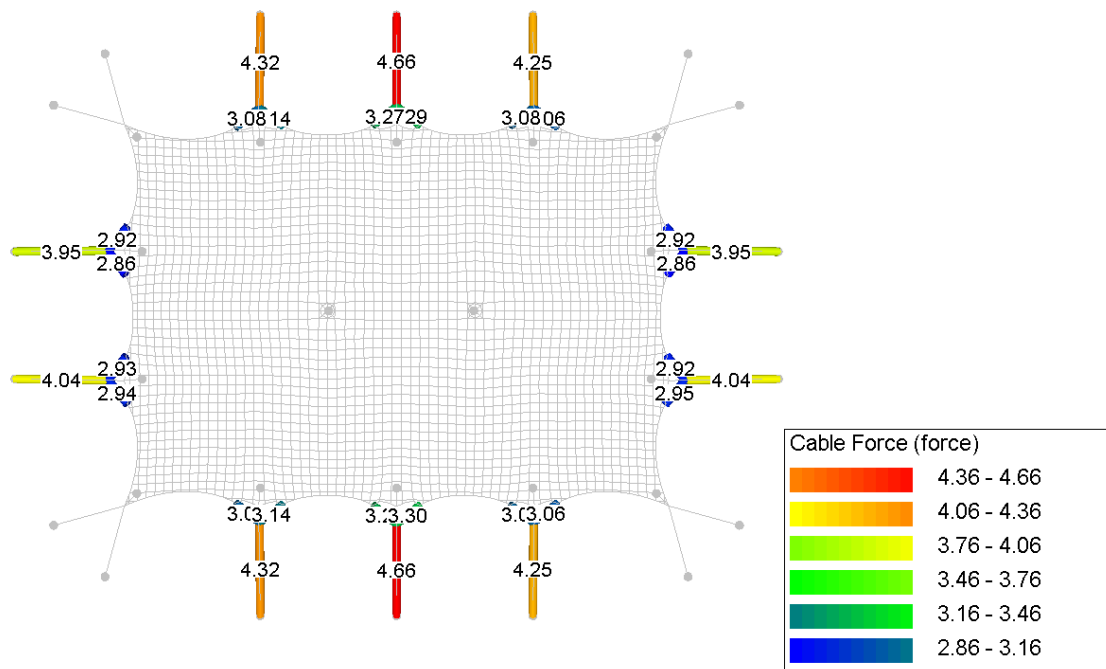
Annex E.5.2. Membrane stress (weft)



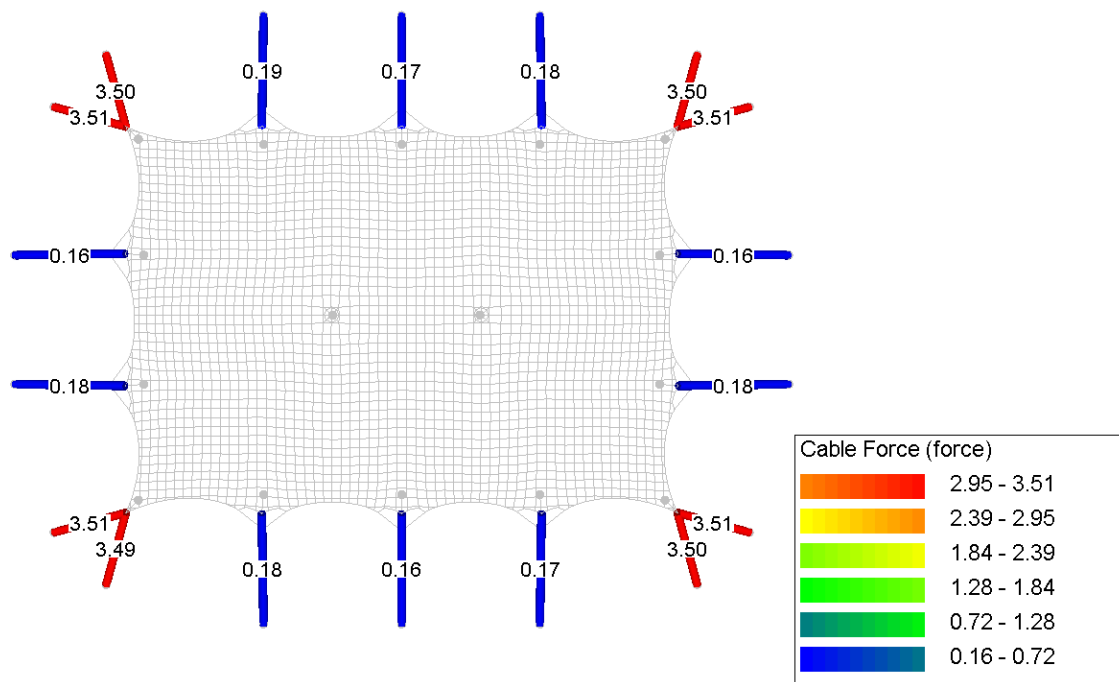
Annex E.5.3. Circumference rope forces



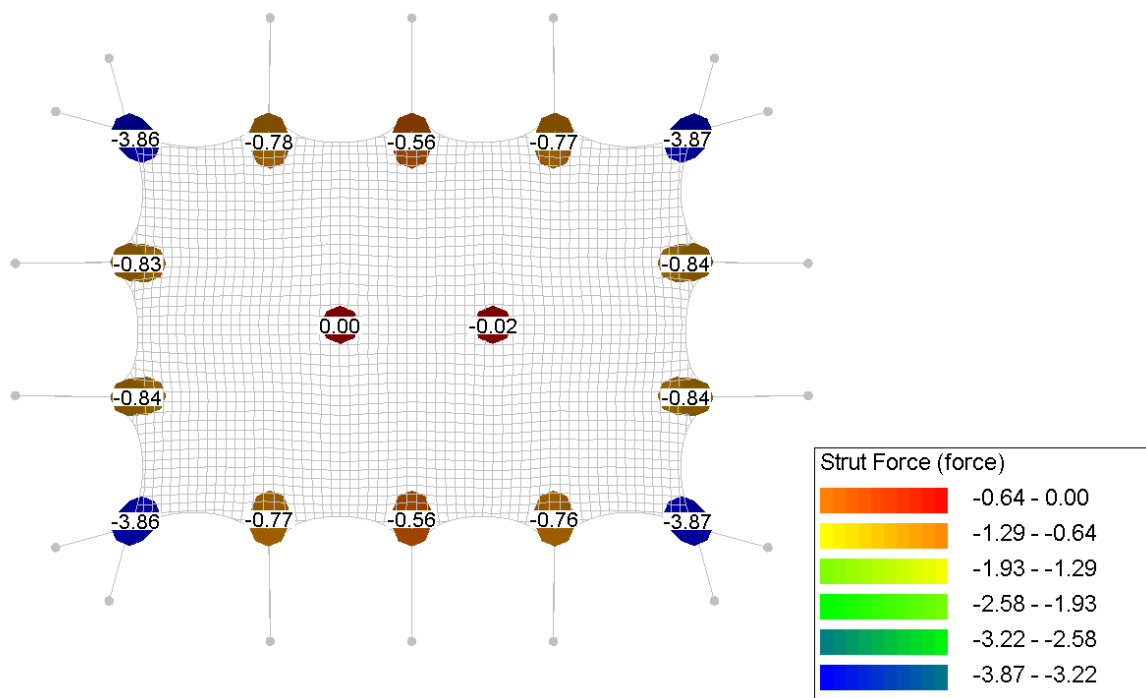
Annex E.5.4. Rope forces



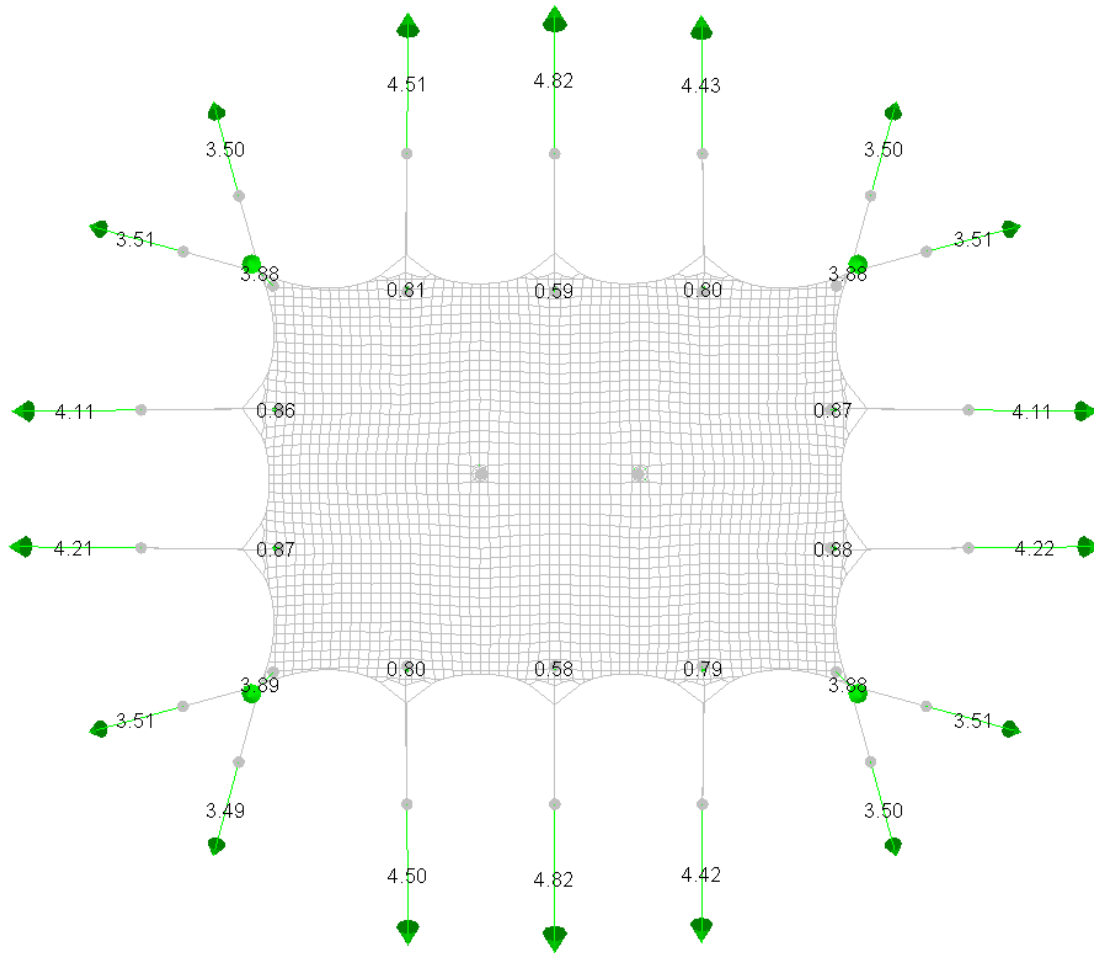
Annex E.5.5. Belt forces



Annex E.5.6. Poles forces

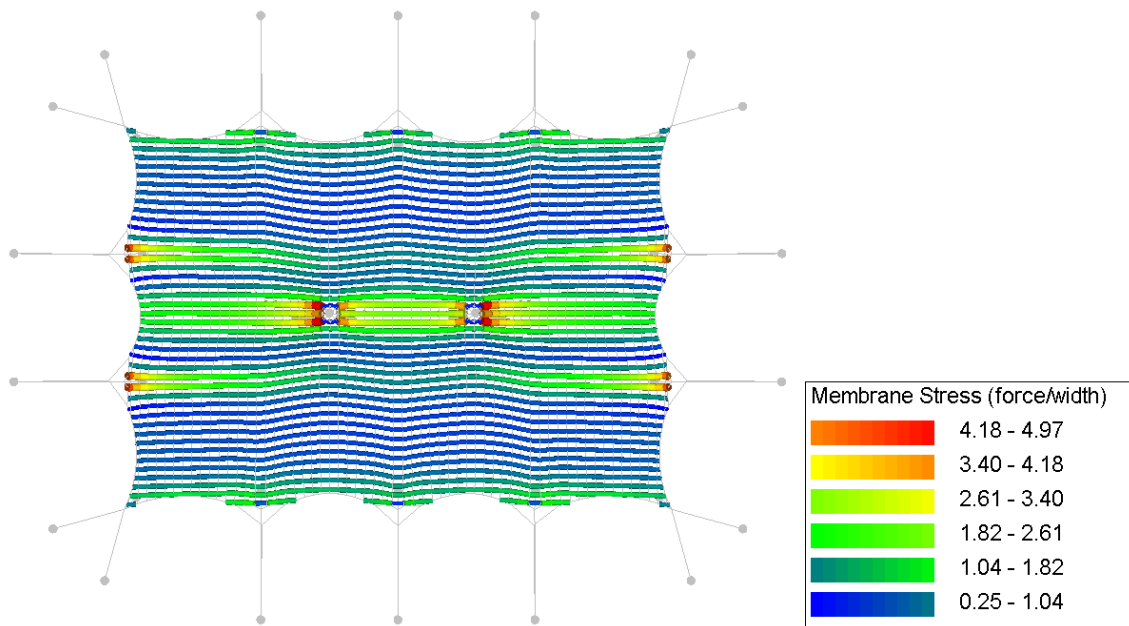


Annex E.5.7. Reaction forces

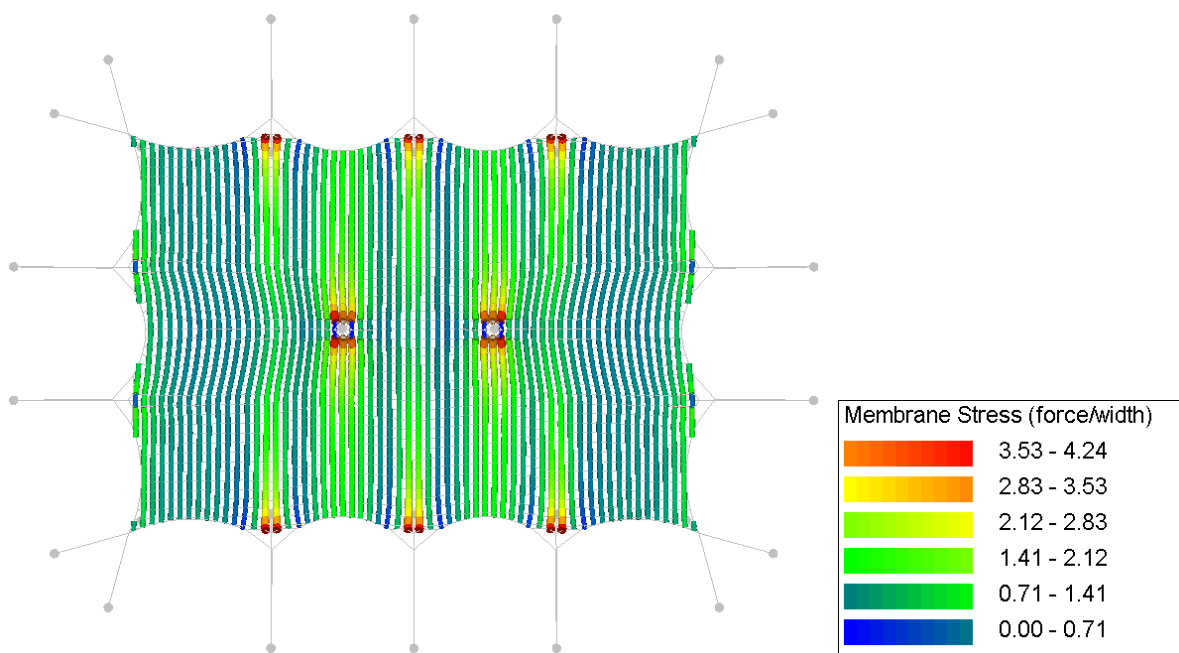


Annex E.6. CO6. Own weight + pretension + wind pressure + snow [FR]

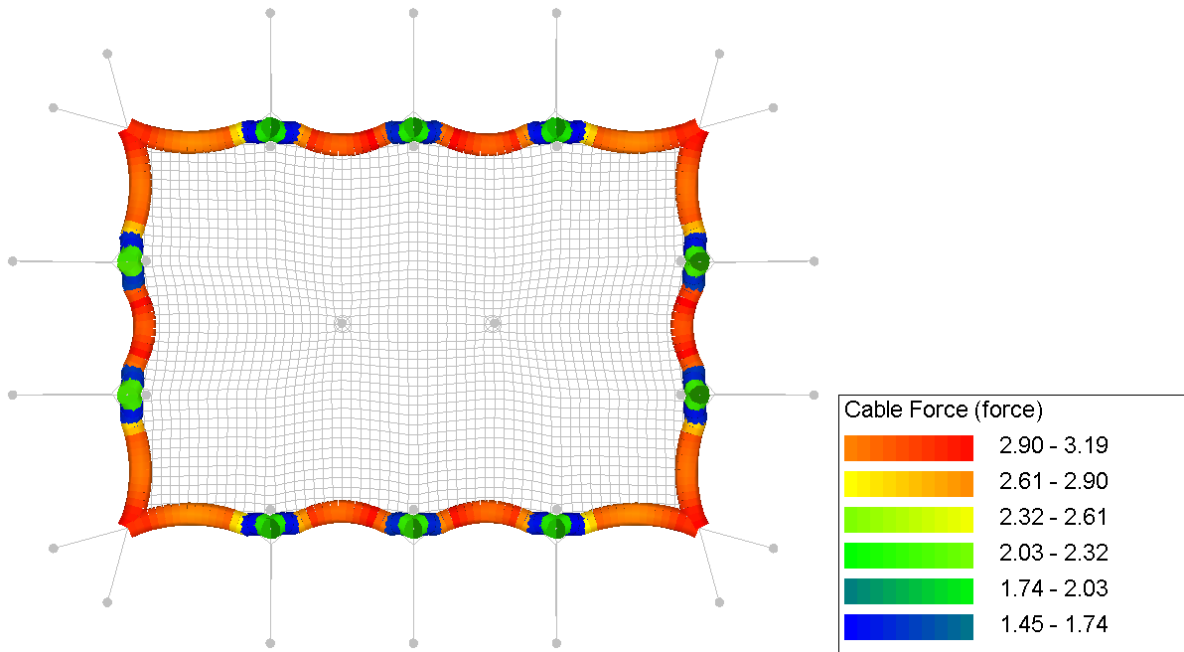
Annex E.6.1. Membrane stress (warp)



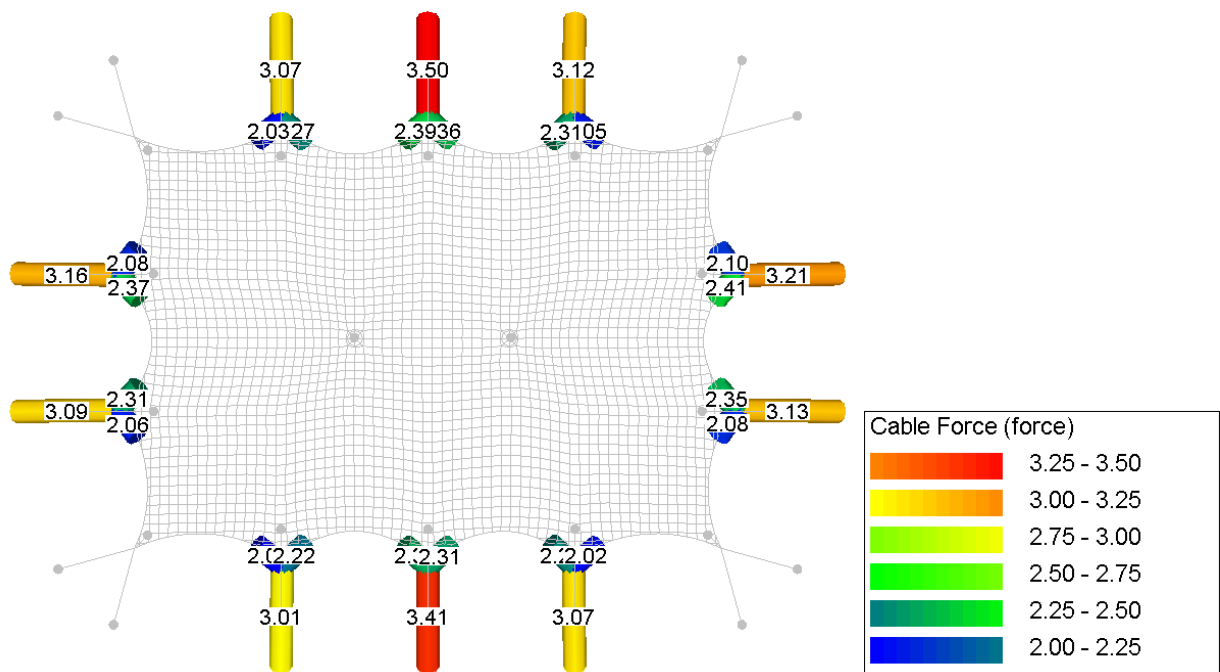
Annex E.6.2. Membrane stress (weft)



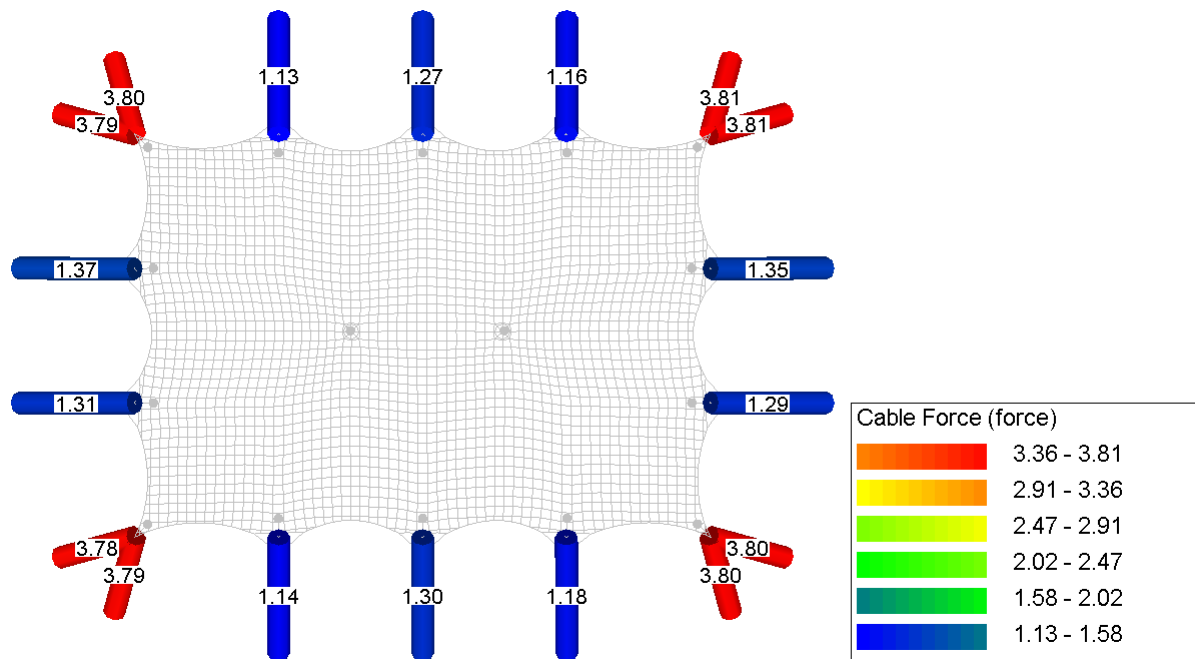
Annex E.6.3. Circumference rope forces



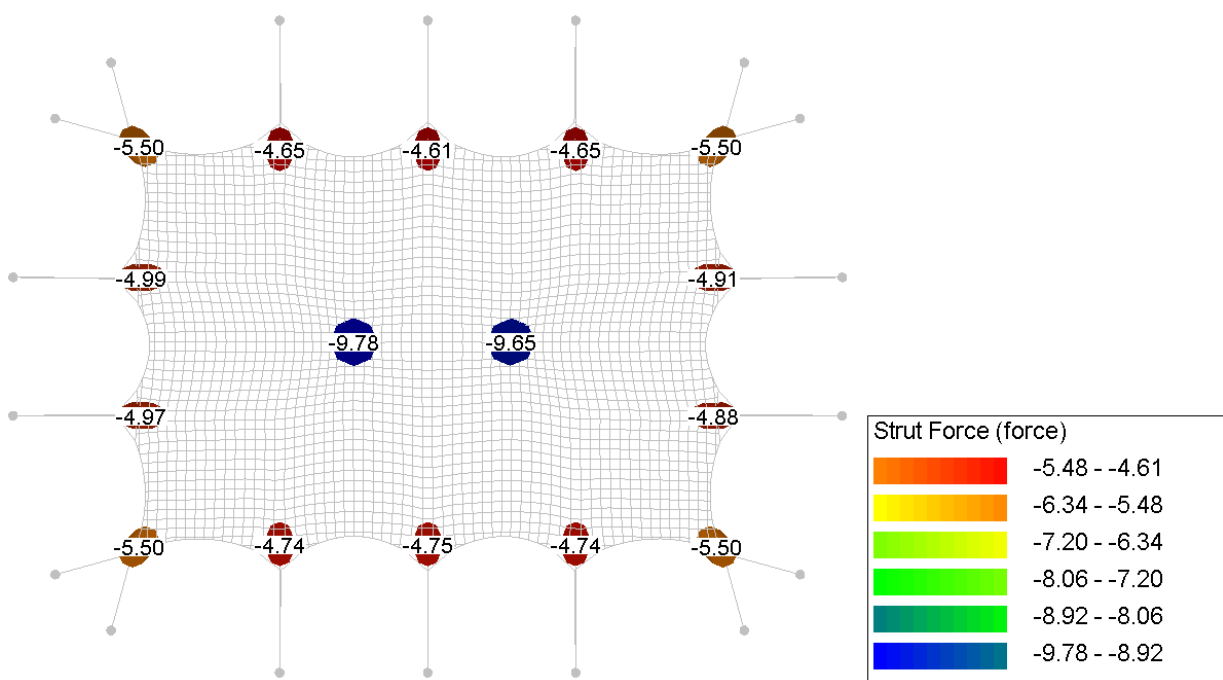
Annex E.6.4. Guy rope forces



Annex E.6.5. Belt forces

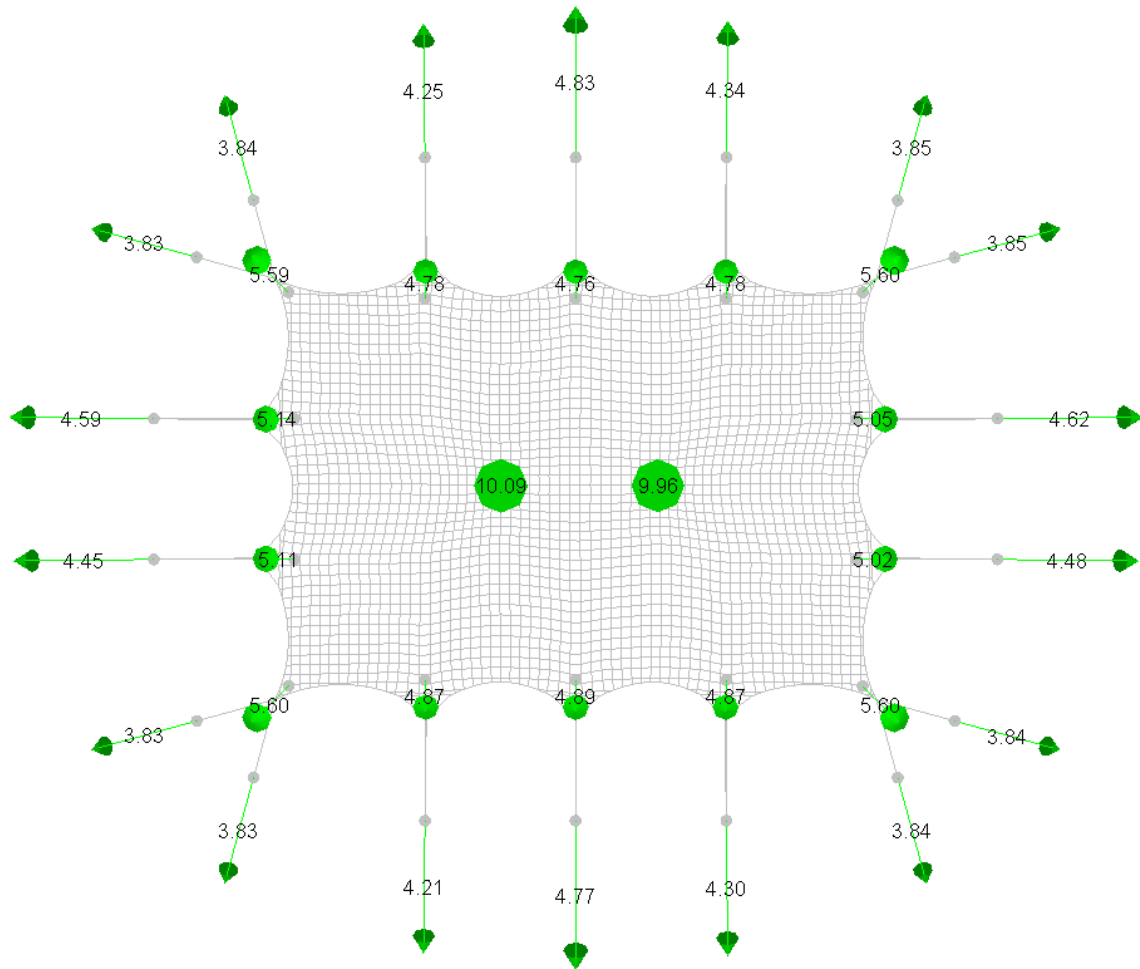


Annex E.6.6. Poles forces



Annex E.6.7. Reaction forces

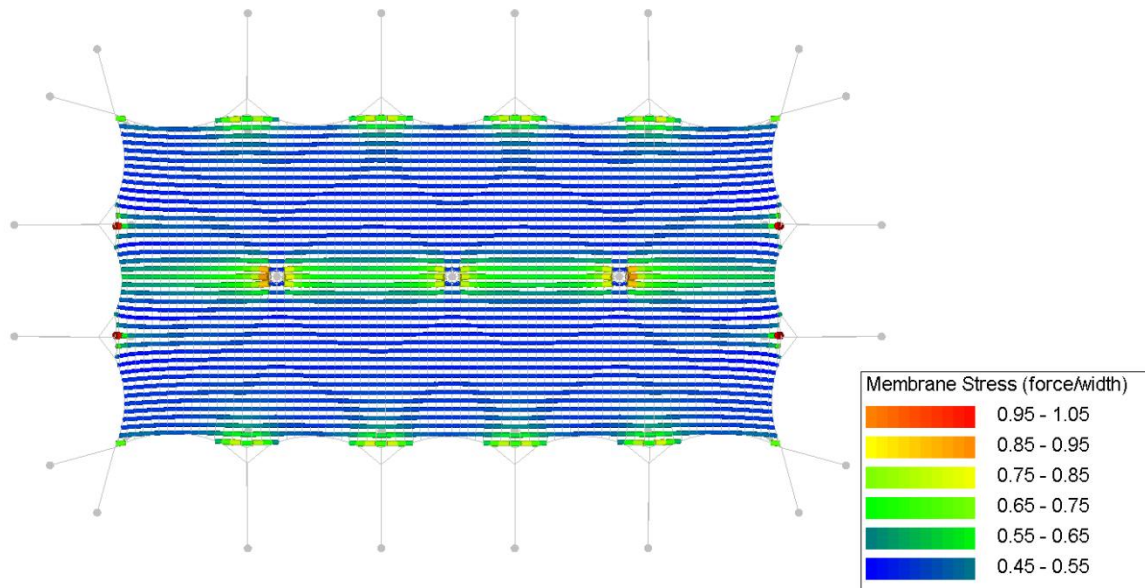
Increased snow load of $0.1 \times 1.5/1.35 = 0.1083 \text{ kN/m}^2$ is applied for ballast and anchor calculation



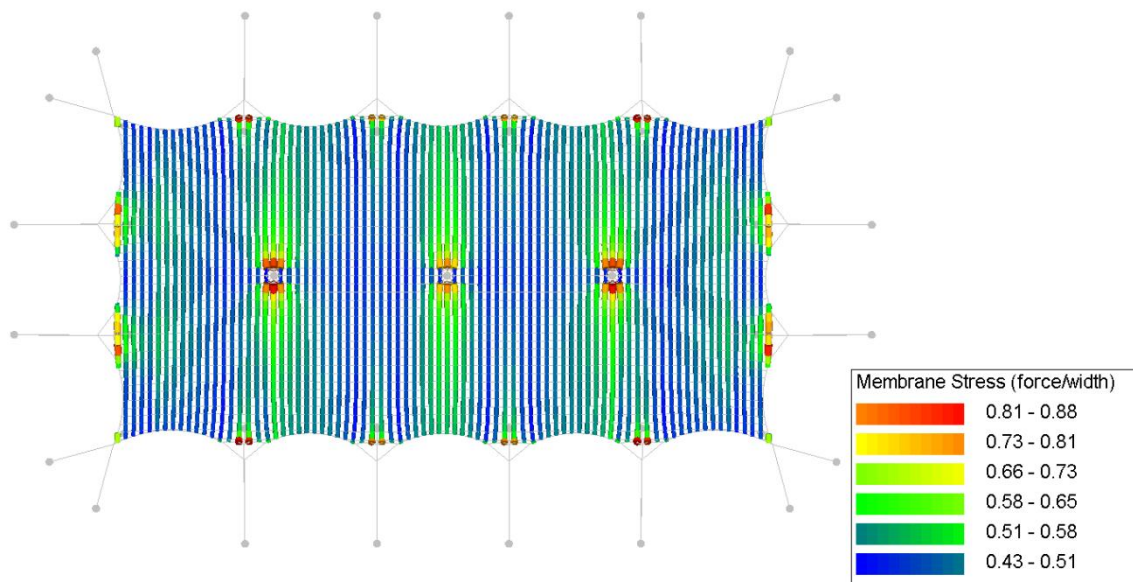
Annex F. Results per load combination – 10x20m model

Annex F.1. CO1. Own weight + pretension

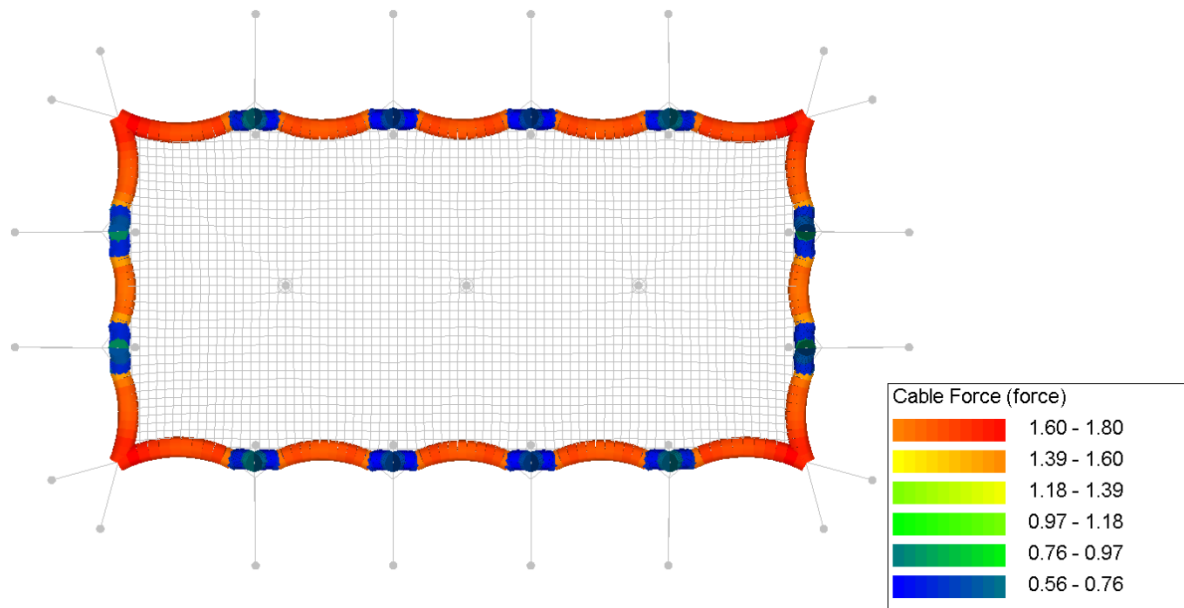
Annex F.1.1. Membrane stress (warp)



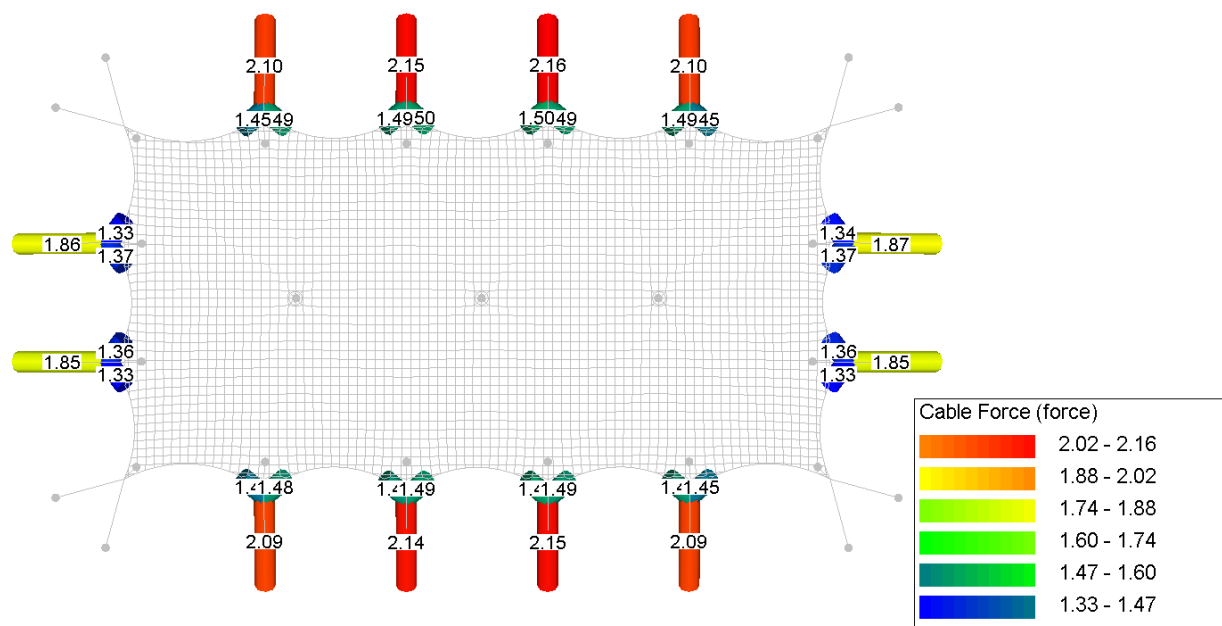
Annex F.1.2. Membrane stress (weft)



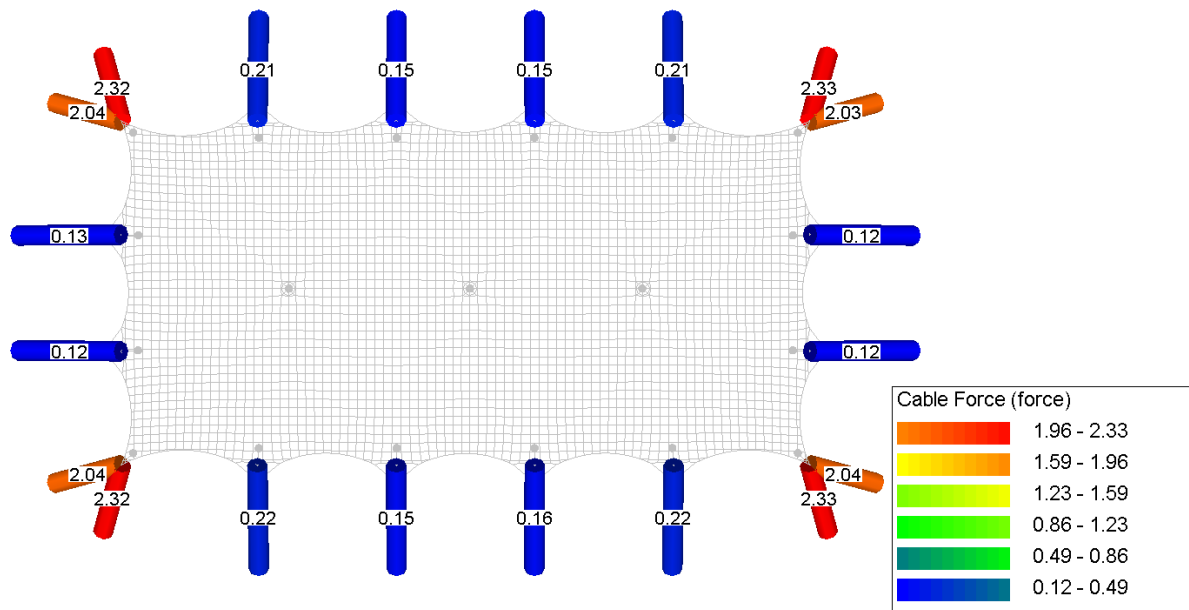
Annex F.1.3. Circumference rope forces



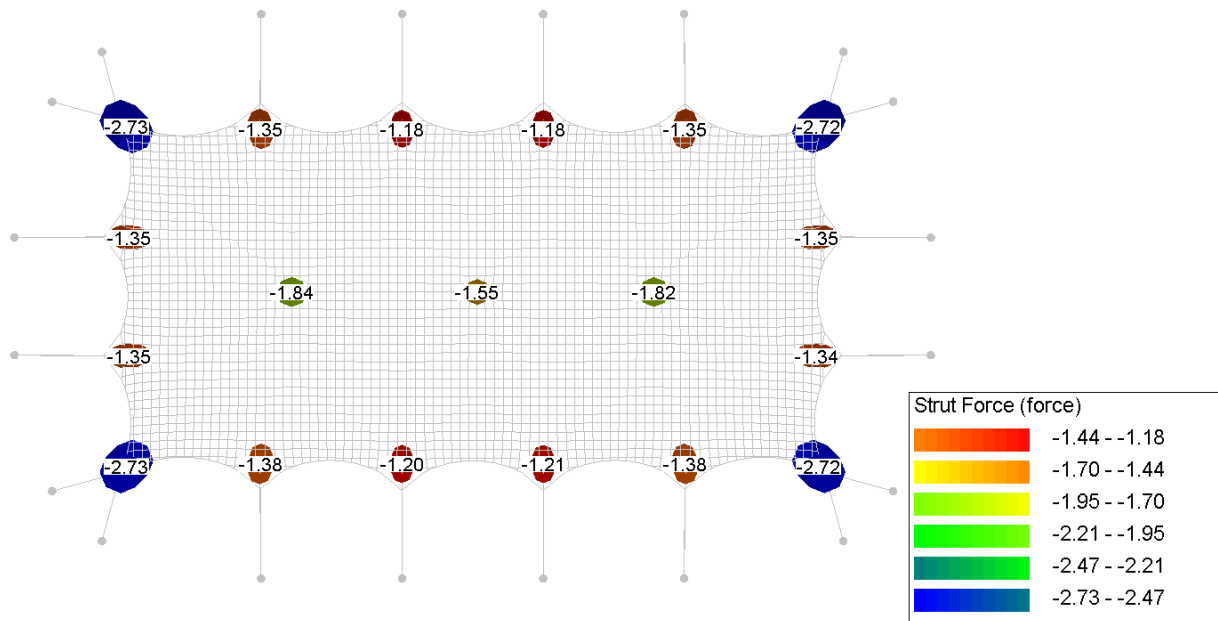
Annex F.1.4. Rope forces



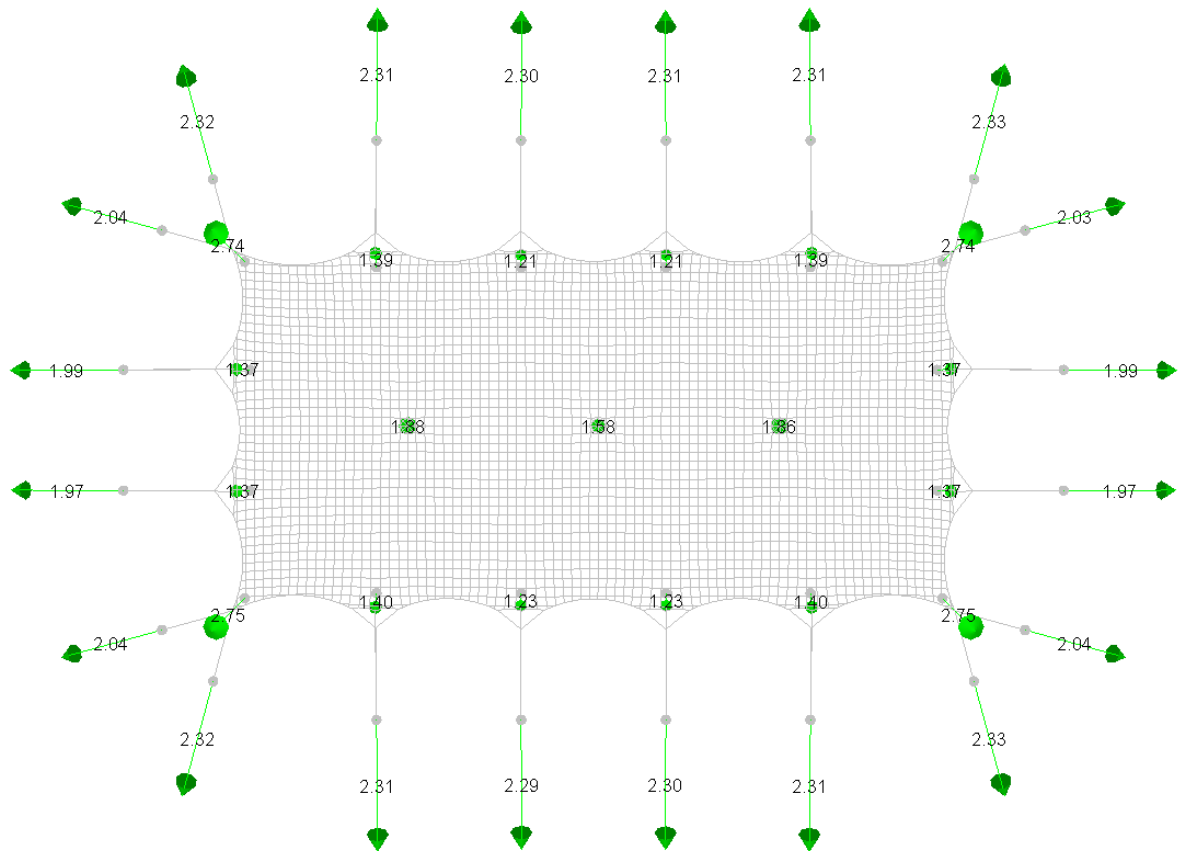
Annex F.1.5. Belt forces



Annex F.1.6. Pole forces

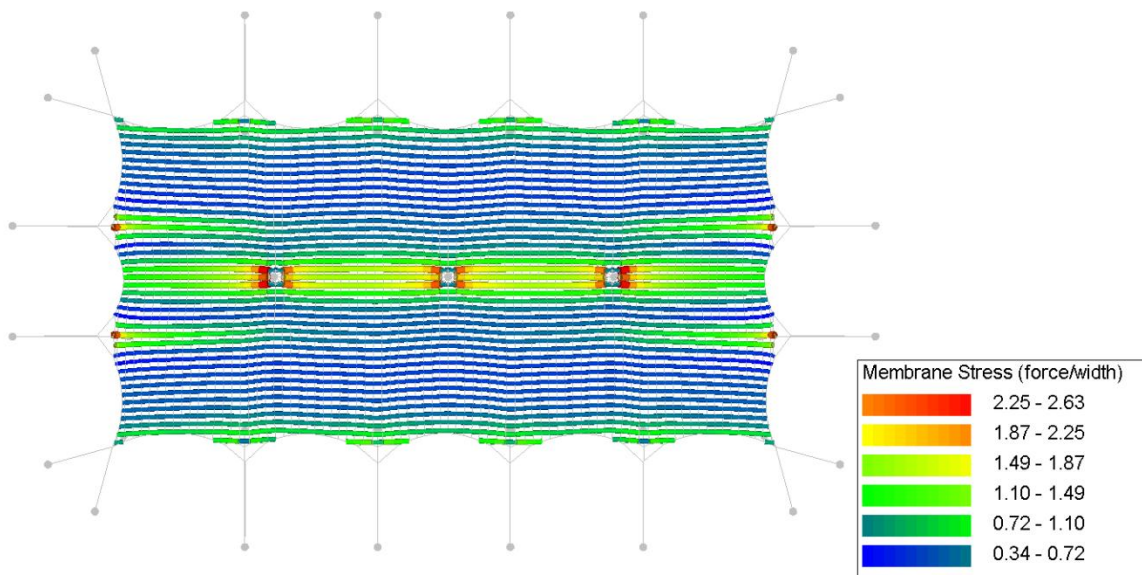


Annex F.1.7. Reaction forces

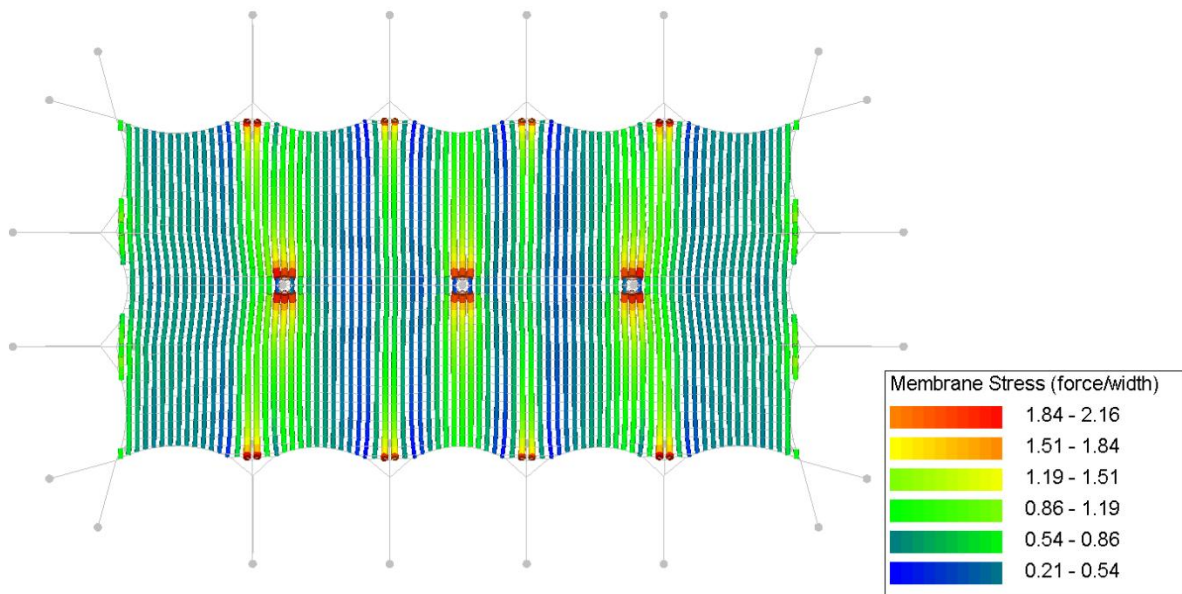


Annex F.2. CO2. Own weight + pretension + snow

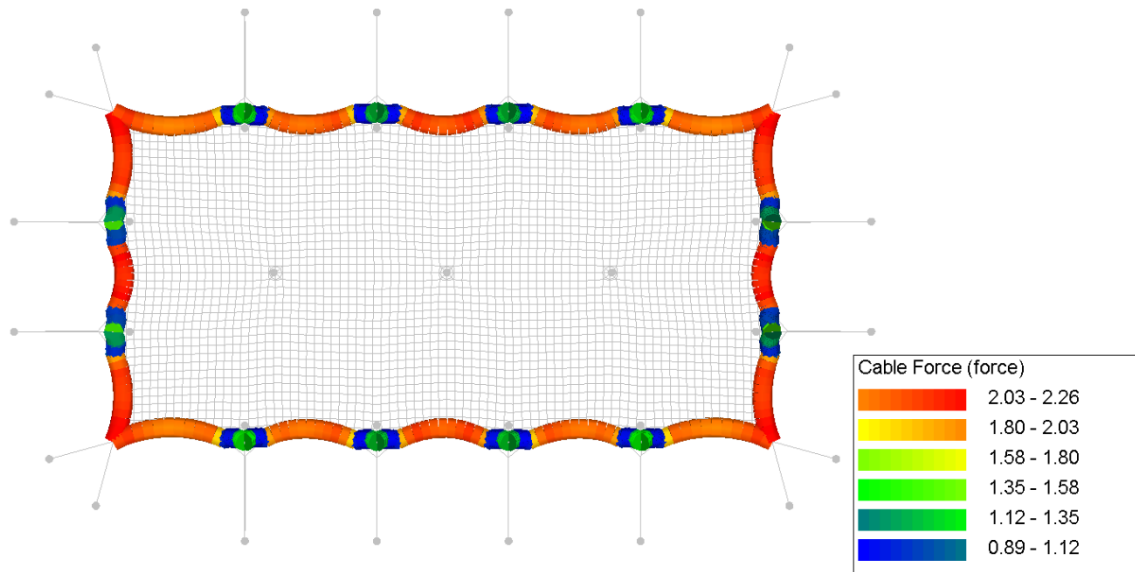
Annex F.2.1. Membrane stress (warp)



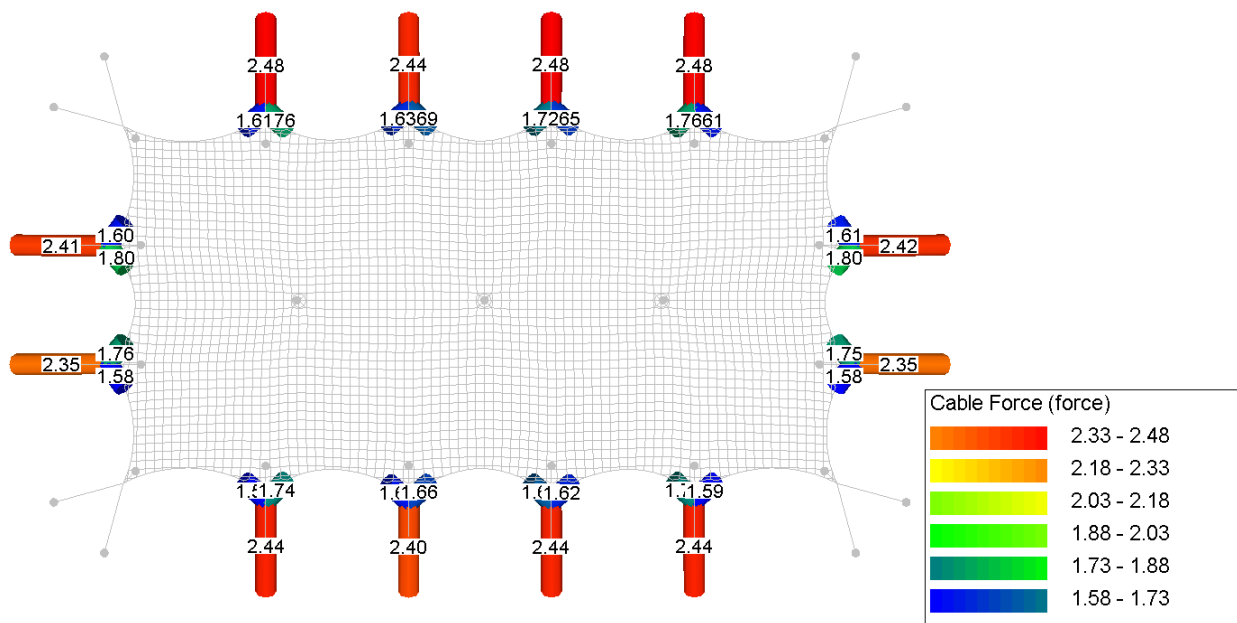
Annex F.2.2. Membrane stress (weft)



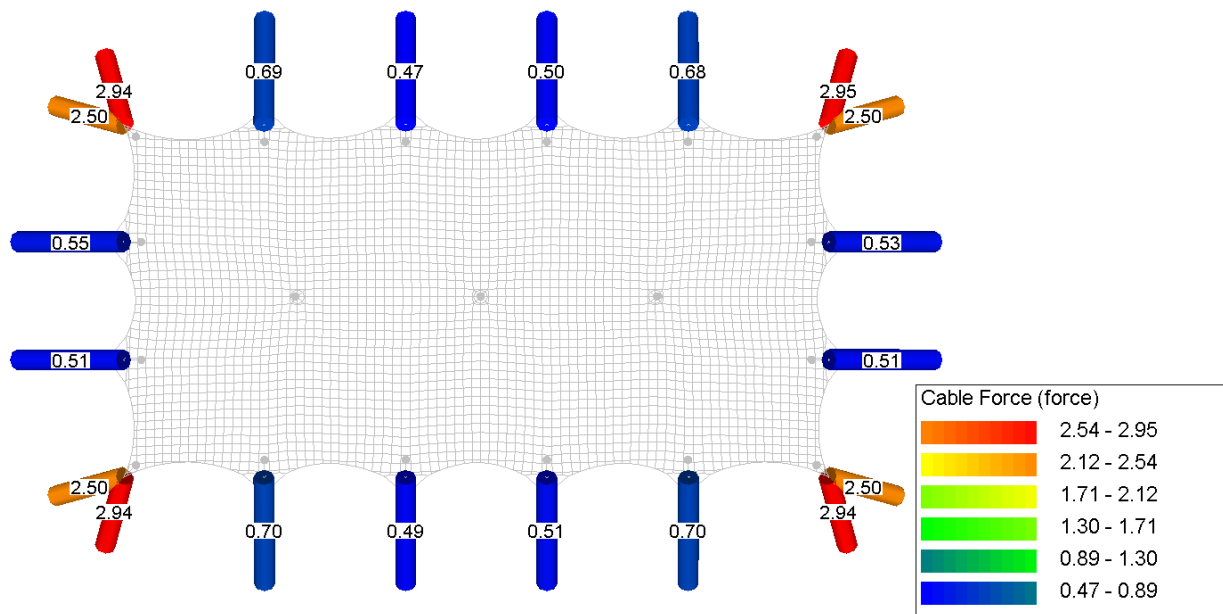
Annex F.2.3. Circumference rope forces



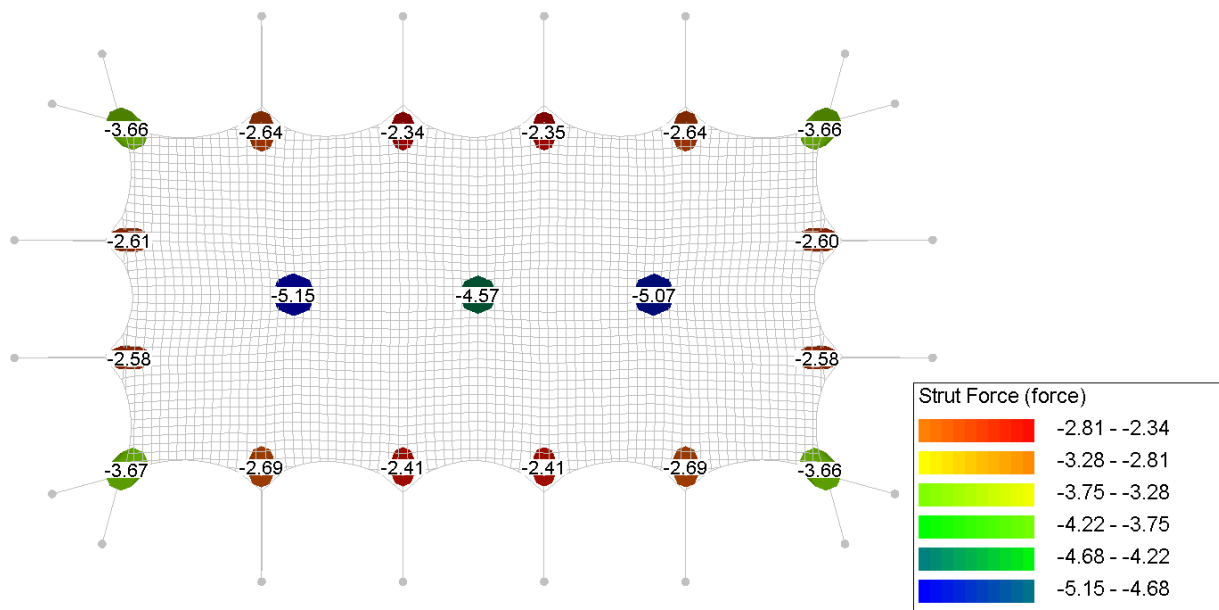
Annex F.2.4. Rope forces



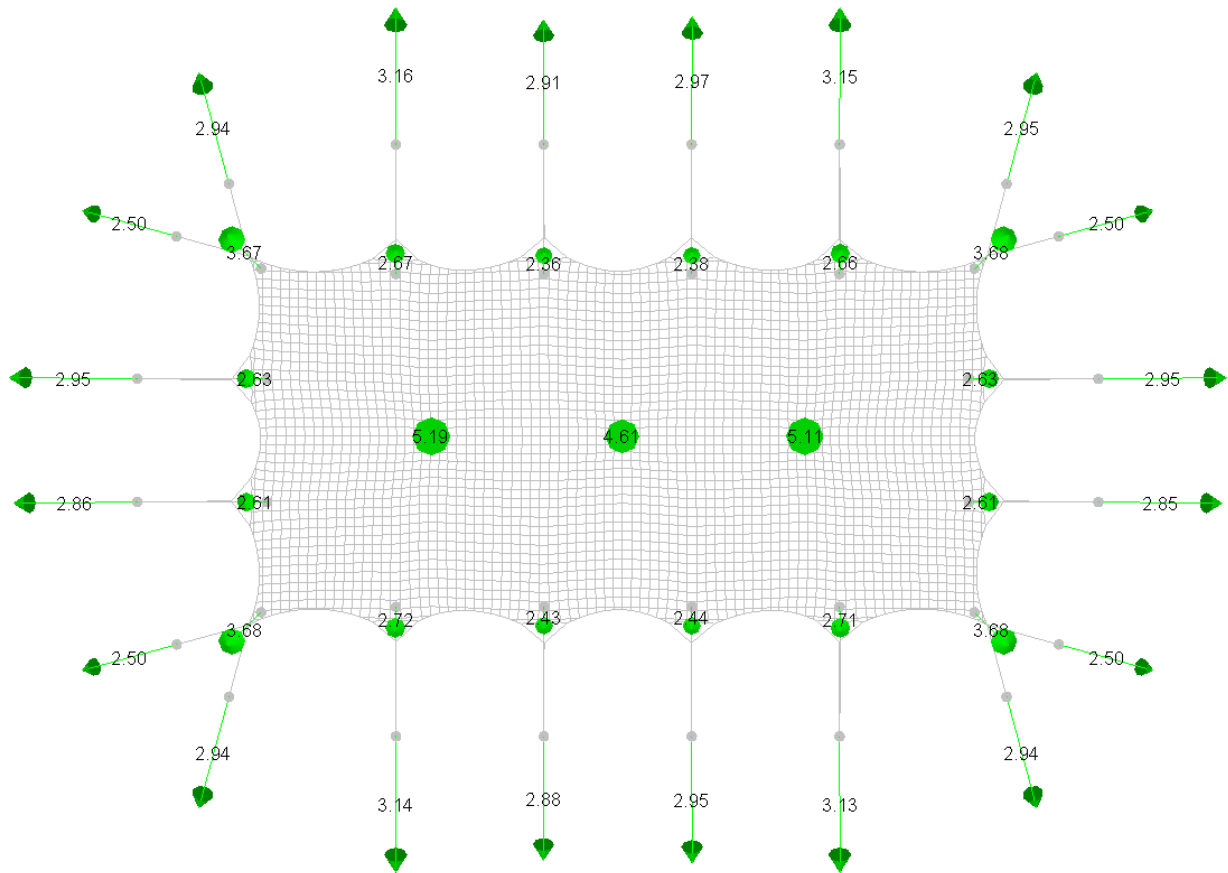
Annex F.2.5. Belt forces



Annex F.2.6. Pole forces

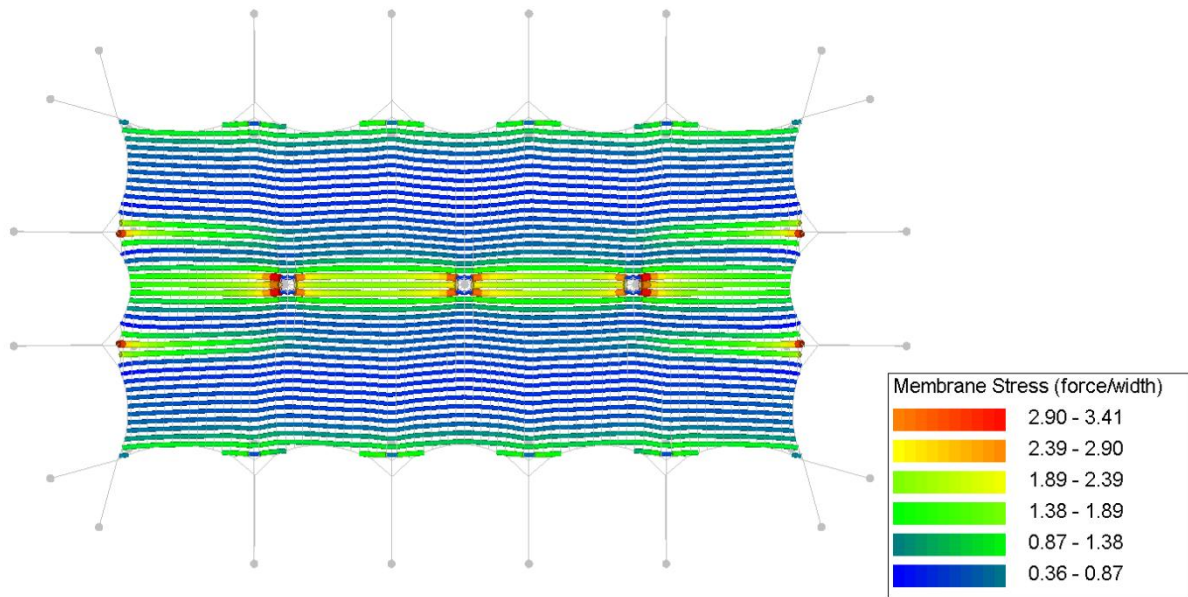


Annex F.2.7. Anchor forces

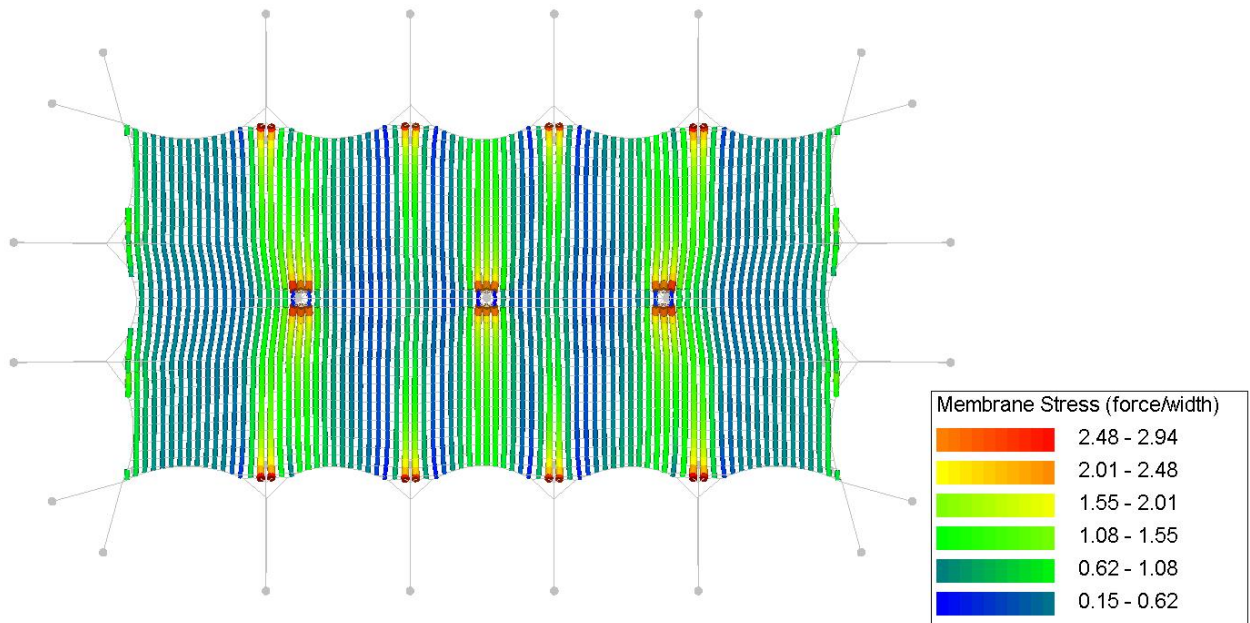


Annex F.3. CO3. Own weight + pretension + wind pressure

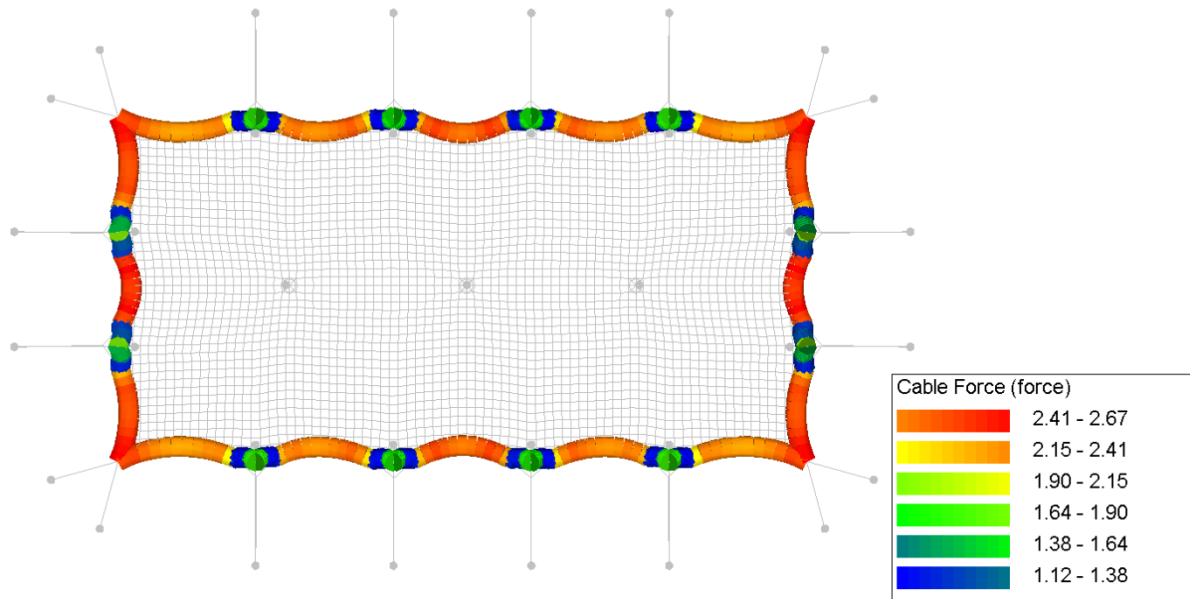
Annex F.3.1. Membrane stress (warp)



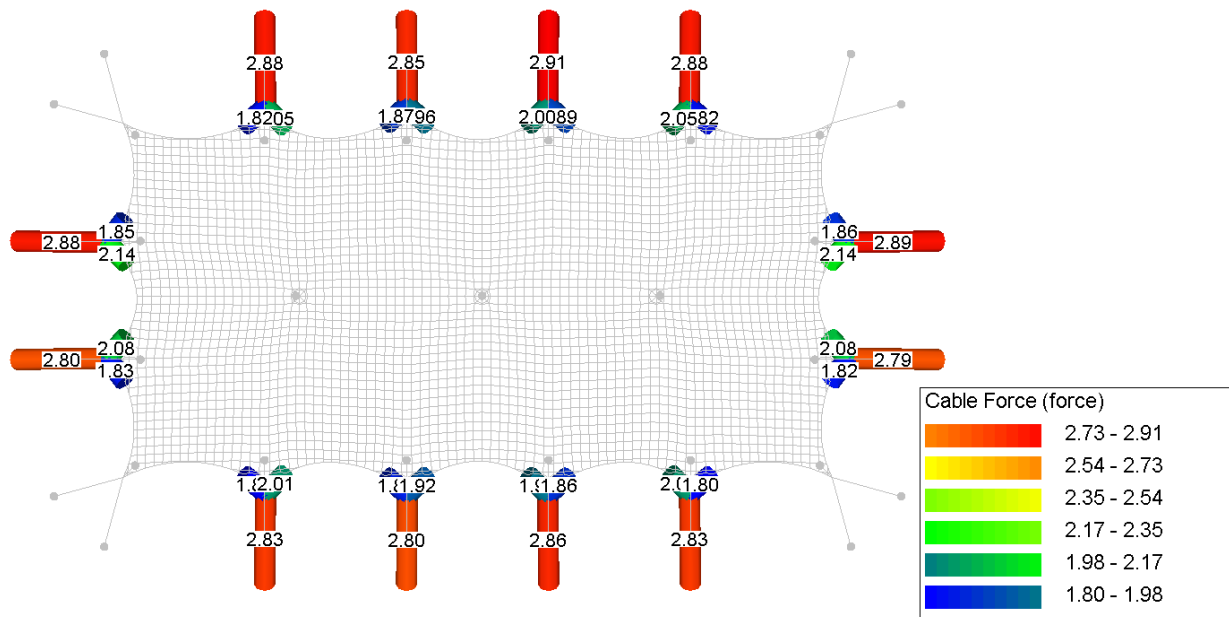
Annex F.3.2. Membrane stress (weft)



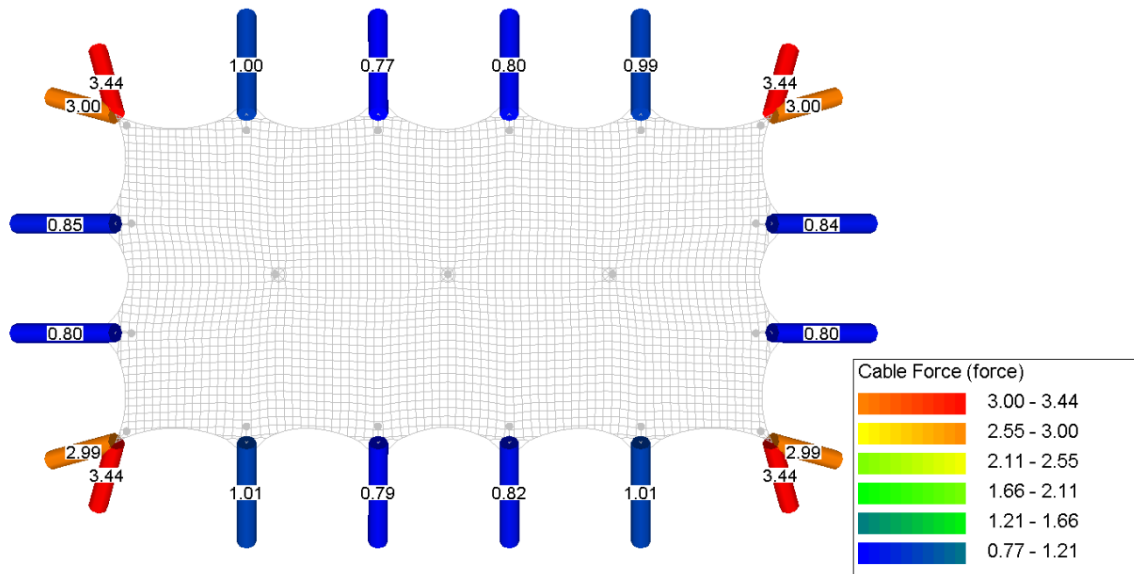
Annex F.3.3. Circumference rope forces



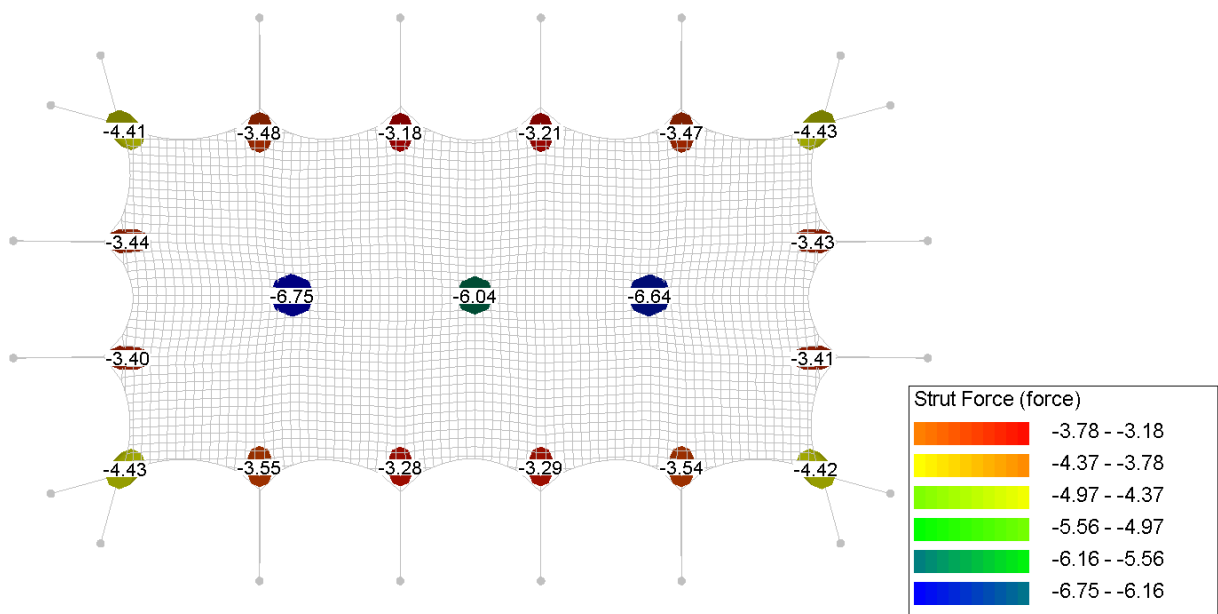
Annex F.3.4. Rope forces



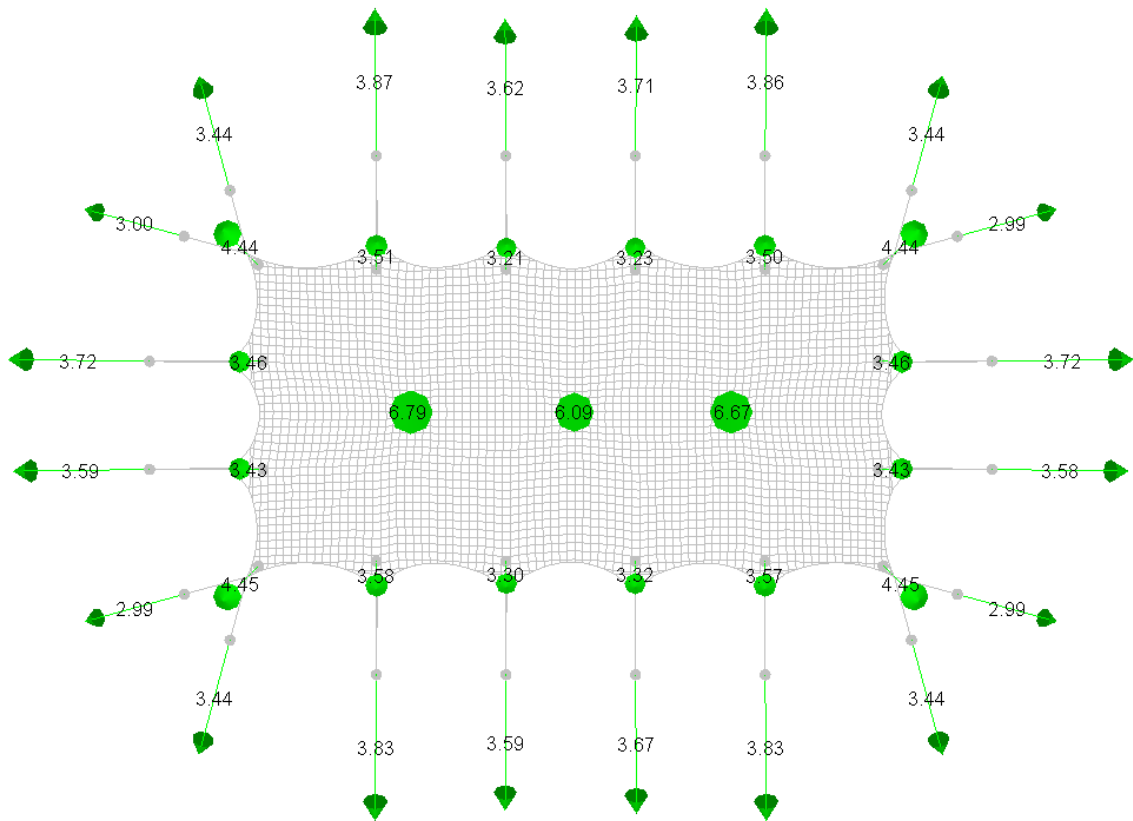
Annex F.3.5. Belt forces



Annex F.3.6. Poles forces

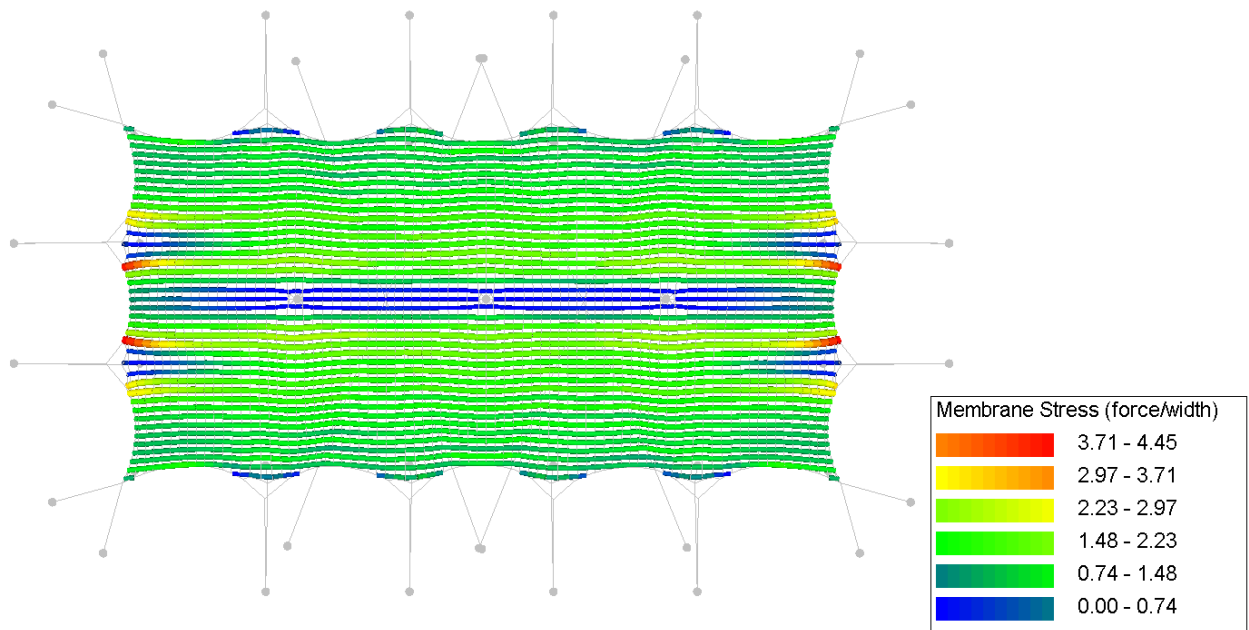


Annex F.3.7. Reaction forces

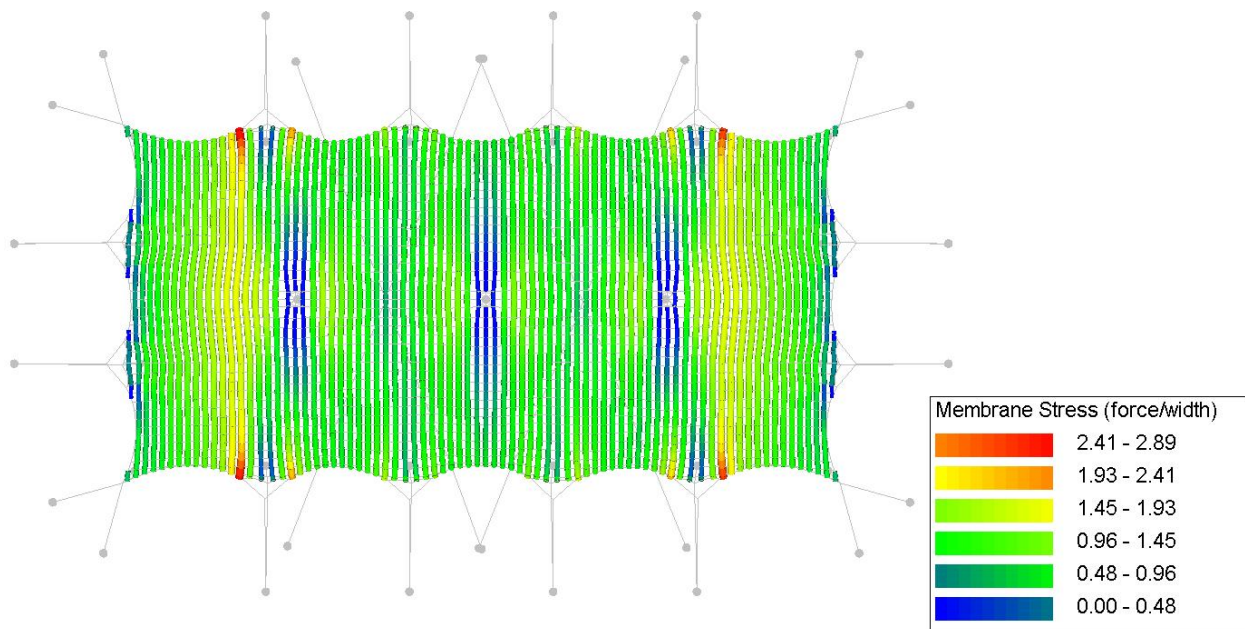


Annex F.4. CO4. Own weight + pretension + wind suction

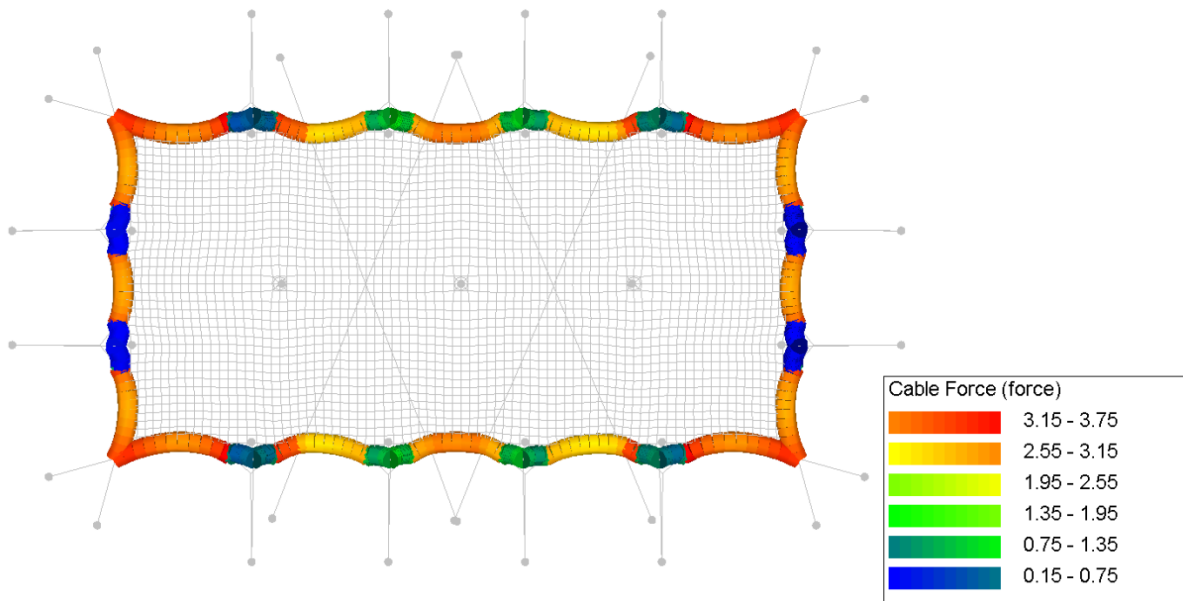
Annex F.4.1. Membrane stress (warp)



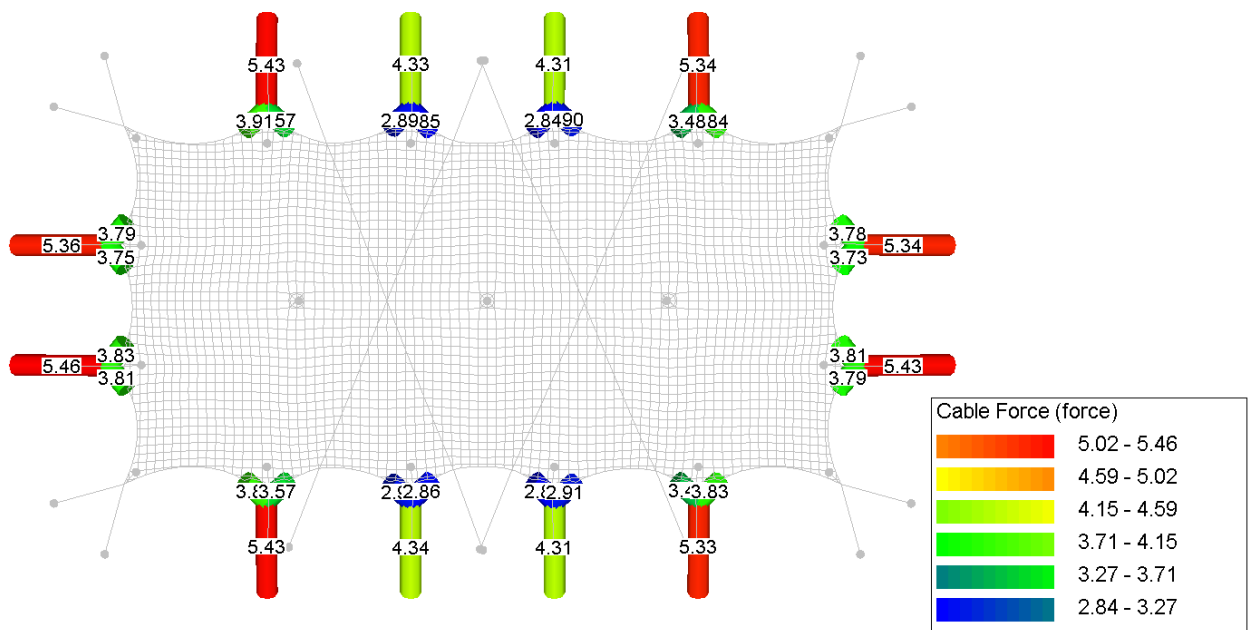
Annex F.4.2. Membrane stress (weft)



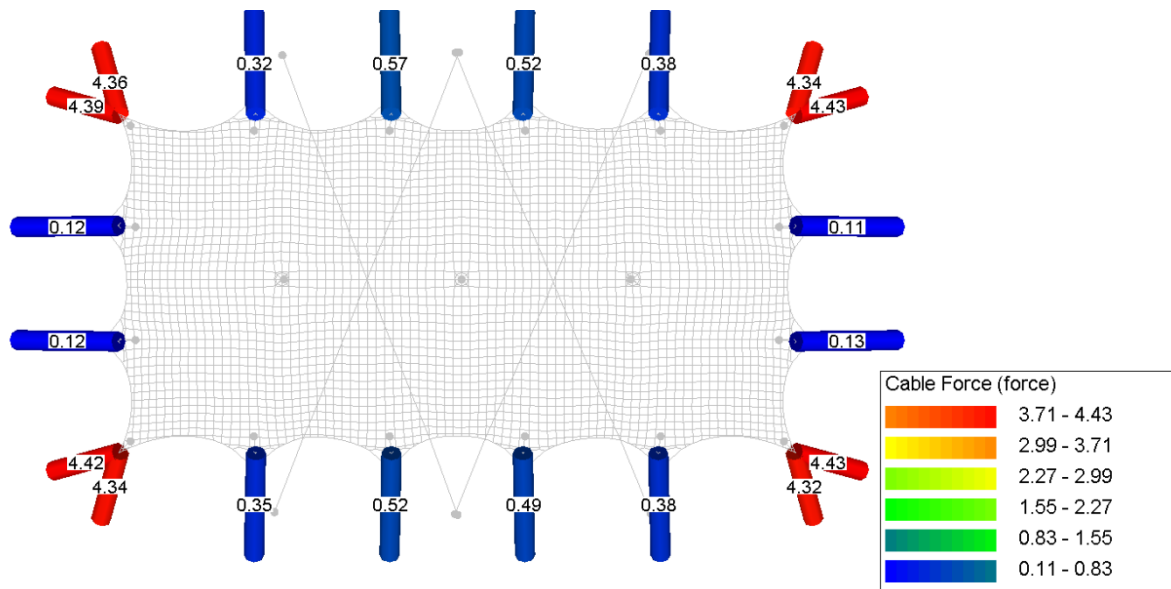
Annex F.4.3. Circumference rope forces



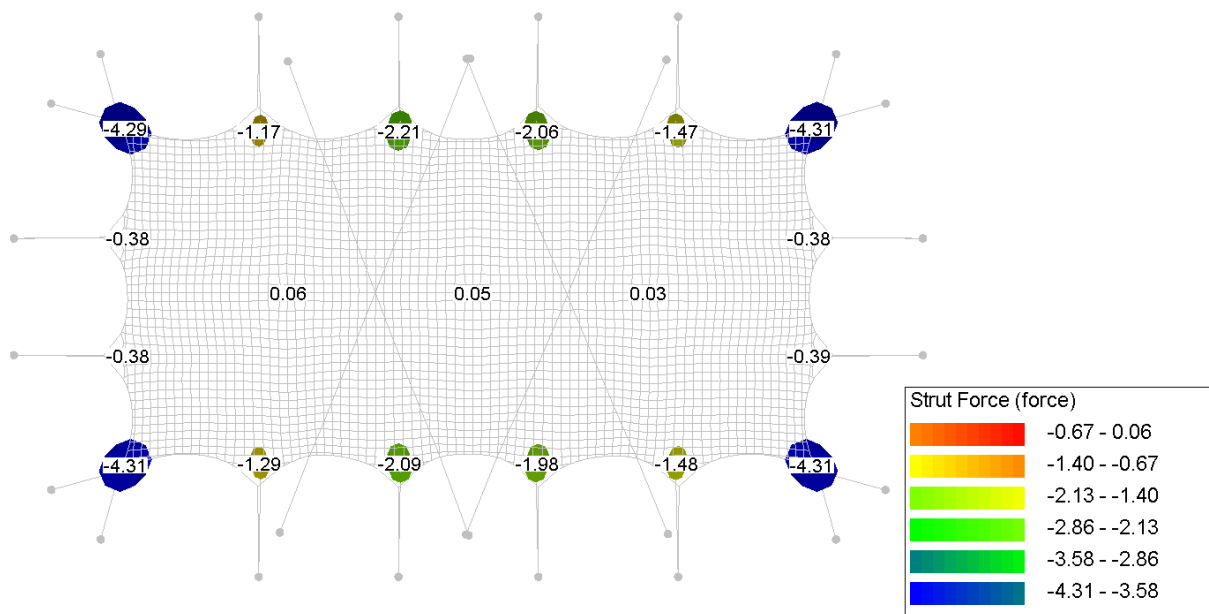
Annex F.4.4. Guy rope forces



Annex F.4.5. Belt forces

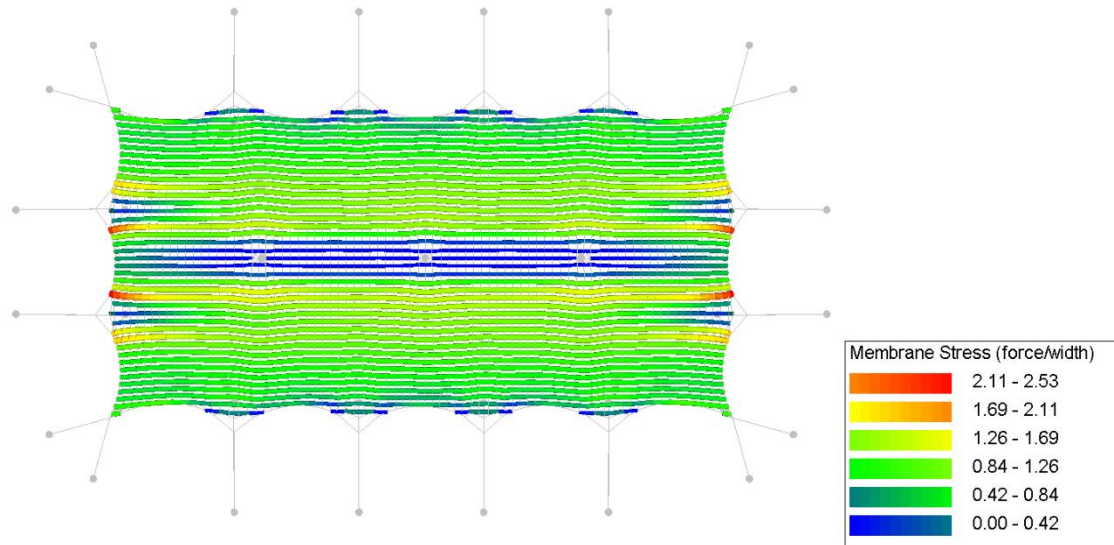


Annex F.4.6. Poles forces

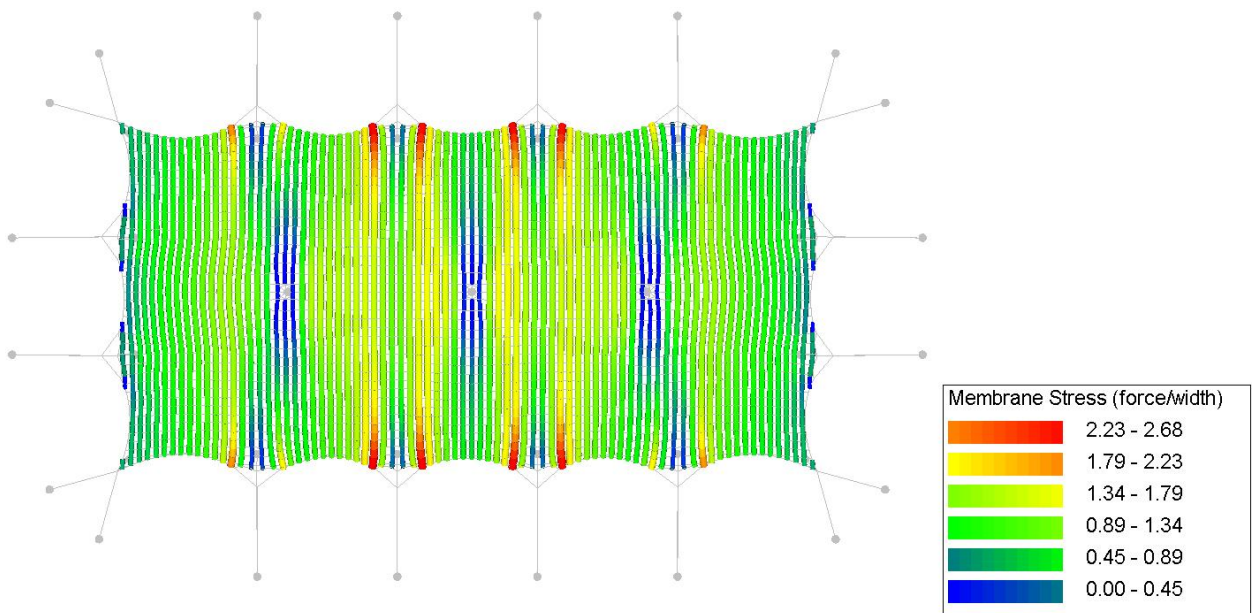


Annex F.5. CO5. Own weight + pretension + wind suction reduced [NL]

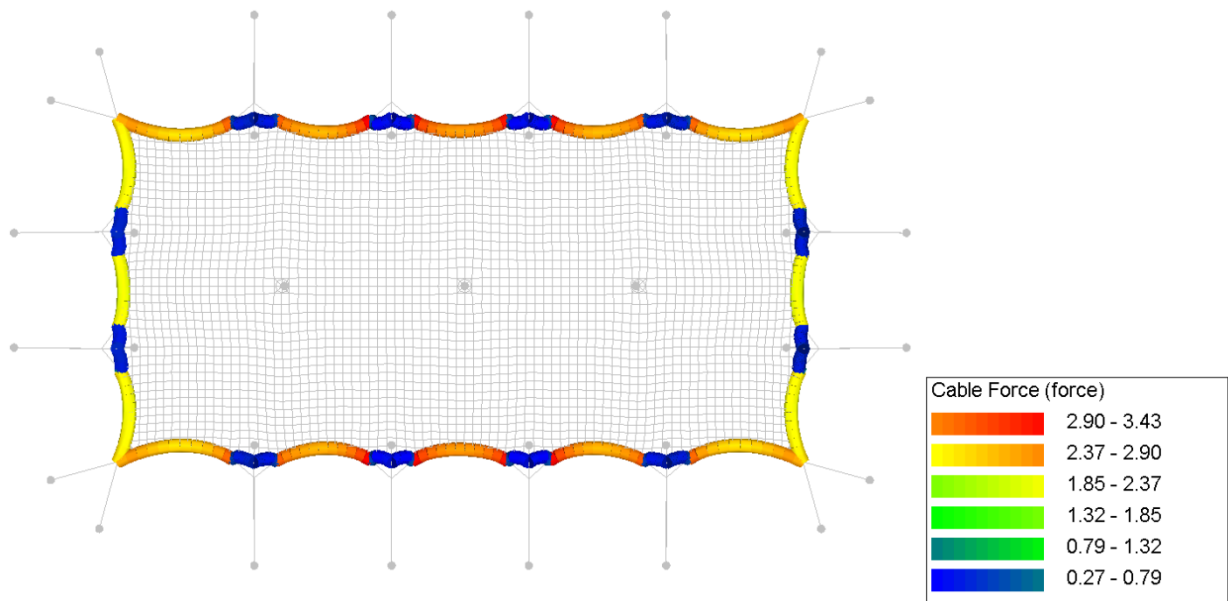
Annex F.5.1. Membrane stress (warp)



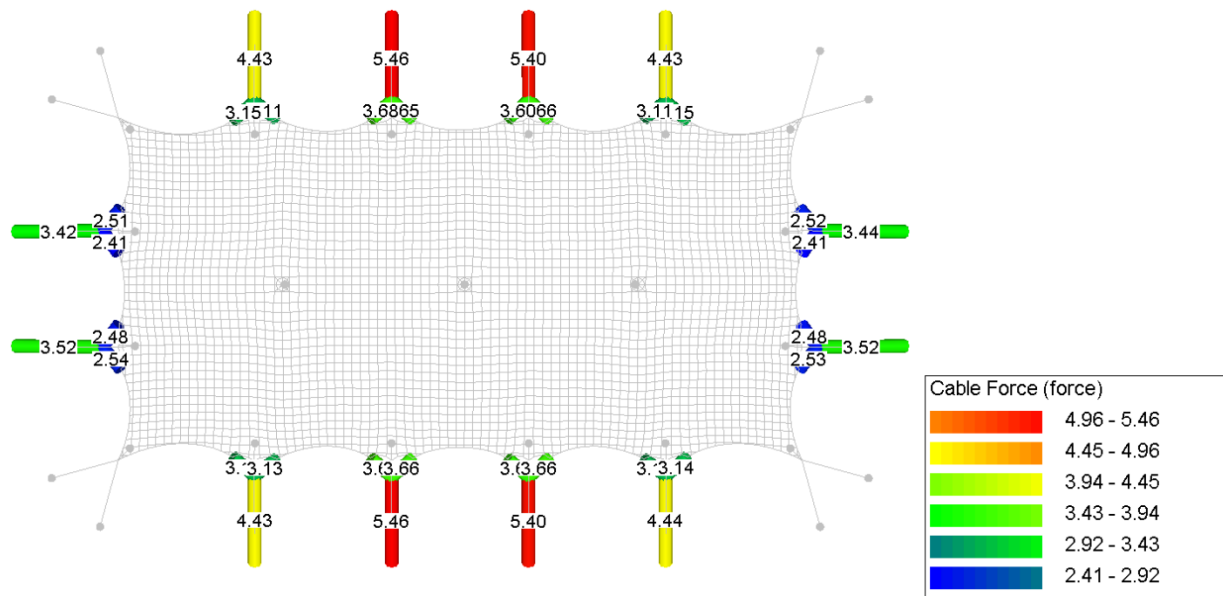
Annex F.5.2. Membrane stress (weft)



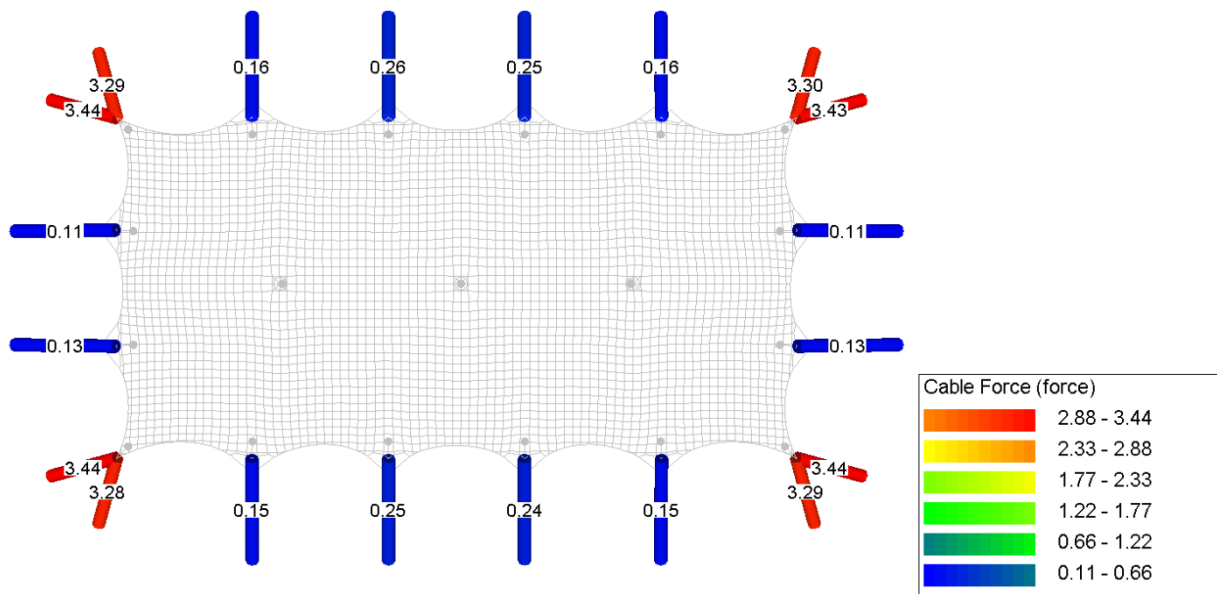
Annex F.5.3. Circumference rope forces



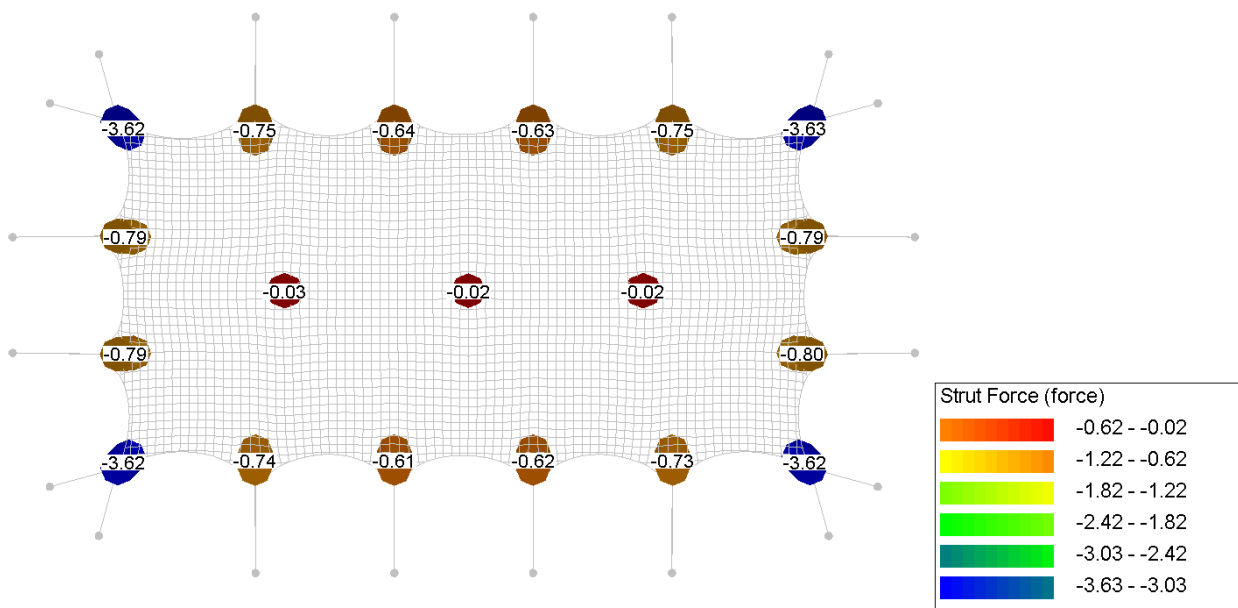
Annex F.5.4. Rope forces



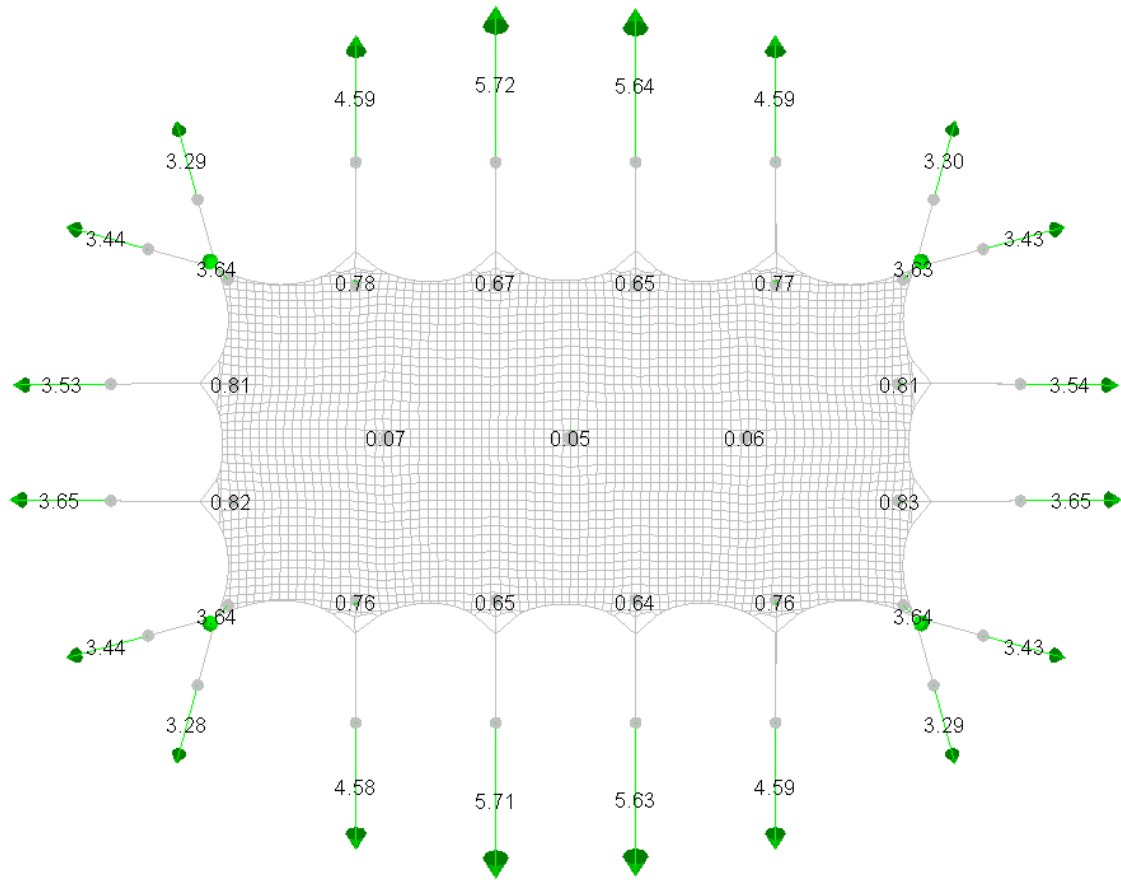
Annex F.5.5. Belt forces



Annex F.5.6. Poles forces

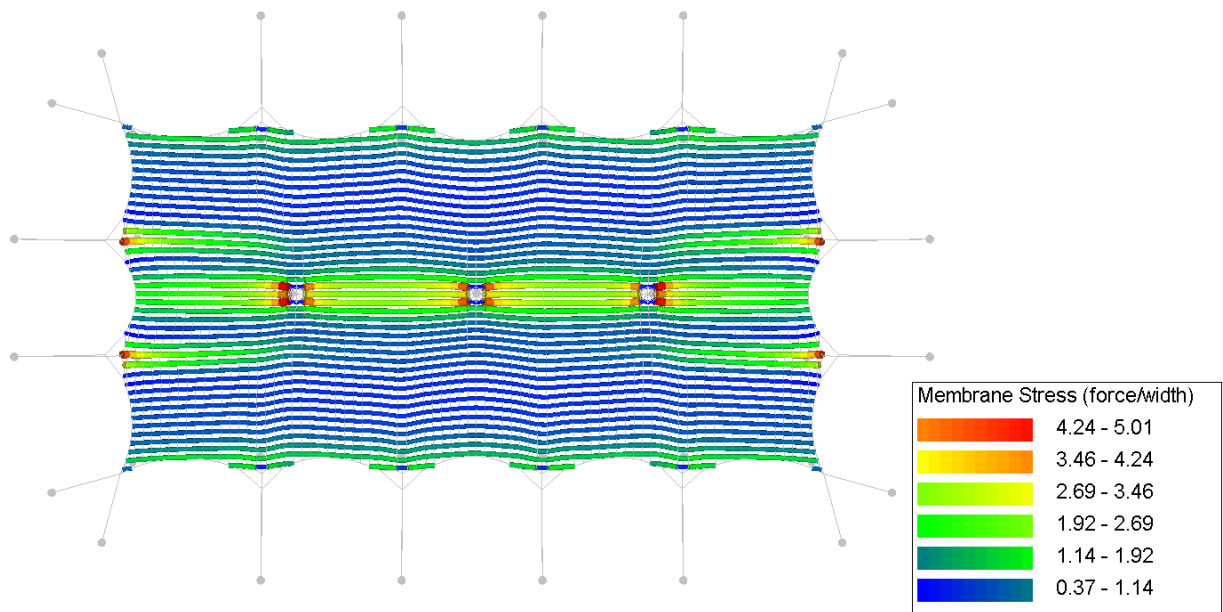


Annex F.5.7. Reaction forces

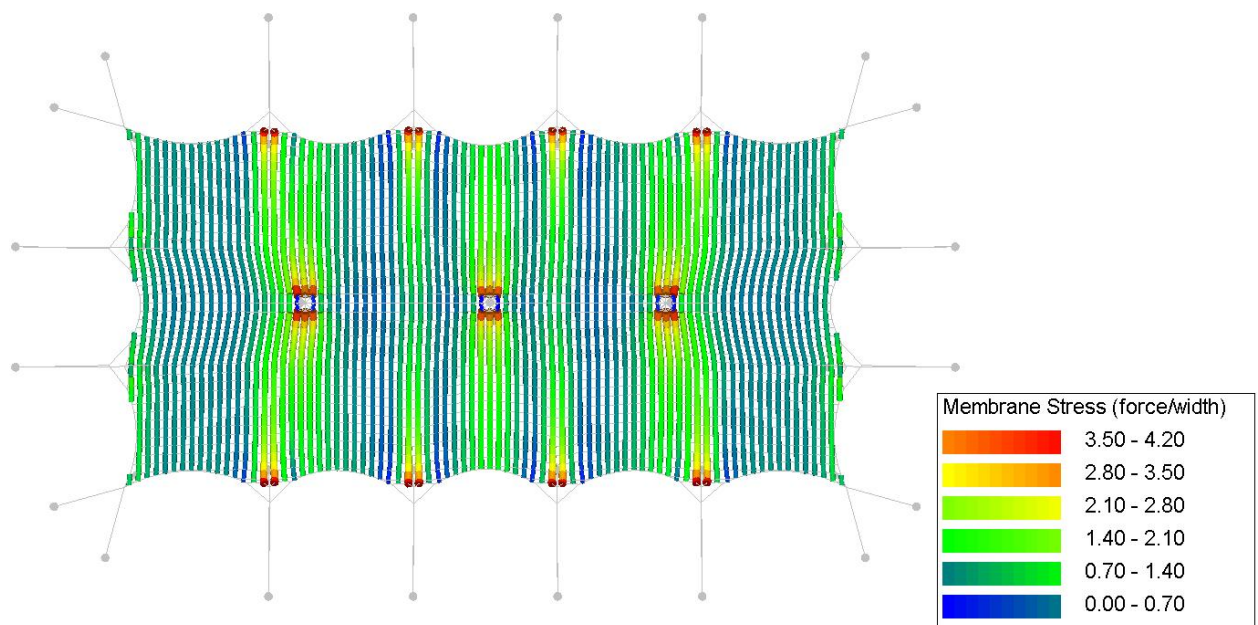


Annex F.6. CO6. Own weight + pretension + wind pressure + snow [FR]

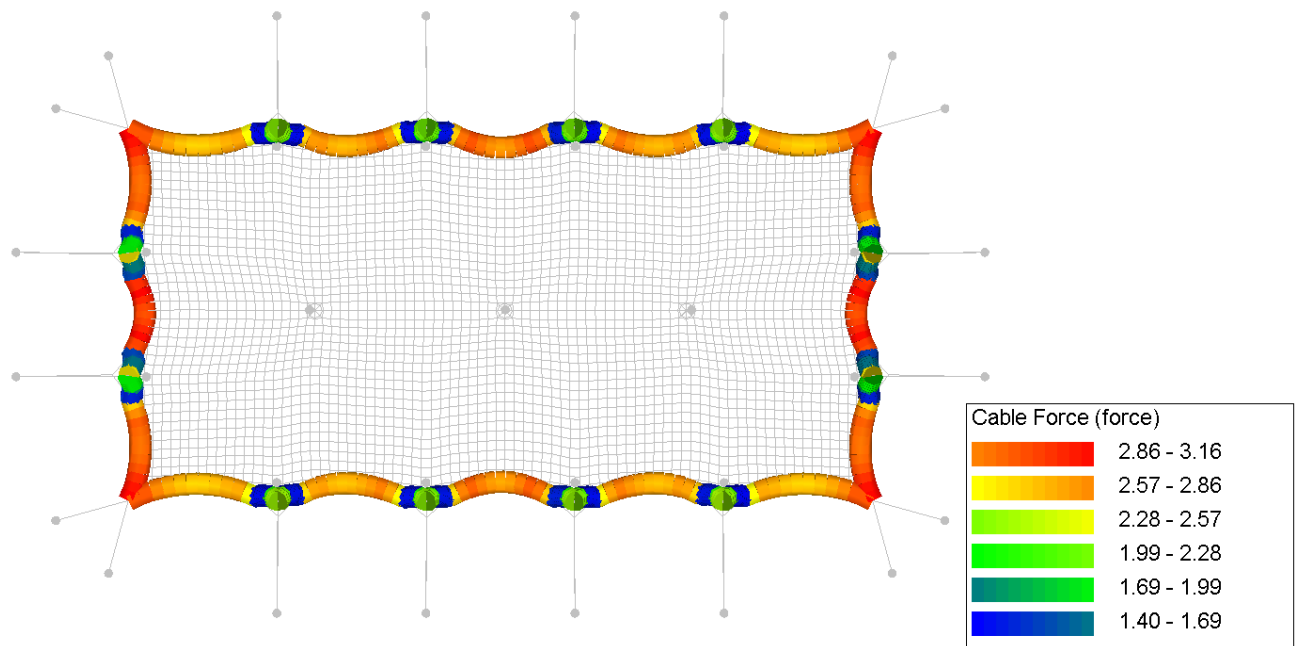
Annex F.6.1. Membrane stress (warp)



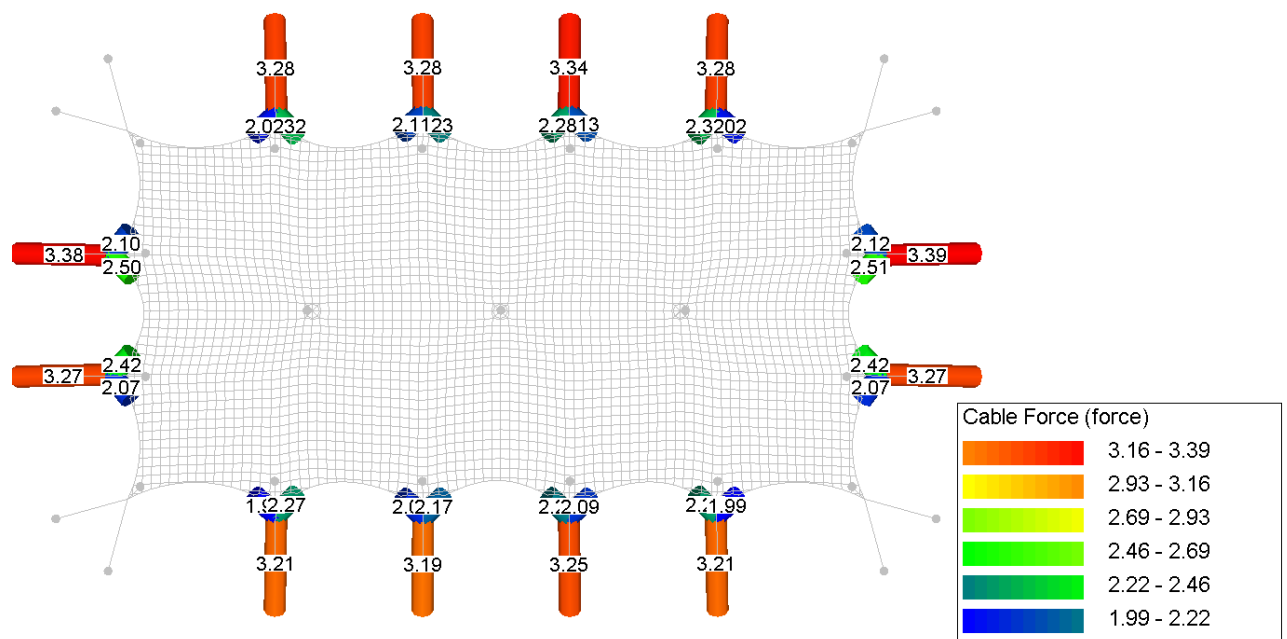
Annex F.6.2. Membrane stress (weft)



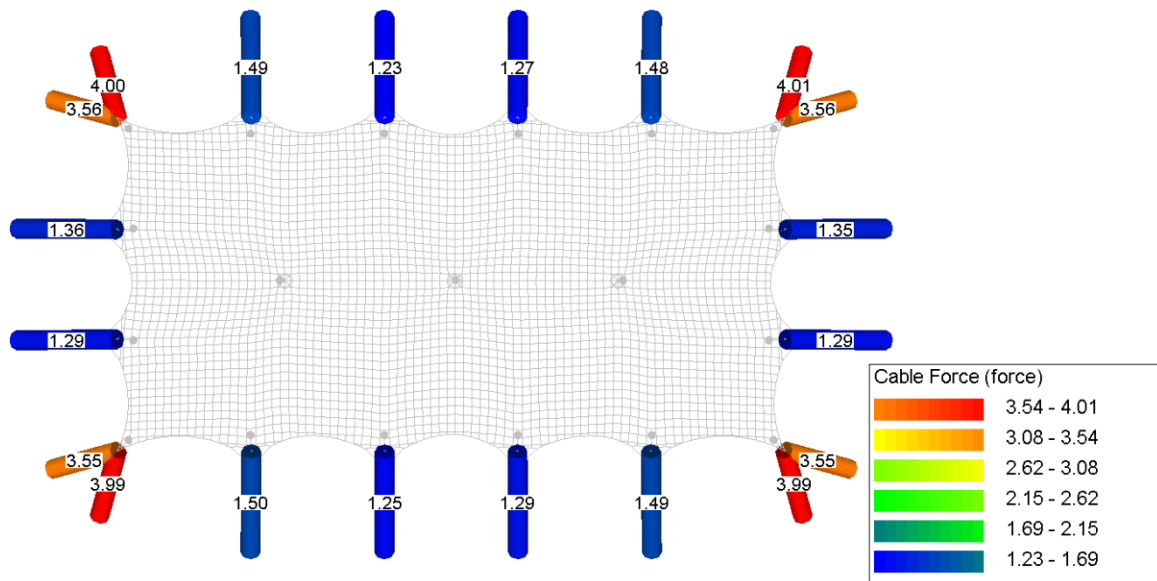
Annex F.6.3. Circumference rope forces



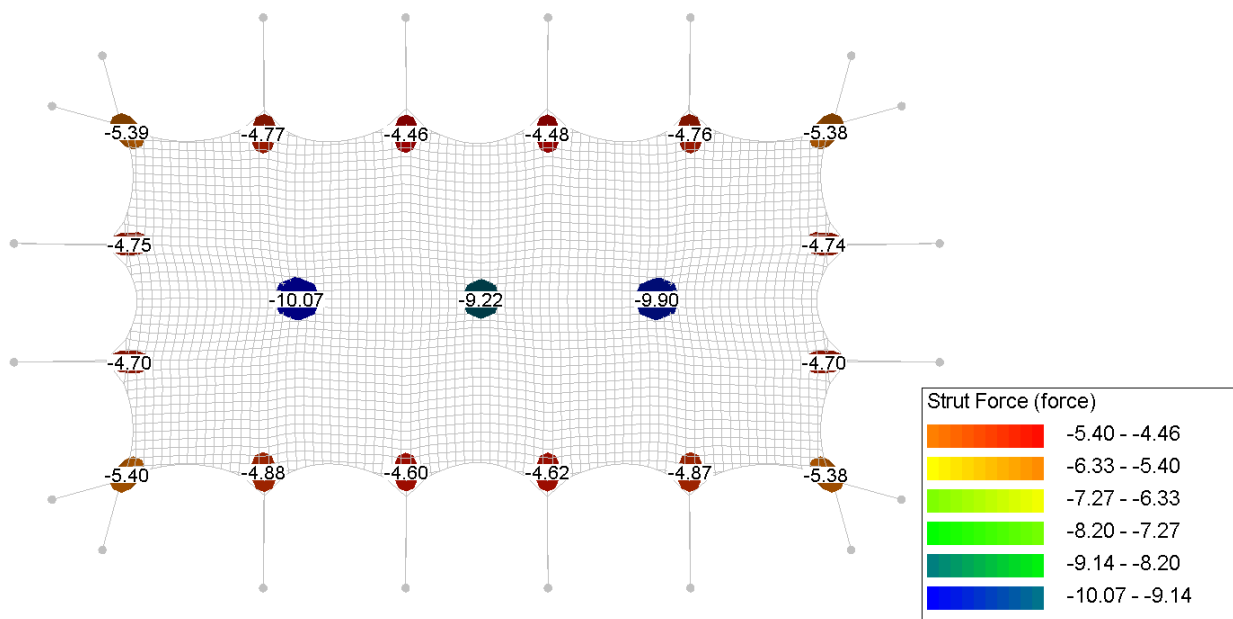
Annex F.6.4. Guy rope forces



Annex F.6.5. Belt forces

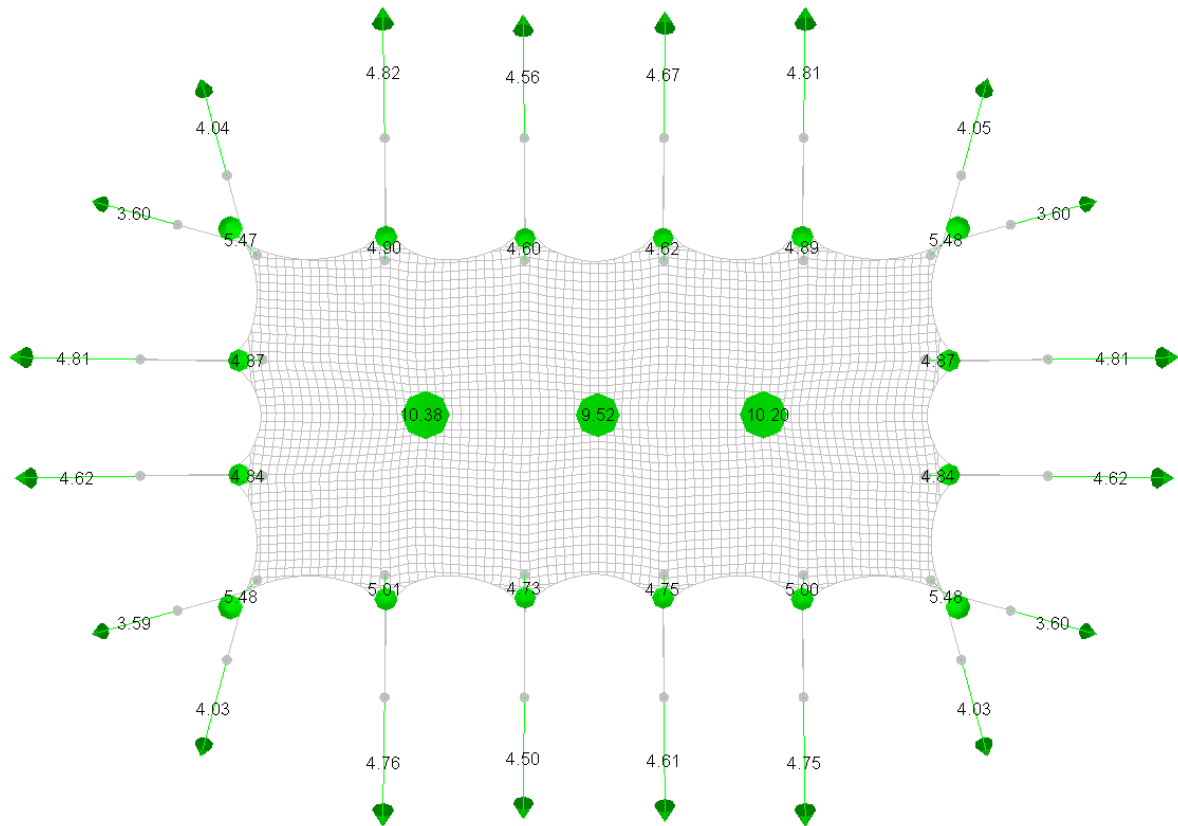


Annex F.6.6. Poles forces



Annex F.6.7. Reaction forces

Increased snow load of $0.1 \times 1.5/1.35 = 0.1083 \text{ kN/m}^2$ is applied for ballast and anchor calculation



Annex G. Member checks

Annex G.1. Main pole 4.0m – Wood, Spruce C18

Woodtype		Spruce	
Strenght type		C18	
Material factor	γ_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	100 mm	
Length (buckling)	l_{bucy}	4 m	
Effective area	A	7853.981634 mm ²	
Moment of innertia	I_y	4908738.521 mm ⁴	
Elastic modulus	W_{ely}	98174.77042 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modulus of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	25.0 mm	
Slenderness	λ_y	160.0	
Relative slenderness	$\lambda_{rel,y}$	2.79	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	4.64	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.120	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	10.76 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cd}	1.37 N/mm ²	
Bending stress	σ_{cd}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cd}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{md}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		0.92	NEN-EN 1995-1-1:2005 equ. 6.23

Annex G.2. Side pole 3m – Wood, Spruce C18

Annex G.2.1. 10x10m model

Woodtype		Spruce	
Strenght type		C18	
Material factor	γ_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	70 mm	
Length (buckling)	l_{buck}	3 m	
Effective area	A	3848.451001 mm ²	
Moment of inertia	I_y	1178588.119 mm ⁴	
Elastic modules	$W_{el,y}$	33673.94626 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	17.5 mm	
Slenderness	λ_y	171.4	
Relative slenderness	$\lambda_{rel,y}$	2.99	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	5.24	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.105	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	5.03 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cd}	1.31 N/mm ²	
Bending stress	σ_{cd}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cd}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{md}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		1.00	NEN-EN 1995-1-1:2005 equ. 6.23

Annex G.2.2. 10x15 and 10x20m model

Woodtype		Spruce	
Strenght type		C18	
Material factor	γ_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	75 mm	
Length (buckling)	l_{buck}	3 m	
Effective area	A	4417.864669 mm ²	
Moment of inertia	I_y	1553155.548 mm ⁴	
Elastic modules	$W_{el,y}$	41417.48127 mm ³	
Charistic pressure strenght	f_{dk}	18 N/mm ²	
Charistic bending strenght	f_{dk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	18.8 mm	
Slenderness	λ_y	160.0	
Relative slenderness	$\lambda_{rel,y}$	2.79	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	4.64	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.120	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	5.74 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{dd}	1.30 N/mm ²	
Bending stress	σ_{dd}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{dd}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{md}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		0.87	NEN-EN 1995-1-1:2005 equ. 6.23

Annex G.3. Corner pole 2.5m - Wood, C18

Annex G.3.1. 10x10 and 10x20m model

Woodtype		Spruce	
Strenght type		C18	
Material factor	γ_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	70 mm	
Length (buckling)	l_{bucy}	2.5 m	
Effective area	A	3848.451001 mm ²	
Moment of innertia	I_y	1178588.119 mm ⁴	
Elastic modules	W_{elky}	33673.94626 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	17.5 mm	
Slenderness	λ_y	142.9	
Relative slenderness	$\lambda_{rel,y}$	2.49	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	3.82	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.149	NEN-EN 1995-1-1:2005 equ. 6.25
Strength check			
Pressure force	F_d	7.12 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cdk}	1.85 N/mm ²	
Bending stress	σ_{cdk}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cdk}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{cdk}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		1.00	NEN-EN 1995-1-1:2005 equ. 6.23

Annex G.3.2. 10x15m model

Woodtype		Spruce	
Strenght type		C18	
Material factor	V_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	75 mm	
Length (buckling)	l_{bucy}	2.5 m	
Effective area	A	4417.864669 mm ²	
Moment of inertia	I_y	1553155.548 mm ⁴	
Elastic modules	W_{ely}	41417.48127 mm ³	
Charistic pressure strenght	f_{ok}	18 N/mm ²	
Charistic bending strenght	f_{ok}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	18.8 mm	
Slenderness	λ_y	133.3	
Relative slenderness	$\lambda_{rel,y}$	2.32	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	3.40	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.170	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	7.49 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cd}	1.70 N/mm ²	
Bending stress	σ_{cd}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cd}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{md}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		0.80	NEN-EN 1995-1-1:2005 equ. 6.23

Annex G.4. Main pole 4.0m – Wood, Spruce C18 – for snow and wind combination [FR]

Woodtype		Spruce	
Strenght type		C18	
Material factor	γ_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	105 mm	
Length (buckling)	l_{bucy}	4 m	
Effective area	A	8659.014751 mm ²	
Moment of innertia	I_y	5966602.352 mm ⁴	
Elastic modules	W_{ely}	113649.5686 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	26.3 mm	
Slenderness	λ_y	152.4	
Relative slenderness	$\lambda_{rel,y}$	2.66	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	4.26	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.132	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	14.27 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cd}	1.65 N/mm ²	
Bending stress	σ_{cd}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cd}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{md}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		1.01	NEN-EN 1995-1-1:2005 equ. 6.23

Annex G.5. Side pole 3.0m – Wood, Spruce C18 – for snow and wind combination [FR]

Annex G.5.1. 10x10m model

Woodtype		Spruce	
Strenght type		C18	
Material factor	γ_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	75 mm	
Length (buckling)	l_{bucy}	3 m	
Effective area	A	4417.864669 mm ²	
Moment of inertia	I_y	1553155.548 mm ⁴	
Elastic modules	$W_{el,y}$	41417.48127 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	18.8 mm	
Slenderness	λ_y	160.0	
Relative slenderness	$\lambda_{rel,y}$	2.79	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	4.64	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.120	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	6.18 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cd}	1.40 N/mm ²	
Bending stress	σ_{cd}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cd}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{md}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		0.94	NEN-EN 1995-1-1:2005 equ. 6.23

Annex G.5.2. 10x15 and 10x20m model

Woodtype		Spruce	
Strenght type		C18	
Material factor	γ_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	80 mm	
Length (buckling)	l_{bucy}	3 m	
Effective area	A	5026.548246 mm ²	
Moment of inertia	I_y	2010619.298 mm ⁴	
Elastic modules	W_{ely}	50265.48246 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	20.0 mm	
Slenderness	λ_y	150.0	
Relative slenderness	$\lambda_{rel,y}$	2.62	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	4.15	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.136	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	7.07 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cd}	1.41 N/mm ²	
Bending stress	σ_{cd}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cd}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{md}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		0.83	NEN-EN 1995-1-1:2005 equ. 6.23

Annex G.6. Corner pole 2.5m – Wood, Spruce C18 – for snow and wind combination [FR]

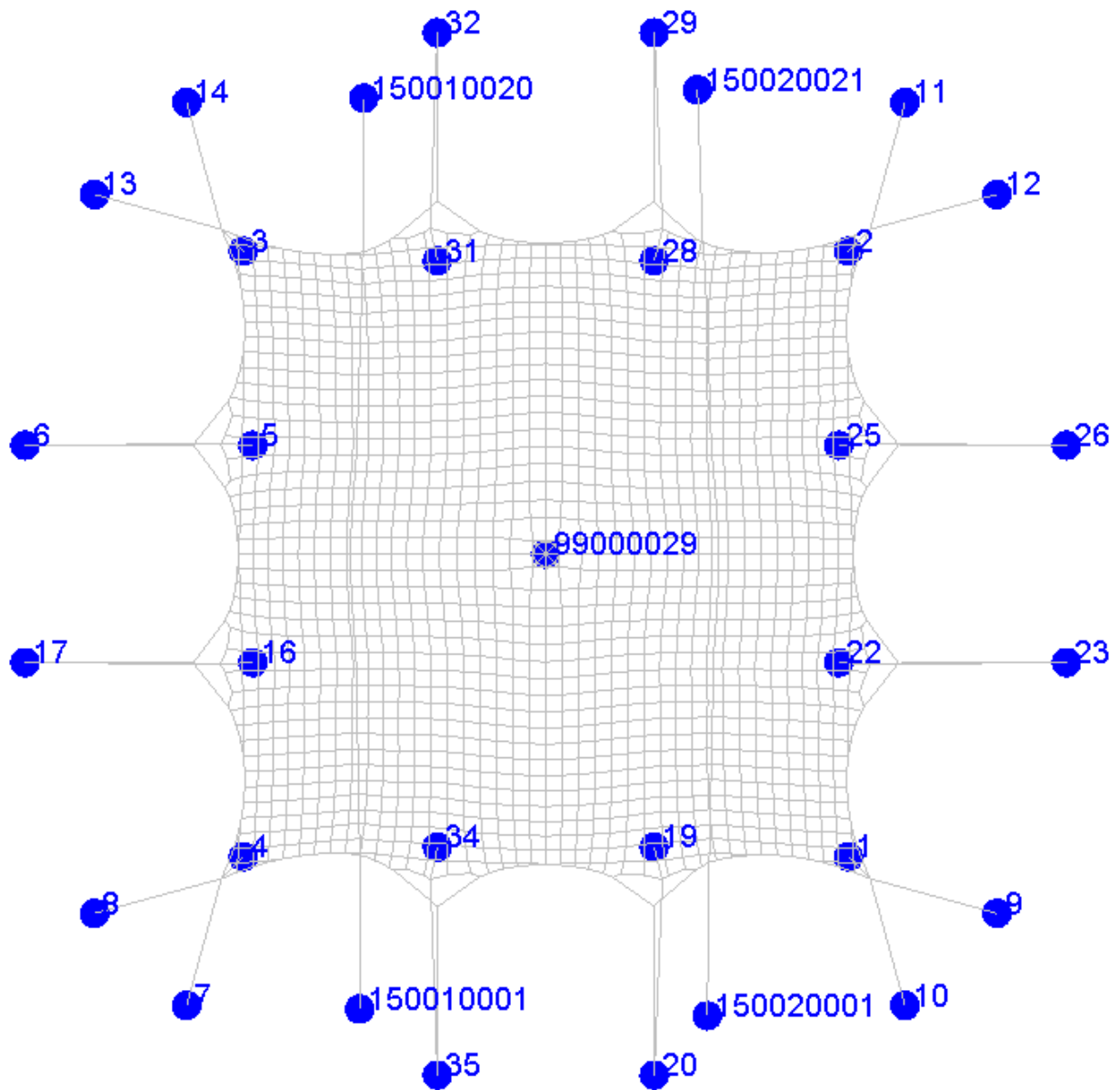
Woodtype		Spruce	
Strenght type		C18	
Material factor	γ_M	1.3	NEN-EN 1995-1-1:2005 table 2.3
Climate class		2	NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2	NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	75 mm	
Length (buckling)	l_{bucy}	2.5 m	
Effective area	A	4417.864669 mm ²	
Moment of inertia	I_y	1553155.548 mm ⁴	
Elastic modules	$W_{el,y}$	41417.48127 mm ³	
Charistic pressure strenght	f_{cdk}	18 N/mm ²	
Charistic bending strenght	f_{cdk}	18 N/mm ²	
Modules of elasticity	$E_{0.05}$	6 kN/m ²	
	i_y	18.8 mm	
Slenderness	λ_y	133.3	
Relative slenderness	$\lambda_{rel,y}$	2.32	NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	3.40	NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.170	NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	7.76 kN	
Bending moment	M_d	0.00 kNm	
Pressure stress	σ_{cdk}	1.76 N/mm ²	
Bending stress	σ_{cdk}	0.00 N/mm ²	
Load duration		short	NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90	NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{cd}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{mod}	12.46 N/mm ²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		0.83	NEN-EN 1995-1-1:2005 equ. 6.23

Annex H. Reaction forces model 10x10m

Annex H.1. Point numbers



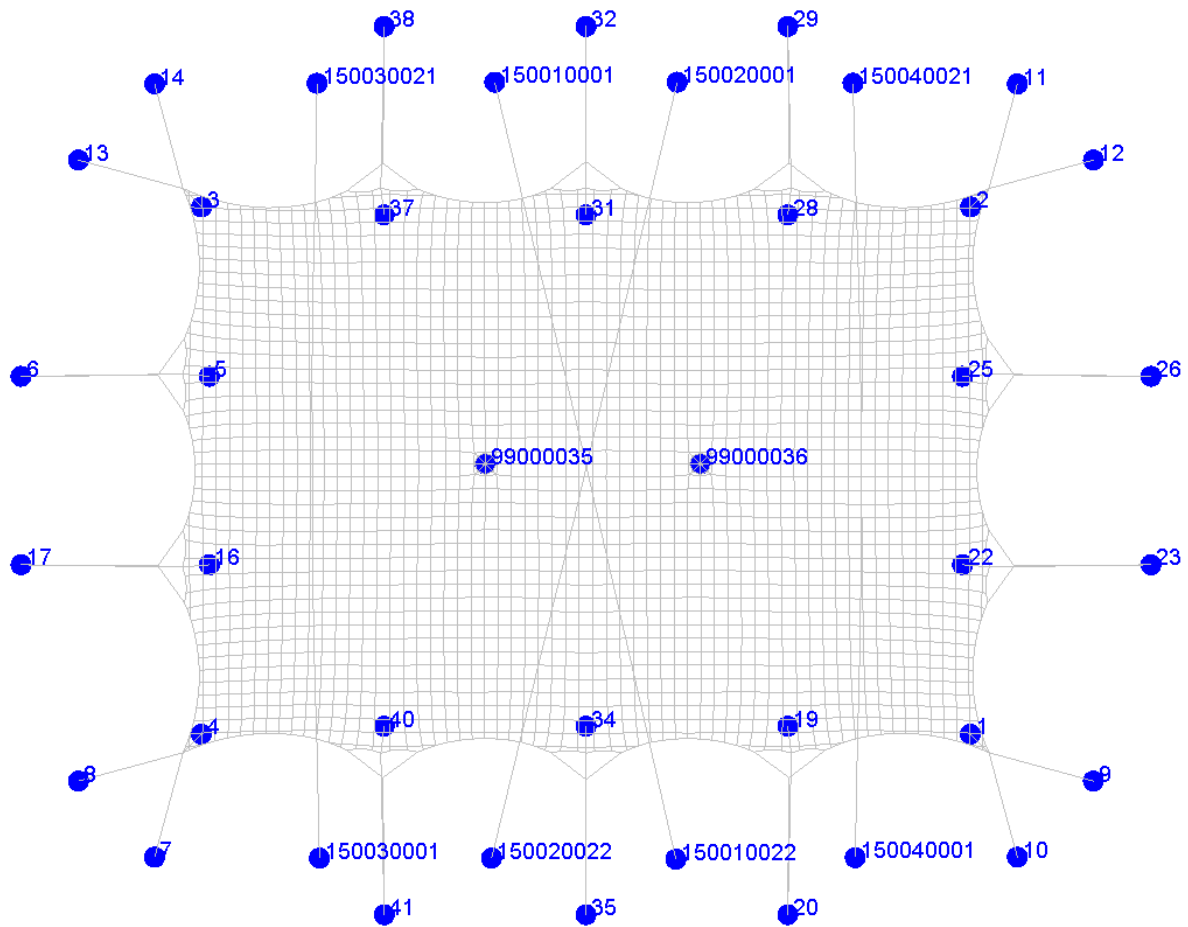
Annex H.2. Reaction forces

Reaction forces are shown in kilonewtons [kN] without any safety factor (1.35, 1.5, 1.2). A load combination especially for anchors and ballast calculation for snow and wind combined is CO6a.

Att	Node	Koor X Koor Y		CO1			CO2			CO3			CO4			CO5			CO6			CO6a		
				Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz
Tie	6	-3.00	6.67	-1.55	-0.01	-1.18	-2.04	0.00	-1.59	-2.44	0.01	-1.91	-3.98	-0.02	-2.97	-3.04	-0.03	-2.27	-3.04	0.03	-2.41	-3.08	0.03	-2.44
Tie	17	-3.00	3.33	-1.55	0.01	-1.18	-2.04	0.00	-1.59	-2.44	-0.01	-1.91	-3.98	0.02	-2.97	-3.04	0.03	-2.28	-3.04	-0.03	-2.41	-3.08	-0.03	-2.44
Tie	20	6.67	-3.00	-0.01	-1.55	-1.18	0.00	-1.98	-1.53	0.01	-2.34	-1.82	0.00	-3.58	-2.66	-0.03	-2.69	-2.02	0.02	-2.84	-2.24	0.02	-2.88	-2.27
Tie	23	13.00	3.33	1.55	0.01	-1.18	2.04	0.00	-1.59	2.44	-0.01	-1.91	4.03	0.02	-3.01	3.04	0.03	-2.27	3.04	-0.03	-2.41	3.08	-0.03	-2.44
Tie	26	13.00	6.67	1.55	-0.01	-1.18	2.04	0.00	-1.59	2.44	0.01	-1.91	4.04	-0.02	-3.02	3.04	-0.03	-2.28	3.04	0.03	-2.41	3.08	0.03	-2.44
Tie	29	6.67	13.00	-0.01	1.55	-1.18	0.00	1.98	-1.53	0.01	2.34	-1.82	0.00	3.60	-2.67	-0.03	2.69	-2.02	0.02	2.84	-2.24	0.02	2.88	-2.27
Tie	32	3.33	13.00	0.01	1.55	-1.18	0.00	1.98	-1.53	-0.01	2.34	-1.82	0.00	3.64	-2.71	0.03	2.69	-2.02	-0.02	2.84	-2.24	-0.02	2.88	-2.27
Tie	35	3.33	-3.00	0.01	-1.55	-1.18	0.00	-1.98	-1.53	-0.01	-2.34	-1.82	0.00	-3.63	-2.70	0.03	-2.69	-2.02	-0.02	-2.84	-2.24	-0.02	-2.88	-2.27
Tie	7	-0.52	-1.93	-0.39	-1.43	-1.47	-0.46	-1.71	-1.76	-0.53	-1.96	-2.02	-0.71	-2.60	-2.68	-0.59	-2.18	-2.25	-0.62	-2.27	-2.34	-0.63	-2.29	-2.36
Tie	8	-1.93	-0.52	-1.43	-0.39	-1.48	-1.71	-0.46	-1.76	-1.96	-0.53	-2.02	-2.67	-0.73	-2.75	-2.04	-0.56	-2.11	-2.28	-0.62	-2.34	-2.30	-0.63	-2.37
Tie	9	11.93	-0.52	1.43	-0.39	-1.48	1.71	-0.46	-1.76	1.96	-0.53	-2.02	2.69	-0.74	-2.77	2.04	-0.56	-2.11	2.28	-0.62	-2.34	2.30	-0.63	-2.37
Tie	10	10.52	-1.93	0.39	-1.43	-1.47	0.46	-1.71	-1.76	0.53	-1.96	-2.01	0.72	-2.61	-2.68	0.59	-2.18	-2.25	0.62	-2.27	-2.34	0.63	-2.29	-2.36
Tie	11	10.52	11.93	0.39	1.43	-1.47	0.46	1.71	-1.76	0.53	1.96	-2.02	0.72	2.61	-2.68	0.59	2.18	-2.25	0.62	2.27	-2.34	0.63	2.29	-2.36
Tie	12	11.93	10.52	1.43	0.39	-1.48	1.71	0.46	-1.76	1.96	0.53	-2.02	2.68	0.74	-2.76	2.04	0.56	-2.11	2.28	0.62	-2.34	2.30	0.63	-2.37
Tie	13	-1.93	10.52	-1.43	0.39	-1.48	-1.71	0.46	-1.76	-1.96	0.53	-2.02	-2.66	0.73	-2.73	-2.04	0.56	-2.11	-2.28	0.62	-2.34	-2.30	0.63	-2.37
Tie	14	-0.52	11.93	-0.39	1.43	-1.47	-0.46	1.71	-1.76	-0.53	1.96	-2.02	-0.71	2.60	-2.67	-0.59	2.18	-2.25	-0.62	2.27	-2.34	-0.63	2.29	-2.36
Pole	1	9.65	0.35	0.44	-0.44	2.58	0.56	-0.56	3.29	0.65	-0.65	3.86	0.72	-0.72	4.30	0.57	-0.56	3.34	0.78	-0.78	4.64	0.79	-0.79	4.70
Pole	2	9.65	9.65	0.44	0.44	2.58	0.56	0.56	3.29	0.65	0.65	3.86	0.71	0.71	4.25	0.57	0.56	3.34	0.78	0.78	4.64	0.79	0.79	4.71
Pole	3	0.35	9.65	-0.44	0.44	2.58	-0.56	0.56	3.29	-0.65	0.65	3.86	-0.73	0.73	4.37	-0.57	0.56	3.34	-0.78	0.78	4.64	-0.79	0.79	4.70
Pole	4	0.35	0.35	-0.44	-0.44	2.58	-0.56	-0.56	3.29	-0.65	-0.65	3.86	-0.74	-0.74	4.41	-0.57	-0.56	3.34	-0.78	-0.78	4.64	-0.79	-0.79	4.71
Pole	5	0.50	6.67	-0.21	0.01	1.31	-0.39	0.01	2.41	-0.51	0.01	3.11	-0.13	0.01	0.82	-0.14	0.01	0.85	-0.70	0.00	4.30	-0.71	-0.01	4.39
Pole	16	0.50	3.33	-0.21	-0.01	1.31	-0.39	-0.01	2.41	-0.51	-0.01	3.11	-0.13	-0.01	0.82	-0.14	-0.01	0.85	-0.70	0.00	4.30	-0.71	0.01	4.39
Pole	19	6.67	0.50	0.01	-0.21	1.31	0.01	-0.38	2.35	0.01	-0.49	3.01	0.08	-0.31	1.87	0.01	-0.12	0.74	0.00	-0.67	4.10	-0.01	-0.68	4.19
Pole	22	9.50	3.33	0.21	-0.01	1.31	0.39	-0.01	2.41	0.51	-0.01	3.11	0.13	-0.01	0.79	0.14	-0.01	0.85	0.70	0.00	4.30	0.71	0.01	4.39
Pole	25	9.50	6.67	0.21	0.01	1.31	0.39	0.01	2.41	0.51	0.01	3.11	0.12	0.01	0.77	0.14	0.01	0.85	0.70	0.00	4.30	0.71	-0.01	4.39
Pole	28	6.67	9.50	0.01	0.21	1.31	0.01	0.38	2.35	0.01	0.49	3.01	0.09	0.33	2.02	0.01	0.12	0.74	0.00	0.67	4.10	-0.01	0.68	4.19
Pole	31	3.33	9.50	-0.01	0.21	1.31	-0.01	0.38	2.35	-0.01	0.49	3.01	-0.05	0.25	1.51	-0.01	0.12	0.74	0.00	0.67	4.10	0.01	0.68	4.19
Pole	34	3.33	0.50	-0.01	-0.21	1.31	-0.01	-0.38	2.35	-0.01	-0.49	3.01	-0.05	-0.24	1.48	-0.01	-0.12	0.74	0.00	-0.67	4.10	0.01	-0.68	4.19
Pole	99000029	5.00	5.00	0.00	0.00	2.15	0.00	0.00	5.45	0.00	0.00	6.98	0.00	0.00	-0.01	0.00	0.00	0.12	0.00	0.00	10.36	0.00	0.00	10.64
Stormbelt	150010001	2.14	-1.99										-0.01	-3.80	-3.35									
Stormbelt	150010020	2.21	12.00										0.02	3.80	-3.34									
Stormbelt	150020001	7.48	-2.09										-0.04	-4.22	-3.72									
Stormbelt	150020021	7.33	12.12										-0.14	4.19	-3.75									

Annex I. Reaction forces model 10x15m

Annex I.1. Point numbers



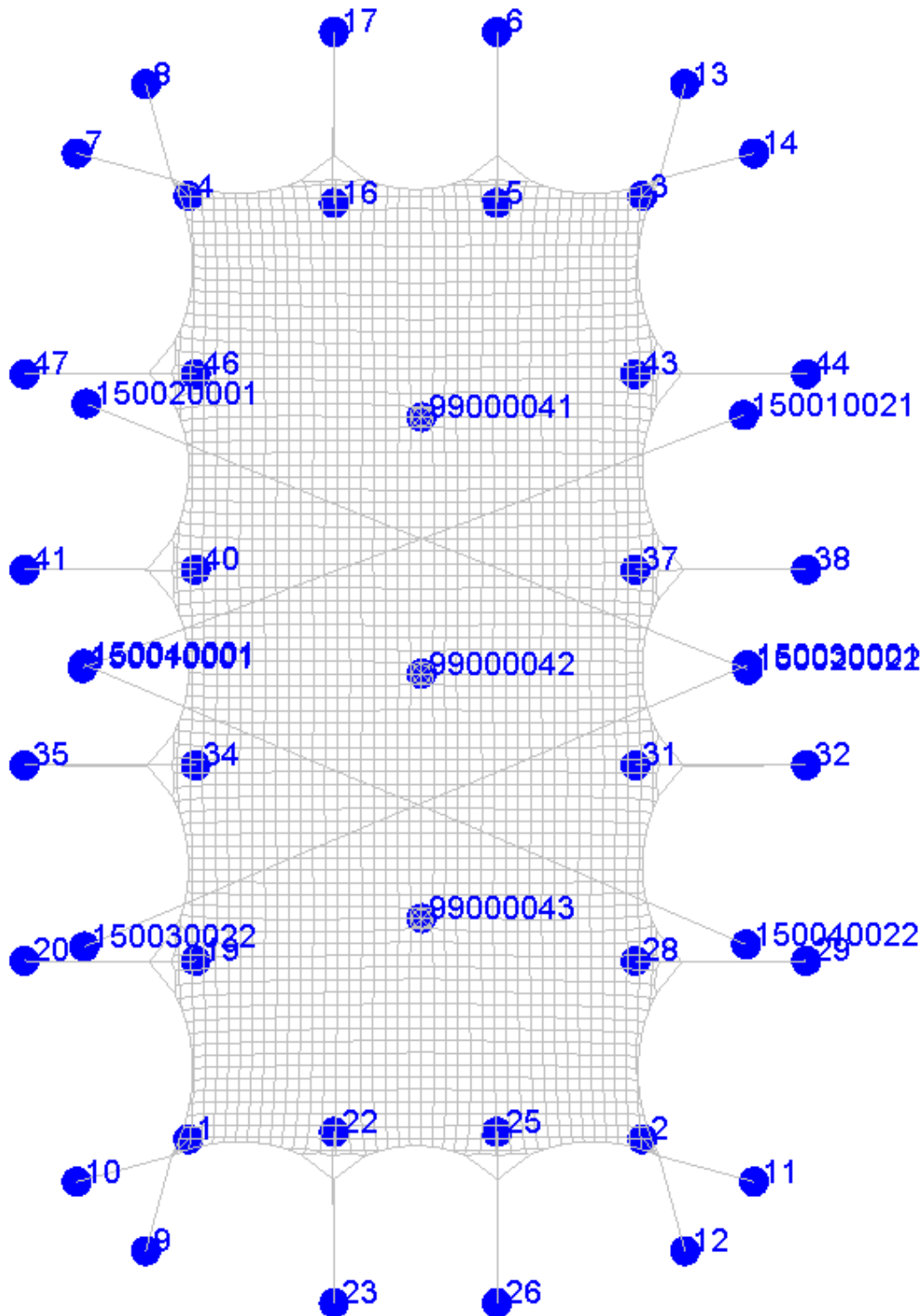
Annex I.2. Reaction forces

Reaction forces are shown in kilonewtons [kN] without any safety factor (1.35, 1.5, 1.2). A load combination especially for anchors and ballast calculation for snow and wind combined is CO6a.

Att	Node	Koor X	Koor Y	CO1			CO2			CO3			CO4			CO5			CO6			CO6a		
				Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz
Tie	6	-3.00	7.00	-1.62	-0.01	-1.24	-2.30	0.00	-1.79	-2.82	0.02	-2.21	-4.07	-0.02	-3.05	-3.29	-0.04	-2.47	-3.54	0.04	-2.80	-3.60	0.04	-2.84
Tie	17	-3.00	3.50	-1.61	0.01	-1.23	-2.24	0.00	-1.74	-2.74	-0.02	-2.14	-4.14	0.02	-3.10	-3.37	0.03	-2.53	-3.44	-0.04	-2.72	-3.49	-0.04	-2.76
Tie	20	11.25	-3.00	-0.02	-1.71	-1.31	0.00	-2.25	-1.75	0.02	-2.72	-2.13	-0.02	-4.05	-3.02	-0.03	-3.52	-2.67	0.04	-3.32	-2.62	0.04	-3.37	-2.66
Tie	23	18.00	3.50	1.62	0.01	-1.23	2.24	-0.01	-1.75	2.75	-0.02	-2.15	4.18	0.02	-3.14	3.38	0.04	-2.53	3.46	-0.04	-2.73	3.52	-0.04	-2.78
Tie	26	18.00	7.00	1.62	-0.01	-1.24	2.31	0.01	-1.80	2.84	0.02	-2.22	4.12	-0.02	-3.09	3.29	-0.04	-2.47	3.57	0.04	-2.82	3.63	0.04	-2.87
Tie	29	11.25	13.50	-0.02	1.72	-1.32	0.00	2.26	-1.76	0.02	2.74	-2.14	-0.02	4.07	-3.04	-0.03	3.52	-2.68	0.04	3.36	-2.65	0.04	3.41	-2.69
Tie	32	7.50	13.50	0.00	1.74	-1.37	0.00	2.41	-1.92	0.00	3.00	-2.38	0.00	3.59	-2.75	0.00	3.78	-3.00	0.00	3.72	-2.97	0.00	3.77	-3.02
Tie	35	7.50	-3.00	0.00	-1.73	-1.37	0.00	-2.39	-1.90	0.00	-2.97	-2.36	0.00	-3.60	-2.74	0.00	-3.78	-2.99	0.00	-3.67	-2.94	0.00	-3.72	-2.98
Tie	38	3.75	13.50	0.02	1.71	-1.31	0.00	2.22	-1.73	-0.02	2.68	-2.09	0.02	4.16	-3.11	0.03	3.59	-2.73	-0.04	3.29	-2.59	-0.04	3.33	-2.63
Tie	41	3.75	-3.00	0.02	-1.71	-1.30	0.00	-2.21	-1.72	-0.02	-2.66	-2.08	0.02	-4.16	-3.10	0.03	-3.59	-2.72	-0.03	-3.26	-2.57	-0.04	-3.30	-2.61
Tie	7	-0.52	-1.93	-0.43	-1.58	-1.63	-0.53	-1.94	-2.00	-0.61	-2.24	-2.30	-0.76	-2.76	-2.84	-0.65	-2.39	-2.46	-0.71	-2.59	-2.67	-0.72	-2.62	-2.70
Tie	8	-1.93	-0.52	-1.52	-0.41	-1.56	-1.87	-0.51	-1.93	-2.21	-0.60	-2.27	-2.89	-0.79	-2.97	-2.40	-0.66	-2.47	-2.59	-0.71	-2.66	-2.62	-0.72	-2.69
Tie	9	16.93	-0.52	1.52	-0.41	-1.56	1.88	-0.51	-1.94	2.21	-0.60	-2.28	2.89	-0.79	-2.97	2.40	-0.66	-2.48	2.60	-0.71	-2.68	2.63	-0.72	-2.71
Tie	10	15.52	-1.93	0.43	-1.58	-1.63	0.53	-1.95	-2.01	0.61	-2.24	-2.31	0.77	-2.78	-2.86	0.65	-2.39	-2.46	0.71	-2.60	-2.68	0.72	-2.63	-2.70
Tie	11	15.52	12.43	0.43	1.58	-1.63	0.53	1.95	-2.01	0.61	2.24	-2.31	0.76	2.78	-2.86	0.66	2.40	-2.47	0.71	2.61	-2.68	0.72	2.63	-2.71
Tie	12	16.93	11.02	1.52	0.41	-1.56	1.88	0.51	-1.94	2.21	0.60	-2.28	2.88	0.79	-2.96	2.40	0.66	-2.48	2.61	0.71	-2.68	2.64	0.72	-2.71
Tie	13	-1.93	11.02	-1.52	0.41	-1.57	-1.88	0.51	-1.94	-2.21	0.60	-2.28	-2.88	0.79	-2.96	-2.40	0.66	-2.47	-2.59	0.71	-2.67	-2.62	0.72	-2.70
Tie	14	-0.52	12.43	-0.43	1.58	-1.63	-0.53	1.94	-2.00	-0.61	2.24	-2.31	-0.76	2.76	-2.84	-0.66	2.39	-2.47	-0.71	2.60	-2.67	-0.72	2.63	-2.70
Pole	1	14.65	0.35	0.48	-0.48	2.81	0.63	-0.63	3.73	0.75	-0.75	4.44	0.78	-0.78	4.68	0.63	-0.64	3.78	0.90	-0.90	5.38	0.91	-0.91	5.45
Pole	2	14.65	10.15	0.48	0.48	2.80	0.63	0.63	3.72	0.75	0.75	4.43	0.77	0.78	4.65	0.63	0.63	3.77	0.90	0.90	5.37	0.91	0.91	5.45
Pole	3	0.35	10.15	-0.48	0.48	2.81	-0.63	0.63	3.73	-0.75	0.75	4.44	-0.78	0.78	4.66	-0.63	0.63	3.78	-0.90	0.90	5.37	-0.91	0.91	5.44
Pole	4	0.35	0.35	-0.48	-0.48	2.81	-0.63	-0.63	3.73	-0.75	-0.75	4.44	-0.77	-0.78	4.65	-0.64	-0.64	3.78	-0.90	-0.90	5.37	-0.91	-0.91	5.45
Pole	5	0.50	7.00	-0.23	0.02	1.39	-0.45	0.02	2.73	-0.58	0.01	3.57	-0.12	0.01	0.78	-0.14	0.01	0.85	-0.80	-0.01	4.96	-0.82	-0.01	5.07
Pole	16	0.50	3.50	-0.23	-0.02	1.39	-0.44	-0.01	2.71	-0.58	0.00	3.55	-0.13	-0.01	0.79	-0.14	-0.01	0.86	-0.80	0.01	4.93	-0.82	0.02	5.04
Pole	19	11.25	0.50	0.02	-0.22	1.37	0.02	-0.43	2.62	0.01	-0.56	3.42	0.06	-0.34	2.10	0.01	-0.13	0.78	-0.02	-0.76	4.70	-0.02	-0.78	4.81
Pole	22	14.50	3.50	0.22	-0.02	1.38	0.44	-0.01	2.67	0.57	0.00	3.49	0.13	-0.01	0.81	0.14	-0.01	0.87	0.78	0.01	4.85	0.80	0.01	4.96
Pole	25	14.50	7.00	0.22	0.02	1.38	0.44	0.02	2.69	0.57	0.01	3.51	0.13	0.01	0.80	0.14	0.01	0.86	0.79	-0.01	4.88	0.81	-0.01	4.99
Pole	28	11.25	10.00	0.02	0.22	1.36	0.02	0.42	2.58	0.01	0.55	3.36	0.06	0.35	2.13	0.01	0.13	0.79	-0.02	0.75	4.62	-0.02	0.76	4.72
Pole	31	7.50	10.00	0.00	0.20	1.21	0.00	0.40	2.46	0.00	0.54	3.30	0.00	0.40	2.46	0.00	0.09	0.59	0.00	0.74	4.59	0.00	0.76	4.69
Pole	34	7.50	0.50	0.00	-0.20	1.23	0.00	-0.41	2.52	0.00	-0.55	3.39	0.00	-0.39	2.40	0.00	-0.09	0.57	0.00	-0.76	4.72	0.00	-0.78	4.83
Pole	37	3.75	10.00	-0.02	0.22	1.36	-0.02	0.42	2.58	-0.01	0.55	3.36	-0.06	0.35	2.15	-0.01	0.13	0.80	0.02	0.75	4.61	0.02	0.77	4.72
Pole	40	3.75	0.50	-0.02	-0.22	1.37	-0.02	-0.43	2.62	-0.01	-0.56	3.42	-0.07	-0.36	2.19	-0.01	-0.13	0.79	0.02	-0.76	4.70	0.02	-0.78	4.80
Pole	99000035	5.63	5.38	-0.01	0.00	1.77	-0.01	-0.01	5.00	0.00	-0.02	6.53	0.00	0.00	-0.01	0.00	0.00	0.04	0.01	-0.05	9.82	0.02	-0.05	10.09
Pole	99000036	9.63	5.38	0.01	0.00	1.77	0.00	-0.01	4.95	-0.03	-0.02	6.46	0.00	0.00	-0.02	0.00	0.00	0.05	-0.08	-0.05	9.69	-0.09	-0.05	9.96
Stormbelt	150010001	5.80	12.46										-1.00	4.25	-3.91									
Stormbelt	150010022	9.17	-1.97										0.96	-4.26	-3.91									
Stormbelt	150020001	9.19	12.46										0.99	4.24	-3.90									
Stormbelt	150020022	5.75	-1.95										-1.01	-4.24	-3.90									
Stormbelt	150030001	2.55	-1.95										0.00	-4.54	-3.95									
Stormbelt	150030021	2.50	12.44										-0.02	4.54	-3.95									
Stormbelt	150040001	12.50	-1.94										0.01	-4.48	-3.91									
Stormbelt	150040021	12.46	12.45										-0.01	4.48	-3.91									

Annex J. Reaction forces model 10x20m

Annex J.1. Point numbers



Annex J.2. Reaction forces

Reaction forces are shown in kilonewtons [kN] without any safety factor (1.35, 1.5, 1.2). A load combination especially for anchors and ballast calculation for snow and wind combined is CO6a.

Att	Node	Koor X Koor Y		CO1			CO2			CO3			CO4			CO5			CO6			CO6a		
				Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz
275	6	-8.00	7.17	-1.57	-0.01	-1.22	-2.32	0.02	-1.83	-2.92	0.04	-2.30	-4.35	-0.03	-3.32	-2.81	-0.03	-2.14	-3.70	0.06	-2.94	-3.76	0.07	-2.99
275	17	-8.00	3.83	-1.56	0.01	-1.21	-2.24	-0.02	-1.77	-2.81	-0.04	-2.22	-4.44	0.02	-3.38	-2.91	0.03	-2.21	-3.56	-0.06	-2.83	-3.61	-0.07	-2.88
275	20	11.00	-2.50	-0.02	-1.83	-1.41	0.00	-2.46	-1.94	0.03	-3.01	-2.38	-0.04	-4.56	-3.43	-0.01	-3.65	-2.78	0.05	-3.66	-2.92	0.05	-3.71	-2.97
275	23	18.00	3.83	1.56	0.01	-1.21	2.24	-0.02	-1.77	2.81	-0.04	-2.22	4.42	0.02	-3.37	2.91	0.03	-2.21	3.56	-0.06	-2.83	3.61	-0.07	-2.88
275	26	18.00	7.17	1.57	-0.01	-1.22	2.32	0.02	-1.82	2.92	0.04	-2.30	4.33	-0.03	-3.31	2.82	-0.03	-2.15	3.70	0.06	-2.94	3.76	0.06	-2.99
275	29	11.00	13.50	-0.02	1.83	-1.41	0.01	2.48	-1.95	0.03	3.03	-2.40	-0.04	4.56	-3.44	-0.01	3.64	-2.79	0.06	3.71	-2.96	0.06	3.76	-3.00
275	32	7.00	13.50	0.00	1.81	-1.44	0.01	2.32	-1.86	0.02	2.88	-2.32	0.01	3.81	-2.97	0.00	4.41	-3.53	0.03	3.58	-2.90	0.03	3.63	-2.94
275	35	7.00	-2.50	0.00	-1.80	-1.43	0.01	-2.30	-1.85	0.02	-2.86	-2.30	0.01	-3.79	-2.94	0.00	-4.40	-3.51	0.03	-3.52	-2.86	0.03	-3.58	-2.90
275	38	3.00	13.50	0.00	1.80	-1.43	-0.01	2.26	-1.82	-0.02	2.82	-2.27	-0.01	3.86	-3.00	0.00	4.47	-3.57	-0.03	3.49	-2.83	-0.03	3.54	-2.87
275	41	3.00	-2.50	0.00	-1.79	-1.43	-0.01	-2.25	-1.81	-0.02	-2.79	-2.25	-0.01	-3.84	-2.97	0.00	-4.46	-3.55	-0.03	-3.45	-2.80	-0.03	-3.50	-2.84
275	44	-1.00	13.50	0.02	1.83	-1.41	-0.01	2.48	-1.96	-0.03	3.03	-2.40	0.04	4.59	-3.47	0.01	3.64	-2.79	-0.05	3.71	-2.96	-0.06	3.77	-3.01
275	47	-1.00	-2.50	0.02	-1.83	-1.41	0.00	-2.46	-1.94	-0.03	-3.01	-2.38	0.04	-4.61	-3.47	0.01	-3.65	-2.78	-0.05	-3.67	-2.93	-0.05	-3.72	-2.97
600	7	-5.52	-1.43	-0.43	-1.59	-1.64	-0.55	-2.01	-2.07	-0.64	-2.35	-2.42	-0.82	-2.97	-3.06	-0.62	-2.25	-2.31	-0.75	-2.73	-2.81	-0.75	-2.76	-2.84
600	8	-6.93	-0.02	-1.39	-0.38	-1.44	-1.71	-0.47	-1.76	-2.05	-0.56	-2.11	-3.03	-0.83	-3.11	-2.35	-0.64	-2.42	-2.43	-0.67	-2.50	-2.46	-0.68	-2.53
600	9	16.93	-0.02	1.39	-0.38	-1.44	1.71	-0.47	-1.76	2.05	-0.56	-2.11	3.03	-0.83	-3.12	2.35	-0.64	-2.42	2.43	-0.67	-2.50	2.46	-0.68	-2.53
600	10	15.52	-1.43	0.43	-1.59	-1.64	0.55	-2.01	-2.07	0.64	-2.35	-2.42	0.81	-2.96	-3.04	0.62	-2.25	-2.32	0.75	-2.73	-2.81	0.75	-2.76	-2.84
600	11	15.52	12.43	0.43	1.59	-1.64	0.55	2.02	-2.08	0.64	2.36	-2.43	0.82	2.97	-3.05	0.62	2.26	-2.32	0.75	2.75	-2.82	0.76	2.77	-2.85
600	12	16.93	11.02	1.39	0.38	-1.44	1.71	0.47	-1.76	2.05	0.56	-2.11	3.03	0.83	-3.12	2.35	0.64	-2.42	2.43	0.67	-2.51	2.46	0.68	-2.54
600	13	-6.93	11.02	-1.39	0.38	-1.44	-1.71	0.47	-1.77	-2.05	0.56	-2.11	-3.01	0.83	-3.09	-2.35	0.64	-2.42	-2.44	0.67	-2.51	-2.47	0.68	-2.54
600	14	-5.52	12.43	-0.43	1.59	-1.64	-0.55	2.01	-2.07	-0.64	2.35	-2.42	-0.82	2.98	-3.07	-0.62	2.25	-2.32	-0.75	2.74	-2.82	-0.76	2.77	-2.85
48160	1	14.65	0.85	0.46	-0.45	2.67	0.61	-0.60	3.58	0.73	-0.73	4.33	0.70	-0.70	4.22	0.60	-0.60	3.54	0.89	-0.88	5.26	0.90	-0.89	5.33
48160	2	14.65	10.15	0.46	0.45	2.66	0.61	0.60	3.57	0.73	0.72	4.32	0.70	0.70	4.21	0.59	0.60	3.54	0.89	0.88	5.26	0.90	0.89	5.33
48160	3	-4.65	10.15	-0.46	0.45	2.66	-0.61	0.60	3.57	-0.73	0.72	4.32	-0.70	0.70	4.20	-0.59	0.60	3.54	-0.89	0.87	5.25	-0.90	0.89	5.33
48160	4	-4.65	0.85	-0.46	-0.45	2.67	-0.61	-0.60	3.58	-0.74	-0.73	4.33	-0.70	-0.70	4.22	-0.60	-0.60	3.54	-0.89	-0.88	5.26	-0.90	-0.89	5.33
48160	5	-4.50	7.17	-0.22	0.01	1.35	-0.42	0.00	2.60	-0.56	-0.01	3.42	-0.06	0.00	0.40	-0.13	0.01	0.80	-0.76	-0.03	4.71	-0.78	-0.03	4.81
48160	16	-4.50	3.83	-0.22	-0.01	1.35	-0.42	0.00	2.57	-0.55	0.01	3.39	-0.06	0.00	0.41	-0.13	-0.01	0.81	-0.76	0.04	4.67	-0.77	0.04	4.78
48160	19	11.00	1.00	0.02	-0.22	1.38	0.01	-0.44	2.68	-0.01	-0.57	3.53	-0.03	-0.24	1.49	0.00	-0.12	0.75	-0.04	-0.78	4.83	-0.04	-0.80	4.93
48160	22	14.50	3.83	0.22	-0.01	1.35	0.42	0.00	2.58	0.55	0.01	3.39	0.06	-0.01	0.42	0.13	-0.01	0.81	0.76	0.04	4.67	0.77	0.04	4.78
48160	25	14.50	7.17	0.22	0.01	1.35	0.42	0.00	2.59	0.56	-0.01	3.41	0.06	0.00	0.41	0.13	0.01	0.80	0.76	-0.04	4.70	0.78	-0.04	4.81
48160	28	11.00	10.00	0.02	0.22	1.37	0.01	0.43	2.62	-0.01	0.56	3.45	-0.03	0.24	1.47	0.00	0.12	0.76	-0.04	0.76	4.72	-0.04	0.78	4.83
48160	31	7.00	10.00	0.00	0.19	1.19	0.00	0.38	2.35	-0.01	0.52	3.19	-0.05	0.34	2.06	0.00	0.10	0.65	-0.02	0.72	4.45	-0.03	0.74	4.56
48160	34	7.00	1.00	0.00	-0.20	1.21	0.00	-0.39	2.41	-0.01	-0.53	3.28	-0.04	-0.32	1.98	0.00	-0.10	0.63	-0.02	-0.74	4.58	-0.03	-0.76	4.69
48160	37	3.00	10.00	0.00	0.19	1.19	0.00	0.38	2.33	0.01	0.52	3.17	0.04	0.36	2.21	0.00	0.10	0.66	0.02	0.72	4.43	0.03	0.73	4.54
48160	40	3.00	1.00	0.00	-0.20	1.21	0.00	-0.39	2.39	0.01	-0.53	3.25	0.04	-0.34	2.09	0.00	-0.10	0.64	0.02	-0.74	4.56	0.03	-0.76	4.67
48160	43	-1.00	10.00	-0.02	0.22	1.37	-0.01	0.43	2.63	0.01	0.56	3.46	0.02	0.19	1.18	0.00	0.12	0.77	0.04	0.76	4.73	0.04	0.78	4.84
48160	46	-1.00	1.00	-0.02	-0.23	1.38	-0.01	-0.44	2.69	0.01	-0.57	3.54	0.02	-0.21	1.31	0.00	-0.12	0.75	0.04	-0.78	4.84	0.04	-0.80	4.95
48160	99000041	-0.12	5.63	-0.01	0.00	1.88	0.06	-0.01	5.19	0.15	-0.02	6.78	0.00	0.00	-0.02	0.00	0.00	0.07	0.32	-0.05	10.10	0.34	-0.05	10.38
48160	99000042	5.13	5.63	0.00	0.00	1.58	-0.01	-0.01	4.61	-0.01	-0.02	6.09	0.00	0.00	-0.01	0.00	0.00	0.05	-0.03	-0.05	9.26	-0.03	-0.06	9.52
48160	99000043	10.13	5.63	0.01	0.00	1.86	-0.05	-0.01	5.11	-0.14	-0.02	6.67	0.00	0.00	0.01	0.00	0.00	0.06	-0.30	-0.05	9.93	-0.32	-0.05	10.20
5900001	150010001	5.01	-1.32										1.52	-4.17	-3.82									
5900001	150010021	-0.16	12.22										-1.51	4.10	-3.90									
5900002	150020001	-0.40	-1.24										-1.59	-4.10	-3.94									
5900002	150020022	5.05	12.30										1.61	4.18	-3.85									
5900003	150030001	4.95	12.30										-1.68	4.17	-3.87									
5900003	150030022	10.71	-1.28										1.67	-4.08	-3.96									
5900004	150040001	4.93	-1.29										-1.66	-4.14	-3.85									
5900004	150040022	10.67	12.27										1.67	4.05	-3.94									